Engineering Protection in Multi-Mode Superconducting Circuits

by

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Superconducting circuits are a promising technology towards realizing a large scale quantum computer. These circuits can protect quantum information against by making the encoded states insensitive to the operators by which noise couples to the circuit. However, single-mode circuits, such as the transmon or fluxonium, face a choice. They cannot be simultaneously protected against both relaxation and dephasing when built using only the standard components of a Josephson junction, capacitor, and inductor. Simultaneous protection requires multi-mode circuits, which are difficult to quantize, simulate, search over, and control. This thesis addresses each of these in turn. First, we give a general procedure to go from a circuit diagram to a multimode Hamiltonian, precisely characterizing the residual coordinate freedoms. Second, we enumerate all unique circuits built from those standard elements (Josephson junction, capacitor, and inductor). Using a consistent choice of coordinates, we sort them by the structure of their quantized Hamiltonian and explicitly model all unique three node circuits as novel qubits. Third, we analyze control of the soft $0-\pi$qubit. We propose the first two qubit gates for the system: a geometric-phase-based CZ and a CNOT that transfers population via conditional splitting on the single qubit gate λ system. Fourth, we consider approximate realizations of a $\cos 2\phi$ system. We derive a symmetry selection rule that determines, from the circuit alone, whether a particular realization forbids first-order flux noise that precludes simultaneous protection. We apply the test to the $0-\pi$, KITE, Harmonium, and rhombus circuits and find that the $0-\pi$and Harmonium pass the test, while the KITE and rhombus do not. Finally, we present GPU-accelerated matrix-free methods for simulating larger multimode circuits. We close with suggestions for future work.

Dedication

To my friends – thank you for listening to me complain and celebrating my triumphs. You've helped me laugh and have fun and made the hard work of a PhD a lot less hard.

To my family – thank you for all you do for me. Dad, Chava, and Fred, thank you for always being so interested in my work and progress these last few years. As you know this process can be a slog, and your encouragement from start to finish has helped me more than you know. Mom, thank you picking up my random phone calls and listening to my rants (I think I got that from you). Sam, Marina, Gaia, and Pietro, thank you for being my family away from home. Stepping into your house lets me take a break from my world of textbooks and pick up a picture book instead. Nathan, thank you for always being willing to lend an ear and for having a unique penchant for making me laugh. Our jokes together are one of life's true joys, and I hope now that I have a PhD you will be willing to mention me at Shul (this is a joke). Hannah & Isaac thank you for always bringing such positive energy and for helping me get out of my way sometimes. My appreciation and love for you is only matched by the amount of fun we have together.

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Contents

Chapter

1 Introduction & Motivation **1**

1.1 Superconducting Qubits 1

1.2 Protected Qubits 3

1.2.1 Decay and Dephasing 3

1.2.2 Single-Mode Qubits 5

1.2.3 The $\cos 2\phi$ Qubit 6

1.3 This Thesis 7

2 Modeling of Multimode Superconducting Circuits: Circuit Quantization, Numerical Modeling, and Mode Preserving Transformations **9**

2.1 Introduction 9

2.2 Multimode Circuit Quantization 10

2.2.1 Constructing the Circuit Lagrangian 10

2.2.2 Legendre Transform, Coordinate Transformations, Mode Types, and Quantization 14

2.2.3 Two Subtleties 19

2.3 Numerical Modeling of Circuit QED Systems 23

2.4 Mode-Preserving Coordinate Transformations 28

2.4.1 Parameter-Independent Symbolic Search 31

2.4.2 Numerical Search . . . 34
2.5 Conclusion . . . 39

3 Circuit Enumeration **41**
3.1 Introduction . . . 41
3.2 Historical Perspective & Related work . . . 43
3.3 Circuit Enumeration Algorithm . . . 46
3.3.1 Combinatorial Enumeration . . . 46
3.3.2 Reduction of Equivalent Configurations . . . 48
3.4 Categorization by Hamiltonian Class . . . 50
3.5 Enumeration Results . . . 55
3.5.1 Inverse Lookup: Find all Circuits in a Class . . . 58
3.5.2 Identification of Delocalized Coordinates . . . 61
3.6 Searching For New Qubits . . . 63
3.6.1 Two Node Circuits . . . 65
3.6.2 Three Node Circuits . . . 67
3.7 Conclusion . . . 71

4 Two-Qubit Gates for the Soft $0-\pi$ Qubit **74**
4.1 Introduction . . . 74
4.2 Basics of the $0-\pi$ Qubit . . . 76
4.3 Single-Qubit X Gates . . . 80
4.3.1 Driven Hamiltonian . . . 80
4.3.2 X gate by driving n_ϕ . . . 81
4.3.3 X gate by driving n_θ . . . 82
4.3.4 Single-Qubit Noise Model . . . 85
4.3.5 Noisy gate fidelity vs. gate time . . . 87

4.4 Coupling for Two $0-\pi$ Qubits . . . 88
4.5 Microwave-activated CZ gate . . . 92
4.5.1 Driven Hamiltonian . . . 92
4.5.2 Gate Concept and Transition Landscape . . . 93
4.6 CZ Gate Performance with Decoherence . . . 97
4.6.1 Two-Qubit Noise Model . . . 97
4.6.2 Optimized Gate . . . 98
4.7 Conclusion . . . 100

References **102**

Appendix

A Code Listings **117**

B Circuit Quantization and Modeling **121**

B.1 Assigning External Flux Using a Spanning Tree . . . 121
B.2 Converting Between Physical and Frequency Units for Circuit Parameters . . . 122
B.3 Product States and Mean-Field Hamiltonians . . . 123
B.3.1 The Hamiltonian in a transformed frame . . . 123
B.3.2 Optimization over S with a product state anzatz . . . 125
B.3.3 Hierarchical diagonalization with mean-field leaves . . . 128
B.4 Notation . . . 130

C Circuit Enumeration **132**

C.1 Subtleties in the Enumeration . . . 132
C.1.1 Ground Placement and Circuit Equivalence . . . 132
C.1.2 Ambiguities Related to SQUIDs and Linearization . . . 133

C.1.3 Charge and Flux Bias . . . 134
C.1.4 Circuit Elements Not Considered . . . 135
C.1.5 Eliminating Single-Node Capacitive and Inductive Islands . . . 136
C.2 Variable Transformation Details . . . 138
C.2.1 Choice of "Reference" Coordinates . . . 138
C.2.2 Generating a "Reference" Transformation . . . 140
C.2.3 Limitations . . . 146
C.3 Pseudocode for Constructing a Nonisomorphic Set of Colored Graphs . . . 147
C.4 Additional Enumeration Results by Number of Junctions, Loops, and Compact Variables 148
C.5 Inductive Skeletons with Delocalized Modes . . . 150
C.5.1 Conditions for Delocalized Modes . . . 152
C.5.2 Catalog and Construction of Skeletons . . . 155
C.6 Modeling Implementation Details . . . 156
C.7 Converting Between Physical and Frequency Units for Circuit Parameters . . . 158
C.8 Gate Time Estimates . . . 158
C.8.1 Direct Transition . . . 159
C.8.2 Raman-Type Gate . . . 162
C.9 Comparing Model with Published Results . . . 163
C.10 Optimization Results by Number of Junctions, Loops, and Compact Variables . . . 168
C.11 Circuit Quantization Details for Top Performing Circuit . . . 168

D Two-Qubit Gates for the Soft $0-\pi$ Qubit **173**

D.1 Single-Qubit Simulation Details . . . 173
D.1.1 Noiseless simulation of single-qubit gates . . . 173
D.1.2 Model reduction for noiseless pulse optimization . . . 174
D.1.3 Single-qubit pulse optimization . . . 175
D.1.4 Noisy simulation of single-qubit gates . . . 177

D.2 Comparison to Experimental Single-Qubit Gate Performance 178
D.3 Quantization of two coupled $0-\pi$ qubits 180
D.3.1 Pure coupling scheme: $n_{\theta 1} n_{\theta 2}$ 181
D.3.2 Mixed coupling scheme: $n_{\theta 1} n_{\theta 2} - n_{\phi 1} n_{\phi 2}$ 186
D.4 CNOT gate via a two-tone Raman pulse 189
D.4.1 Driven Hamiltonian 190
D.4.2 Transition selection and gate principle 190
D.4.3 Gate operation and noiseless fidelity 192
D.4.4 Single-qubit Z corrections for the CNOT gate 194
D.4.5 Level Crossing in Coupled Circuits 195
D.5 Two-Qubit Gate Simulation Details 196
D.5.1 Noiseless simulation of two-qubit gates 196
D.5.2 Noisy simulation of two-qubit gates 198
D.6 Sensitivity Analysis of Circuit Parameters 199

Tables

Table

2.1 Passive branch types and their branch-flux dependent energy contributions $U_k(\Phi_{b,k})$. Capacitance contributes to the kinetic energy $T_k = \frac{1}{2}C_k\dot{\Phi}_{b,k}^2$ for every branch type. Ideal voltage are treated separately and are fixed at $\dot{\Phi}_{b,k} = V_k(t)$. Energies are in SI units, with $\Phi_0 = h/(2e)$ being the superconducting flux quantum. 11

2.2 **Non-standard symbols.** N is the number of internal nodes, N_b the number of branches, N_ℓ the number of independent loops, and N_V the number of voltage sources. N_L and N_J count the inductive and Josephson branches. All symbols are collected in Table B.1. 15

2.3 Central finite-difference stencil of order $p = 2k$ for the second derivative and its weights w_j, with error $\mathcal{O}(\Delta\phi^p)$ [66]. The weights are symmetric, $w_{-j} = w_j$. 25

3.1 Number of circuits and Hamiltonian classes for circuits with two to five nodes. The first column gives the number of nodes. The second gives the naive circuit count, 7^{N_E}, summed over all base graphs for that node number (see Figure 3.3). For three-node circuits, this gives $7^3 + 7^2 = 392$ from the two base graphs. The third column gives the number of unique circuits containing at least one Josephson junction. Because the naive count includes many equivalent circuits, the reduction from column 2 to 3 is large. The fourth column gives the number of Hamiltonian classes, namely distinct combinations of mode types and inter-mode couplings identified from the unique circuits. 56

3.2 Hamiltonian classes (top) and corresponding circuits (bottom) up to 3 nodes in size. Compact variables are denoted by n, θ, while extended and harmonic variables are denoted by q, φ. Grayed-out circuit elements may be included or excluded without changing the class. Because each Josephson junction includes a small intrinsic capacitance, most optional elements correspond to capacitors placed in parallel with junctions. Optional inductors, such as those in classes $3-5$, $3-6$, $3-11$, and $3-12$, are present when the circuit has no compact modes and already contains a series inductor across the specified branch. Likewise, the optional capacitors in classes $3-2$, $3-4$, and $3-16$ occur when that branch already contains a series capacitor. 57

3.3 Allowed ranges for circuit parameters. The junction capacitance C_J is taken to be fixed and matches the smallest capacitance allowed. For details on how to convert between physical and frequency units see Section C.7. 65

3.4 Optimized number of gates for all two-node circuits. The maximum value, $N_{\text{gate}} = 187,000 \approx 1.9 \times 10^5$, for the optimized fluxonium serves as a benchmark for larger circuits. 65

3.5 Four highest-scoring three-node circuits for each base graph. The "Base Graph" column shows the circuit topology, and the "Circuit" column lists the elements on each edge, separated by commas. Note that J edges include an intrinsic capacitance. 68

3.6 Reoptimized versions of the top four circuits from Table 3.5, including inductor parasitic capacitance. In addition to the parameter variations considered in the original Monte Carlo analysis, we individually vary inductor capacitances, junction capacitances, and ground capacitances with the same Gaussian sampling as before. The inductor, junction, and ground capacitance distributions are centered on $C_L = 0.6$ fF, $C_J = 1.9$ fF, and $C_g = 1$ fF, respectively. 70

4.1 Soft $0-\pi$ qubit parameters. 78

4.2 Optimized X gate parameters. 87

4.3 Capacitive connections coupling $0-\pi$ modes. 92

4.4 Optimized CZ pulse parameters. 97

B.1 **Notation of Section 2.2.** The circuit has N internal nodes (excluding ground and voltage-source nodes), N_b branches, N_ℓ independent loops, and N_V voltage sources. N_L and N_J count the inductive and Josephson branches, N_c the compact coordinates, and N_d the dynamical coordinates. The last column gives the behavior under $\Phi = Z\tilde{\Phi}$ if applicable. 131

C.1 Number of enumerated circuits as a function of node count for circuits built from `C`, `L`, `J`, and `Q`. Column 1 gives the number of nodes. Column 2 gives the naive count, 15^{N_E}, summed over all base graphs for that node count (see Figure 3.3). Column 3 gives the number of unique nonlinear circuits containing at least one Josephson junction or one QPS element. 136

C.2 Number of unique circuits and Hamiltonian classes, grouped by node count and number of Josephson junctions. 149

C.3 Number of unique circuits and Hamiltonian classes for circuits with the specified numbers of nodes and compact variables. As node count increases, compact variables appear in fewer unique circuits but in a larger fraction of Hamiltonian classes. 150

C.4 Number of unique circuits and Hamiltonian classes by number of flux loops. Circuits with different numbers of flux loops can belong to the same Hamiltonian class. 151

C.5 Transmon: estimates of device parameters shown in Figure C.7, extracted from published results. If incomplete parameters are given, we infer missing values by approximating $\eta \approx -E_C$ and $f_{01} \approx \sqrt{8E_C E_J} - E_C$. For tunable qubits, if no idle frequency is given, we use the middle of the given range. N_{gate} is calculated by $N_{\text{gate}} = T_2/T_{\text{gate}}$. If no T_2 is provided, we take $1/T_2 = 1/2T_1$ – which implies (incorrectly) that T_ϕ is infinite. For α, we choose device D6, for β we choose device q0, and for γ we use the mean device parameters. 167

C.6 Fluxonium: device parameters shown in Figure C.7, extracted from published results. N_{gate} is calculated by $N_{\text{gate}} = T_2/T_{\text{gate}}$. $T_{\text{gate,est}}$ represents our estimated gate time, calculated using the model described in Section C.8. For ϵ, we choose device I, for η we choose device A, for θ we choose device 3, and for κ we choose device S. 167

C.7 Optimized number of gates for all three-node circuits, split by number of junctions N_{JJ}. Circuits with fewer junctions perform better than those with more junctions. . 168

C.8 Optimized number of gates for all three-node circuits, split by number of flux loops N_{loop}. Circuits with one or two loops score the highest. 169

C.9 Optimized number of gates for all three-node circuits, split by number of compact modes N_{compact}. Circuits with fewer compact modes perform better. 169

D.1 Optimized CNOT pulse parameters. 190

D.2 Sensitivity of the CZ gate to parameter variations. 199

Figures

Figure

1.1 Single-mode superconducting qubits and their protection 5

2.1 **Example circuit.** A grounded SQUID (junctions J_1, J_2) is coupled through C_c to a resonator (L_r, C_r), which is driven by the voltage source $V(t)$ through C_d. The resonator is treated as a single branch. External fluxes Φ_x, Φ_y, and Φ_z thread three loops, with the latter two broken by capacitors. 13

2.2 **Six mode types.** Top: Euler diagram of the mode bases, where each basis is a distinct region. C is contained in J because we assume every junction has positive intrinsic capacitance. Thus any vector in null(C) must also lie in null(ω_J^T). Note that regions may be empty for a particular circuit. Bottom: which terms of the Lagrangian $\mathcal{L}$ each mode enters. Columns of Z which define the specified mode type are spanned by the specified bases, built from the null spaces J, L, and C. Here $A + B$ is the subspace sum, $A \cap B$ the intersection, and $A^\perp$ the orthogonal complement. 18

2.3 **Mode types of the floating LCLC box.** (a) The circuit, with nodes 1–4. (b)–(e) The columns of Z in Equation (2.10), with the sign of each node's entry. Grey branches carry no flux difference in that column, so they contribute no energy. 19

2.4 **Charge basis of two transmons** (a) coupled by the capacitor C_c. (b) Unit cell in the the reduced node flux (φ_1, φ_2) (orange) and sum/difference (φ_+, φ_-) (blue) coordinates. Grey dots mark the centers of adjacent unit cells in the node coordinates. (c) Charge lattice, with the (n_1, n_2) basis containing all charge combinations, and the (n_+, n_-) (blue) containing only values for which $n_1 \pm n_2 = 0 \bmod 2$. 20

2.5 Numerical diagonalization of a fluxonium (top) with $E_J = 10$, $E_C = 0.5$, $E_L = 0.4$ and $\phi_{\text{ext}} = \pi$, and a transmon (bottom) with $E_J = 5$ and $E_C = 0.2$. (a,d) Potential, lowest eigenvalues and wavefunctions in the phase basis, from Listing A.1 (201 grid points) and Listing A.5 (160 grid points, stencil of order 8, $n_g = 0$). (b) Error of the lowest four eigenvalues in the Fock basis versus the number of levels, relative to a 140-level reference. (e) Error of the lowest three eigenvalues in the charge basis (Listing A.4) versus the charge cutoff n_{cut}, relative to $n_{\text{cut}} = 40$, at $n_g = 0.25$. (c,f) Largest eigenvalue error in the phase basis versus grid spacing, for finite-difference stencils of order 2 to 8 (Listing A.3), with $n_g = 0.25$ for the transmon. Dotted lines show $\Delta\phi^p$ for each order p. 26

2.6 S_{hc} **twists the compact boundary condition.** For the circuit of Figure 2.1, the transmon phase φ_1 is periodic and the resonator phase φ_2 is real, so the configuration space is a cylinder (a). The blue path makes one turn at fixed φ_2 and closes at $(0, 0)$. In the coordinates $\varphi_1 = \varphi_u$, $\varphi_2 = \varphi_v + \alpha\varphi_u$ (b), a turn at fixed φ_v (orange) ends at the distinct point $(2\pi, 0)$, which is $(0, 2\pi\alpha)$ on the cylinder. (c) The transmon's integer charge is $n_u - \alpha n_v$, so the allowed charges lie on tilted lines while n_v remains continuous. 30

2.7 **The Hamiltonian class depends on the cost.** Circuit (b) adds a capacitor C_{12} in parallel with the inductor between nodes 1 and 2 of circuit (a). Adapted from Ref. [71]. 33

2.8 **Mode types of the 0-π qubit** [32, 35, 36]. (a) The circuit, with nodes 1–4. The diagonals carrying C_1 and C_2 cross without connecting. (b)–(e) The columns of Z in Section 2.4.2, with the sign of each node's entry. Grey branches carry no flux difference in that column, so they contribute no energy. 36

2.9 **Product-state coordinates of the** $\cos(2\varphi)$ **KITE.** (a) The $\cos(2\varphi)$ KITE circuit [34, 41]. Only the shunted node carries capacitance to ground, so three dynamical modes remain: the compact phase φ across C_S, plus the extended sum and difference $\varphi_\Sigma, \varphi_\Delta$ of the two upper inductor phases. (b) Largest error of the four lowest levels of hierarchical diagonalization with m mean-field eigenstates per mode, one set per well, in three frames. We compare the published coordinates $(\varphi, \varphi_\Sigma, \varphi_\Delta)$ of Ref. [41], the normal modes, and the product-state optimum $S^\star$ of Equation (2.32). Details of the diagonalization are given in Section B.3. 37

3.1 Visual representations of circuit enumeration, synthesis, and equivalence. The aim of our work is to enumerate all unique nonlinear quantum circuits with up to five nodes. The results of our enumeration leverage circuit equivalence and can be used in circuit synthesis tasks to find new qubits and amplifiers. In the circuit synthesis example, we desire a low-pass filter with a certain roll-off and search for an LC circuit that has this property. The illustrated circuit equivalence is defined with respect to the impedance Z_i. 42

3.2 Example circuit enumeration and reduction for three-node circuits. **Enumeration:** (Step 1) For a given number of nodes, select a base graph. For $n = 3$ there are two options, and for this figure, we choose the Delta topology. (Step 2) Given the Delta topology, we allow each edge to be one of m components. In this example, the edge can be either a capacitor, inductor, or Josephson junction. In general, we also allow combinations of the three basic elements so the edges could be in the set {`C`, `J`, `L`, `CL`, `JL`, `JC`, `JCL`}. (Step 3) We enumerate all possible m-ary strings. In this example, there are 27 possible configurations. **Reduction:** Circuits that are eliminated at each step are grayed out. (Step 1) We remove any circuit that does not contain a `J`. (Step 2) We remove circuits that have linear components in series because there is an equivalent circuit with one fewer node. (Step 3) Of the remaining circuits, several are equivalent, so we perform a graph isomorphism check to remove equivalent circuits. (Step 4) Four unique nonlinear circuits remain out of the original 27. 47

3.3 The set of "base graphs," or non-isomorphic, connected, undirected graphs with the specified number of nodes. Each represents a unique circuit topology. All except for the fully connected five-node graph can be drawn as a planar graph. Acyclic graphs, or "trees," are highlighted in blue. 48

3.4 Two equivalent circuits (top), along with their component graph representations (bottom). Each circuit component and node in the circuit diagram is represented by a colored vertex. Different circuit components are assigned different colors (capacitor → blue, Josephson junction → purple). Each node in the circuit is assigned a color based on its degree (1 connection → pink, 3 connections → orange). An isomorphism exists from $G_1 \to G_2$ with vertex mapping $\{1 \to 5, 2 \to 4, 3 \to 3, 4 \to 2, 5 \to 1, 6 \to 6, 7 \to 7\}$. 50

3.5 Three circuits with the same Hamiltonian class under the coordinates $\varphi_a = \phi_2 - \phi_1$, $\varphi_b = \phi_2 - \phi_3$, and $\varphi_c = \phi_4 - \phi_3 + \phi_2 - \phi_1$. With $C = C_1C_2/(C_1 + C_2)$ and $L = L_1 + L_2$, they also have identical equations of motion. 53

3.6 Types of dynamical (top) and non-dynamical (bottom) modes present in superconducting circuits of pure capacitors, pure inductors, and Josephson junctions with intrinsic capacitance. Cartoons of potential energy are shown for each mode type, with a corresponding "particle" moving within the potential. Here and throughout we will denote harmonic and extended modes using q, φ and compact modes using n, θ. The compact mode has periodic boundary conditions, shown by the particle split between $\theta = -\pi$ and $\theta = \pi$. 54

3.7 Two circuits from classes (a) 3-11 and (b) 3-12 in Table 3.2 that have equivalent mode structures if a different choice of coordinates is used. 55

3.8 (Left) Topologically equivalent circuits to the Difluxmon circuit [141], up to trivial rotations and reflections. (Right) All circuits in the same Hamiltonian class as the original Difluxmon circuit. Gray elements can be added or removed without changing the Hamiltonian class and represent additional capacitances and inductances between nodes with existing capacitive or inductive paths. 59

3.9 (a) All circuits in the same Hamiltonian class as the original $0 - \pi$ circuit. Gray capacitors must be included or excluded in pairs across identical elements to preserve the required symmetry. For the second class, which contains the Bifluxon circuit [156], the series inductors may be excluded if the diagonal inductor is included. (b) Parameter regions accessible to four different capacitive layouts of the class containing the original $0 - \pi$ circuit, assuming junction charging energies of 5–10 GHz and capacitor charging energies of 0.05–5 GHz. Only the original $0 - \pi$ circuit (top left) achieves the small E_{C1} and large E_{C2} required for protection. The top right circuit has no dedicated capacitors, but both modes still have charging energy because we include the junction's intrinsic capacitance. The Bifluxon-class circuits, not depicted in (b), can achieve every combination of E_{C1} and E_{C2}, except for the protected small E_{C1} and large E_{C2} case. 60

3.10 Topological origins of localized and delocalized extended coordinates. (a) The three-node Josephson junction tree and (b–d) three possible single-branch inductive modifications, with each added branch shown in teal. (b) A loop-completing junction nonlinearly couples two compact coordinates. (c) An inductive path in parallel with a single junction converts one compact coordinate into an extended coordinate that can remain localized to that junction. (d) An inductor completing a mixed loop instead creates an extended coordinate that must be delocalized across both junctions. This is the inductive skeleton of Hamiltonian class $3-16$. 61

3.11 Catalog of irreducible inductive skeletons up to five nodes in size with at least one necessarily delocalized extended coordinate. No displayed junction is shunted by an inductor-only path, so all extended coordinates are delocalized (see Section C.5). The skeletons are separated by their numbers of compact and extended modes. There is one three-node skeleton, four four-node skeletons, and seventeen five-node skeletons. We omit skeletons containing shunted junctions, nodes connected only to inductors, or all-junction or all-inductor loops. We also omit circuits that become disconnected when one node is removed. Extra capacitors may create harmonic modes and couplings but do not modify the nonlinear coordinate structure. 61

3.12 Dependence of the first three energy levels on external flux (top), optimized fluxonium (left) and integer fluxonium (right), using parameters from [160] device D. The first three eigenstates are plotted for both circuits (bottom), with global phase calibrated to make the peak value of the $|0\rangle$ state real and positive. The optimized fluxonium has $E_C = 2.63$ GHz, $E_J = 2.85$ GHz, and $E_L = 0.51$ GHz. The integer fluxonium has $E_C = 1.46$ GHz, $E_J = 6.86$ GHz, and $E_L = 0.23$ GHz. Both circuits operate at zero external flux. 66

3.13 The top-performing circuit "J, L, CL". Panel (a) shows the circuit diagrams with parameter values. All junctions also include an intrinsic capacitance $E_{C_J} = 10$ GHz. (b) shows the flux dispersion about the optimized operating point. (c) shows the wavefunctions of the lowest three energy eigenstates, multiplied by a global phase so that the peak of the $|0\rangle$ state is positive and real. Green denotes positive amplitude and pink negative amplitude. The gray background shows contours of the potential energy landscape, with white for low energy and dark gray for high energy. In the optimized "J, L, CL" circuit, the wavefunctions appear fluxonium-like in one direction and harmonic-oscillator-like in the other. 68

3.14 (a) Proposed coupling scheme for driving gates on the top-performing circuit "J, L, CL." In the ideal case, the voltage source couples only to the harmonic mode, but capacitance to ground also induces coupling to the extended mode. (b) Magnitudes of the drive-Hamiltonian matrix elements H_d for the lowest five energy levels. Transitions near the qubit frequency (0-2 and 1-3) are forbidden, allowing fast gates with little leakage. (c) Fidelity versus gate time for an optimized monotone pulse with a Gaussian envelope under coherent dynamics. A fidelity of 99.9% is reached at about 16.4 ns, compared with 18.6 ns predicted by the gate-time model. (d) Dependence of the first and second excited-state energies on the external charge offset. Both energies are insensitive to this parameter. 69

4.1 Overview of the soft $0-\pi$ qubit. 76

4.2 Electrical connections for coupling to the $0-\pi$ modes. 79

4.3 Optimized X gate driving n_ϕ. 83

4.4 Optimized X gate driving n_θ. 84

4.5 Simulated X gate error versus gate time. 86

4.6 Two capacitively coupled $0-\pi$ qubits. 90

4.7 CZ gate implemented by driving $n_{\theta 2}$. 95

4.8 Population transfer during the CZ gate. 98

4.9 CZ gate performance. 99

B.1 **The two nested minimizations of the product-state search.** The inner loop converges a product state in a fixed frame: each mode in turn is replaced by the ground state of its mean-field Hamiltonian h_i, the full Hamiltonian averaged over the other modes, until E_{prod} stops changing, and the lowest of the solutions reached from the standard starts is kept. The outer loop minimizes the cost of Equation (B.12) over the frame S by BFGS. 126

C.1 Illustration of how the number and placement of ground nodes can induce circuit isomorphism. Circuit (a) has three nodes, while circuit (b) has four. The two circuits become equivalent when the indicated one node in (a) and two nodes in (b) are grounded. 133

C.2 Flux-tunable elements not included explicitly in our enumeration. 134

C.3 Examples of (a) a charge-bias circuit and (b) a flux-bias circuit, neither of which is included explicitly in the enumeration. 134

C.4 QPS elements and multiport/multiterminal devices. 135

C.5 Types of modes in superconducting circuits made of pure capacitors, pure inductors, and Josephson junctions with intrinsic capacitance. The columns of Z_1^m for a given mode type m are placed within the null spaces of the incidence matrices ω_J^T, ω_L^T, and ω_C^T. Light indicates a small capacitance or "mass" from only the intrinsic junction capacitance. 143

C.6 Transitions and leakage channels included in the gate-time estimate. (a) A weakly anharmonic qubit driven directly on the $|0\rangle \to |1\rangle$ transition, with leakage through $|1\rangle \to |2\rangle$. (b) A qubit with a nearly degenerate second excited state, again driven directly on $|0\rangle \to |1\rangle$, with leakage through $|0\rangle \to |2\rangle$. (c) A Raman-type transition $|0\rangle \to |2\rangle \to |1\rangle$, where population in $|2\rangle$ is treated as leakage. 159

C.7 Qualitative estimate of N_{gate} calculated for the `JC` (a-b) and `JCL` (c-d) circuits. The left column (a,c) uses the `SQcircuit` noise model, which uses a second-order approximation for pure dephasing, while the right column (b,d) uses the `scQubits` noise model, which uses a first-order approximation for pure dephasing. For the `JC` circuit, $n_g = 0.25$ is chosen to represent a worst-case when a DC offset is not included. For the `JCL` circuit, the standard $\varphi_{\text{ext}} = \pi$ is used, and E_C is fixed at 500 MHz. The black dot marks the maximum point, and the black contour shows the region that is at least 50% of the maximum value. The maximum value of N_{gate} is approximately 12,500 for the transmon (`JC`) and 54,500 for fluxonium (`JCL`). The streak with low N_{gate} for the `JC` circuit is due to a zero crossing of the anharmonicity. The bottom right represents the Transmon regime ($E_J/E_C \gg 1$), while the top left represents the Cooper Pair Box, or charge qubit, regime (small E_J/E_C). To gauge the discrepancies between our model and experimental data, we extracted the circuit parameters from various published qubit designs. Each letter corresponds to parameters from a published paper: transmon α–[28], β–[201], γ–[171], and δ–[202]; fluxonium ϵ–[170], ζ–[203], η–[204], θ–[205], ι–[18], and κ–[206]. Qubits from the literature fall near or within the best performing regions for both the transmon and fluxonium circuits. 164

C.8 Underlying coherence times and gate times estimates for the number of gates N_{gate} estimates in Figure C.7. Noise estimates from `SQcircuit` are shown for (a) transmon and (b) fluxonium. Channels not shown did not contribute meaningfully for the respective systems. Gate time estimates based on the analysis given in Section C.8 are shown in (c). The gate time estimates for the transmon seem consistent with gate times from the literature. It is important to note that removing noises or reducing the noise magnitude will change the conclusion of our numerics. 166

C.9 Circuit diagram for the top performing circuit, coupled to an AC voltage source. Each circuit node is labeled 1, 2, or 3, corresponding to node flux variables ϕ_i. 170

D.1 Optimized single-qubit pulse parameters. 175

D.2 Population transfer in a simulated gate comparable to experiment. 179

D.3 CNOT gate implemented by driving $n_{\theta 1}$. 191

D.4 Population transfer and error of the CNOT gate. 193

D.5 Level crossing versus coupling rate. 196

D.6 Optimized two-qubit pulse parameters. 197

Chapter 1

Introduction & Motivation

Quantum computers represent information as quantum states, and for specific problems they may asymptotically outperform known classical algorithms [1]. Quantum computation was first suggested as an efficient method to simulate quantum systems [2, 3]. The premise is simple: to exactly represent N two-state quantum systems requires memory scaling of $\mathcal{O}(2^N)$, but it would require only N physical two state quantum systems. Simulating quantum systems remains one of the most promising applications, with problems of interest across chemistry and materials science [4]. That being said, the most famous application of a quantum computer is Shor's algorithm, which offers an exponential speedup for factoring numbers and would break widely used public-key cryptography [5].

Realizing these algorithms at a useful scale requires a large-scale, fault-tolerant quantum computer, and such active error correction comes with substantial overhead. The threshold theorem guarantees that logical errors can be suppressed arbitrarily, provided the physical error rate p lies below a threshold p_{th} [6, 7]. How far below matters. In the surface code, for example, the logical error rate falls as $(p/p_{\text{th}})^{(d+1)/2}$ with code distance d, so better physical qubits reach a target logical error rate with a smaller code and far fewer physical qubits per logical qubit [8]. Reducing this overhead, through better physical qubits and more efficient codes, is central to the field.

1.1 Superconducting Qubits

One of the most promising technologies to realize a fault tolerant quantum computer is superconducting Josephson junction circuits. Cooled to millikelvin temperatures, these circuits

behave as quantum systems, and two of their energy levels can be isolated and used as a qubit [9, 10]. They are often called artificial atoms, because their spectrum is set by the circuit design rather than fixed by nature. Superconducting circuits also host bosonic encodings in the modes of microwave cavities [11], which are not the focus of this thesis.

We now quickly review the development of these systems and establish where they stand today. The first superconducting qubits were the Cooper pair box [12, 13] and the persistent-current flux qubit [14, 15], which were highly sensitive to charge and flux noise, respectively. Indeed, the first coherent oscillations of a Cooper pair box were observed for only about 2 ns [13]. Two designs that reduce these sensitivities underlie most qubits in use today (Figure 1.1a). The transmon shunts the junction with a large capacitance, which suppresses its sensitivity to charge noise exponentially at the cost of a small anharmonicity [16]. The fluxonium instead shunts the junction with a large inductance [17], which gives a large anharmonicity and low qubit frequency, at the cost of flux noise sensitivity. Both now reach millisecond coherence, with $T_2^* = 1.48$ ms in a fluxonium [18] and T_1 up to 1.68 ms in transmons [19]. For a more thorough introduction to circuit quantum electrodynamics (cQED), including gates, readout, and experimental considerations, many excellent reviews and tutorials have been published [20–24].

The last few years have additionally seen substantial improvements in scaling and quantum error correction of superconducting systems. Bosonic encodings had already reached memory break-even with a cat code in 2016 [25]. Subsequent experiments demonstrated logical lifetimes exceeding those of the best available physical encodings by factors of 2.27 and approximately 1.16 using Gottesman–Kitaev–Preskill (GKP) and binomial codes, respectively [26, 27]. Progress with arrays of superconducting qubits has followed a similar route. Repeated extraction of both bit- and phase-error syndromes was first demonstrated in a distance-three surface code in 2022 [28]. In 2025, a beyond threshold surface code demonstration showed decreasing error rates with increasing surface code distance [29].

Despite these recent successes, it may pay off to investigate superconducting qubit systems

with lower physical gate error rates [30]. There are two complementary paths towards fault tolerance: lower-error physical qubits and better codes. The lower the physical error rate or the better the code, the fewer physical qubits are needed. This thesis concerns analyzing and identifying lower-error superconducting qubits.

1.2 Protected Qubits

Protected qubits suppress errors at the hardware level and represent a promising path towards lower error qubits [31]. The key is that the qubit states are insensitive to the local operators through which noise couples to the circuit. We now examine this criteria in more detail.

1.2.1 Decay and Dephasing

Energy-relaxation (and excitation) arise when a quantum system couples to a noisy environmental bath. To first order, the system-bath interaction has the form $H_{\text{int}} = \lambda(t)\hat{O}_\lambda$, where $\lambda(t)$ represents a noisy parameter, and

$$\hat{O}_\lambda = \frac{\partial \hat{H}}{\partial \lambda} \tag{1.1}$$

is the operator to which the noise couples. Fermi's golden rule gives the downward transition rate between two eigenstates $|i\rangle$ and $|j\rangle$ of the Hamiltonian $\hat{H}$:

$$\Gamma^{(\lambda)}_{j\to i} = \frac{1}{\hbar^2}\left|\langle i|\,\hat{O}_\lambda\,|j\rangle\right|^2 S_\lambda(\omega_{ij}), \tag{1.2}$$

where $S_\lambda(\omega) = \int_{-\infty}^{\infty} dt\, e^{-i\omega t}\langle\lambda(0)\lambda(t)\rangle$ is the noise spectral density and $\omega_{ij} = (E_j - E_i)/\hbar$ is the transition frequency [32]. The logical states are protected against energy relaxation if, for every logical state $|j\rangle$ and every state $|i\rangle$ below it, logical or not,

$$\langle i|\,\hat{O}_\lambda\,|j\rangle = \mathcal{O}\big(e^{-\xi}\big) \tag{1.3}$$

is exponentially suppressed in a system design parameter ξ that can be made large. Decay into a non-logical state is leakage, so it must be suppressed as well as decay between logical states. In the soft $0-\pi$ qubit, for example, the logical states are $|0\rangle$ and $|2\rangle$, so $|2\rangle$ must decay slowly into both $|0\rangle$ and $|1\rangle$ (see Chapter 4). Such suppression typically follows from nearly disjoint support of the logical wavefunctions, because the common noise operators, such as $\hat{n}$ and $\hat{\phi}$, act locally [31].

Dephasing arises from slow fluctuations of a noisy parameter, which leads to shifting energy eigenvalues. First-order sensitivity of the $|i\rangle \leftrightarrow |j\rangle$ transition to dephasing is set by the derivative of the transition frequency with respect to the noisy parameter [33]

$$D_{ij}^{(\lambda)} \equiv \frac{\partial \omega_{ij}}{\partial \lambda} = \frac{1}{\hbar}\left(\langle j|\,\hat{O}_\lambda\,|j\rangle - \langle i|\,\hat{O}_\lambda\,|i\rangle\right) . \tag{1.4}$$

The second equality follows from the Feynman–Hellmann theorem. If $\partial\omega_{ij}/\partial\lambda = 0$, the qubit is at a sweet spot and the leading sensitivity is determined by $\partial_\lambda^2\omega_{01}$. The dephasing rate is proportional to the sensitivity squared $\Gamma_{\varphi,ij}^{(\lambda)} \propto |D_{ij}^{(\lambda)}|^2$. A qudit is protected against dephasing if for all combinations of logical states $|i\rangle, |j\rangle$,

$$\frac{\partial \omega_{ij}}{\partial \lambda} = \mathcal{O}\big(e^{-\eta}\big) \tag{1.5}$$

is exponentially suppressed in a system design parameter η that can also be made large. Equivalently, the total dispersion of ω_{ij} over the range of λ is exponentially small. Following Gyenis et al. [31], we call protection exponential when these conditions hold, and sweet-spot protection when only the first derivative of ω_{ij} vanishes at the operating point.

Relaxation and dephasing together set the total coherence time $T_2 = 1/(2\Gamma_\varphi + \Gamma_1) = [1/(2T_1) + 1/T_\varphi]^{-1}$, for $T_1 = 1/\Gamma$ and $T_\varphi = 1/2\Gamma_\varphi$.

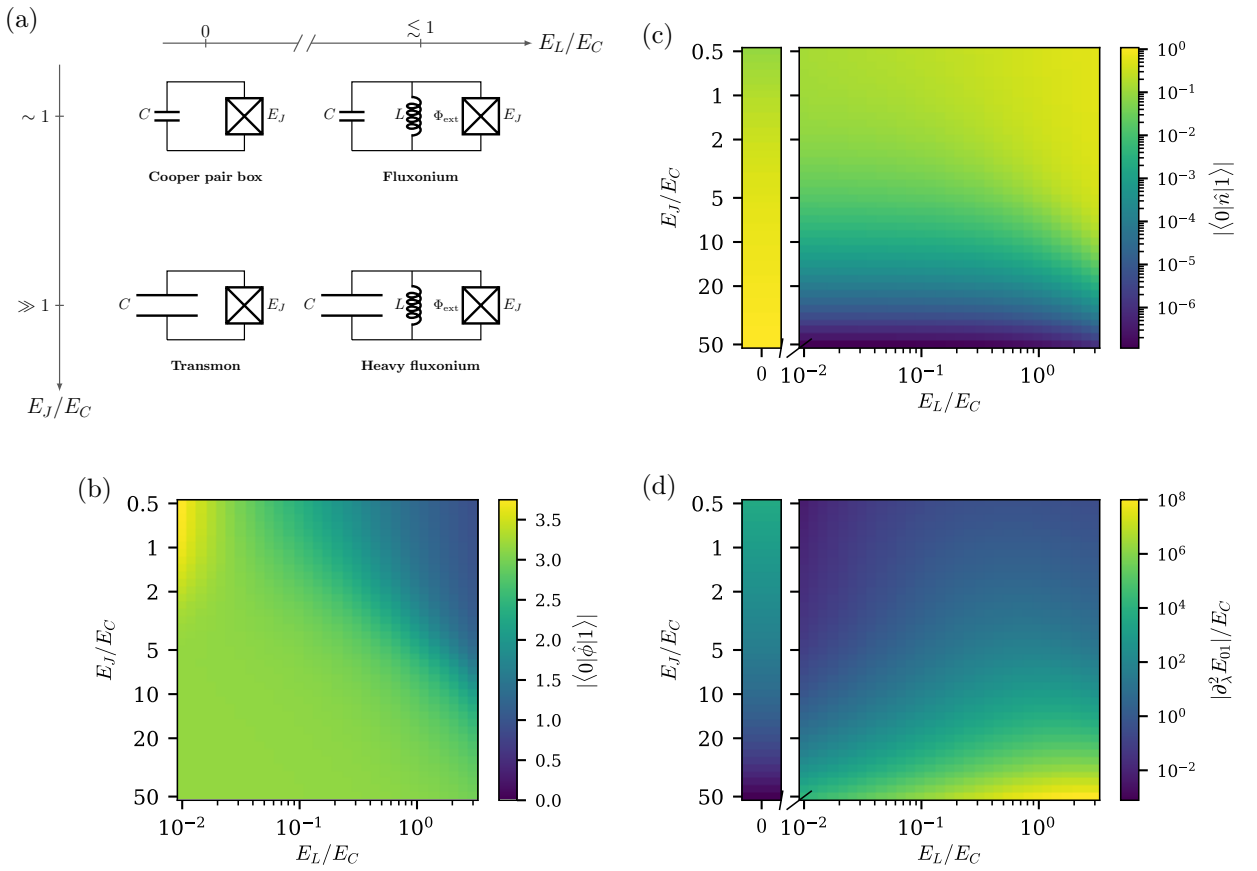


Figure 1.1: (a) Single-mode superconducting qubits, arranged by E_J/E_C and E_L/E_C as in (b)–(d). Adding capacitance moves a circuit down, and adding an inductance moves it to the left (an infinitely large inductor is an open circuit). (b), (c) Transition matrix elements, which set the coupling to flux and charge noise that causes decay. (d) Sweet-spot curvature, which sets the second-order sensitivity of the qubit frequency to that noise. Strips at $E_L = 0$ are the Cooper pair box and transmon at $n_g = 0$ ($\lambda = n_g$), and maps are the fluxonium at $\phi_{\text{ext}} = \pi$, with energies in units of E_C.

1.2.2 Single-Mode Qubits

The single-mode qubits of Figure 1.1a are each built from one junction, a capacitor, and at most one inductor. Figures 1.1b and 1.1c show the transition matrix elements that set relaxation, and Figure 1.1d shows the sweet-spot curvature that sets dephasing. None of these circuits are protected against both relaxation and dephasing. Any region in parameter space where one quantity is suppressed is unsuppressed in another.

The key is that a single mode offers only one coordinate in which to arrange the two logical

wavefunctions. Exponentially small matrix elements require disjoint support, which corresponds to localized (and non-overlapping) logical states in the relevant coordinate. Exponentially small dispersion instead requires the logical states to be delocalized so that a small shift in the coordinate has little effect. Such states typically have large overlap, so they are connected strongly by local matrix elements. Protection against both requires either a second mode, so that the logical states can be disjoint in one coordinate and delocalized in another [31], or additional circuit elements, which we examine next.

1.2.3 The $\cos 2\phi$ Qubit

The argument above assumes that the circuit is built from Josephson junctions, capacitors, and inductors. A single mode can be protected against both relaxation and dephasing if its junction is replaced by an element that tunnels only pairs of Cooper pairs [31, 34]:

$$\hat{H} = 4E_C \left(\hat{n} - n_g\right)^2 - E_J \cos 2\hat{\phi}. \tag{1.6}$$

The $\cos 2\hat{\phi}$ term changes $\hat{n}$ by ± 2, so $\hat{H}$ conserves the charge parity $(-1)^{\hat{n}}$. Its two lowest eigenstates are supported on even and odd charge states, respectively, so any noise operator that is a function of $\hat{n}$ has an exactly zero matrix element. This is disjoint support enforced by symmetry. In the phase basis, the potential has minima at $\phi = 0$ and π, and for $E_J \gg E_C$ the two eigenstates are the even and odd superpositions of states localized in these wells. Tunneling splits the pair by $E_1 - E_0 \approx 4t|\cos(\pi n_g)|$, where $t \propto e^{-\sqrt{2E_J/E_C}}$ is the tunneling amplitude along each of the two paths between the wells.

No single element tunnels only pairs of Cooper pairs, so $\cos 2\phi$ qubits are approximated by multi-mode circuits whose low-energy dynamics approximate Equation (1.6). For example, the $0-\pi$ qubit has an effective $\cos 2\phi$ potential in one mode [32, 35–37]. Only a "soft" version has been realized experimentally [38], which remains sensitive to dephasing. Other circuits, like the Josephson rhombus [39] and the KITE [34], cancel single Cooper pair tunneling by interference at a particular

flux bias. Both have seen recent experimental implementations [40, 41], but the demonstrations fall short of simultaneous protection against both decay and dephasing. We examine approximate $\cos 2\phi$ implementations in more detail in **??**, where we show that circuit symmetries lead to a selection rule for first order flux noise protection, which limits the performance of both of the aforementioned experiments.

1.3 This Thesis

Protection requires multimode circuits, and multimode circuits are difficult to quantize, simulate, search over, and control. This thesis addresses each of these in turn. Chapter 2 gives a general procedure to go from a circuit diagram to a multimode Hamiltonian, which every later chapter starts from. This is not as straightforward as writing down the energy stored in each element. The procedure must separate modes with periodic coordinates from those with real-valued coordinates, identify non-dynamical modes, and handle time dependent external flux. The final choice of coordinates represents a residual freedom that decides how the Hamiltonian presents symbolically and how quickly it can be diagonalized numerically. The chapter also summarizes this numerical treatment. Chapter 3 searches over circuits, by enumerating all unique circuits built from a given set of elements and sorting them by the structure of their Hamiltonian, using a consistent choice of the final coordinates. Earlier work on circuit equivalence compares circuits at a port, through an impedance or a transfer function. Here two circuits are equivalent when they have the same quantized Hamiltonian, which fixes their internal dynamics rather than their behavior at a port. Chapter 4 considers control for a protected qubit, the soft $0-\pi$. It proposes two-qubit CZ and CNOT gates and analyzes the performance of single qubit gates in the same system. **??** returns to the $\cos 2\phi$ qubit and gives a symmetry selection rule that determines, from the circuit alone, whether an approximate realization forbids first order flux noise. It applies the rule to the $0-\pi$, KITE, Harmonium, and rhombus circuits. The section also gives a short quantum error correction perspective on which noise operators distinguish the logical states. **??** presents GPU-accelerated

matrix-free methods for efficiently simulating larger multimode circuits, followed by suggestions for future work and the conclusion of the thesis.

Chapter 2

Modeling of Multimode Superconducting Circuits: Circuit Quantization, Numerical Modeling, and Mode Preserving Transformations

2.1 Introduction

Superconducting circuits with many interacting modes are increasingly common, and the first step in analyzing any such circuit is writing down its Hamiltonian. Lumped-element circuit quantization provides a route from a circuit diagram to that Hamiltonian [23, 42], which several software packages now automate [43–47].

Whether by hand or in software, circuit quantization begins from the topology of a circuit, which sorts its degrees of freedom by the energies they carry. A compact mode enters only through Josephson cosines, so its phase coordinate is periodic and its charge is quantized, as in the transmon. An extended mode also carries inductive energy quadratic in phase, so its phase coordinate is real-valued, as in the fluxonium. A harmonic mode carries only quadratic inductive and charging energy, as in an LC resonator. The remaining non-dynamical modes lack kinetic and/or potential energy and do not contribute to the dynamics of the circuit.

The topology dictates how many modes of each type there are, but it does not fix the coordinates that describe them. Any invertible change of basis that respects the mode types gives an equally valid Hamiltonian, and automated tools generally pick one by a fixed rule. This choice decides which couplings appear in the Hamiltonian, whether the Hamiltonians of different circuits can be easily compared, and how quickly a numerical diagonalization converges. This chapter precisely

characterizes transformations that preserve every mode type and poses the choice of coordinates as an optimization over them.

Section 2.2 reviews flux-based quantization in a matrix formulation [20, 21, 23, 48], with particular care for the cases of multiple compact modes and time-dependent external flux. We quantize several compact modes on their joint lattice and give the irrotational flux assignment [49] in closed form, including loops broken by capacitors. Section 2.3 summarizes how the resulting Hamiltonians are diagonalized numerically. Section 2.4 uses the framework of the previous section to define the class of mode-preserving transformations. The choice of coordinates is then an optimization problem, which can be solved symbolically with general circuit parameters or numerically at fixed ones. Section 3.7 concludes.

2.2 Multimode Circuit Quantization

We begin by reviewing circuit quantization of multimode superconducting circuits made of capacitors, inductors, and Josephson junctions. Each ingredient appears in at least one existing review or software package [20–24, 43–46, 48, 50, 51]. Our goal is to present a concise but complete treatment that quantizes any multimode circuit of capacitors, inductors, and Josephson junctions. Less common symbols used in this section are collected in Table 2.2, with Table B.1 collecting all symbols for reference. We retain time-dependent external flux, including through capacitively broken loops. Which choices are conventions and which are required by the physics are clearly distinguished. It should be noted that our treatment is restricted to the standard node flux formalism and does not allow for nonlinear capacitors or nonreciprocal elements. More advanced methods [52–57] are required for these systems, but none have been implemented in an automated fashion.

2.2.1 Constructing the Circuit Lagrangian

We take a circuit to be fully specified by:

Branch type	Parameters	$U_k(\Phi_{b,k})$
Linear capacitor	$C_k > 0$	0
Linear inductor	$L_k > 0,\ C_k \geq 0$	$\Phi_{b,k}^2/(2L_k)$
Josephson junction	$E_{J_k} > 0,\ C_k > 0$	$-E_{J_k}\cos(2\pi\Phi_{b,k}/\Phi_0)$

Table 2.1: Passive branch types and their branch-flux dependent energy contributions $U_k(\Phi_{b,k})$. Capacitance contributes to the kinetic energy $T_k = \frac{1}{2}C_k\dot{\Phi}_{b,k}^2$ for every branch type. Ideal voltage are treated separately and are fixed at $\dot{\Phi}_{b,k} = V_k(t)$. Energies are in SI units, with $\Phi_0 = h/(2e)$ being the superconducting flux quantum.

(1) **Circuit branches**. N_b circuit branches, each specifying a head and tail node, along with circuit elements and their parameters.

(2) **A choice of ground node** or a floating circuit.

(3) **Connections to voltage sources**. Classical voltage sources driving the circuit through coupling capacitors. Each voltage source connects to ground.

(4) **Flux biases**. Loops threaded by external flux.

Each passive branch carries a capacitance $C_k \geq 0$ and at most one inductor or Josephson junction. Its contribution to the Lagrangian is

$$\mathcal{L}_k = \frac{1}{2}C_k\dot{\Phi}_{b,k}^2 - U_k(\Phi_{b,k})\,, \tag{2.1}$$

where $\dot{\Phi}_{b,k} = V_{b,k}(t)$ is the branch voltage and $\Phi_{b,k} = \int_{-\infty}^{t} dt V_{b,k}(t)$ is the generalized branch flux. Table 2.1 specifies the three passive branch types and their potentials.

The branch fluxes Φ_b are the physical circuit variables, but they are not independent when the circuit forms loops. We describe the circuit loops by the loop matrix G, whose ℓ-th row has entries ± 1 on the branches around loop ℓ, with signs set by each branch's orientation relative to the loop. Every loop obeys Faraday's law, $\mathrm{G}\dot{\Phi}_b = \dot{\Phi}_{\mathrm{ext}}$, where Φ_{ext} collects the external fluxes threading the loops. Loops whose branches all carry an inductor or junction, the rows G_{JL}, additionally obey

fluxoid quantization, $\mathrm{G}_{JL}\Phi_b = \Phi_{\mathrm{ext}}$, with any integer fluxoid absorbed into Φ_{ext}. For loops broken by a capacitor we fix the integration constant of Faraday's law in the same way, so all loops satisfy $\mathrm{G}\Phi_b = \Phi_{\mathrm{ext}}$.

To solve this constraint, we collect the fluxes of the N internal circuit nodes in Φ, excluding ground and prescribed voltage-source nodes. The incidence matrix Ω maps node fluxes to branch differences. Column k corresponds to branch k and has entries $+1$ at its head node and -1 at its tail node, with only the $+1$ for branches connecting to ground or a voltage source. Node differences sum to zero around any loop, so $\mathrm{G}\Omega^T = 0$ by definition, and $\Phi_b = \Omega^T\Phi$ is the general solution without external flux. Adding a particular solution gives the full vector of branch variables

$$\Phi_b = \Omega^T\Phi + \mathrm{B}\Phi_{\mathrm{ext}}\,, \qquad \mathrm{GB} = I\,. \tag{2.2}$$

The flux assignment matrix B places the external fluxes on branches, with $[\mathrm{B}\Phi_{\mathrm{ext}}]_k$ being the flux assigned to branch k. The constraint $\mathrm{GB} = I$ ensures that each external flux offset is fully allocated and leaves B free up to $\mathrm{B} \to \mathrm{B} + \Omega^T M$ for any $N \times N_\ell$ matrix M. This shift in B is equivalent to an overall shift in the node variables $\Phi \to \Phi + M\Phi_{\mathrm{ext}}$. We later use this freedom to simplify the Hamiltonian when external flux depends on time.

To build the lagrangian, we take the sum over branch variables. With the diagonal matrix C_b containing the per-branch capacitance, the total capacitive energy in the circuit is $\frac{1}{2}\dot{\Phi}_b^T C_b \dot{\Phi}_b$. It is typical to define the capacitance matrix $C = \Omega C_b \Omega^T$ to capture the node-flux dependent term of Equation (2.2). Voltage sources and the external-flux dependent term both contribute to an offset charge

$$Q_g(t) = -C_V V(t) + \Omega C_b \mathrm{B}\dot{\Phi}_{\mathrm{ext}}(t)\,, \tag{2.3}$$

where the source-coupling matrix C_V has entry (i, s) equal to the total capacitance connecting node i to voltage source s, and $V(t)$ gives the voltages of the classical source nodes. The Lagrangian is

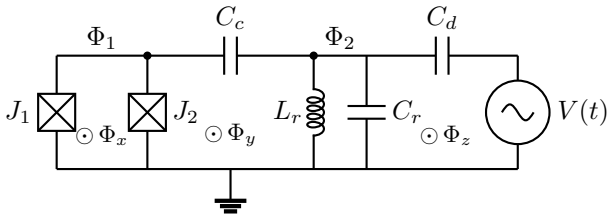


Figure 2.1: **Example circuit.** A grounded SQUID (junctions J_1, J_2) is coupled through C_c to a resonator (L_r, C_r), which is driven by the voltage source $V(t)$ through C_d. The resonator is treated as a single branch. External fluxes Φ_x, Φ_y, and Φ_z thread three loops, with the latter two broken by capacitors.

the sum of these terms

$$\mathcal{L}(\Phi, \dot{\Phi}) = \frac{1}{2}\dot{\Phi}^T C \dot{\Phi} + \dot{\Phi}^T Q_g(t) - \mathcal{U}(\Phi)\,, \tag{2.4}$$

plus the flux-dependent potential energy,

$$\begin{aligned}\mathcal{U}(\Phi) =& \frac{1}{2}\Phi^T L^* \Phi + \Phi^T \omega_L L_b^{-1} \mathrm{B}_L \Phi_{\mathrm{ext}} \\ &- \sum_{k \in \mathcal{S}_J} E_{J_k} \cos\left(2\pi \frac{[\omega_J^T \Phi + \mathrm{B}_J \Phi_{\mathrm{ext}}]_k}{\Phi_0}\right)\,.\end{aligned} \tag{2.5}$$

Here $L^* = \Omega L_b^{-1} \Omega^T$ is equivalent to C for linear inductance and $\mathcal{S}_J$ is the set of branches containing Josephson junctions. The sub-matrices ω_L and ω_J are type-restricted incidence matrices for only the branches containing inductors and junctions, respectively. Table 2.2 summarizes these matrices.

As an example, consider the circuit of Figure 2.1, which contains each of the four ingredients listed above. It has $N = 2$ internal nodes $\Phi = (\Phi_1, \Phi_2)^T$ and $N_b = 5$ branches, ordered $(J_1, J_2, C_c, L_r, C_d)$. The incidence and loop matrices are

$$\Omega = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 1 \end{pmatrix}\,, \quad \mathrm{G} = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{pmatrix}\,,$$

and we can explicitly confirm that $G\Omega^T$ is the 3x2 zero matrix. The orientation of each element

is arbitrary as all elements are reciprocal (i.e., any column of Ω can be multiplied by -1), but the orientations must be kept consistent with G to maintain $G\Omega^T = 0$. The type-restricted matrices are the corresponding columns of Ω. We assign Φ_x to J_1, Φ_y to C_c, and Φ_z to C_d, so that

$$\omega_J = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \omega_L = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathrm{B} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which satisfies $\mathrm{GB} = I$. The first loop belongs to G_{JL} and obeys fluxoid quantization, while the other two are broken by C_c and C_d. Only J_1 carries external flux in the potential, so Equation (2.5) becomes

$$\mathcal{U} = \frac{\Phi_2^2}{2L_r} - E_{J_1} \cos\left(2\pi \frac{\Phi_1 + \Phi_x}{\Phi_0}\right) - E_{J_2} \cos\left(2\pi \frac{\Phi_1}{\Phi_0}\right). \tag{2.6}$$

The capacitively broken loops instead enter through the offset charge. For the single source coupled through C_d, the capacitance matrices and Equation (2.3) give

$$C_V = \begin{pmatrix} 0 \\ C_d \end{pmatrix}, Q_g = \begin{pmatrix} C_{J_1}\dot{\Phi}_x + C_c\dot{\Phi}_y \\ -C_c\dot{\Phi}_y + C_d(\dot{\Phi}_z - V) \end{pmatrix}, \tag{2.7}$$
$$C = \begin{pmatrix} C_{J_1} + C_{J_2} + C_c & -C_c \\ -C_c & C_c + C_r + C_d \end{pmatrix}.$$

2.2.2 Legendre Transform, Coordinate Transformations, Mode Types, and Quantization

A Hamiltonian is obtained through a Legendre transform, which requires an invertible capacitance matrix. This is not always the case. A non-invertible capacitance corresponds to non-dynamical degrees of freedom, which we can identify and eliminate with a change of variables.

Symbol	Name	Dimension	Domain
Ω	branch incidence	$N \times N_b$	$\{0, \pm 1\}$
ω_L	inductor incidence	$N \times N_L$	$\{0, \pm 1\}$
ω_J	junction incidence	$N \times N_J$	$\{0, \pm 1\}$
$\mathcal{S}_J$	junction branches	N_J	$\subseteq \{1, \ldots, N_b\}$
G	loop matrix	$N_\ell \times N_b$	$\{0, \pm 1\}$
B	flux assignment	$N_b \times N_\ell$	$\mathbb{R}$
C_b	branch capacitance	$N_b \times N_b$	diagonal, ≥ 0
L_b^{-1}	branch inverse inductance	$N_b \times N_b$	diagonal, ≥ 0
C	node capacitance	$N \times N$	symmetric, $\succeq 0$
L^*	node susceptance	$N \times N$	symmetric, $\succeq 0$
C_V	source coupling	$N \times N_V$	≥ 0
Z	transformation	$N \times N$	$\mathrm{GL}(N, \mathbb{R})$

Table 2.2: **Non-standard symbols.** N is the number of internal nodes, N_b the number of branches, N_ℓ the number of independent loops, and N_V the number of voltage sources. N_L and N_J count the inductive and Josephson branches. All symbols are collected in Table B.1.

We can identify new coordinates $\tilde{\Phi}$ through a transformation Z on the flux coordinates $\tilde{\Phi}$

$$\Phi = Z\tilde{\Phi}, \qquad \tilde{\Phi} = Z^{-1}\Phi\,. \tag{2.8}$$

The circuit matrices in turn transform as

$$\begin{aligned} \tilde{C} &= Z^T C Z\,, \quad \tilde{L}^* = Z^T L^* Z\,, \\ \tilde{Q}_g &= Z^T Q_g\,, \quad \tilde{\omega}_\alpha^T = \omega_\alpha^T Z\,, \end{aligned} \tag{2.9}$$

where $\alpha = L, J$. In the language of symplectic transformations, Z is a real-valued point transformation [54], which defines the transformed conjugate charges $\tilde{Q} = Z^T Q$ up to a total derivative.

The columns of Z can be chosen to expose which transformed coordinates enter the kinetic, inductive, and Josephson terms. These dependences define the mode types illustrated in Figure 2.2. Non-dynamical variables contain either only potential energy, only kinetic energy, or neither, with the latter two causing C to be non-invertible. To illustrate how non-dynamical modes manifest, consider an ideal floating transmon. It is described by two node fluxes but only their difference enters the junction and its capacitance.

The same reasoning holds for any circuit. In general, a column of Z carries kinetic energy unless it lies in $\mathsf{C} = \text{null}(C)$, inductive energy unless it lies in $\mathsf{L} = \text{null}(L^*)$, and Josephson energy unless it lies in $\mathsf{J} = \text{null}(\omega_J^T)$. Six distinct mode types emerge based on whether the column carries each of these terms. A full description is given in Figure 2.2, as a table and as an Euler diagram. For the transmon, $\mathsf{J} = \mathsf{C} = \text{span}\{(1,1)^T\}$, and L spans the whole plane. This leaves one compact and one cyclic mode.

As a full example, consider the floating LCLC box of Figure 2.3(a), which consists of alternating capacitors and inductors in one loop. From the dimensions of the respective bases, we can identify one harmonic mode, one free mode, one frozen mode, and one cyclic mode, which define a transformation

$$\begin{pmatrix} \Phi_1 \\ \Phi_2 \\ \Phi_3 \\ \Phi_4 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}}_{Z=\left(Z_h\, Z_{fr}\, Z_{fz}\, Z_{cy}\right)} \begin{pmatrix} \tilde{\Phi}_{\text{h}} \\ \tilde{\Phi}_{\text{fr}} \\ \tilde{\Phi}_{\text{fz}} \\ \tilde{\Phi}_{\text{cy}} \end{pmatrix}, \tag{2.10}$$

with columns in this order. Each branch flux is the difference of two rows of Z,

$$\begin{aligned} \Phi_2 - \Phi_1 &= 2\big(\tilde{\Phi}_{\text{fr}} - \tilde{\Phi}_{\text{h}}\big) && (C_1)\,, \\ \Phi_4 - \Phi_3 &= 2\big(\tilde{\Phi}_{\text{h}} + \tilde{\Phi}_{\text{fr}}\big) && (C_2)\,, \\ \Phi_3 - \Phi_1 &= 2\big(\tilde{\Phi}_{\text{fz}} - \tilde{\Phi}_{\text{h}}\big) && (L_1)\,, \\ \Phi_2 - \Phi_4 &= -2\big(\tilde{\Phi}_{\text{h}} + \tilde{\Phi}_{\text{fz}}\big) && (L_2)\,. \end{aligned}$$

A coordinate drops out of a branch exactly when its column has equal entries on the two nodes of that branch. The free column has equal entries on nodes 1 and 3 and on nodes 2 and 4, so neither inductor sees it, and the frozen column likewise drops out of both capacitors. The cyclic column has equal entries on every node and drops out of every branch. We depict the columns graphically in cyclic (Figure 2.3(b)–(e)), showing that each one corresponds to a positive/negative combination of

node variables that create differences across capacitors, inductors, both, or neither.

After Equation (2.10), the non-dynamical modes remain coupled to the harmonic mode, but we can construct a secondary transformation to remove this coupling. Begin with a block C and L^*,

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}, \quad L^* = \begin{pmatrix} L^*_{11} & L^*_{12} \\ L^*_{21} & L^*_{22} \end{pmatrix}, \tag{2.11}$$

with block 1 containing all dynamical variables and block 2 containing all non-dynamical variables. The off-diagonal blocks can be made zero by

$$Z_{\text{decouple}} = \begin{pmatrix} I & 0 \\ M & I \end{pmatrix}, \tag{2.12}$$

with $M = -(L^*_{22})^{-1}L^*_{21}$ for the frozen block and $M = -C_{22}^{-1}C_{21}$ for the free block [58]. The resulting dynamical blocks are the Schur complements $C/C_{22} = \tilde{C}_{11} - \tilde{C}_{12}\tilde{C}_{22}^{-1}\tilde{C}_{21}$ and $L^*/L^*_{22} = \tilde{L}^*_{11} - \tilde{L}^*_{12}\tilde{(L^*_{22})}^{-1}\tilde{L}^*_{21}$.

C/C_{22} is invertible, so we can complete the Legendre transform. In the following equations, all circuit matrices are assumed to be the dynamical block, post-decoupling transformation. The full circuit Hamiltonian is given by,

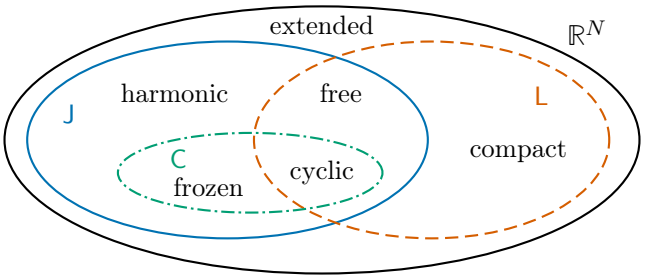
$$\mathcal{H}(\tilde{Q}, \tilde{\Phi}) = \frac{1}{2}\big(\tilde{Q} - \tilde{Q}_g(t)\big)^T \tilde{C}^{-1}\big(\tilde{Q} - \tilde{Q}_g(t)\big) + \mathcal{U}(\tilde{\Phi}), \tag{2.13}$$

with the transformed potential energy and offset charge

$$\begin{aligned} \mathcal{U}(\Phi) = &\frac{1}{2}\tilde{\Phi}^T\tilde{L}^*\tilde{\Phi} + \tilde{\Phi}^T\tilde{\omega}_L L_b^{-1}\mathrm{B}_L\Phi_{\text{ext}}(t) \\ &- \sum_{k\in\mathcal{S}_J} E_{J_k}\cos\left(2\pi\frac{[\tilde{\omega}_J^T\tilde{\Phi} + \mathrm{B}_J\Phi_{\text{ext}}(t)]_k}{\Phi_0}\right), \end{aligned} \tag{2.14}$$

$$\tilde{Q}_g(t) = -Z^T C_V V(t) + Z^T\Omega C_b \mathrm{B}\dot{\Phi}_{\text{ext}}(t). \tag{2.15}$$

Canonical quantization promotes Equation (2.13) to a quantum Hamiltonian, and different



mode	$\dot{\Phi}^2$	Φ^2	$\cos\Phi$	basis
extended	✓	✓	✓	$(\mathsf{J}+\mathsf{L})^\perp$
compact	✓	×	✓	$\mathsf{L}\cap(\mathsf{J}\cap\mathsf{L})^\perp$
harmonic	✓	✓	×	$\mathsf{J}\cap\left[\mathsf{C}+(\mathsf{J}\cap\mathsf{L})\right]^\perp$
		non-dynamical		
free	✓	×	×	$(\mathsf{J}\cap\mathsf{L})\cap(\mathsf{C}\cap\mathsf{L})^\perp$
frozen	×	✓	×	$\mathsf{C}\cap(\mathsf{C}\cap\mathsf{L})^\perp$
cyclic	×	×	×	$\mathsf{C}\cap\mathsf{L}$

Figure 2.2: **Six mode types.** Top: Euler diagram of the mode bases, where each basis is a distinct region. C is contained in J because we assume every junction has positive intrinsic capacitance. Thus any vector in null(C) must also lie in null(ω_J^T). Note that regions may be empty for a particular circuit. Bottom: which terms of the Lagrangian $\mathcal{L}$ each mode enters. Columns of Z which define the specified mode type are spanned by the specified bases, built from the null spaces J, L, and C. Here $A+B$ is the subspace sum, $A\cap B$ the intersection, and $A^\perp$ the orthogonal complement.

coordinate types require different treatment. Extended and Harmonic modes are real valued coordinates and obtain standard conjugate flux (position) and charge (momentum) operators. These are cleanest in terms of the dimensionless phase $\varphi = \tilde{\Phi}/\phi_0$ and Cooper-pair number $n = \tilde{Q}/2e$, with $\phi_0 = \hbar/2e$, which satisfy

$$[\hat{\varphi}_i, \hat{n}_j] = i\delta_{ij}$$

with no residual $\hbar$. Section B.2 gives the full Hamiltonian in these variables, with circuit matrices in frequency units. A compact mode enters only through the periodic junction potential and is thus an angular coordinate. This requires special care when several are present, which we examine in the next subsection.

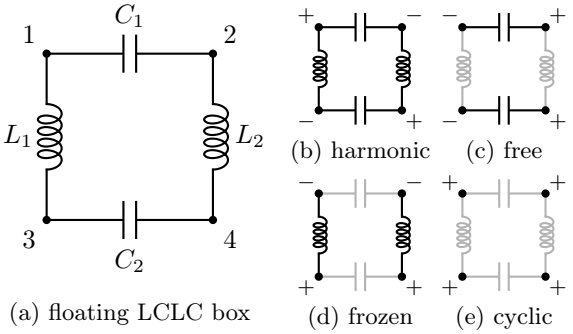


Figure 2.3: **Mode types of the floating LCLC box.** (a) The circuit, with nodes 1–4. (b)–(e) The columns of Z in Equation (2.10), with the sign of each node's entry. Grey branches carry no flux difference in that column, so they contribute no energy.

2.2.3 Two Subtleties

Two subtleties of this procedure are easy to get wrong, and both grow in complexity for larger multimode circuits. The first is the quantization of several compact modes at once, and the second is the assignment of time-dependent external flux.

2.2.3.1 Quantization with Multiple Compact Modes

With several compact coordinates, it is tempting to treat each one as an independent 2π-periodic rotor with a quantized charge variable. The example of two capacitively coupled transmons (Figure 2.4) shows when this picture fails. Treating the sum and difference phases $\varphi_\pm = \varphi_1 \pm \varphi_2$ as independent rotors gives the wrong answer. The true unit cell in these coordinates is rescaled and rotated relative to the node coordinates, and the independent-rotor charges have support on only half of the integer Cooper-pair number states. Ultimately this miscounting corresponds to not correctly enforcing the periodic boundary conditions.

The correct boundary condition follows from the junctions. For a system with N_c compact coordinates, the Hamiltonian is periodic on the lattice $\mathcal{P} = \{a \in \mathbb{R}^{N_\mathrm{c}} : \tilde{\omega}_{J,\mathrm{c}}^T a \in 2\pi\mathbb{Z}^{N_J}\}$, where $\tilde{\omega}_{J,\mathrm{c}}$ is restricted to compact coordinates. The wavefunctions must be periodic on the same lattice, with

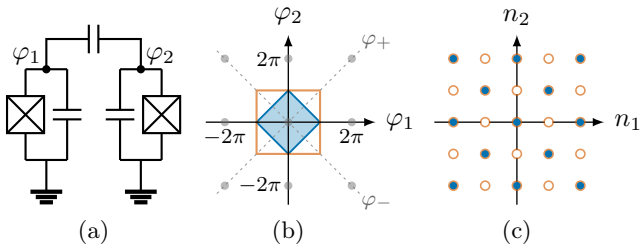


Figure 2.4: **Charge basis of two transmons** (a) coupled by the capacitor C_c. (b) Unit cell in the the reduced node flux (φ_1, φ_2) (orange) and sum/difference (φ_+, φ_-) (blue) coordinates. Grey dots mark the centers of adjacent unit cells in the node coordinates. (c) Charge lattice, with the (n_1, n_2) basis containing all charge combinations, and the (n_+, n_-) (blue) containing only values for which $n_1 \pm n_2 = 0 \bmod 2$.

the reciprocal (charge) lattice defining the units of charge quantization

$$\begin{aligned} \psi(\varphi_c + a) = \psi(\varphi_c) \text{ for all } a \text{ with } \tilde{\omega}_{J,c}^T a \in 2\pi\mathbb{Z}^{N_J}\,, \\ \hat{n}_c \in \left\{\tilde{\omega}_{J,c}\, m : m \in \mathbb{Z}^{N_J}\right\}. \end{aligned} \tag{2.16}$$

For the two transmons in the coordinates $\varphi_\pm$, $\tilde{\omega}_{J,c}^T = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, so $\mathcal{P}$ is generated by $(2\pi, \pm 2\pi)$ and a shift of 2π in φ_+ alone is not a symmetry. Correspondingly, $n_\pm = (m_1 \pm m_2)/2$ take half-integer values with an integer sum.

We choose compact coordinates in which $\mathcal{P}$ is the standard lattice, so that Equation (2.16) holds axis by axis. This choice is a convention rather than a requirement. Other compact coordinates are valid, but the charge basis must then be rebuilt from the original integer charges rather than taken as a product of independent rotors. To identify these coordinates, select N_c junctions $\mathcal{K}$ whose rows of $\tilde{\omega}_{J,c}^T$ are linearly independent and replace the compact columns of Z by

$$Z^c \to Z^c \left(\tilde{\omega}_{J,\mathcal{K}}^T\right)^{-1}. \tag{2.17}$$

Each selected junction now depends on a single 2π-periodic compact phase φ_j. The conjugate

Cooper-pair number $\hat{n}_j$ is an integer, and Equation (2.16) reduces to

$$\psi(\varphi_j + 2\pi) = \psi(\varphi_j)\,, \quad \hat{n}_j \in \mathbb{Z}\,. \tag{2.18}$$

In general, Equation (2.18) may carry a quasimomentum q_j, $\psi(\varphi_j + 2\pi) = e^{2\pi i q_j}\psi(\varphi_j)$, as for a periodic potential [59, 60]. The gauge transformation $\psi \to e^{-iq_j\varphi_j}\psi$ restores the periodic boundary condition and shifts $\hat{n}_j \to \hat{n}_j + q_j$, so q_j combines with the explicit offset $\tilde{n}_{g,j}$ from $\tilde{Q}_g$ into $\tilde{n}_{g,j} - q_j$. Because q_j is static, this adds no further terms. These choices together simplify the commutator for compact modes

$$[e^{i\hat{\varphi}_i}, \hat{n}_j] = -\delta_{ij}\, e^{i\hat{\varphi}_i}\,, \tag{2.19}$$

which otherwise involves simultaneous displacements across multiple compact variables.

2.2.3.2 Time-Dependent External Flux

In the presence of time dependent external flux, extra care must be taken with the second term of Equation (2.15), which represents induced EMFs of capacitive branches. This term gives the offset charge an external flux dependence, which we saw explicitly in Equation (2.7) for the circuit of Figure 2.1. For time dependent problems with nonzero $\dot{\Phi}_{\text{ext}}$, the term must be tracked throughout. Failure to do so results in different predictions for different external flux assignments B [49, 61, 62]. Alternatively, one can work in the "irrotational gauge" [49], which choses B such that $Z^T\Omega C_b \mathrm{B} = 0$, leaving the Hamiltonian with no explicit $\dot{\Phi}_{\text{ext}}$ dependence. Beyond the lumped-element model, an accurate treatment of external flux depends further on the circuit geometry [61].

We can solve for such a B by combining the constraints $Z^T\Omega C_b \mathrm{B} = 0$ and $\mathrm{GB} = I$. They are solved by

$$\mathrm{B}^{\text{irr}} = \mathrm{G}^{+} - \Omega^T Z \tilde{C}^{-1} Z^T \Omega C_b\, \mathrm{G}^{+}\,, \tag{2.20}$$

which can also be interpreted as a shift $\mathrm{B} \to \mathrm{B}^{\mathrm{irr}} = \mathrm{G}^{+} + \Omega^{T} M$ using the freedom described in Section 2.2, for an initial $\mathrm{B} = \mathrm{G}^{+}$. This solution is unique up to $\mathrm{B}^{\mathrm{irr}} \to \mathrm{B}^{\mathrm{irr}} + \Omega^{T} W$ for W satisfying $Z^{T} C W = 0$, which shifts only the discarded non-dynamical coordinates. For example, moving the flux of the LCLC box (Figure 2.3) from L_1 to L_2 shifts nodes 1 and 2 together, $W \propto (1, 1, 0, 0)^{T}$, a combination of the frozen and cyclic columns of Equation (2.10).

Returning to the circuit of Figure 2.1 shows that the irrotational assignment can be messy even for a simple circuit. All 15 entries of $\mathrm{B}^{\mathrm{irr}}$ are nonzero, so each of Φ_x, Φ_y, and Φ_z reaches every branch. The junction phases, for example, acquire the offsets

$$\mathrm{B}_{J}^{\mathrm{irr}} \Phi_{\mathrm{ext}} = \frac{1}{C_{J_1} + C_{J_2} + C_s} \begin{pmatrix} (C_{J_2} + C_s)\Phi_x - C_s \Phi' \\ -C_{J_1} \Phi_x - C_s \Phi' \end{pmatrix},$$

where $C_s = C_c(C_r + C_d)/(C_c + C_r + C_d)$ is the capacitance from node 1 to ground through the coupling capacitor and $\Phi' = \Phi_y + C_d \Phi_z / (C_r + C_d)$. The SQUID flux is split by a capacitive voltage divider, in which C_s adds to C_{J_2} because the coupling path and J_2 bound loop y. If the capacitively broken loops enclose no flux and $C_s \ll C_{J_2}$, this reduces to the standard ratio of junction capacitances [49]. The capacitively broken loops instead shift both junction phases equally, acting as an effective drive that for loop z is equivalent to replacing V by $V - \dot{\Phi}_z$. The inductor also takes a share of every flux, so the SQUID flux drives the resonator directly even without flux in the capacitively broken loops. Such distributed effects, in which a time-dependent flux through one loop shifts the phases of elements elsewhere in the circuit, have recently been studied as inherent flux crosstalk [63].

It should be noted that only the time-dependent part of the flux needs this assignment. Splitting $\Phi_{\mathrm{ext}}(t) = \Phi_{\mathrm{ext}}^{\mathrm{DC}} + \Phi_{\mathrm{ext}}^{\mathrm{AC}}(t)$ into a static DC part and a time-dependent AC part, the two parts may be assigned separately in Equation (2.2),

$$\mathrm{B} \Phi_{\mathrm{ext}}(t) \;\to\; \mathrm{B}^{\mathrm{DC}} \Phi_{\mathrm{ext}}^{\mathrm{DC}} + \mathrm{B}\, \Phi_{\mathrm{ext}}^{\mathrm{AC}}(t)\,, \tag{2.21}$$

with $\mathrm{GB}^{\mathrm{DC}} = \mathrm{GB} = I$ forming two independent requirements. Only B reaches the EMF term

of Equation (2.15), so the irrotational condition constrains it alone. The DC part may be freely assigned to simplify the Hamiltonian. The AC part includes any time dependence, such as drives or flux noise, and need not be small or sinusoidal.

As an example, consider a fluxonium, or a junction shunted by an inductor L between a single node Φ and ground. With branches ordered (J, L), the incidence and loop matrices are $\Omega = (1, 1)$ and $\mathrm{G} = (1, -1)$. Equation (2.20) gives the capacitive divider $\mathrm{B}^{\mathrm{irr}} = (C_L, -C_J)^T/(C_J + C_L)$, where C_L is the capacitance of the inductor branch. For $C_L = 0$ this is $\mathrm{B}^{\mathrm{irr}} = (0, -1)^T$, so the time-dependent flux sits entirely on the inductor. At the half-flux point $\Phi_{\mathrm{ext}}^{\mathrm{DC}} = \Phi_0/2$, we can instead assign the DC flux to the junction, $\mathrm{B}^{\mathrm{DC}} = (1, 0)^T$, which flips the sign of the junction cosine, giving a potential of

$$\mathcal{U} = \frac{\Phi^2}{2L} + E_J \cos\left(2\pi\frac{\Phi}{\Phi_0}\right) - \frac{\Phi}{L}\Phi_{\mathrm{ext}}^{\mathrm{AC}}(t)\,.$$

The static potential is then even in Φ, with a double well at $2\pi\Phi/\Phi_0 = \pm\pi$, and the drive entering as a linear coupling to Φ. Assigning the DC flux to the inductor describes the same physics, but it centers the double well at $\Phi = \Phi_0/2$, rather than at zero.

2.3 Numerical Modeling of Circuit QED Systems

Section 2.2 ends with a Hamiltonian written in compact, extended, and harmonic coordinates. Beyond a single harmonic mode, few such Hamiltonians can be diagonalized analytically, so their spectra and dynamics are computed numerically. This section summarizes how we do so rather than surveying numerical methods, and packages such as scqubits and SQcircuit automate these steps [43–45, 64]. The mode type fixes the basis: an extended or harmonic mode is represented on a discretized phase grid or in a truncated Fock basis, while a compact mode is represented in its integer charge basis or on a periodic phase grid. **??** treats single modes and their convergence, and **??** combines modes by tensor products and hierarchical diagonalization. How fast these truncations converge depends on the coordinates chosen in Section 2.2.2, which motivates the transformations of Section 2.4.

Extended and harmonic modes are real-valued and typically modeled represented on a discretized phase grid or in a truncated Fock basis. These bases are chosen because in the infinite limit they span possible square-normalizable eigenfunctions $\psi(x)$, such that, $\int_{-\infty}^{\infty} dx|\psi(x)|^2 = 1$.

The discrete phase basis is built of idealized phase eigenstates that obey

$$\hat{n}\,|\phi\rangle = -i\frac{\partial}{\partial\phi}\,|\phi\rangle\,, \quad \hat{\phi}\,|\phi\rangle = \phi\,|\phi\rangle\ . \tag{2.22}$$

In practice the basis is constructed by sampling at a set of points $\{\phi_1, \phi_2, \ldots, \phi_n\}$ with a fixed finite difference $\phi_i - \phi_{i-1} = \Delta\phi$ between adjacent points. Care must be taken to extend the bounds and sample densely enough to accurately represent the eigenstates of the system. Operators ϕ, $\cos\phi$ are then represented as diagonal matrices, with $\hat{n}$ given by a finite difference derivative. For example, the standard three-point formula gives $\hat{n}^2 = -\partial_\phi^2 \approx (2\psi_i - \psi_{i+1} - \psi_{i-1})/\Delta\phi^2$, which is tridiagonal in the phase basis. Higher-order stencils are given in Table 2.3. In Listing A.1, numpy code is provided to diagonalize a fluxonium qubit with Hamiltonian $\hat{H} = 4E_C\hat{n}^2 - E_J\cos\hat{\phi} + \frac{1}{2}E_L(\hat{\phi} + \phi_{\text{ext}})^2$.

Another common choice a the harmonic oscillator basis of the quadratic part of the Hamiltonian, $H_0 = 4E_C\hat{n}^2 + \frac{1}{2}E_L\hat{\phi}^2$. We define creation and annihilation operators,

$$\hat{n} = in_{\text{ZPF}}(\hat{a}^\dagger - \hat{a})\,, \quad \hat{\phi} = \phi_{\text{ZPF}}(\hat{a}^\dagger + \hat{a}), \quad \phi_{\text{ZPF}} = \left(\frac{2E_C}{E_L}\right)^{1/4}, \quad n_{\text{ZPF}} = \left(\frac{E_L}{32E_C}\right)^{1/4}, \tag{2.23}$$

so that $H_0 = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right)$ and $\hbar\omega = \sqrt{8E_CE_L}$. In the fock basis $|m\rangle$, $a^\dagger\,|m\rangle = \sqrt{m+1}\,|m+1\rangle$ and $a\,|m\rangle = \sqrt{m}\,|m-1\rangle$, which are truncated at the top of the ladder. The cosine is written using displacement operator $D(\alpha) = e^{\alpha a^\dagger - \alpha^* a}$. Since $e^{i\phi} = e^{i\phi_{\text{ZPF}}(a^\dagger + a)} = D(i\phi_{\text{ZPF}})$, we have

$$\cos(\phi - \phi_{\text{ext}}) = \tfrac{1}{2}\left[e^{-i\phi_{\text{ext}}}D(i\phi_{\text{ZPF}}) + e^{i\phi_{\text{ext}}}D(-i\phi_{\text{ZPF}})\right]\ . \tag{2.24}$$

For the Hamiltonian above the flux is in the inductive term, so the cosine has $\phi_{\text{ext}} = 0$ and the phase factors drop out. The matrix elements of $D(\alpha)$ are known in closed form, for $n \geq m$,

$\langle n|\, D(\alpha)\, |m\rangle = \sqrt{\frac{m!}{n!}}\, \alpha^{n-m} e^{-|\alpha|^2/2} L_m^{(n-m)}\big(|\alpha|^2\big)$, with associated Laguerre polynomials $L_m^{(k)}$, and the elements with $n < m$ follow from $D(\alpha)^\dagger = D(-\alpha)$ [65]. Listing A.2 diagonalizes the same fluxonium Hamiltonian in a truncated Fock basis. The listing builds $D(\alpha)$ by exponentiating the truncated a and $a^\dagger$ rather than from the closed form, so its matrix elements differ from the closed form near the cutoff.

Both bases must be truncated, so we study how the lowest four eigenvalues converge. As a reference we use the Fock basis with 140 levels, which is converged to $\sim 10^{-13}$. In the Fock basis, the error falls exponentially with the number of levels until it reaches machine precision. In the phase basis, the error is set by the finite-difference approximation of $n^2 = -\partial_\phi^2$. A central stencil of order $p = 2k$ uses the $2k+1$ grid points nearest to ϕ_i, with error $\mathcal{O}(\Delta\phi^p)$. The stencil and its weights w_j are given in Table 2.3 [66]. Listing A.3 compares the orders 2, 4, 6 and 8. Figure 2.5c shows that the error scales as $\Delta\phi^p$ for a stencil of order p, until it reaches the floor set by the finite phase range.

$$\partial_\phi^2 \psi(\phi_i) \approx \frac{1}{\Delta\phi^2}\left[w_0\psi_i + \sum_{j=1}^{k} w_j\left(\psi_{i+j} + \psi_{i-j}\right)\right]$$

p	w_0	$w_{\pm1}$	$w_{\pm2}$	$w_{\pm3}$	$w_{\pm4}$
2	-2	1			
4	$-\frac{5}{2}$	$\frac{4}{3}$	$-\frac{1}{12}$		
6	$-\frac{49}{18}$	$\frac{3}{2}$	$-\frac{3}{20}$	$\frac{1}{90}$	
8	$-\frac{205}{72}$	$\frac{8}{5}$	$-\frac{1}{5}$	$\frac{8}{315}$	$-\frac{1}{560}$

Table 2.3: Central finite-difference stencil of order $p = 2k$ for the second derivative and its weights w_j, with error $\mathcal{O}(\Delta\phi^p)$ [66]. The weights are symmetric, $w_{-j} = w_j$.

Similarly, for compact systems, we require a basis that spans possible 2π periodic wavefunctions. The typical choice is the Fourier dual charge basis $|n\rangle$,

$$|n\rangle = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} d\phi\, e^{in\phi}\, |\phi\rangle\,, \quad |\phi\rangle = \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} e^{-in\phi}\, |n\rangle\,, \tag{2.25}$$

which takes on integer values for each mode if coordinates are chosen as in Section 2.2.3. In the

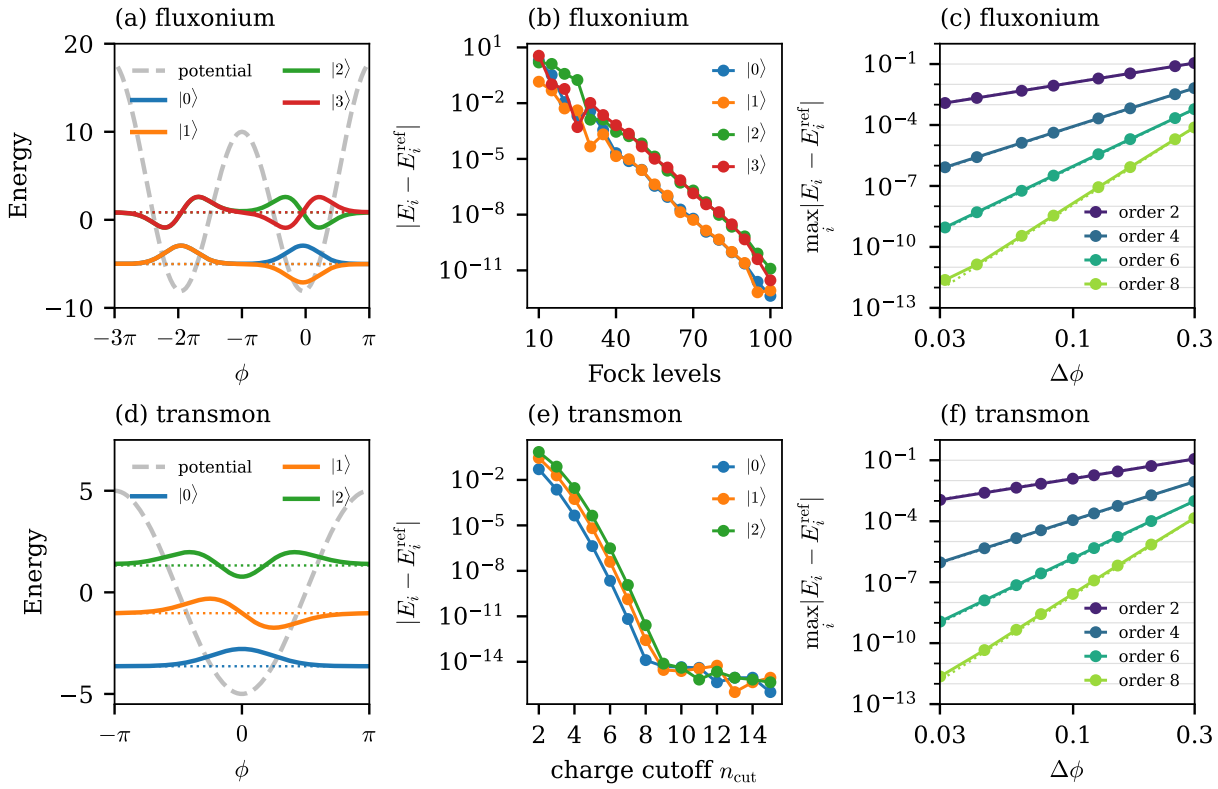


Figure 2.5: Numerical diagonalization of a fluxonium (top) with $E_J = 10$, $E_C = 0.5$, $E_L = 0.4$ and $\phi_{\mathrm{ext}} = \pi$, and a transmon (bottom) with $E_J = 5$ and $E_C = 0.2$. (a,d) Potential, lowest eigenvalues and wavefunctions in the phase basis, from Listing A.1 (201 grid points) and Listing A.5 (160 grid points, stencil of order 8, $n_g = 0$). (b) Error of the lowest four eigenvalues in the Fock basis versus the number of levels, relative to a 140-level reference. (e) Error of the lowest three eigenvalues in the charge basis (Listing A.4) versus the charge cutoff n_{cut}, relative to $n_{\mathrm{cut}} = 40$, at $n_g = 0.25$. (c,f) Largest eigenvalue error in the phase basis versus grid spacing, for finite-difference stencils of order 2 to 8 (Listing A.3), with $n_g = 0.25$ for the transmon. Dotted lines show $\Delta\phi^p$ for each order p.

charge basis, the charge operator n is diagonal, and the $\cos\phi$ operator is a sum of translations,

$$e^{i\hat{\phi}} = \int d\phi\, e^{i\phi}\, |\phi\rangle\langle\phi| = \sum_{n=-\infty}^{\infty} |n+1\rangle\langle n| \,. \tag{2.26}$$

It is possible to use a 2π periodic phase basis directly. Here the operators are qualitatively similar to the continuous case. The $\cos\phi$ operator is diagonal and the n operator is a finite difference derivative. The offset charge can enter in two equivalent ways. One is to impose a twisted

boundary condition $\psi(\phi + 2\pi) = e^{i2\pi q}\psi(\phi)$ on the wavefunction, with a quasimomentum q. Here the derivative wraps around with an additional phase depending on which side of the domain is being considered. The other is to gauge the twist out of the wavefunction and into the Hamiltonian. Writing $\psi(\phi) = e^{iq\phi}\chi(\phi)$, the function χ is 2π periodic, and the charge operator $n = -i\partial_\phi$ acts as $n\,\psi = -i\partial_\phi\left(e^{iq\phi}\chi\right) = e^{iq\phi}\,(n+q)\,\chi$. A Hamiltonian $4E_C(n - n_g)^2$ with twist q is thus the periodic problem with offset charge $n_g - q$. Here the finite difference derivative wraps around without any additional phases, and the whole offset charge appears explicitly in the Hamiltonian.

In Listing A.4 and Listing A.5 we show numpy code to diagonalize the transmon Hamiltonian $\hat{H} = 4E_C(\hat{n} - n_g)^2 - E_J\cos(\hat{\phi})$ in the charge and periodic phase bases. The charge basis exhibits much faster convergence with the number of states included, and the phase basis solution is limited by the error of the finite difference derivative (Figure 2.5).

An isolated transmon, fluxonium, or resonator is a single quantum mode, but we require methods to consider the multi-mode circuits quantized in the previous section. As in standard quantum mechanics, we can combine two or more Hilbert spaces via a tensor product $\otimes$. For example two systems H_1, H_2 can be represented by $H = H_1 \otimes I + I \otimes H_2 + g_1 \otimes g_2$, where $g_1 \otimes g_2$ captures any coupling between the two systems. The cost to naively represent a composite system grows with the product of the individual basis sizes.

A technique used to mitigate this cost is hierarchical diagonalization, which is available in scqubits [43, 44]. Rather than represent the full tensor product basis, each individual subsystem is diagonalized individually, and the coupling operator is then expressed in the eigenbases of the individual modes. The full Hamiltonian is then

$$H = \mathrm{diag}(E_0^{(1)}, E_1^{(1)}, \ldots, E_{n-1}^{(1)}) \otimes I + I \otimes \mathrm{diag}(E_0^{(2)}, E_1^{(2)}, \ldots, E_{m-1}^{(2)}) + V_1^\dagger g_1 V_1 \otimes V_2^\dagger g_2 V_2\,, \quad (2.27)$$

with $E_i^{(1)}$, V_1 and $E_i^{(2)}$, V_2 the eigenvalues and column-stacked eigenvector matrix of the 1st and 2nd system, respectively. Rather than keep all eigenvalues/states, only the lowest n and m are kept. The matrices V_1 and V_2 are thus rectangular with n and m columns, and the inverse is replaced by

the conjugate transpose (the two are equivalent if all eigenvectors are kept). The reduced size basis can still accurately represent lower energy dynamics in the composite system, although convergence must be studied on a case by case basis to ensure that enough states are being retained. Listing A.6 implements hierarchical diagonalization for a capacitively coupled fluxonium and transmon. Kerman [67] provides a study on convergence and efficient hierarchical diagonalization techniques.

2.4 Mode-Preserving Coordinate Transformations

The construction of Section 2.2.2 fixes the retained coordinates only up to an invertible change of basis among the dynamical modes. This section characterizes that freedom, identifies the compact block of the transformation as discrete, and poses the choice of basis as an optimization. Sections 2.4.1 and 2.4.2 solves the problem both symbolically and numerically, illustrated by three tasks.

To characterize the remaining coordinate freedom, we ask which invertible transformation S keeps the existing dynamical coordinates as compact, extended, and harmonic? Following the index sets of Section 2.2.3.1, we write the transformation as a 3×3 block matrix, S_{xy}, with $x, y \in \{\mathrm{c}, \mathrm{e}, \mathrm{h}\}$. Block multiplication gives the junction content of the new harmonic columns and the inductive content of the new compact columns,

$$\left(\tilde{\omega}_J^T S\right)_{\mathrm{h}} = \tilde{\omega}_{J,\mathrm{c}}^T S_{\mathrm{ch}} + \tilde{\omega}_{J,\mathrm{e}}^T S_{\mathrm{eh}}\,,$$
$$\left(\tilde{\omega}_L^T S\right)_{\mathrm{c}} = \tilde{\omega}_{L,\mathrm{e}}^T S_{\mathrm{ec}} + \tilde{\omega}_{L,\mathrm{h}}^T S_{\mathrm{hc}}\,.$$

A harmonic coordinate must carry no junction term and a compact coordinate no inductive term, so both expressions must vanish. This forces $S_{\mathrm{ch}} = S_{\mathrm{eh}} = 0$ and $S_{\mathrm{ec}} = S_{\mathrm{hc}} = 0$. Extended coordinates already carry every term, so nothing restricts what they absorb. The general type-preserving

transformation is therefore

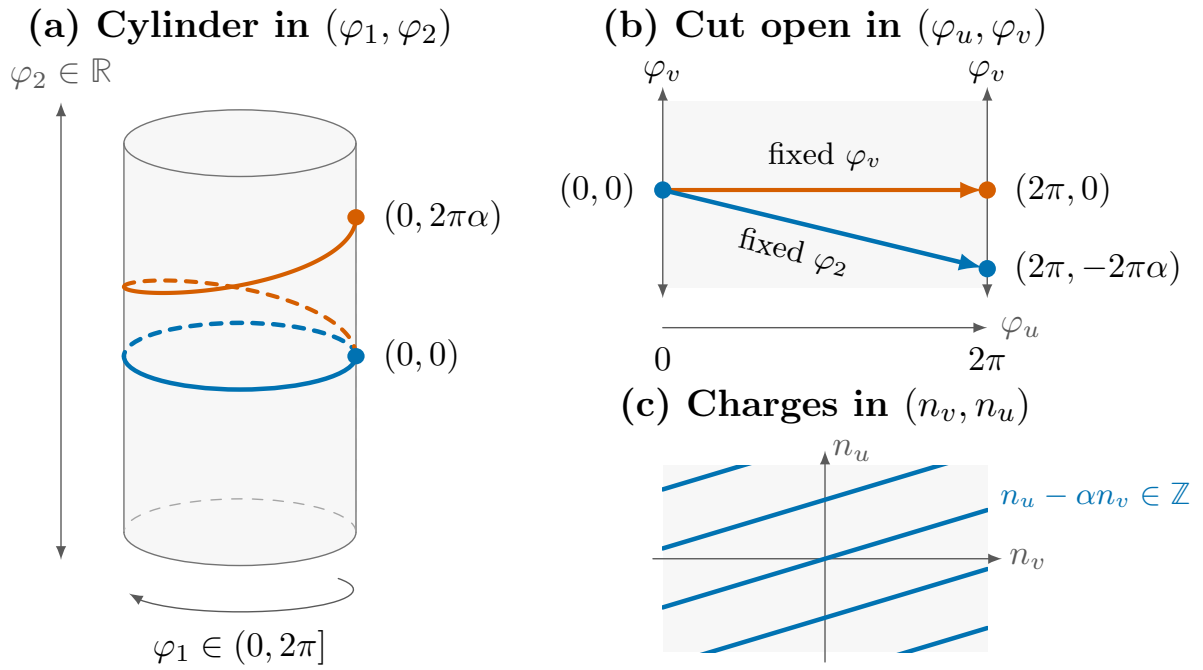
$$S = \begin{pmatrix} S_{\mathrm{cc}} & S_{\mathrm{ce}} & 0 \\ 0 & S_{\mathrm{ee}} & 0 \\ 0 & S_{\mathrm{he}} & S_{\mathrm{hh}} \end{pmatrix} ,$$
$$S_{\mathrm{relaxed}} = \begin{pmatrix} S_{\mathrm{cc}} & S_{\mathrm{ce}} & S_{\mathrm{ch}} \\ 0 & S_{\mathrm{ee}} & S_{\mathrm{eh}} \\ 0 & S_{\mathrm{he}} & S_{\mathrm{hh}} \end{pmatrix} \equiv \begin{pmatrix} S_{\mathrm{cc}} & S_{\mathrm{cn}} \\ 0 & S_{\mathrm{nn}} \end{pmatrix} , \tag{2.28}$$

with rows and columns of S ordered (compact, extended, harmonic). Its determinant is $\det S_{\mathrm{cc}} \det S_{\mathrm{ee}} \det S_{\mathrm{hh}}$, so S is invertible if and only if its three diagonal blocks are. S_{relaxed} combines the extended and harmonic blocks into the noncompact block $\mathrm{n} = \mathrm{e} \oplus \mathrm{h}$ and allows $S_{\mathrm{ch}}, S_{\mathrm{eh}} \neq 0$. This still gives a valid set of coordinates, although it gives the harmonic coordinates junction content and relabels them extended. This relabeling is innocuous, as extended and harmonic variables are both real valued variables.

The compact block columns are more restricted, and Figure 2.6 shows what can go wrong by relaxing the zeros in the first column. Consider $S_{\mathrm{hc}} = \alpha$ for the example of Figure 2.1. This gives the transformed inductive incidence matrix $\tilde{\omega}_L^T S = \begin{pmatrix} \alpha & 1 \end{pmatrix}$, which has inductive content in the compact column. The new coordinates follow from S^{-1} and their conjugate momenta from S^T,

$$\begin{pmatrix} \varphi_u \\ \varphi_v \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\alpha & 1 \end{pmatrix} \begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix} , \quad \begin{pmatrix} n_u \\ n_v \end{pmatrix} = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} ,$$

so φ_u enters the inductive energy of L_r through $\varphi_2 = \varphi_v + \alpha\varphi_u$. The periodic boundary condition becomes $\psi(\varphi_u, \varphi_v) = \psi(\varphi_u + 2\pi, \varphi_v - 2\pi\alpha)$, and the quantized charge of the compact mode sits in $n_1 = n_u - \alpha n_v$. Like in Section 2.2.3.1, the variable φ_u cannot be treated as an independent rotor. Such coordinates can still be quantized with careful bookkeeping, but $\mathcal{A}$ excludes them so that Equation (2.18) continues to hold axis by axis. The remaining entry of the compact column, S_{cc}, corresponds to a different choice of linearly independent rows in Equation (2.17), so S_{cc} has a set of discrete choices.



$$\varphi_1 = \varphi_u, \qquad \varphi_2 = \varphi_v + \alpha\varphi_u, \qquad \psi(\varphi_u, \varphi_v) = \psi(\varphi_u + 2\pi, \varphi_v - 2\pi\alpha)$$

Figure 2.6: S_{hc} **twists the compact boundary condition.** For the circuit of Figure 2.1, the transmon phase φ_1 is periodic and the resonator phase φ_2 is real, so the configuration space is a cylinder (a). The blue path makes one turn at fixed φ_2 and closes at $(0, 0)$. In the coordinates $\varphi_1 = \varphi_u$, $\varphi_2 = \varphi_v + \alpha\varphi_u$ (b), a turn at fixed φ_v (orange) ends at the distinct point $(2\pi, 0)$, which is $(0, 2\pi\alpha)$ on the cylinder. (c) The transmon's integer charge is $n_u - \alpha n_v$, so the allowed charges lie on tilted lines while n_v remains continuous.

The admissible set of transformations $\mathcal{A}$ therefore consists of Equation (2.28), with $S_{\text{cc}} \in \left(\tilde{\omega}^T_{J,\mathcal{K}}\right)^{-1}$ derived from a linearly independent set of rows $\mathcal{K}$ and S_{ee}, S_{hh} invertible. The set $\mathcal{A}_{\text{relaxed}}$ is the same with S_{ch} and S_{eh} free.

We now frame the specific choice of S as an optimization. We aim to identify an optimal $S^\star \in \mathcal{A}$ or $\mathcal{A}_{\text{relaxed}}$ that minimizes a cost $\mathcal{F}$

$$S^\star = \arg\min_{S \in \mathcal{A}} \mathcal{F}\big(\mathcal{H}_S;\ \lambda\big)\,, \tag{2.29}$$

where $\mathcal{H}_S$ is the Hamiltonian in the coordinates defined by S, parametrized by circuit parameters λ.

It depends on S through $S^T\tilde{C}S$, $S^T\tilde{L}^*S$, and $\tilde{\omega}_J^T S$, together with the transformed offsets $S^T\tilde{Q}_g$ and $S^T\tilde{\omega}_L L_b^{-1}\mathrm{B}_L\Phi_{\mathrm{ext}}$. A cost may act on these matrices directly, or on quantities that require solving for the states of $\mathcal{H}_S$. The accompanying code [68] takes $\mathcal{F}$ as a user-supplied function. The two routes below differ in whether λ is kept symbolic and parameter independent.

2.4.1 Parameter-Independent Symbolic Search

The first route keeps the circuit parameters $\lambda = \{C_k, L_k, E_{J,k}, n_g, \Phi_{\mathrm{ext}}\}$ symbolic and seeks a coordinate transformation $S^\star \in \mathcal{A}$ (or $\mathcal{A}_{\mathrm{relaxed}}$) whose entries are rational numbers independent of λ. The dynamical coordinates are then fixed by the circuit topology alone.

Every element of the transformed circuit matrices is a polynomial in the transformation entries and circuit parameters, which limits what can be solved. The only conditions that hold irrespective of circuit parameters must exclusively constrain the transformation entries. We represent these conditions as a set of polynomial equalities $\mathcal{T}$, for which any given S fulfills a subset $\mathcal{G}(S) \subseteq \mathcal{T}$. The symbolic route solves Equation (2.29) for any cost of the form $\mathcal{F}\big(\mathcal{G}(S)\big)$, which depends on the subset of conditions fulfilled. Because there are finitely many choices for the compact block S_{cc} (Equation (2.17)), we iterate over each option and search the remaining blocks.

For a fixed S_{cc}, a condition requires some expression $E(\mathbf{s}, \lambda)$ to vanish for all λ, where E is built from S and the circuit matrices by products and inverses, such as an entry of $(S^T\tilde{C}S)^{-1}$ or of the junction matrix. Such an expression is a rational function of the unknown entries $\mathbf{s}$ of the remaining blocks and of λ. Clearing denominators and expanding the numerator in monomials of the circuit parameters, $\mathrm{num}(E) = \sum_\alpha f_{E,\alpha}(\mathbf{s})\lambda^\alpha$ vanishes for all λ exactly when every coefficient $f_{E,\alpha}(\mathbf{s})$ does. We therefore allow in $\mathcal{T}$ any condition that reduces to a polynomial system in $\mathbf{s}$ alone.

A pattern $\mathcal{Z} \subseteq \mathcal{T}$ is feasible when its systems have a common solution, and we structure a search to find the highest scoring pattern. Candidate patterns are considered in non-decreasing order of $\mathcal{F}$, skipping any that contain a subset already found to be infeasible. The search stops at the first level where a feasible solution is found.

The solutions of a feasible pattern generally form under-constrained systems of polynomials, which we solve using algebraic geometry techniques. Using `Singular` [69], we decompose each into irreducible components and compute its Gröbner cover [70], which parameterizes every valid transformation by a maximal independent set of free entries $\mathbf{s}_{\text{free}}$. The remaining entries are rational functions of them. Every choice of $\mathbf{s}_{\text{free}}$ fulfills the same pattern and obtains the same cost. Any ties under $\mathcal{F}$, or decisions on free entries, are left to the task. Section 2.4.1.1 fixes both for the case of circuit enumeration.

2.4.1.1 Application: Hamiltonian classes for circuit enumeration

As an example of the symbolic search, we consider circuit enumeration, or generating every distinct circuit built from a fixed set of elements, such as capacitors, inductors, and Josephson junctions. In Ref. [71], we cataloged these circuits by writing down one representative set of coordinates per circuit, chosen by the same rule for every circuit, so that the Hamiltonians of different circuits could be compared directly. The functional form of the resulting Hamiltonian, with the parameter values stripped, served as a "Hamiltonian class." Here we place that procedure concretely within the admissible set $\mathcal{A}$, as the minimizer of a pattern cost.

The cost is a two stage procedure that seeks to minimize intermode coupling. The targets of the first stage are the off-diagonal entries of the junction matrix $\sum_k E_{J,k}(\tilde{\omega}_{J,k}^T S)^T(\tilde{\omega}_{J,k}^T S)$, which are nonzero exactly when any junction phase involves both modes. This stage maximizes the number of vanishing entries. The second stage takes the off-diagonal entries of $S^T\tilde{L}^*S$ and $(S^T\tilde{C}S)^{-1}$ and again maximizes the number of zero entries, subject to the constraints imposed by the first stage. Among ties, fewer inductive couplings are chosen.

The winning family still has free entries $\mathbf{s}_{\text{free}}$, which the pattern cost cannot see. They set the phase arguments of the extended modes, so they must be fixed the same way for every circuit to give a consistent class label. We fix them by a second, discrete optimization that requires integer junction coefficients on the extended columns (e.g., $\cos(2\varphi)$ rather than $\cos(1.45\varphi)$) and minimizes

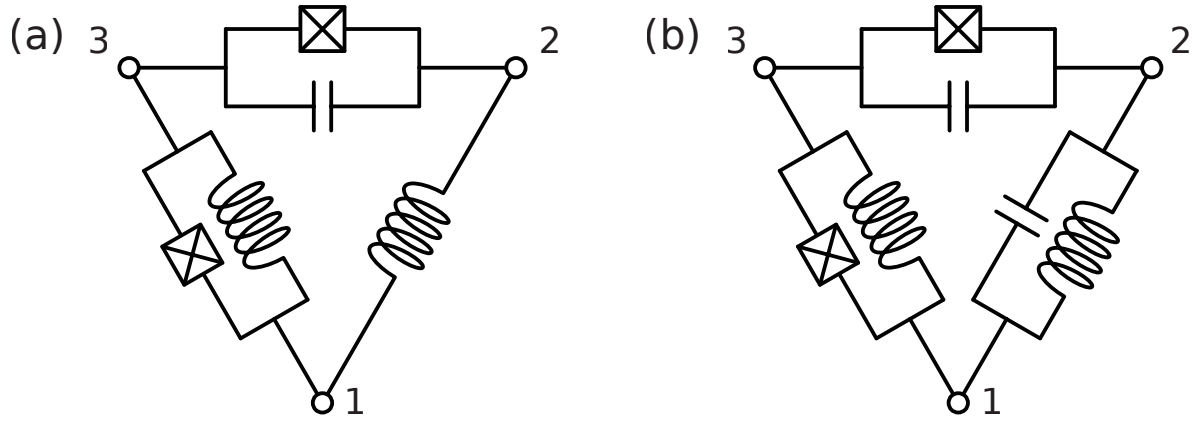


Figure 2.7: **The Hamiltonian class depends on the cost.** Circuit (b) adds a capacitor C_{12} in parallel with the inductor between nodes 1 and 2 of circuit (a). Adapted from Ref. [71].

their ℓ_1 norm. We solve this optimization using the `Z3` SMT solver [72] to obtain a provably optimal answer. Free parameters absent from $\tilde{\omega}_J^T S$, such as harmonic scalings in S_{hh}, take the simplest rational values that leave S invertible, yielding a concrete $S^\star$.

As a worked example, consider the circuits of Figure 3.7, with node 3 as reference and node coordinates $\Phi_1 - \Phi_3$ and $\Phi_2 - \Phi_3$. Both coordinates are extended, so any invertible $S = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is admissible. Each junction acts on a single node coordinate, so $\tilde{\omega}_J^T S = S$ and the off-diagonal entry of the junction matrix is $E_{J,31}\, ab + E_{J,32}\, cd$. It vanishes for all parameters exactly when $ab = cd = 0$. The second-stage targets multiply out to

$$\begin{aligned} \left(S^T \tilde{L}^* S\right)_{12} &= \frac{ab}{L_{31}} + \frac{(a-c)(b-d)}{L_{21}}\,, \\ \left(S^T \tilde{C} S\right)_{12} &= C_{31}\, ab + C_{32}\, cd + C_{12}\, (a-c)(b-d)\,, \end{aligned}$$

where C_{31} and C_{32} include the junction capacitances and $C_{12} = 0$ in circuit (a). With two modes, the off-diagonal entry of $(S^T \tilde{C} S)^{-1}$ is proportional to $(S^T \tilde{C} S)_{12}$, so both give the same target. Collecting the coefficient of each parameter, inductive decoupling requires $ab = 0$ and $(a-c)(b-d) = 0$, while charge decoupling requires $ab = cd = 0$, together with $(a-c)(b-d) = 0$ in circuit (b). The first stage fixes $ab = cd = 0$, so S is diagonal up to a swap of columns and keeps the node coordinates. The

inductive entry is then $-ad/L_{21} \neq 0$, while the charge entry $-ad\,C_{12}$ vanishes only in circuit (a). The resulting classes are $q_1^2 + q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos\varphi_1 + \cos\varphi_2$ for (a), with an additional $q_1 q_2$ for (b). Inductive decoupling instead forces $b = 0$ and $c = a$, which places junction 32 on both coordinates, so the two patterns are mutually exclusive. A cost that ranks the inductive entry first selects this family, and the integer-coefficient step sets $a = 1$ and $d = \pm 1$, and with $d = -1$ the second coordinate is $\Phi_1 - \Phi_2$. Both circuits then fall into the same class, $q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_2^2 + \cos(\varphi_1 - \varphi_2) + \cos\varphi_1$, trading the linear coupling $\varphi_1\varphi_2$ for a nonlinear one.

2.4.2 Numerical Search

The symbolic route sees only conditions that hold for every parameter value, but transformations that depend on circuit parameter values can be very useful. For such tasks we can fix λ at numerical values and minimize $\mathcal{F}$ over the continuous blocks of S for each discrete choice of the compact block.

To optimize over invertible matrices, we use the polar decomposition to parametrize $\mathcal{A}$ with unconstrained real variables [73, 74] . The polar decomposition writes any real invertible matrix D as $D = QP$, with Q orthogonal and P symmetric positive definite. Each factor is the exponential of a simpler matrix. Any symmetric positive-definite $P = e^B$ for a unique symmetric B, and any orthogonal Q with $\det Q = +1$ equals e^A for some antisymmetric A. Every D with $\det D > 0$ can therefore be written as $D = e^A e^B$, and negative determinants can be reached by flipping the sign of a column. Antisymmetry forces the diagonal of A to vanish, so the free entries are those of A above the diagonal and of B on and above it. For each discrete choice of S_{cc}, we thus substitute

$$S_{\mathrm{ee}} = e^{A_{\mathrm{ee}}} e^{B_{\mathrm{ee}}},\ S_{\mathrm{hh}} = e^{A_{\mathrm{hh}}} e^{B_{\mathrm{hh}}},\ S_{\mathrm{nn}} = e^{A_{\mathrm{nn}}} e^{B_{\mathrm{nn}}} \tag{2.30}$$

into Equation (2.28) and leave the remaining blocks other than S_{cc} unconstrained. The optimization is then over the independent entries of $(A_{\mathrm{ee}}, B_{\mathrm{ee}}, A_{\mathrm{hh}}, B_{\mathrm{hh}}, S_{\mathrm{ce}}, S_{\mathrm{he}})$ for $\mathcal{A}$, or of $(A_{\mathrm{nn}}, B_{\mathrm{nn}}, S_{\mathrm{cn}})$ for $\mathcal{A}_{\mathrm{relaxed}}$. Any cost $\mathcal{F}$ that returns a real scalar can be used, and it can be minimized with any

standard optimization routine.

One thing to be aware of is that coordinates found by the numerical search depend on the circuit parameters, so they require care when those parameters change in time. As for a standard rotating frame, the unitary $U(t)$ implementing it adds the generator term $-i\hbar\, U^\dagger(t)\,\partial_t U(t)$ to the transformed Hamiltonian, which must be kept or avoided by fixing $S^\star$ at a single operating point. The irrotational assignment of Section 2.2.3.2 is an instance of the same construction. Changing B is a time-dependent translation of the node fluxes, and the irrotational choice is the one whose generator term cancels the explicit $\dot{\Phi}_{\text{ext}}$ dependence of the offset charge.

As a concrete example, consider normal mode diagonalization based on Williamson's theorem [75], which has been widely used in superconducting circuits to identify coordinates that remove all quadratic coupling in the Hamiltonian [45, 47, 58, 67, 76–81]. We can express the transformation as the minimum of a cost on the residual coordinate freedom. For $C' = S^T \tilde{C} S$ and $L' = S^T \tilde{L}^* S$, the normal mode transformation is the minimum of

$$\mathcal{F}_{\text{NM}}(S) = \sum_{\substack{i<j \\ j\notin \text{c}}} \left(C'_{ij}\right)^2 + \sum_{\substack{i<j \\ \{i,j\}\in \text{n}}} \left(L'_{ij}\right)^2 . \tag{2.31}$$

The first sum omits the compact column blocks, and the second sum considers only the non-compact modes.

Of course, this solution can also be obtained analytically. In compact/noncompact blocks, choose $S_{\text{cn}} = -\tilde{C}_{\text{cc}}^{-1}\tilde{C}_{\text{cn}} S_{\text{nn}}$ to remove any charge coupling between the compact and non-compact modes. In other words, this shear sets $[S^T\tilde{C}S]_{\text{ce}} = [S^T\tilde{C}S]_{\text{ch}} = 0$. The retained noncompact block of $S^T\tilde{C}S$ is then $S_{\text{nn}}^T \bar{C}_{\text{nn}} S_{\text{nn}}$, with the Schur complement $\bar{C}_{\text{nn}} = \tilde{C}_{\text{nn}} - \tilde{C}_{\text{nc}}\tilde{C}_{\text{cc}}^{-1}\tilde{C}_{\text{cn}}$. For the remaining noncompact block, S_{nn}, choose the columns to solve the eigenvalue problem $\bar{C}_{\text{nn}}^{-1}\tilde{L}_{\text{nn}}^* v_i = \lambda_i v_i$. Because S_{ch} and S_{eh} are nonzero, normal modes fall in the relaxed family of transformations. If the basis is chosen to be $\bar{C}_{\text{nn}}$-orthonormal, then both sums vanish. Conversely, a transformation satisfying $\mathcal{F}_{\text{NM}}(S) = 0$ simultaneously diagonalizes the two noncompact quadratic forms, so its

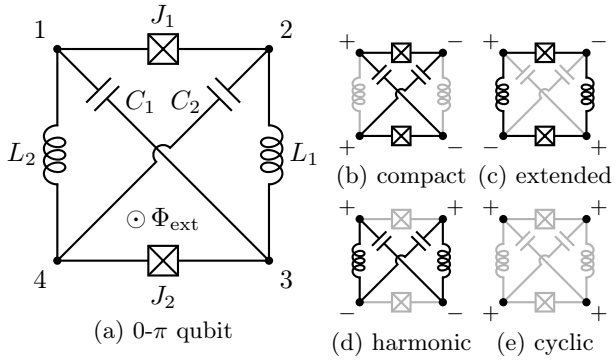


Figure 2.8: **Mode types of the 0-π qubit** [32, 35, 36]. (a) The circuit, with nodes 1–4. The diagonals carrying C_1 and C_2 cross without connecting. (b)–(e) The columns of Z in Section 2.4.2, with the sign of each node's entry. Grey branches carry no flux difference in that column, so they contribute no energy.

columns solve the generalized eigenvalue problem $\tilde{L}^*_{\text{nn}} v_i = \lambda_i \bar{C}_{\text{nn}} v_i$.

The 0-π qubit [32, 35, 36] shows when the normal modes leave $\mathcal{A}$. Its circuit (Figure 2.8) alternates the junctions J_1 (nodes 1–2) and J_2 (3–4) with the inductors L_1 (2–3) and L_2 (4–1), and the capacitors C_1 and C_2 join the opposite nodes 1–3 and 2–4. The null spaces give one compact, one extended, one harmonic, and one cyclic mode,

$$Z = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & -1 & 1 \end{pmatrix},$$

with columns in this order depicted in Figure 2.8 (b)–(e). The standard coordinates (θ, ϕ, ζ) of Refs. [32, 36] follow from $S = \text{diag}(-\frac{1}{2}, -\frac{1}{2}, \frac{1}{2})$, which lies in $\mathcal{A}$. In them, the junction phases are $\phi \pm \theta$, the inductor phases $\phi \pm \zeta$, and the capacitor phases $\theta \pm \zeta$. With identical elements, the cross terms cancel within each pair, so $\tilde{C}$ and $\tilde{L}^*$ are diagonal and (θ, ϕ, ζ) are the normal modes.

Disorder breaks this cancellation pair by pair. With the relative deviations $dX = (X_1 - X_2)/X$ of Ref. [32], dC_J couples θ and ϕ through $\tilde{C}$, dC couples θ and ζ through $\tilde{C}$, and dE_L couples ϕ and ζ through $\tilde{L}^*$, while dE_J enters neither quadratic form. To first order in the disorder, the normal

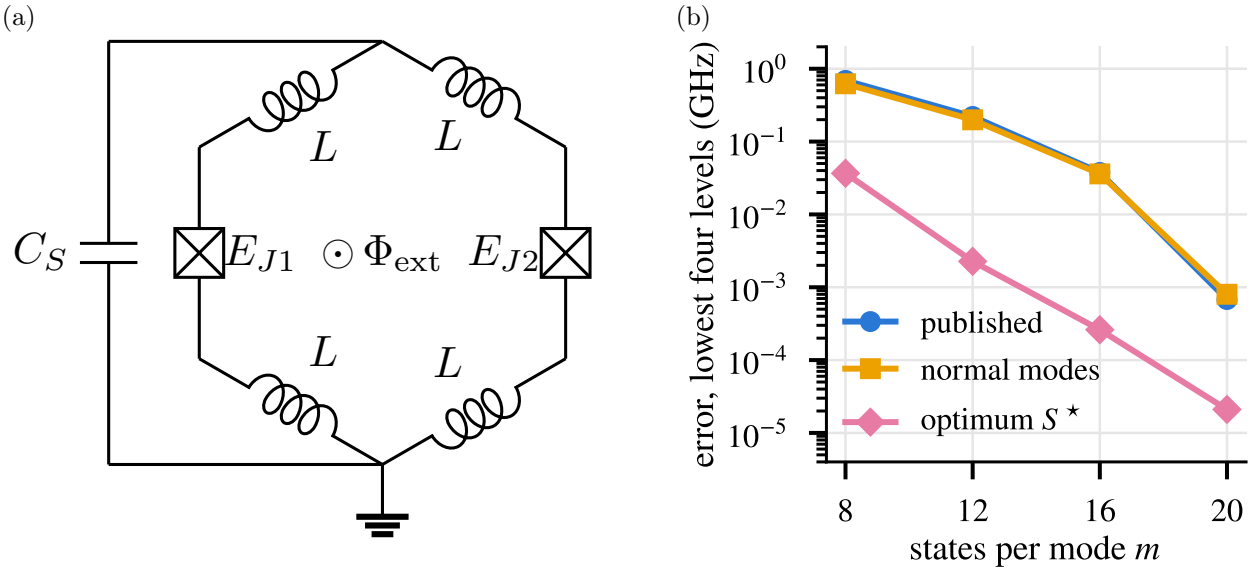


Figure 2.9: **Product-state coordinates of the** $\cos(2\varphi)$ **KITE.** (a) The $\cos(2\varphi)$ KITE circuit [34, 41]. Only the shunted node carries capacitance to ground, so three dynamical modes remain: the compact phase φ across C_S, plus the extended sum and difference $\varphi_\Sigma, \varphi_\Delta$ of the two upper inductor phases. (b) Largest error of the four lowest levels of hierarchical diagonalization with m mean-field eigenstates per mode, one set per well, in three frames. We compare the published coordinates $(\varphi, \varphi_\Sigma, \varphi_\Delta)$ of Ref. [41], the normal modes, and the product-state optimum $S^\star$ of Equation (2.32). Details of the diagonalization are given in Section B.3.

modes follow from (θ, ϕ, ζ) by

$$S = \begin{pmatrix} 1 & -\frac{C_J\, dC_J}{2(C+C_J)} & \frac{C\, dC}{2(C+C_J)} \\ 0 & 1 & -\frac{C\, dE_L}{2(C-C_J)} \\ 0 & \frac{C_J\, dE_L}{2(C-C_J)} & 1 \end{pmatrix},$$

which lies in $\mathcal{A}_{\text{relaxed}}$ if dC or dE_L are nonzero. The normal-mode ζ thus enters both junctions with a weight on the order of the disorder, so the coupling moves from the quadratic terms into the cosines.

2.4.2.1 Application: product-state coordinates

One motivation for normal modes is to make the hierarchical diagonalization of the previous section efficient. Hierarchical diagonalization solves each mode separately and then diagonalizes

the full Hamiltonian in a truncated product of the low-lying single-mode states. The truncation is accurate when the full system eigenstates are close to the products of single-mode states. Normal modes remove the quadratic coupling but leave the junction coupling terms, so for a circuit with strong nonlinear coupling, it does not generally achieve this goal.

The $\cos(2\varphi)$ KITE [34, 41] is such a circuit, and we can use an optimization over variable transformations to find an efficient basis for diagonalization. The cost we consider is the energy of a product of single-mode ground states, evaluated in the full Hamiltonian. Minimizing it over S is a proxy for how well products of single-mode basis states represent the full ground state. For a fixed S, each mode's ground state is found in a mean-field Hamiltonian, in which the operators of the other modes are replaced by their expectation values. These expectation values are updated and the single-mode problems re-solved until the product state converges. An outer optimizer then minimizes the converged energy over $S \in \mathcal{A}_{\text{relaxed}}$, parametrized as in Equation (2.30). This procedure is inspired by [82], which identifies optimized coordinates for anharmonic vibrational structures. Section B.3 gives the full construction.

For the KITE at half flux, the optimized frame gives a much better product state than either the published coordinates or the normal modes, and the gain carries over to hierarchical diagonalization. We take the parameters of Ref. [41], $E_J = 16.83$, $E_{C_J} = 4.82$, $E_{C_S} = 0.072$ GHz, an inductive energy of 2.54 GHz per inductor, and junction asymmetry $\epsilon = 0.03$, with the flux shared equally over the four inductors. The ground state is then the doublet of the two φ wells, split by 13.5 MHz. We measure the gap $E_{\text{prod}} - E_0$ and the weight of the product state outside the doublet. Both are evaluated against a reference sparse eigensolve in the published coordinates on 49 charge states for φ and 160 grid points for each extended mode, 1,254,400 states in all. The same solve in the optimized coordinates agrees with it to within 10^{-5} GHz on the four lowest levels. In the frame of Figure 2.9(a) both junctions span all three coordinates, and the product state is a poor representation of the ground state, despite the small quadratic coupling, with a gap of 1.44 GHz and 18% of the weight outside the doublet. The normal modes remove the quadratic coupling but gain

little, with a gap of 1.06 GHz and 17% of the weight outside the doublet.

The optimized coordinates greatly improve the picture. Taken relative to $(\varphi, \varphi_\Sigma, \varphi_\Delta)$ with the noncompact columns C-normalized,

$$S^\star = \begin{pmatrix} 1 & 0.24 & 0.23 \\ 0 & 0.70 & 0.73 \\ 0 & -0.57 & 0.63 \end{pmatrix}, \tag{2.32}$$

reaches a gap of 0.09 GHz with 0.5% of the weight outside the doublet. It rotates $(\varphi_\Sigma, \varphi_\Delta)$ by about 40° back toward the two arm phases and shears φ into both. Each junction depends on φ and essentially one new extended coordinate, with the arguments $\cos(\varphi - 1.03\varphi_2' + 0.13\varphi_3')$ and $\cos(\varphi + 0.11\varphi_2' - 1.13\varphi_3')$. The price is the quadratic coupling that the normal modes had removed. For hierarchical diagonalization, we keep m eigenstates of each mode's mean-field Hamiltonian, drawn from both wells as described in Section B.3, and diagonalize the full Hamiltonian in their product. This basis has m^3 states, so at $m = 20$ it holds 8000 states, about 160 times fewer than the reference solve. The largest error of the four lowest levels is about 2×10^{-5} GHz in the optimized coordinates, against about 10^{-3} GHz in the other two, and the optimized frame stays one to two orders of magnitude ahead at every m considered.

2.5 Conclusion

The topology of a circuit fixes the number and types of modes present but leaves a large family of valid coordinates. The choice within that family is a modeling decision with consequences. We have characterized all valid mode-preserving transformations and posed the choice of coordinates as an optimization over them, which can be solved symbolically or numerically for any scalar cost. The examples show that no single choice serves every task. The reference coordinates of circuit enumeration, the normal modes, and the product-state coordinates of the KITE all differ, and the last gives up the quadratic decoupling of the normal modes in exchange for a hierarchical basis that is one to two orders of magnitude more accurate. More generally, the coordinates can and should be

tuned to specific applications by supplying the corresponding cost.

Our treatment is restricted to circuits of linear capacitors, inductors, and Josephson junctions, whose potential energy depends on flux alone. Nonlinear capacitive elements such as quantum phase slip junctions, multiport devices, and non-reciprocal elements such as gyrators require a more general treatment [52–55, 81]. Their energies depend on both flux and charge, and the flux-charge symmetric formalism obtains the Hamiltonian by a symplectic reduction in both sets of variables [53, 54]. The coordinate freedom is then a family of linear symplectic transformations of fluxes and charges together, rather than the point transformations considered here, for which $\tilde{Q} = Z^T Q$ follows from the choice of Z. Characterizing the type-preserving subset of that larger family is a natural extension of this work.

Chapter 3

Circuit Enumeration

3.1 Introduction

Superconducting quantum circuits have gained widespread use over the last 25 years as qubits, amplifiers, and transducers [20]. However, most devices have been built from a small library of circuits. The most widely used superconducting qubit today, the transmon, consists of a Josephson junction in parallel with a capacitor. Large-scale industrial efforts in superconducting quantum computing rely on this basic design, building complex systems by repeating the transmon circuit. With the advancements of the last two decades, we are now in a position to design larger multimode circuits, which are necessary to realize systems with intrinsic protection against both decay and dephasing [31].

Classical nonlinear circuits have been studied for over a century [83, 84]. Large, complex analog and digital circuits are constructed by connecting smaller well-understood modules [85]. These smaller circuits or modules, usually containing fewer than 10 nonlinear elements, are designed for specific functions. In the discovery and design of these modules, three key concepts are closely connected (depicted in Figure 3.1):

(1) CIRCUIT ENUMERATION uses a combinatorial approach to list all unique circuit configurations, given a set of components.

(2) CIRCUIT SYNTHESIS constructs a circuit that meets performance requirements, subject to constraints.

(3) CIRCUIT EQUIVALENCE determines if different circuits perform the identical function.

In circuit design, the goal is to synthesize high-performing circuits. This process becomes inefficient if an algorithm suggests circuits equivalent to previously examined ones or cannot explore new topologies.

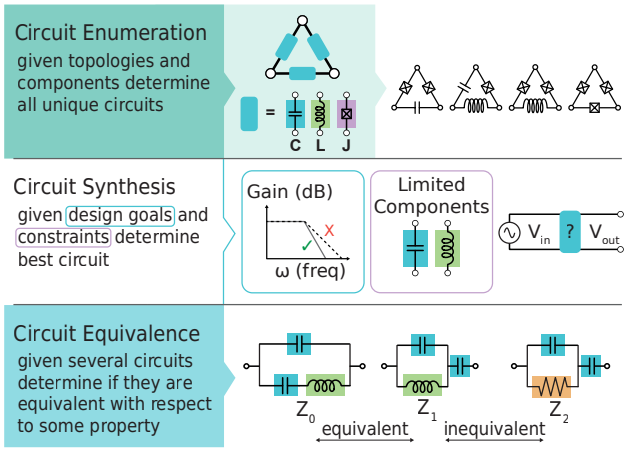


Figure 3.1: Visual representations of circuit enumeration, synthesis, and equivalence. The aim of our work is to enumerate all unique nonlinear quantum circuits with up to five nodes. The results of our enumeration leverage circuit equivalence and can be used in circuit synthesis tasks to find new qubits and amplifiers. In the circuit synthesis example, we desire a low-pass filter with a certain roll-off and search for an LC circuit that has this property. The illustrated circuit equivalence is defined with respect to the impedance Z_i.

This chapter enumerates all superconducting quantum circuits with up to five nodes built from capacitors, inductors, and Josephson junctions.[1] We consider circuits up to five nodes in size due to computational constraints, though the procedure could be applied to limited subsets of larger circuits. Our enumeration framework has two related components. First, circuit enumeration maps the design space and identifies equivalent circuits, providing a systematic basis for circuit synthesis. Second, we group the quantized circuits into Hamiltonian classes according to their mode and coupling structure. As a proof of concept for enumeration, we search all unique three-node circuits for new superconducting qubits. As a proof of concept for the Hamiltonian classification,

[1]It is straightforward to include other two-terminal devices such as the quantum phase-slip junction, but it is difficult to include three- or more-terminal devices; see Section C.1.4 for more details.

we start from a target Hamiltonian and then identify all circuits that realize it, and we use the classification to find circuits whose nonlinear coordinates are necessarily delocalized.

The remainder of this chapter is structured as follows. In Section 3.2, we give brief historical context for circuit enumeration and summarize related work. In Section 3.3, we describe our circuit enumeration algorithm, which leans on the work of Herber et al. [86–88] to create a set of topologically unique circuits. In Section 3.4, we compute the Hamiltonian of each unique circuit and categorize the circuits into Hamiltonian equivalence classes relative to a particular choice of reference coordinates. In Section 3.5, we present the enumeration and classification results and emphasize that distinct circuits can belong to the same Hamiltonian class. In Section 3.6, we develop a numerical procedure to estimate and optimize circuit performance as a qubit, then apply it to all unique three-node circuits. This numerical work leverages several existing circuit quantization software packages [43–47, 64, 89, 90]. Finally, in Section 3.7, we conclude with a discussion of open problems.

3.2 Historical Perspective & Related work

In the late 1800s, circuit enumeration became important due to the rapid expansion of telephone networks. As Darlington recounts [91]: Foster and Campbell [92] were exploring four-port transformer circuits and initially aimed to patent several promising designs. However, Bell Labs advised against patenting designs for which equivalent circuits existed. Upon realizing the vast number of possible circuits (83,539), they instead published the enumeration, which kept these designs open for use.

Circuit enumeration has a long and active history [86, 87, 93–107]. Most work has focused on circuits with two-terminal, passive linear elements, although there has been progress on nonlinear and multiport devices [108–110]. For example, one set of studies identified previously unknown amplifier circuits [111–114]. Other work has explored the integration of machine learning and advanced optimization methods into more traditional approaches to enumeration and synthesis [115–119], with a recent focus on non-redundant representations of the circuit design space [120].

The systematic study of quantum circuit theory, now often called circuit quantum electrodynamics (circuit QED) [20], was initiated by Yurke and Denker [121]. This body of work is largely disconnected from classical circuit theory, as analyzing quantum circuits requires the use of Lagrangian and Hamiltonian mechanics, a formalism rarely used in classical electrical engineering [122–124]. The field of circuit QED has grown rapidly, driven by experimental breakthroughs in the early 2000s [12, 13, 125–128] and by continued progress over the past two decades [28, 29, 129].

The field has mainly focused on superconducting qubits, driven by their importance in quantum computing. Over the past few decades, the most widely used superconducting-qubit circuit has been a Josephson junction in parallel with a capacitor, namely the Cooper-pair box or transmon [12, 16]. The transition from the Cooper-pair box to the transmon kept the circuit unchanged but moved to a parameter regime with much lower noise. By optimizing the capacitance and Josephson energy, coherence times have increased from nanoseconds [12] to median values of T_1 and T_2^{echo} of about 0.5 milliseconds [130]. Of course, improvements in materials and fabrication have also contributed substantially to this increase [130–132].

The majority of superconducting quantum circuits built today are small two-node circuits made of capacitors (`C`), inductors (`L`), and Josephson junctions (`J`). Important examples include the resonator (`LC`), the transmon (`JC`), and fluxonium (`JCL`) [17, 18]. While these circuits are often modeled as lumped elements, their physical realization is more complex. For example, the inductor in fluxonium is typically constructed from an array of Josephson junctions, which behaves like a linear inductor only at certain frequencies [133].[2] In recent years, larger multimode circuits have gained interest, driven primarily by the subfield of "protected qubits" [31]. Identifying the most promising multimode superconducting systems remains an open question. Our work provides a systematic way to catalog these possibilities.

Finally, we place our work in the context of related work in superconducting quantum circuit design. Early approaches generally optimized the parameters of a prescribed circuit. For example,

[2]Developing large inductors, not built from junction arrays, is an active area of research [134–136].

Yan et al. [137] consider a generalized flux qubit and analytically explore the design space using approximations. While their analysis captures many previous single-loop qubit designs, it does not straightforwardly apply to general multi-loop, multimode circuits. In Ref. [138], Liu et al. use a variational quantum algorithm to optimize the component values of a parallel `JCL` circuit, where the cost function is a weighted sum of the qubit frequency and anharmonicity.

More recent work has incorporated circuit structure into the design problem by integrating machine learning and other advanced optimization methods into circuit design. Menke et al. [139] use a closed-loop swarm optimization algorithm to optimize desired circuit properties, given a circuit topology. Cárdenas-López et al. [140] employ genetic algorithms to design noise-resilient qubits with particular selection rules. In related work by some of the same authors, this algorithm is used to design a new four-node qubit called the "Difluxmon" [141]. Kumar et al. [142] use graph-based machine learning methods to address the circuit synthesis problem. These methods demonstrate the value of searching over circuit structure, but they generally explore restricted or heuristically generated subsets of the possible circuit space.

Parallel research has developed increasingly powerful methods for optimizing circuits once a topology has been selected, including gradient-based and differentiable optimization of circuit parameters [64, 143], circuit decomposition and classification [81], and automated optimization or inverse design of physical device geometries [144–147]. Recent work has also approached the inverse design directly at the Hamiltonian level by constructing circuits that approximate a desired nonlinear potential [148].

Circuit enumeration addresses a distinct and complementary part of this inverse-design problem. Previous work has also considered the uniqueness and organization of superconducting circuit design spaces [64, 81], but generally without attempting a complete enumeration. Our approach seeks to catalog the design space itself in the style of Foster and Campbell: we construct the complete set of unique circuits with a specified number of nodes and organize them into Hamiltonian classes. The resulting database provides a discrete search space that can be combined with existing optimization

methods, giving future circuit synthesis efforts a concrete starting point for identifying and comparing candidate designs.

3.3 Circuit Enumeration Algorithm

Our enumeration procedure consists of two main steps:

(1) *Combinatorial enumeration.*

We represent circuit topologies using a set of unique undirected graphs. Each edge represents a location for a two-terminal circuit component, like a capacitor. For every topology, we enumerate all possible assignments of components to these edges.

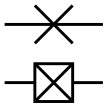
(2) *Reduction of equivalent configurations.*

The combinatorial set generated in step 1 contains many identical circuits. We construct a minimal set of "unique" circuits.

The remainder of this section details these two steps, with a graphical summary provided in Figure 3.2. Our procedure builds on the work of Herber et al. [86–88].

3.3.1 Combinatorial Enumeration

Our combinatorial enumeration generates all circuits with up to n nodes built from linear capacitors (`C`), linear inductors (`L`), and Josephson junctions (`J`). We choose `C`, `J`, and `L` because most nonlinear superconducting circuits are built exclusively from these three elements. Throughout this chapter, we will assume that Josephson junctions always have an accompanying intrinsic capacitance. While not strictly necessary for the combinatorial enumeration, this assumption is physically accurate and simplifies the analysis of multimode circuits. In superconducting-circuit diagrams, Josephson junctions with the junction capacitance omitted or shown explicitly are conventionally denoted, respectively, by



(3.1)

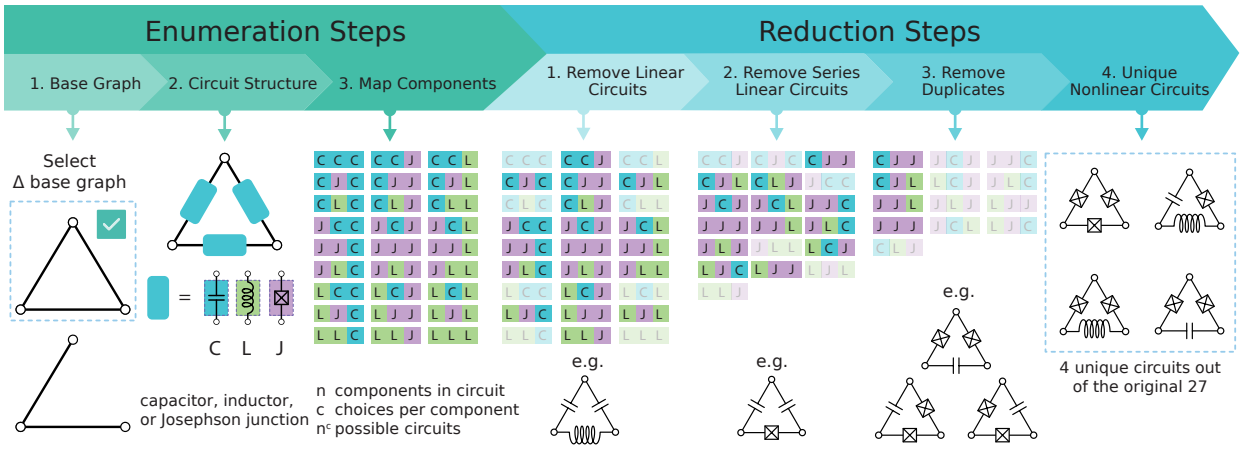


Figure 3.2: Example circuit enumeration and reduction for three-node circuits. **Enumeration:** (Step 1) For a given number of nodes, select a base graph. For $n = 3$ there are two options, and for this figure, we choose the Delta topology. (Step 2) Given the Delta topology, we allow each edge to be one of m components. In this example, the edge can be either a capacitor, inductor, or Josephson junction. In general, we also allow combinations of the three basic elements so the edges could be in the set {`C`, `J`, `L`, `CL`, `JL`, `JC`, `JCL`}. (Step 3) We enumerate all possible m-ary strings. In this example, there are 27 possible configurations. **Reduction:** Circuits that are eliminated at each step are grayed out. (Step 1) We remove any circuit that does not contain a `J`. (Step 2) We remove circuits that have linear components in series because there is an equivalent circuit with one fewer node. (Step 3) Of the remaining circuits, several are equivalent, so we perform a graph isomorphism check to remove equivalent circuits. (Step 4) Four unique nonlinear circuits remain out of the original 27.

We only use the latter in this work.

We define unique circuit topologies by the minimal set of connected, undirected, nonisomorphic graphs, which we obtain from Brendan McKay's combinatorial database [149]. We refer to these graphs as "base graphs" and show them in Figure 3.3. There are 1, 2, 6, and 21 such topologies for circuits with two, three, four, and five nodes, respectively. The count rises to 112 for six nodes, 853 for seven nodes, and 11,117 for eight nodes.

Next, we generate all possible assignments of circuit elements to the edges of each base graph. For a base graph with N_E edges and m two-terminal components, there are m^{N_E} such assignments, with each configuration naturally represented by a base-m string of length N_E. Allowing for parallel combinations of the three fundamental elements, we define $m = 7$ allowed edge types. Their string

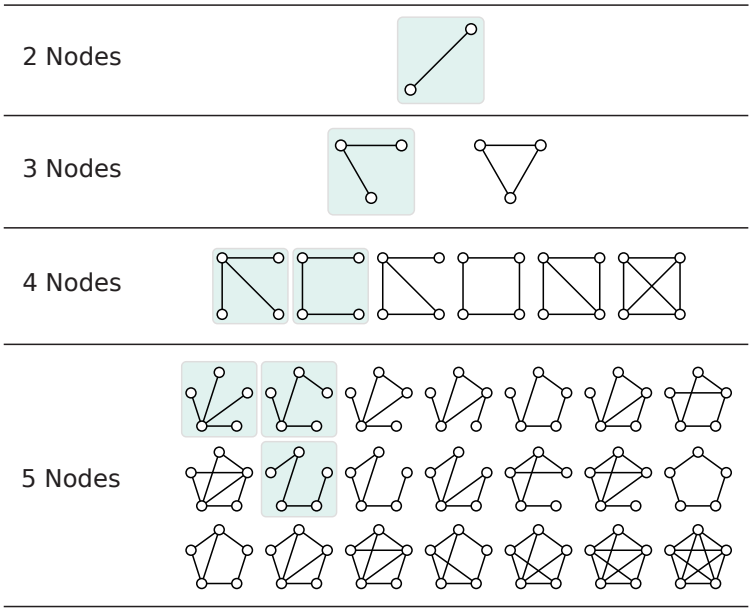


Figure 3.3: The set of "base graphs," or non-isomorphic, connected, undirected graphs with the specified number of nodes. Each represents a unique circuit topology. All except for the fully connected five-node graph can be drawn as a planar graph. Acyclic graphs, or "trees," are highlighted in blue.

representations are

$$\texttt{C},\ \texttt{J},\ \texttt{L},\ \texttt{JC},\ \texttt{CL},\ \texttt{JL},\ \text{and}\ \texttt{JCL}. \tag{3.2}$$

For example, three-node circuits utilize two base graphs (containing two and three edges, respectively), yielding $7^3 + 7^2 = 392$ total configurations. Similarly, summing the contributions across all base graphs for four and five nodes yields 139,944 and 338,332,113 raw assignments. Many of these configurations are equivalent; reducing them to a minimal set of "unique" circuits is discussed in Section 3.3.2.

3.3.2 Reduction of Equivalent Configurations

To construct a minimal set of "unique" nonlinear circuits, we remove configurations that are fully linear, contain reducible series linear components, or are otherwise topologically equivalent to another configuration. These three steps are illustrated in Figure 3.2.

3.3.2.1 Fully Linear Circuits

Linear circuits are well studied, so our focus is only on nonlinear circuits. Although quantum information can be encoded in a linear circuit, like an LC resonator, nonlinear interactions are needed for universal gates or for encoding into a bosonic code [25–27]. We therefore discard any configuration that lacks a Josephson junction — see Figure 3.2 step (1).

3.3.2.2 Series Linear Components

Two identical linear elements, such as capacitors or inductors, connected in series are equivalent at the outer terminals to a single element. A simple example appears in reduction step (2) of Figure 3.2, where a circuit with two series capacitors is discarded because it is equivalent to the transmon circuit.

This extends naturally to n linear elements of the same type meeting at a single node. They can be reduced to an equivalent network on the outer terminals alone. The case $n = 3$ is the familiar star-mesh, or Δ-Y, transformation [150, 151]. In the Lagrangian formalism, these star configurations appear as non-dynamical modes. As shown in Section C.1.5, eliminating those modes from the circuit Lagrangian exactly recovers the generalized star-mesh transformation.

We therefore discard any circuit that contains one of these reducible star configurations, since its essential dynamics are already captured by a circuit with fewer nodes.

3.3.2.3 Component-like Isomorphism

Two circuit diagrams represent the same physical system if they differ only by a permutation of node labels. For example, in reduction step (3) of Figure 3.2, the combinations `CJJ`, `JCJ`, and `JJC` all describe the same circuit because of the rotational symmetry of the underlying Δ base graph. We therefore retain one configuration, `CJJ`.

To identify these topological equivalences systematically, we test for isomorphism using a "component graph" representation [86]. In a component graph, each circuit element and each circuit node is mapped to a colored vertex. Elements of the same type share a color, while nodes are colored

by the number of elements connected to them. If two circuits are topologically equivalent, their component graphs are isomorphic [86–88].

While graph isomorphism can be computationally expensive in general, the small graphs considered here make this approach practical. As illustrated in Figure 3.4, isomorphism between component graphs provides a mathematical description of the underlying circuit symmetries.

We perform pairwise isomorphism checks with the Python package `NetworkX` [152]. To reduce the search cost, we compare only circuits with the same base graph and the same component counts, for example, 2 `C`, 1 `L`, and 1 `J`. Section C.3 gives pseudocode for constructing the resulting nonisomorphic set.

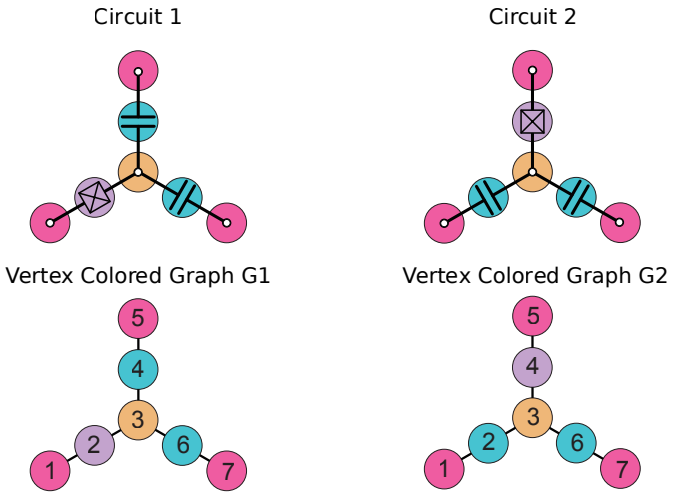


Figure 3.4: Two equivalent circuits (top), along with their component graph representations (bottom). Each circuit component and node in the circuit diagram is represented by a colored vertex. Different circuit components are assigned different colors (capacitor → blue, Josephson junction → purple). Each node in the circuit is assigned a color based on its degree (1 connection → pink, 3 connections → orange). An isomorphism exists from $G_1 \to G_2$ with vertex mapping $\{1 \to 5, 2 \to 4, 3 \to 3, 4 \to 2, 5 \to 1, 6 \to 6, 7 \to 7\}$.

3.4 Categorization by Hamiltonian Class

We want to study the quantum properties of our enumerated circuits, so we need a Hamiltonian for each circuit. We follow the standard procedure: begin with a circuit Lagrangian and obtain the

Hamiltonian using a Legendre transform [22, 23, 153]. Crucially, this procedure requires a choice of coordinates.

The circuit Lagrangian is most straightforwardly expressed in terms of node flux variables $\Phi = (\phi_1, \phi_2, \ldots, \phi_n)^T$:

$$\mathcal{L}(\Phi, \dot{\Phi}) = \frac{1}{2}\dot{\Phi}^T C \dot{\Phi} - \frac{1}{2}\Phi^T L^* \Phi - E_{JJ}(\Phi)\,, \tag{3.3}$$

where $\dot{\Phi}$ is the time derivative of Φ, C is the capacitance matrix, L^* is the inverse inductance (susceptance) matrix, and $E_{JJ}(\Phi)$ is the Josephson energy. When defining Hamiltonian classes, we treat charge offsets and external fluxes as operating-point parameters. Including their continuous values would turn the finite structural classification into a continuous one. We therefore set them to zero for the classification and reintroduce them in the numerical analysis.

While node-flux variables provide a straightforward basis for writing the Lagrangian, they are rarely the best choice for understanding the dynamics. To understand why, we can compare a grounded and an ideal floating transmon. A grounded transmon has one non-ground node flux, which directly gives the phase difference across the Josephson junction. A floating transmon is instead described by two node fluxes, ϕ_1 and ϕ_2, even though its internal dynamics depend only on their difference, $\theta = \phi_1 - \phi_2$. Introducing the common coordinate $\phi_+ = (\phi_1 + \phi_2)/2$ separates the internal transmon coordinate from collective motion of the two islands. Thus, even in this simple example, changing coordinates isolates the physically relevant degree of freedom. Below, we apply this idea systematically to separate mode types and minimize intermode coupling, which provides the coordinates used to define Hamiltonian classes.

In general, we can define a new set of coordinates $\tilde{\Phi}$ via a linear transformation of node flux variables, $\tilde{\Phi} = Z^{-1}\Phi$. Substituting $\Phi = Z\tilde{\Phi}$ into Equation (3.3) gives the Lagrangian in the new

coordinates.

$$\mathcal{L}(\tilde{\Phi}, \dot{\tilde{\Phi}}) = \frac{1}{2}\dot{\tilde{\Phi}}^T \tilde{C} \dot{\tilde{\Phi}} - \frac{1}{2}\tilde{\Phi}^T \tilde{L}^* \tilde{\Phi} - E_{JJ}(Z\tilde{\Phi})\,. \tag{3.4}$$

The quadratic terms are described by the transformed matrices $\tilde{C} = Z^T C Z$ and $\tilde{L}^* = Z^T L^* Z$. This Lagrangian can then be transformed to a Hamiltonian via the Legendre transform.

A single Hamiltonian can describe different operating regimes of the same circuit. For example, the Hamiltonian $E_C\, n^2 - E_J \cos\theta$ describes both the transmon and the Cooper-pair box for different ratios E_J/E_C.

Distinct circuits, however, can also share the same Hamiltonian structure. Consider the topologically distinct circuits in Figure 3.5, which reduce to the same Hamiltonian. With an appropriate choice of coordinates, each circuit has one dynamical degree of freedom, a kinetic term proportional to q^2, and a potential term proportional to φ^2. When circuits share this structure, relative to a specific choice of "reference" coordinates, we say that they belong to the same "Hamiltonian class".

Definition 1 (Hamiltonian Class) A set of Hamiltonians that share the same functional form, up to parameter values.

Consequently, we denote Hamiltonian classes without parameters, e.g., for the transmon the class is $n^2 - \cos\theta$. The Hamiltonian class of a circuit depends on the choice of "reference coordinates".

Definition 2 (Reference Coordinates) For a set of circuits, reference coordinates are a choice of valid coordinates for every circuit in the set.

The reference coordinates thus establish a mapping from circuits to Hamiltonian classes. Different coordinate constructions can merge or split classes, and thus yield fewer or more classes.

Hamiltonian classes can aid circuit design by providing a bias-independent classification of mode and coupling structure. We present two applications. First, they enable an inverse lookup from a desired Hamiltonian structure to the circuit topologies that realize it. Circuits within the

same class can nevertheless access different parameter regimes and present different implementation tradeoffs, as explored in Section 3.5.1. Second, as shown in Section 3.5.2, the classification reveals topology-level constraints that can force genuinely collective, delocalized nonlinear coordinates.

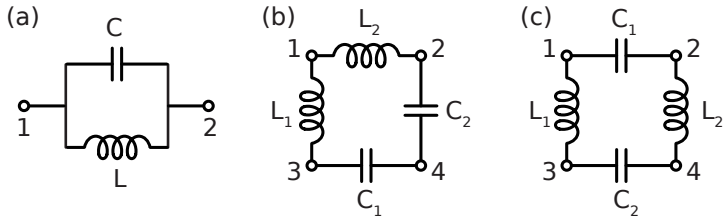


Figure 3.5: Three circuits with the same Hamiltonian class under the coordinates $\varphi_a = \phi_2 - \phi_1$, $\varphi_b = \phi_2 - \phi_3$, and $\varphi_c = \phi_4 - \phi_3 + \phi_2 - \phi_1$. With $C = C_1C_2/(C_1 + C_2)$ and $L = L_1 + L_2$, they also have identical equations of motion.

In order to categorize circuits into Hamiltonian classes, we introduce an algorithm to generate reference coordinates. We require it to return the same coordinates for circuits that differ only by a relabeling of nodes, that is, for circuits with isomorphic component graphs. Otherwise, Hamiltonian classes would depend on which circuit diagram is chosen to represent a given unique circuit. Additionally, we choose to consider only coordinates that are independent of circuit parameters. This ensures each unique circuit has a single choice of coordinates. The procedure consists of three steps, described in more detail in Section C.2:

(1) *Mode separation.*

For a circuit containing capacitors, inductors, and Josephson junctions (with intrinsic capacitance[3]), we separate coordinates into six mode types, shown in Figure 3.6. Compact variables (those with a periodic potential) are required to form an integer lattice with charge quantized in units of Cooper pairs.

(2) *Non-dynamical mode removal.*

We eliminate coordinates corresponding to constraints or redundancies, yielding an invertible capacitance matrix and a minimal set of dynamical variables.

[3]We discuss the open question of treating circuits without intrinsic junction capacitance in Section C.2.

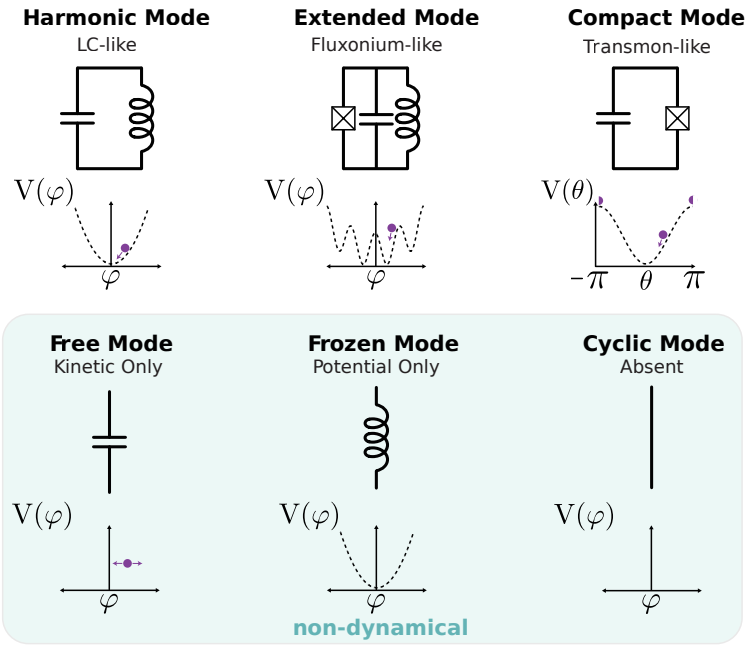


Figure 3.6: Types of dynamical (top) and non-dynamical (bottom) modes present in superconducting circuits of pure capacitors, pure inductors, and Josephson junctions with intrinsic capacitance. Cartoons of potential energy are shown for each mode type, with a corresponding "particle" moving within the potential. Here and throughout we will denote harmonic and extended modes using q, φ and compact modes using n, θ. The compact mode has periodic boundary conditions, shown by the particle split between $\theta = -\pi$ and $\theta = \pi$.

(3) *Coupling minimization.*

Within the constraints of the previous two steps, we minimize intermode coupling, while keeping dynamical coordinates independent of circuit parameters.[4] We define "minimizing coupling" as maximizing the number of strictly zero intermode coupling terms, first for nonlinear terms and then for linear terms. We compute a provably optimal solution using a custom solver built on SINGULAR [69] and the Z3 SMT solver [72]. This step can be viewed as a linear change of coordinates within each dynamical subspace.

We now illustrate how the reference-coordinate selection minimizes coupling. For the two circuits in Figure 3.7, the algorithm produces $\varphi_1 = \phi_1 - \phi_3$ and $\varphi_2 = \phi_2 - \phi_3$, where ϕ_i is the flux

[4] Allowing parameter-dependent transformations can eliminate linear coupling at the cost of increased nonlinear coupling [45].

variable at node i. The resulting Hamiltonian classes for circuits (a) and (b) are

$$q_1^2 + \qquad q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos\varphi_1 + \cos\varphi_2 \tag{3.5a}$$

$$q_1^2 + q_1q_2 + q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos\varphi_1 + \cos\varphi_2. \tag{3.5b}$$

The extra coupling in circuit (b) comes directly from the extra capacitor between nodes 1 and 2. If we had instead chosen $\varphi_2 = \phi_2 - \phi_1$, both circuits would fall into the same Hamiltonian class

$$q_1^2 + q_1q_2 + q_2^2 + \varphi_1^2 + \varphi_2^2 + \cos(\varphi_1 - \varphi_2) + \cos\varphi_1\,. \tag{3.6}$$

This alternative choice replaces linear coupling, $\varphi_1\varphi_2$, with nonlinear coupling, $\cos(\varphi_1 - \varphi_2)$. Our procedure prioritizes minimizing nonlinear coupling and then, within that constraint, minimizing linear coupling.

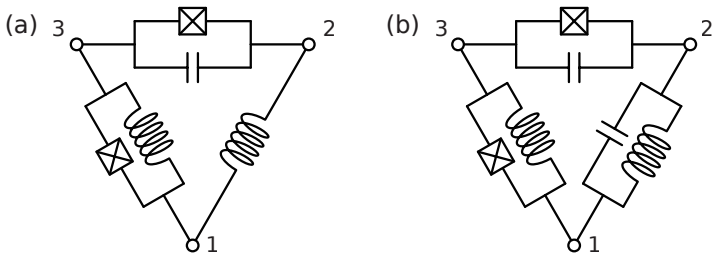


Figure 3.7: Two circuits from classes (a) 3-11 and (b) 3-12 in Table 3.2 that have equivalent mode structures if a different choice of coordinates is used.

3.5 Enumeration Results

Table 3.1 summarizes our enumeration results by number of nodes. The second column shows a combinatorial explosion of the naive circuit count from Section 3.3.1. The third column counts unique circuits using methods from Section 3.3.2, showing that the numbers are far smaller than naive counting suggests. We show similar summary statistics in Section C.4 as a function of the number of junctions, compact variables, and phase loops in a circuit.

Nodes	Circuits	Unique Circuits	Classes
2	7	4	2
3	392	80	19
4	139,944	10,982	232
5	338,332,113	7,976,949	4893

Table 3.1: Number of circuits and Hamiltonian classes for circuits with two to five nodes. The first column gives the number of nodes. The second gives the naive circuit count, 7^{N_E}, summed over all base graphs for that node number (see Figure 3.3). For three-node circuits, this gives $7^3 + 7^2 = 392$ from the two base graphs. The third column gives the number of unique circuits containing at least one Josephson junction. Because the naive count includes many equivalent circuits, the reduction from column 2 to 3 is large. The fourth column gives the number of Hamiltonian classes, namely distinct combinations of mode types and inter-mode couplings identified from the unique circuits.

This combinatorial enumeration defines the circuit design space. However, unique circuits can behave similarly. In the fourth column of Table 3.1, we use the transformation developed in Section 3.4 to group circuits into Hamiltonian classes that capture these shared behaviors. Recall that the number of Hamiltonian classes depends on the choice of reference coordinates, so the counts reported here correspond to our specific construction. For circuits with up to five nodes, this technique reduces millions of unique circuits to fewer than five thousand Hamiltonian classes. A full database of these circuits is available through [154].

In Table 3.2, we list the Hamiltonian classes for two- and three-node circuits.[5] Each class is labeled by its node number and class number, [Nodes] – [Class]. The "Hamiltonian Class" column gives the Hamiltonian with coefficients omitted, since the coefficients vary across circuits. The "Compact," "Extended," and "Harmonic" columns list the mode types explicitly, although they can also be inferred from the Hamiltonian.

Two-node circuits are either fluxonium-like, with one extended mode, or transmon-like, with one compact mode. Three-node circuits fall into 19 classes. Most of them, classes 3-1 through 3-14,

[5] A similar table of Hamiltonian classes for four- and five-node circuits is available on our repository [155].

Nodes	Class	Hamiltonian Form	Compact	Extended	Harmonic	#Circuits
2	1	$n_1^2 + \cos(\theta_1)$	1	0	0	2
	2	$q_1^2 + \varphi_1^2 + \cos(\varphi_1)$	0	1	0	2
3	1	$n_1^2 + q_2^2 + \varphi_2^2 + \cos(\theta_1)$	1	0	1	2
	2	$n_1^2 + n_1 q_2 + q_2^2 + \varphi_2^2 + \cos(\theta_1)$	1	0	1	4
	3	$q_1^2 + q_2^2 + \varphi_1^2 + \varphi_2^2 + \cos(\varphi_1)$	0	1	1	2
	4	$q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_2^2 + \cos(\varphi_1)$	0	1	1	4
	5	$q_1^2 + q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos(\varphi_1)$	0	1	1	4
	6	$q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos(\varphi_1)$	0	1	1	4
	7	$n_1^2 + n_2^2 + \cos(\theta_1) + \cos(\theta_2)$	2	0	0	3
	8	$n_1^2 + n_1 n_2 + n_2^2 + \cos(\theta_1) + \cos(\theta_2)$	2	0	0	3
	9	$q_1^2 + q_2^2 + \varphi_1^2 + \varphi_2^2 + \cos(\varphi_1) + \cos(\varphi_2)$	0	2	0	3
	10	$q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_2^2 + \cos(\varphi_1) + \cos(\varphi_2)$	0	2	0	3
	11	$q_1^2 + q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos(\varphi_1) + \cos(\varphi_2)$	0	2	0	7
	12	$q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos(\varphi_1) + \cos(\varphi_2)$	0	2	0	7
	13	$n_1^2 + q_2^2 + \varphi_2^2 + \cos(\theta_1) + \cos(\varphi_2)$	1	1	0	4
	14	$n_1^2 + n_1 q_2 + q_2^2 + \varphi_2^2 + \cos(\theta_1) + \cos(\varphi_2)$	1	1	0	4
	15	$n_1^2 + n_1 n_2 + n_2^2 + \cos(\theta_1) + \cos(\theta_2) + \cos(\theta_1 + \theta_2)$	2	0	0	4
	16	$n_1^2 + n_1 q_2 + q_2^2 + \varphi_2^2 + \cos(\theta_1) + \cos(\theta_1 + \varphi_2)$	1	1	0	6
	17	$q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_2^2 + \cos(\varphi_1) + \cos(\varphi_2) + \cos(\varphi_1 + \varphi_2)$	0	2	0	6
	18	$q_1^2 + q_1 q_2 + q_2^2 + \varphi_1^2 + \varphi_1\varphi_2 + \varphi_2^2 + \cos(\varphi_1) + \cos(\varphi_2) + \cos(\varphi_1 + \varphi_2)$	0	2	0	4
	19	$n_1^2 + n_1 q_2 + q_2^2 + \varphi_2^2 + \cos(\theta_1) + \cos(\varphi_2) + \cos(\theta_1 + \varphi_2)$	1	1	0	6

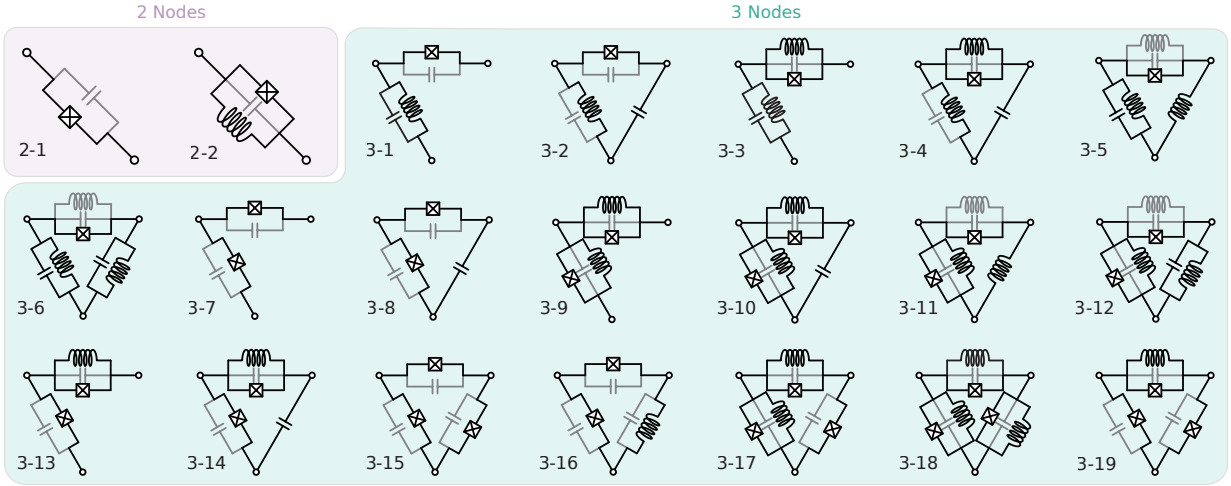


Table 3.2: Hamiltonian classes (top) and corresponding circuits (bottom) up to 3 nodes in size. Compact variables are denoted by n, θ, while extended and harmonic variables are denoted by q, φ. Grayed-out circuit elements may be included or excluded without changing the class. Because each Josephson junction includes a small intrinsic capacitance, most optional elements correspond to capacitors placed in parallel with junctions. Optional inductors, such as those in classes $3-5$, $3-6$, $3-11$, and $3-12$, are present when the circuit has no compact modes and already contains a series inductor across the specified branch. Likewise, the optional capacitors in classes $3-2$, $3-4$, and $3-16$ occur when that branch already contains a series capacitor.

can be understood as individual or linearly coupled transmon-like, fluxonium-like, or LC-like systems. The remaining classes exhibit nonlinear coupling. They include all circuits with three Josephson junctions, as well as the two-junction Bifluxon circuit 3-16 [156].

The "# Circuits" column gives the number of circuits in each class. The corresponding diagrams appear in the bottom panel, with gray elements indicating optional elements that do not change the class. Because we include a small intrinsic capacitance for every Josephson junction, most optional elements are capacitors in parallel with junctions. The optional inductors in classes $3-5$, $3-6$, $3-11$, and $3-12$ arise when the circuit has no compact modes and already contains a series inductance across the specified branch. The optional capacitors in classes $3-2$, $3-4$, and $3-16$ similarly arise when there is already a series capacitance across the specified branch.

The three-node circuits in Table 3.2 suggest three design principles. First, the low-node design space is far smaller than raw combinatorics suggests. Many apparently different circuit diagrams collapse to the same behavior. Second, the types of modes present are determined by the inductive portion of each circuit (e.g., `J`, `L`). In three-node circuits, capacitors only modify the "masses" of the individual modes and alter the intermode charge coupling, although for larger circuits they can also convert frozen modes into harmonic modes. Finally, most classes can be viewed as coupled transmon-like, fluxonium-like, or LC-like modes. Only one three-node class, $3-16$, features a Hamiltonian that cannot be straightforwardly decomposed into these component modes, with respect to our reference coordinates.

3.5.1 Inverse Lookup: Find all Circuits in a Class

As an application of Hamiltonian classification, we consider an inverse-lookup problem. Given an existing circuit (or equivalently a target Hamiltonian), find all circuits within the same Hamiltonian class. We consider two four-node examples, the Difluxmon circuit [141] and the $0-\pi$ circuit [38], which is a protected qubit. The Difluxmon class has similar membership rules to the three-node circuits, while $0-\pi$ is an example of a system that exhibits emergent dynamics when imposing

circuit symmetry.

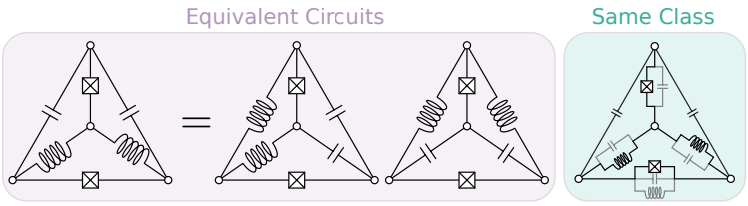


Figure 3.8: (Left) Topologically equivalent circuits to the Difluxmon circuit [141], up to trivial rotations and reflections. (Right) All circuits in the same Hamiltonian class as the original Difluxmon circuit. Gray elements can be added or removed without changing the Hamiltonian class and represent additional capacitances and inductances between nodes with existing capacitive or inductive paths.

In Figure 3.8, we show all circuits in the same Hamiltonian class as the Difluxmon circuit. Our framework identifies two distinct inductive circuits, with many possible capacitor placements. As with the three-node Hamiltonian classes, the optional elements represent capacitances and inductances across branches that already have a capacitive or inductive path, respectively. In this sense, the Difluxmon can be viewed as a specific instantiation within the specified Hamiltonian class, with deliberately small capacitances on all but two circuit branches.

We now turn our attention to the $0-\pi$ circuit, which has the same base graph as the Difluxmon, but with one inductor and one capacitor interchanged. Assuming identical pairs of capacitors, inductors, and junctions, its Hamiltonian is

$$H_{0-\pi} = E_{C1} n_1^2 + E_{C2} q_2^2 + E_L \varphi_2^2 - 2E_J \cos\theta_1 \cos\varphi_2 \,, \tag{3.7}$$

ignoring a completely decoupled harmonic mode. This system is intrinsically protected against decay and dephasing when E_{C1}/E_J is small and E_{C2}/E_L is large [37]. In Figure 3.9, we examine all three- and four-node circuits that belong to this Hamiltonian class.

The standard $0-\pi$ circuit is shown in Figure 3.9(a). Through our enumeration, we find two distinct inductive circuits in the same Hamiltonian class. Each inductive circuit can have many possible capacitor placements. The second inductive circuit corresponds to the Bifluxon circuit

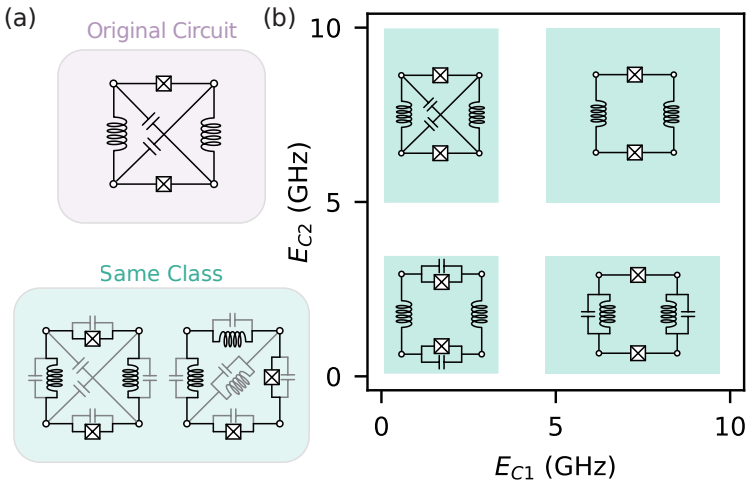


Figure 3.9: (a) All circuits in the same Hamiltonian class as the original $0-\pi$ circuit. Gray capacitors must be included or excluded in pairs across identical elements to preserve the required symmetry. For the second class, which contains the Bifluxon circuit [156], the series inductors may be excluded if the diagonal inductor is included. (b) Parameter regions accessible to four different capacitive layouts of the class containing the original $0-\pi$ circuit, assuming junction charging energies of 5–10 GHz and capacitor charging energies of 0.05–5 GHz. Only the original $0-\pi$ circuit (top left) achieves the small E_{C1} and large E_{C2} required for protection. The top right circuit has no dedicated capacitors, but both modes still have charging energy because we include the junction's intrinsic capacitance. The Bifluxon-class circuits, not depicted in (b), can achieve every combination of E_{C1} and E_{C2}, except for the protected small E_{C1} and large E_{C2} case.

[156]. While these circuits share the same mode structure, their additional capacitors set drastically different effective parameters E_{C1} and E_{C2}. Moreover, depending on the capacitive configuration, an additional harmonic mode can also emerge, as in the original $0-\pi$ circuit. As shown in Figure 3.9(b), only the original $0-\pi$ circuit reaches the regime of small E_{C1} and large E_{C2} required for protection.

This example also highlights a broader methodological point. Many proposed multimode protected qubits, including the $0-\pi$ circuit, rely on highly symmetric parameter regimes in which pairs of capacitances, inductances, and Josephson energies are equal, causing coupling terms that depend on the difference of those parameters to go to zero [34, 38, 156, 157]. In the general enumeration, each circuit element is treated as an independent symbolic parameter. Here, however, we imposed exact symmetry between matching capacitors, inductors, and junctions to capture this decoupling. Our coordinate-generation algorithm is capable of identifying these special regimes

where symmetric parameters simplify the dynamics by decoupling certain modes.

3.5.2 Identification of Delocalized Coordinates

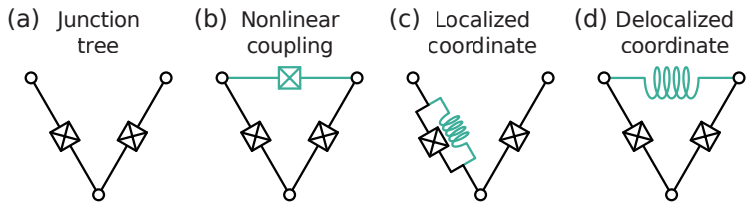


Figure 3.10: Topological origins of localized and delocalized extended coordinates. (a) The three-node Josephson junction tree and (b–d) three possible single-branch inductive modifications, with each added branch shown in teal. (b) A loop-completing junction nonlinearly couples two compact coordinates. (c) An inductive path in parallel with a single junction converts one compact coordinate into an extended coordinate that can remain localized to that junction. (d) An inductor completing a mixed loop instead creates an extended coordinate that must be delocalized across both junctions. This is the inductive skeleton of Hamiltonian class $3 - 16$.

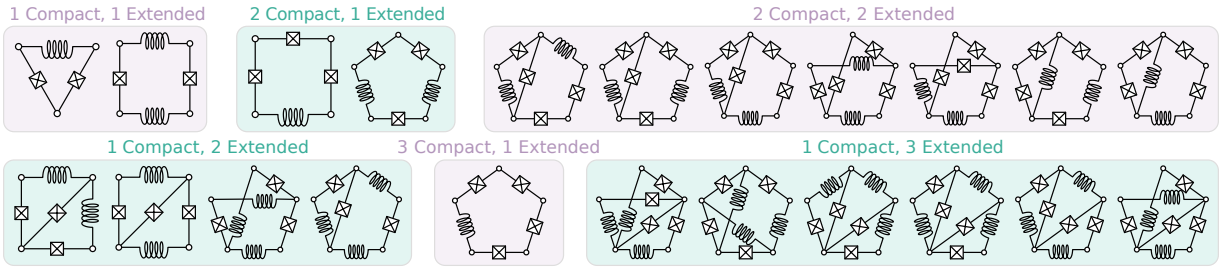


Figure 3.11: Catalog of irreducible inductive skeletons up to five nodes in size with at least one necessarily delocalized extended coordinate. No displayed junction is shunted by an inductor-only path, so all extended coordinates are delocalized (see Section C.5). The skeletons are separated by their numbers of compact and extended modes. There is one three-node skeleton, four four-node skeletons, and seventeen five-node skeletons. We omit skeletons containing shunted junctions, nodes connected only to inductors, or all-junction or all-inductor loops. We also omit circuits that become disconnected when one node is removed. Extra capacitors may create harmonic modes and couplings but do not modify the nonlinear coordinate structure.

The Hamiltonian classes provide a compressed catalog of the enumerated circuits, but they can reveal more than redundancy. Some circuit topologies force an extended coordinate to be *delocalized* over several circuit components. Remarkably, this can be identified directly from the *inductive skeleton*—the subcircuit formed only by junctions and inductors—before any capacitances

or component values are chosen.

We call an extended nonlinear coordinate *localized* if it can be chosen as the phase difference across a single Josephson junction. The three-node circuits in Figure 3.10 illustrate when this is and is not possible. Starting from the two-junction tree in (a), adding a junction closes a junction-only loop and couples the two compact coordinates, as shown in (b). Adding an inductor in parallel with one junction instead converts one compact coordinate into an extended coordinate that remains localized to that junction, as in (c). In contrast, closing a loop containing two junctions and an inductor, as in (d), produces an extended coordinate that necessarily involves both junctions. Hence the coordinate is delocalized. This is the inductive skeleton of Hamiltonian class $3-16$, which contains the Bifluxon circuit [156].

The same loop structure gives a general criterion. A necessarily delocalized coordinate requires a mixed loop containing multiple junctions and at least one inductor. If no individual junction is shunted by an inductor-only path, such a loop is also sufficient. In Section C.5, we derive the general criterion for any circuit to have a delocalized extended nonlinear coordinate (or coordinates). We give a procedure to determine the exact count δ for a circuit with any number of nodes. Importantly, δ depends only on the inductive skeleton.

After removing inductively shunted junctions and other redundancies (see Section C.5), we obtain one three-node, four four-node, and seventeen five-node irreducible-skeletons with necessarily delocalized coordinates. These are circuits where all extended modes are delocalized, and they are shown in Figure 3.11. Thus the Hamiltonian categorization can highlight circuits with collective nonlinear degrees of freedom without constructing and diagonalizing every capacitive realization.

Delocalization does not by itself imply noise protection. We have already seen that the Bifluxon is a three-node example of a protected qubit with coordinate delocalization. It turns out that the $0-\pi$ circuit is a four-node example as well. In symmetric coordinates, its Josephson potential is $-2E_J \cos\theta_1 \cos\varphi_2$. Equivalently, the two physical junction phases involve the combinations $\theta_1 \pm \varphi_2$. Its inductive skeleton is shown in Figure 3.11 and Figure 3.9. This collective nonlinear structure

is one reason that delocalized skeletons provide a natural subset of circuits to investigate when searching for protected qubits.

3.6 Searching For New Qubits

The main use for our superconducting circuit enumeration is to enhance circuit synthesis methods. As a proof of concept, we search for new three-node qubit circuits. The search performed here uses the enumerated set of unique circuits directly; it does not use Hamiltonian classes to reduce the optimization space. A class-level optimization could substantially reduce this search space, but a Hamiltonian class does not uniquely determine either the physically accessible parameter region or circuit-specific noise couplings. We therefore optimize explicit circuit realizations.

We begin by developing a framework to optimize the performance of an arbitrary circuit as a qubit and then apply it to each unique three-node circuit. Any circuit can be used as a superconducting qubit, but some perform better than others. We quantify this performance as the number of gates N_{gate} that can be performed within the coherence time T_2

$$N_{\text{gate}} = \frac{T_2}{T_{\text{gate}}} \,. \tag{3.8}$$

Here T_{gate} estimates the duration of a single qubit X gate, and we define $1/T_2 = 1/2T_1 + 1/T_\phi$ to include both energy relaxation T_1 and dephasing T_ϕ.

Because we consider many previously unstudied circuits, our modeling requires several simplifying assumptions, which we describe here and in Section C.6. Our objective function N_{gate} is best viewed as a *qualitative* architecture-agnostic screening tool for rapid evaluation, not as a quantitative predictor of gate count. It highlights circuits and operating points that balance coherence and control.

To estimate T_2, we use the `SQcircuit` package to analyze our circuits [45, 64]. Energy-relaxation rates are calculated using Fermi's golden rule, while dephasing rates depend on the sensitivity of the qubit frequency to noise parameters. We use the default noise amplitudes, which

agree with the lower bound presented in [158]. For more details on how these rates are calculated, see [45, 64]. Moreover, for circuits with external flux, `SQcircuit` assigns the branch fluxes using its gauge-consistent irrotational construction [45, 49, 62]. Flux-noise estimates therefore use the resulting circuit-specific flux distribution.

To estimate T_{gate}, we consider a simplified three-level system and estimate the time required to complete either a direct transition $|0\rangle \to |1\rangle$ or a transition via the higher-lying state $|0\rangle \to |2\rangle \to |1\rangle$. Our model, detailed in Section C.8, uses the level spacing to compute a lower bound on the gate time, independent of any specific drive.

While estimating gate time from the matrix elements of an explicit drive operator is more precise, constructing a physical drive model for every candidate is computationally prohibitive. Identifying and optimizing these circuit-specific couplings therefore lies beyond the scope of this enumeration study. We instead rely on our analytical speed limit for efficient optimization. Then, for the top identified circuit, we construct an explicit drive circuit and show that the simulated gate speed closely matches our analytical model.

We optimize Equation (3.8) for each circuit; the optimization dimension is given by the number of circuit elements and offset biases. We execute this search using a Python implementation of the differential evolution algorithm [159], a widely used gradient-free genetic algorithm. Our optimization returns the optimal circuit-element parameters and offset biases, which we call the "optimized point". We also perform a sensitivity analysis to model imperfections in fabrication. Good candidate qubits are robust to small changes in the optimized parameters. We do this by randomly sampling 100 nearby points in circuit parameter space from a normal distribution with a 2.5% standard deviation about the optimized point.

A key choice in the optimization is the allowed values for circuit parameters. We choose ranges that extend somewhat beyond present-day superconducting qubits (see Table 3.3). This choice reflects the possibility that design, materials, and fabrication advances may make such devices feasible. We note that, in the first-pass optimization over all circuits, every Josephson junction

includes the fixed intrinsic capacitance $C_J = 1.9\,\text{fF}$.

Parameter	Min (Freq)	Max (Freq)	Min (SI)	Max(SI)
$E_C,\ C$	0.05 GHz	10.0 GHz	1.9 fF	0.39 pF
$E_L,\ L$	0.05 GHz	5.0 GHz	33 nH	3.3 μH
$E_J,\ L_J$	1.0 GHz	30.0 GHz	5.4 nH	0.16 μH
$E_{C_J},\ C_J$	10.0 GHz	10.0 GHz	1.9 fF	1.9 fF

Table 3.3: Allowed ranges for circuit parameters. The junction capacitance C_J is taken to be fixed and matches the smallest capacitance allowed. For details on how to convert between physical and frequency units see Section C.7.

To assess this method, we compare our model's predictions to devices from the literature in Section C.9. Accordingly, in Section C.9 we restrict the circuit parameters to values that can be realized with current technology. We sweep the circuit parameter space for transmon and fluxonium-like circuits and find that high-performing transmon devices from the literature lie within the best performing regions predicted by our model. We find similarly good agreement in fluxonium if we slightly modify the noise model. Furthermore, our gate time estimate is able to predict the relative gate times of the considered devices.

3.6.1 Two Node Circuits

Base Graph	Circuit	Class	N_{gate} (Mean)	N_{gate} (Std)
	J	2-1	$3.75 \cdot 10^3$	$1.8 \cdot 10^2$
	JL	2-2	$8.99 \cdot 10^4$	$1.4 \cdot 10^3$
	JC	2-1	$4.14 \cdot 10^4$	$8.6 \cdot 10^2$
	JCL	2-2	$1.87 \cdot 10^5$	$6.0 \cdot 10^3$

Table 3.4: Optimized number of gates for all two-node circuits. The maximum value, $N_{\text{gate}} = 187,000 \approx 1.9 \times 10^5$, for the optimized fluxonium serves as a benchmark for larger circuits.

There are four unique two-node circuits. We optimize their performance over the parameter ranges given in Table 3.3. As shown in Table 3.4, the fluxonium-like `JCL` circuit scores highest among the two-node circuits, with an average $N_{\text{gate}} = 187,000$. This value is obtained for $E_C = 2.63$ GHz, $E_J = 2.85$ GHz, $E_L = 0.51$ GHz, and $\varphi_{\text{ext}} = 0$. As we consider three-node circuits, $N_{\text{gate}} = 187,000 \approx 1.9 \times 10^5$ will serve as a benchmark.

In Figure 3.12, we show the energy spectrum and wave functions of the optimized fluxonium. It is similar to the "Integer Fluxonium" qubit. The ground state $|0\rangle$ lies mainly in the central well, while $|1\rangle$ and $|2\rangle$ lie mostly in a superposition of the wells at $\varphi = -\pi, \pi$ [160]. Our optimized fluxonium acts as a "light" version of the integer fluxonium, with the large inductor flattening the flux dispersion. Neither system appears in the parameter sweep of Section C.9, because both operate at $\varphi_{\text{ext}} = 0$.

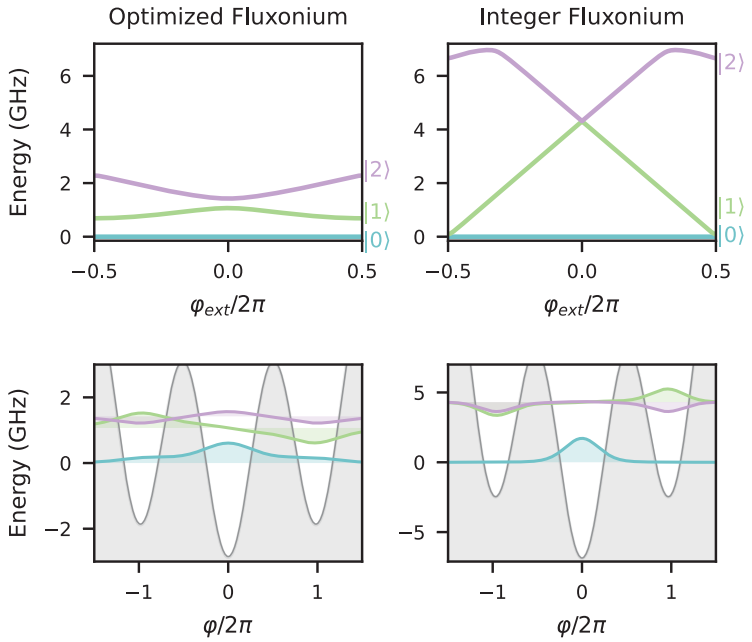


Figure 3.12: Dependence of the first three energy levels on external flux (top), optimized fluxonium (left) and integer fluxonium (right), using parameters from [160] device D. The first three eigenstates are plotted for both circuits (bottom), with global phase calibrated to make the peak value of the $|0\rangle$ state real and positive. The optimized fluxonium has $E_C = 2.63$ GHz, $E_J = 2.85$ GHz, and $E_L = 0.51$ GHz. The integer fluxonium has $E_C = 1.46$ GHz, $E_J = 6.86$ GHz, and $E_L = 0.23$ GHz. Both circuits operate at zero external flux.

While the integer fluxonium is a compelling candidate, our optimization ultimately ranked it below our top-performing circuit for two reasons. First, integer fluxonium enjoys greater disjoint support than the optimized fluxonium, but it is more sensitive to external flux, which penalizes its performance within our noise model. Second, our spectral gate-time estimate penalizes circuits with nearly degenerate first and second excited states ($E_1 \approx E_2$), a defining feature of the integer fluxonium. This second penalty illustrates a key limitation of our gate-time model. In the integer fluxonium, the $|0\rangle \to |2\rangle$ transition is physically forbidden, meaning direct $|0\rangle \to |1\rangle$ transitions can be driven much faster than the near-degeneracy of the $|1\rangle$ and $|2\rangle$ states would imply. Because our architecture-agnostic model does not capture this matrix-element selection rule, it artificially penalizes the integer fluxonium in favor of circuits with nondegenerate $|1\rangle$ and $|2\rangle$ states.

3.6.2 Three Node Circuits

There are 80 unique three-node circuits. For each base graph, Table 3.5 lists the four highest-scoring circuits. In Section C.10, we also group the highest-scoring circuits by number of junctions, number of flux loops, and number of compact modes.

Circuits belonging to the Δ base graph are the highest performing. Their optimized N_{gate} values exceed the two-node benchmark, $N_{\text{gate}} \approx 1.9 \times 10^5$, by factors of approximately 1.2–1.5, corresponding to a 20%–50% improvement. However, their larger Monte Carlo variance indicates greater sensitivity to variations in the optimized component parameters.

We now examine the highest-performing circuit, `J`, `L`, `CL`, in more detail. It belongs to Hamiltonian class 3-5 and consists of a light extended mode coupled to a heavy harmonic mode. As shown in Figure 3.13(a-c), its wavefunctions appear fluxonium-like along the extended coordinate and harmonic-oscillator-like along the other. Its flux dispersion is also fluxonium-like, but with reduced sensitivity near the optimized operating point.

An explicit coupling scheme for driving single-qubit gates in the top-performing circuit, "`J`, `L`, `CL`," is constructed and shown in Figure 3.14(a). In principle, the voltage source could be replaced

Base Graph	Circuit	Class	N_{gate} (Mean)	N_{gate} (Std)
	JL, JCL	3-9	$1.87 \cdot 10^5$	$5.6 \cdot 10^3$
	JCL, JCL	3-9	$1.86 \cdot 10^5$	$5.0 \cdot 10^3$
	JC, CL	3-1	$1.19 \cdot 10^5$	$2.3 \cdot 10^3$
	J, JCL	3-13	$9.53 \cdot 10^4$	$1.2 \cdot 10^3$
	J, L, CL	3-5	$2.77 \cdot 10^5$	$1.6 \cdot 10^4$
	J, CL, CL	3-6	$2.50 \cdot 10^5$	$1.1 \cdot 10^4$
	J, JL, L	3-11	$2.39 \cdot 10^5$	$1.8 \cdot 10^4$
	JC, CL, CL	3-6	$2.38 \cdot 10^5$	$1.1 \cdot 10^4$

Table 3.5: Four highest-scoring three-node circuits for each base graph. The "Base Graph" column shows the circuit topology, and the "Circuit" column lists the elements on each edge, separated by commas. Note that J edges include an intrinsic capacitance.

We also give the corresponding circuit class from Table 3.2.

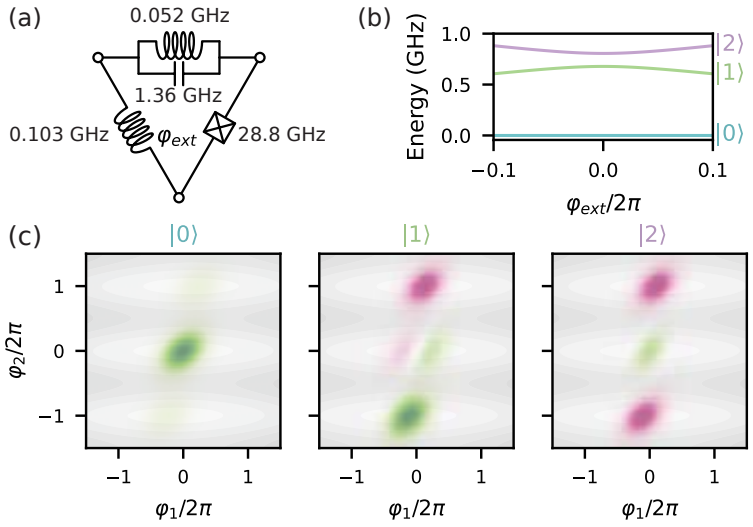


Figure 3.13: The top-performing circuit "J, L, CL". Panel (a) shows the circuit diagrams with parameter values. All junctions also include an intrinsic capacitance $E_{C_J} = 10$ GHz. (b) shows the flux dispersion about the optimized operating point. (c) shows the wavefunctions of the lowest three energy eigenstates, multiplied by a global phase so that the peak of the $|0\rangle$ state is positive and real. Green denotes positive amplitude and pink negative amplitude. The gray background shows contours of the potential energy landscape, with white for low energy and dark gray for high energy. In the optimized "J, L, CL" circuit, the wavefunctions appear fluxonium-like in one direction and harmonic-oscillator-like in the other.

by a resonator for dispersive readout. In the ideal case, with no capacitance to ground, the source couples only to the harmonic mode q_1. The coupling capacitors add mass to this already heavy mode and do not meaningfully change the system. When capacitance to ground is included, however, the drive also couples to the extended mode. With dimensionless charge operators $q_i = Q_i/2e$ and charge offset $n_g = C_c V/2e$, the drive Hamiltonian is

$$H_{\text{drive}} = n_g(g_1 q_1 + g_2 q_2), \tag{3.9}$$

with $g_1 \approx 2.45$ GHz and $g_2 \approx 0.87$ GHz. Here q_1 and q_2 are the charge operators for the harmonic and extended modes, respectively. See Section C.11 for details of the circuit quantization.

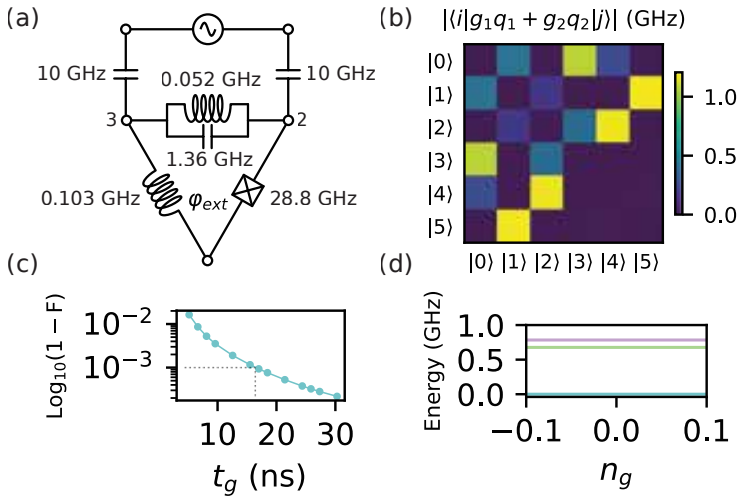


Figure 3.14: (a) Proposed coupling scheme for driving gates on the top-performing circuit "J, L, CL." In the ideal case, the voltage source couples only to the harmonic mode, but capacitance to ground also induces coupling to the extended mode. (b) Magnitudes of the drive-Hamiltonian matrix elements H_d for the lowest five energy levels. Transitions near the qubit frequency (0-2 and 1-3) are forbidden, allowing fast gates with little leakage. (c) Fidelity versus gate time for an optimized monotone pulse with a Gaussian envelope under coherent dynamics. A fidelity of 99.9% is reached at about 16.4 ns, compared with 18.6 ns predicted by the gate-time model. (d) Dependence of the first and second excited-state energies on the external charge offset. Both energies are insensitive to this parameter.

Figure 3.14(b) shows the direct transition $|0\rangle \leftrightarrow |1\rangle$ has a large matrix element, while nearby

transitions in frequency (0-2 and 1-3) are forbidden. This allows fast gates with little leakage. We optimize single-qubit X gates with durations between 5 and 30 ns using a monotone carrier modulated by a Gaussian envelope. In Figure 3.14(c), we plot the infidelity $1 - F$ versus gate time t_g, where the coherent fidelity F is the average gate fidelity [161]. A fidelity of 99.9% is reached at about 16.4 ns, close to the 18.6 ns predicted by our naive gate-time model. The first and second excited-state energies are insensitive to the new control parameter, see Figure 3.14(d). A key limitation of this design is the requirement for a large inductor and high energy junction with small parallel capacitances.

Our circuit-screening procedure makes several simplifying assumptions. In particular, the original optimization neglected inductor parasitic capacitance, and the Monte Carlo analysis did not vary junction capacitance or capacitance to ground. We therefore added a nominal 0.6 fF inductor capacitance [133] and reoptimized the four leading three-node circuits in Table 3.5. We then repeated the fabrication-variation analysis for these circuits. The new analysis includes variations in the inductor, junction, and ground capacitances, together with all previously considered parameter variations; see Table 3.6.

Base Graph	Circuit	Class	N_{gate} (Mean)	N_{gate} (Std)
	J, L, CL	3-5	$2.70 \cdot 10^5$	$1.5 \cdot 10^4$
	J, CL, CL	3-6	$2.51 \cdot 10^5$	$1.1 \cdot 10^4$
	J, JL, L	3-11	$1.90 \cdot 10^5$	$1.1 \cdot 10^4$
	JC, CL, CL	3-6	$2.36 \cdot 10^5$	$0.9 \cdot 10^4$

Table 3.6: Reoptimized versions of the top four circuits from Table 3.5, including inductor parasitic capacitance. In addition to the parameter variations considered in the original Monte Carlo analysis, we individually vary inductor capacitances, junction capacitances, and ground capacitances with the same Gaussian sampling as before. The inductor, junction, and ground capacitance distributions are centered on $C_L = 0.6$ fF, $C_J = 1.9$ fF, and $C_g = 1$ fF, respectively.

Three of the four leading circuits perform comparably. The third-ranked J, JL, L circuit does not contain an explicit capacitor. Unsurprisingly, it is most sensitive to the newly included variations

in the inductor, junction, and ground capacitances, which reduce performance by approximately 20%. Had they been included in the original optimization, this circuit would likely not have ranked among the top four candidates. The broader lesson is that relevant effects should be included before optimization. If omitted, they introduce sensitivities and degrade the performance of the optimized design.

3.7 Conclusion

We have enumerated all nonlinear superconducting circuits of up to five nodes built from capacitors, inductors, and Josephson junctions. These circuits could serve as amplifiers [20], current mirrors [157, 162], or qubits [20].

As a proof of concept, we used a naive optimization metric to search over all unique three-node circuits to find a good qubit. The best three-node circuit offered an approximate 50% improvement in the number of gates performed within a coherence time relative to our optimized fluxonium circuit. Because three-node circuits have at most two modes, the range of possible Hamiltonians is limited, and so is the improvement in gate count. A more flexible objective function (e.g., considering a variety of logical states) may reveal better three-node qubits.

Four-node circuits, with richer mode structure and couplings, may outperform three-node circuits, but at the cost of greater complexity. Extending our search for the best qubit to four-node circuits will require more efficient modeling and optimization. `SQcircuit` now includes automatic differentiation, enabling gradient-based methods [64]. Other efforts have added automatic differentiation or GPU support to superconducting-qubit modeling [163, 164], and additional methods, such as tensor networks [165], may also be needed. More advanced numerical methods will be crucial for studying larger circuits. Accurate noise models for larger circuits, such as those proposed by Dravid et al. [166], would improve predictions as well.

Our enumeration procedure extends readily to other two-terminal devices. For example, we have enumerated all four-node circuits that include a quantum phase-slip junction (QPS) `Q`, built

from the components

C, J, L, Q, JC, CL, QC, JL, JQ, QL,

JCL, JQC, QCL, JQL, and JQCL.

See Section C.1 for the resulting circuit counts. At present, systematic circuit quantization with QPS elements remains an active area of research, and no software exists to carry it out automatically [52, 55]. Thus we cannot construct Hamiltonian classes or search over circuits that include QPS devices. Similarly, extending the enumeration to three- or higher-terminal or non-reciprocal devices (e.g., gyrators) would be much harder because the underlying graph theory is more complex.

Perhaps the most novel contribution of this work is the development of Hamiltonian classes. Enumeration identifies which unique circuits exist, while Hamiltonian classes organize which circuits realize the same underlying dynamical structure. Hamiltonian classes extend the notion that a single circuit can behave differently across parameter regimes by showing that different circuits can also behave similarly. This allowed us to reduce hundreds of millions of circuit configurations down to a few thousand Hamiltonian classes. Moreover, Hamiltonian classes allowed us to identify all circuits up to five nodes whose nonlinear coordinates are necessarily delocalized. Such collective mode structure appears in several protected-qubit designs and provides a natural criterion for identifying promising circuit families.

Our Hamiltonian enumeration results suggest a Mendeleev-like table for multimode circuits. The Mendeleev table, or periodic table, organizes elements by atomic number and reveals trends: elements in the same group often show similar chemical behavior. In superconducting qubits, the family of JCL circuits can be represented as points in the two-dimensional "table" defined by E_L/E_C and E_J/E_C [167, 168]. In this picture, each point represents a different qubit. Such plots provide a simple framework for understanding the differences among most superconducting qubits built today [10]. The key idea is that varying circuit parameters produces distinct behaviors, and this

analogy works well for single-mode circuits.

Our framework suggests a different way to classify multimode circuits. Harmonic, compact (i.e., periodic), and extended modes can be viewed as "atoms," and different circuits as "molecules" built from them. Varying circuit parameters then produces different "species" or isotopes of those atoms. In this expanded analogy, the atoms (modes), bonds (inter-mode couplings), and species (circuit parameters) together determine the behavior of a superconducting circuit "molecule."

Chapter 4

Two-Qubit Gates for the Soft $0-\pi$ Qubit

This chapter is adapted from Ref. [169], written with Zhenxing Liu and Joshua Combes.

4.1 Introduction

The transmon and fluxonium are the workhorses of superconducting quantum computing. They combine relatively simple fabrication with high performance [16, 17, 170, 171]. Their coherence times, however, are limited by a trade-off between relaxation and dephasing. The transmon suppresses charge-noise dephasing but is susceptible to energy relaxation. Heavy fluxonium is comparatively robust against relaxation but more sensitive to flux-noise dephasing. The standard approach to overcome this issue is to use error correction. The relaxation/dephasing trade-off is generic to single-mode superconducting qubits, which cannot be strongly protected against both relaxation and dephasing [31].

Multimode circuits offer an alternative route beyond this limitation. They can realize *protected qubits* [31] that are resistant to both energy relaxation and dephasing, and thus can lower error rates. However, this protection comes at the cost of increased circuit and fabrication complexity. Replacing transmons and fluxoniums with such protected qubits could reduce the error-correction overhead for large-scale quantum computation [30].

Among protected-qubit proposals, the $0-\pi$ qubit is one of the most widely studied [157]. Since its introduction [35, 157], studies have examined disorder [36], coherence [32], readout and control [37, 38], and tunable circuit designs [34, 172–175]. The first experimental realization operated

in a partially protected "soft" regime [38], with exponential protection against relaxation but only first-order protection against dephasing. Full protection requires large inductances and small stray capacitances, which pose major fabrication challenges. Although recent advances have moved experimental parameters closer to the ideal regime [176, 177], near-term $0-\pi$ devices are likely to remain soft, with steadily increasing protection.

Long-lived qubits alone are not enough for quantum computation. Universal control also requires high-fidelity single- and two-qubit gates. Fully protected gates are possible [178] but are difficult to realize experimentally. Near-term $0-\pi$ devices will likely rely on unprotected gates that do not preserve the qubit's protection. Indeed, an unprotected single-qubit gate has been experimentally demonstrated in the soft 0-π regime [38]. Sufficiently fast and high-fidelity gates could help make these approaches more viable for larger-scale quantum computation.

This leaves an important gap. To our knowledge, no microwave-activated, unprotected two-qubit gate has been proposed for experimentally relevant soft $0-\pi$ devices. Such gates are challenging because coupling two $0-\pi$ qubits requires multi-node interactions, the computational states depend on external flux, and matrix elements are suppressed within the logical subspace due to protection. As a result, many gate mechanisms used in transmon architectures do not directly apply.

In this chapter, we address this gap by proposing an unprotected controlled-Z (CZ) gate for capacitively coupled soft $0-\pi$ qubits. Since accurately simulating two coupled $0-\pi$ circuits requires a large Hilbert space, we focus on the idealized zero-disorder limit. This assumption substantially reduces the required Hilbert-space dimension but eliminates an important source of device imperfections. The resulting gate performance should therefore be interpreted as a best-case bound for realistic devices.

The organization of this chapter is as follows. In Section 4.2, we introduce the basics of a single $0-\pi$ qubit. In Section 4.3, we optimize single-qubit X gates using a voltage drive that capacitively couples to the ϕ or θ mode. In Section 4.4, we introduce a capacitive coupling scheme between two $0-\pi$ circuits. In Section 4.5, we introduce the CZ gate. In Section 4.6, we develop a two-qubit noise

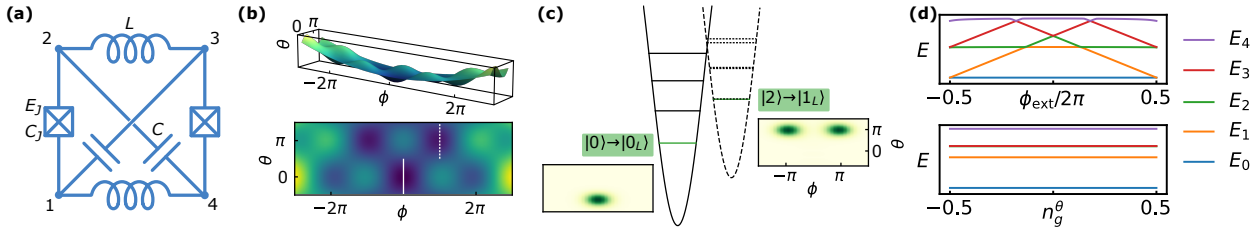


Figure 4.1: An overview of a soft $0-\pi$ qubit. (a) Circuit diagram. The circuit consists of two capacitors with capacitance C, two inductors with inductance L, and two Josephson junctions with Josephson energy E_J and junction capacitance C_J. Each circuit node, labeled by $i \in \{1,2,3,4\}$, is associated with a flux variable φ_i shown in Equation (4.1). (b) Three-dimensional surface plot and two-dimensional map of potential energy $V(\theta,\phi)$ at $\phi_{\text{ext}} = 0$, with parameters of qubit 1 from Table 4.1. (c) Lowest ten eigenvalues (horizontal lines) are plotted in the double-well potential with a solid-line well ($\phi = 0$) and a dashed-line well ($\phi = \pi$), corresponding to the solid and dashed white lines in (b). The two logical states $|0_L\rangle = |0\rangle$ and $|1_L\rangle = |2\rangle$ are highlighted in green and are strongly localized inside the two valleys. (d) Energy spectrum of the lowest five eigenvalues versus external flux $\phi_{\text{ext}}/2\pi$ in flux quanta at $n_g^\theta = 0$ in the top panel, and versus offset-charge n_g^θ at $\phi_{\text{ext}} = 0$ in the bottom panel.

model and evaluate the performance of the CZ gate in the presence of noise. Finally, in Section 4.7, we conclude with open questions.

4.2 Basics of the $0-\pi$ Qubit

In this section, we introduce the basics of the $0-\pi$ qubit. Readers interested in more details should consult references [32, 36–38]. The four-node $0-\pi$ circuit is built using pairs of capacitors, inductors, and Josephson junctions. As shown in Figure 4.1(a), each circuit node is connected to one element of each type.

Because the $0-\pi$ circuit contains multiple modes, the node flux variables φ_i are not the easiest way to understand the circuit. To obtain variables that are easier to interpret, we use the variable

transformation from Ref. [36]

$$\begin{pmatrix} \theta \\ \phi \\ \zeta \\ \Sigma \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -1 & 1 & 1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \\ \varphi_4 \end{pmatrix} . \tag{4.1}$$

The θ mode is transmon-like, as it depends on the phase differences across junctions and capacitors. Similarly, the ϕ mode is fluxonium-like, as it depends on the phase differences across junctions and inductors.

We make the simplifying assumption that all the capacitances, inductances, and junctions are the same (zero-disorder). Under this assumption, the LC-like ζ mode decouples exactly from other modes (and can be ignored), as can the cyclic Σ mode [179]. The energy stored in any individual circuit element is thus parameterized by $E_C = e^2/2C$, $E_L = \phi_0^2/L$, and $E_J = I_c\phi_0$, where $\phi_0 = \hbar/2e$ is the reduced flux quantum, I_c is the critical current of the Josephson junction, and C and L are the capacitance and inductance, respectively. Because the zero-disorder assumption is not experimentally realistic, our work represents an upper bound on the performance for more realistic devices.

In the zero-disorder limit, we consider only the θ and ϕ modes. Circuit quantization yields generalized flux operators $\hat{\theta}$ and $\hat{\phi}$ and conjugate charge operators $\hat{n}_\theta$ and $\hat{n}_\phi$ such that, e.g., $[\hat{\phi}, \hat{n}_\phi] = i$ [36]. For the remainder of the chapter, we will omit hats on operators for simplicity. We also introduce an offset charge n_g^θ and an external flux bias ϕ_{ext}. Using these operators, the two-mode quantum Hamiltonian for the $0-\pi$ circuit is expressed as

$$\begin{aligned} H_{0\text{-}\pi} = & 4E_C^\theta(n_\theta - n_g^\theta)^2 + 4E_C^\phi n_\phi^2 \\ & + E_L\phi^2 - 2E_J \cos\theta \cos\left(\phi - \frac{\phi_{\text{ext}}}{2}\right) , \end{aligned} \tag{4.2}$$

where $\phi_{\text{ext}} = \Phi_{\text{ext}}/\phi_0$. The charging energies of the θ and ϕ modes are

$$E_C^\theta = \frac{e^2}{2C_\theta} \quad \text{and} \quad E_C^\phi = \frac{e^2}{2C_\phi}\,, \tag{4.3}$$

where $C_\theta = 2C + 2C_J + C_0$ and $C_\phi = 2C_J + C_0$. Here C_J is the junction capacitance and C_0 is the capacitance between the nodes and ground [not shown in Figure 4.1(a)].

We now explain why the $0-\pi$ is a protected qubit. Protection against energy relaxation is achieved by choosing logical states with disjoint support [31]. Any charge matrix elements (and corresponding decay rates) between them are thus suppressed. The selection of the computational states of the $0-\pi$ circuit can thus be understood from the potential energy $V(\theta,\phi) = E_L\phi^2 - 2E_J\cos\theta\cos\phi$, corresponding to the second line of Equation (4.2), evaluated at the working point $\phi_{\text{ext}} = 0$. In Figure 4.1(b) we plot $V(\theta,\phi)$ using experimentally realized parameters of the $0-\pi$ qubit. The exact values are shown as qubit 1 in Table 4.1. As seen in Figure 4.1(b)–(c), the logical states of the soft $0-\pi$, $|0_L\rangle = |0\rangle$ and $|1_L\rangle = |2\rangle$, are chosen to be the lowest states in the $\phi = 0$ and $\phi = \pi$ valleys. To ensure disjoint support, the effective barrier height between the two valleys ($\approx 4E_J$) must be much larger than the qubit frequency ($\approx E_L\pi^2$) and the kinetic energy (E_C^θ) along the θ direction, i.e., $E_J \gg E_L, E_C^\theta$. Note that the first excited state is in the $\phi = 0$ valley, so selecting $|1_L\rangle = |1\rangle$ would not yield a protected qubit.

Q	E_J	E_L	E_C^θ	E_C^ϕ	E_{C_c}	E_{C_0}	g_θ
1	6.013	0.377	0.092	1.14	1.0	1.491	0.031
2	5.412						

Table 4.1: Parameters for two soft $0-\pi$ qubits in units of $h\cdot$GHz. Qubit 1 is the experimentally realized soft $0-\pi$ from [38], and qubit 2 is an otherwise identical system with E_J reduced by 10%. We consider qubit 1 for all single-qubit studies.

The dephasing protection of the $0-\pi$ is realized by suppressing the charge and flux dispersions. In Figure 4.1(d) we plot the eigenvalues of Equation (4.2) as a function of the offset charge n_g^θ and

external flux ϕ_{ext}. The charge dispersion is exponentially suppressed via a large ratio of E_J/E_C^θ, following the same mechanism as the transmon qubit [16]. As for the flux dispersion, the symmetric and antisymmetric states in the two local minima of the $\theta = \pi$ valley give a hyperbolic dispersion as a function of ϕ_{ext}, so first-order protection can be achieved at $\phi_{\text{ext}} = 0$. Further flattening of the flux dispersion requires delocalization of wavefunctions along the ϕ direction, which entails larger kinetic energy E_C^ϕ. To summarize, an ideal hard $0-\pi$ qubit requires $E_L, E_C^\theta \ll E_J, E_C^\phi$ in order to achieve full protection against both energy relaxation noise and pure dephasing noise from flux and charge fluctuations.

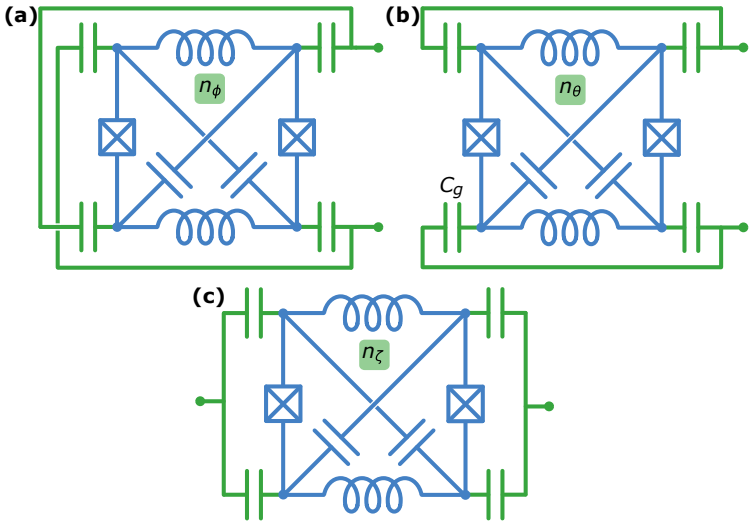


Figure 4.2: Electrical connections required to couple to the ϕ, θ, and ζ modes of the $0-\pi$ qubit [37]. The $0-\pi$ and coupling circuit are shown in blue and green respectively. A microwave source is connected across the two green nodes to drive a specific mode.

To control the $0-\pi$ qubit, we must couple to it. We consider only capacitive coupling, as inductive connections break the periodicity of the qubit potential – except in the $L \to \infty$ limit [37]. We are primarily interested in coupling to the θ and ϕ modes, but coupling to the ζ mode may be necessary to cool the system and suppress photon shot noise [37]. To selectively couple to a single mode of the $0-\pi$ qubit, individual connections to all four circuit nodes are required (see Figure 4.2), which is difficult to achieve in practice. Recent work has aimed to enable access to all four nodes [176, 177] through the use of parallel-plate instead of interdigitated capacitors. In

this paper, we assume that each node of the $0-\pi$ circuit can be electrically accessed and that the individual modes can be addressed.

In the experimental implementation of $0-\pi$, the authors use a higher non-logical state $|9\rangle$ to transform the qubit into an effective Λ system and transfer population between the two logical states [38]. Since certain nodes of the circuit were inaccessible, it was not possible to employ the coupling schemes in Figure 4.2. The experimental device induces transitions primarily by coupling to the n_ϕ operator, with residual coupling to n_θ. This scheme is known to have problems, such as leakage into even higher energy levels [180, 181]. We will discuss these limitations in detail in the next section. There are several alternative proposals for single-qubit gates for the $0-\pi$ qubit, but they are primarily focused on the more strongly protected "hard" $0-\pi$ regime [37, 178].

4.3 Single-Qubit X Gates

In this section, we investigate the limitations of unprotected single-qubit X gates in the soft $0-\pi$ qubit that use population transfer via a non-logical state. Assuming the capacitive coupling shown in Figure 4.2, we consider driving the ϕ and θ modes individually. Because both gates are driven through the charge operators, we refer to them as the n_ϕ-driven and n_θ-driven gates. We show that shorter, higher-fidelity operations are achieved by driving n_θ, which is consistent with previous studies [182].

4.3.1 Driven Hamiltonian

All single-qubit gate schemes in this work are implemented using two-tone drives. The full Hamiltonian of the qubit with drive can be expressed as

$$H_{\text{tot}} = H_{0\text{-}\pi} + H_{\text{drive}}, \tag{4.4}$$

where the parameters for $H_{0\text{-}\pi}$ come from the first qubit in Table 4.1. The microwave drives are parametrized as

$$H_{\text{drive}} = \sum_i A_i f(t) \cos(\omega_{di} t) n_{\text{op}} \,, \tag{4.5}$$

where A_i is the drive amplitude, ω_{di} is the drive frequency, and $n_{\text{op}} \in \{n_\theta, n_\phi\}$ is a charge operator. For a pulse of length t_g, we choose $f(t)$ to be a Gaussian drive envelope[1]

$$f(t) = \begin{cases} 0, & t < 0 \text{ or } t > t_g \\ e^{-8t(t-t_g)/t_g^2} - 1, & 0 \le t \le t_g \end{cases} . \tag{4.6}$$

Below we identify an effective Λ system for each charge operator by finding an intermediate state $|\lambda\rangle$ with achievable transitions $|0_L\rangle \leftrightarrow |\lambda\rangle$ and $|1_L\rangle \leftrightarrow |\lambda\rangle$. The transition frequency between levels $|a\rangle$ and $|b\rangle$ is denoted as $\omega_{a\text{-}b}$. Each transition is driven by a separate pulse with amplitude A_i and drive frequency

$$\omega_{di} = \omega_{i\text{-}\lambda} + \delta_i \,, \tag{4.7}$$

where $i \in \{0_L, 1_L\}$ and δ_i is the detuning. Similarly, we denote the charge matrix element between levels $|a\rangle$ and $|b\rangle$ as

$$n_{\text{op}}^{a\text{-}b} = |\langle a| n_{\text{op}} |b\rangle| \,. \tag{4.8}$$

4.3.2 X gate by driving n_ϕ

We first consider directly coupling to the ϕ mode. The n_ϕ operator induces an effective Λ system involving $|0\rangle$, $|2\rangle$, and $|9\rangle$, as illustrated in Figure 4.3(a). The colored arrows denote the target transitions. Numerous additional transitions give rise to significant leakage channels. Black dashed arrows denote "level-1" leakage transitions that go from states in the Λ system to level-1 states. Gray dashed arrows show level-2 leakage (from level-1 to level-2 states), and gray dotted lines

[1]Because a Gaussian $g(t) = \exp[-(t - t_g/2)^2/2(t_g/4)^2]$ has $\mu = t_g/2$ and $\sigma = t_g/4$, the envelope $f(t) = \exp[-8t(t - t_g)/t_g^2] - 1 = e^2(g(t) - g(0)) = e^2(g(t) - g(t_g))$ has a 4σ width. Note that $f(t)$ has a maximum of $e^2 - 1$, so the maximum amplitude of the drive is $A_i f(t) = A_i(e^2 - 1)$.

indicate higher-level leakage. We use these connections to infer leakage pathways. For example, state $|37\rangle$ couples to states $(|9\rangle, |10\rangle)$, which in turn couple back to the computational space $(|0\rangle, |2\rangle)$.

Figure 4.3(b) shows the population of several levels during an X gate of fidelity 99.659% with gate time $t_g = 180.0$ ns. The populations of logical states and leakage states are plotted separately, with $|9\rangle$ shown in both subplots for scale. We show only the most-populated leakage states, while the total population of all remaining states is grouped into the category "other." Multiple leakage states have comparable population to state $|9\rangle$ during gate operation, with the top three being $|10\rangle$, $|19\rangle$, and $|33\rangle$.

To identify the associated leakage transitions, we plot charge matrix elements $n_\phi^{i\text{-}j}$ for these states as a function of transition frequency $\omega_{i\text{-}j}$ in Figure 4.3(c). The two transitions forming the Λ system are shown in blue and orange, with corresponding drive frequencies marked by red arrows. These optimized drives are off-resonant due to the presence of nearby transitions. The dashed lines indicate levels of leakage using the same color scheme as Figure 4.3(a). The dominant level-1 leakage transitions appear to be $|0\rangle$-$|10\rangle$, $|2\rangle$-$|19\rangle$, and $|9\rangle$-$|33\rangle$. Importantly, the $|10\rangle \leftrightarrow |37\rangle$ transition is nearly resonant with the $|0\rangle \leftrightarrow |9\rangle$ transition but has a much larger matrix element. Leakage to such a transition can be reduced by limiting population in the intermediate and level-1 leakage states. Due to frequency crowding, the drive strength must be kept small compared to the detuning from neighboring transitions to suppress leakage, resulting in long gate times.

4.3.3 X gate by driving n_θ

Now we perform an X gate by driving the θ mode. The n_θ operator induces a Λ-like system involving $|0\rangle$, $|2\rangle$, and $|7\rangle$. Figure 4.4(b) shows the population transfer for an X gate with a fidelity of 99.758% and gate time $t_g = 25.0$ ns. The two drive frequencies ω_{d1} and ω_{d2} are indicated by red arrows in Figure 4.4(c). Unlike the n_ϕ case, there is a single most prominent leakage state $|25\rangle$, which is populated through the $|7\rangle$-$|25\rangle$ transition. The next most prominent leakage states are $|1\rangle$, $|4\rangle$, and $|5\rangle$. The transitions that populate these states are strongly detuned, but the relevant matrix

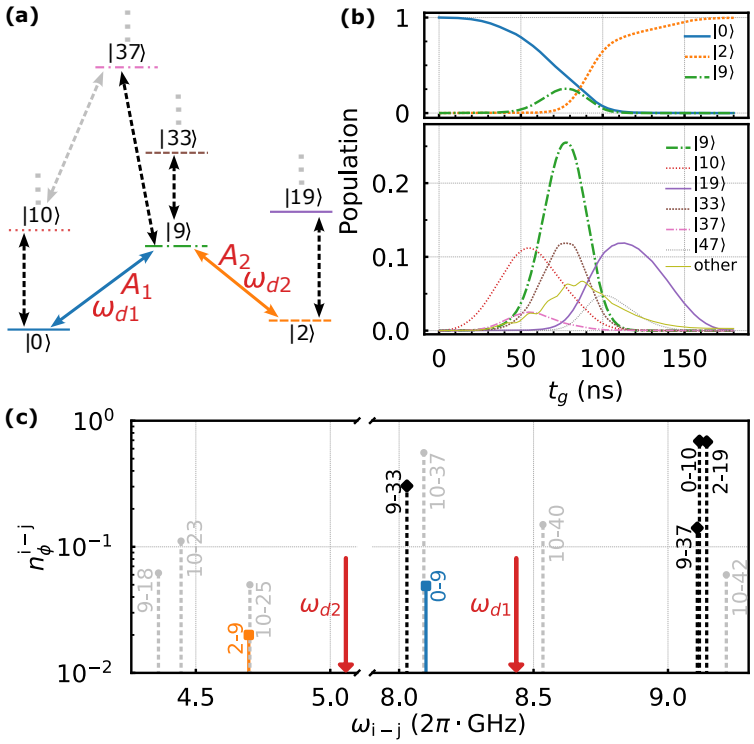


Figure 4.3: Numerically optimized X gate by driving n_ϕ, with $F = 99.659\%$ and $t_g = 180.0$ ns. (a) Energy level diagram. The desired transitions, $|0\rangle$-$|9\rangle$ and $|2\rangle$-$|9\rangle$, are shown with blue and orange arrows. These transitions are driven with amplitudes A_i and frequencies ω_{di} for $i \in \{1, 2\}$. Leakage pathways are denoted with dashed arrows. Black arrows represent "level-1" leakage transitions, while gray arrows represent "level-2" leakage transitions that involve level-1 states. (b) Population transfer. "other" denotes the total population of states that are not explicitly shown. (c) Charge matrix elements $n_\phi^{i\text{-}j}$ versus transition frequency $\omega_{i\text{-}j}$. The line colors and styles match those in (a). The desired transitions have $\omega_{0\text{-}9}/2\pi = 8.100$ GHz and $\omega_{2\text{-}9}/2\pi = 4.697$ GHz, with $n_\phi^{0\text{-}9} = 0.049$ and $n_\phi^{2\text{-}9} = 0.020$. Simulations use the lowest 1000 levels; see Section D.1 and our open-source GitHub repository [183] for numerical implementation details. Pulse parameters can be found in Table 4.2.

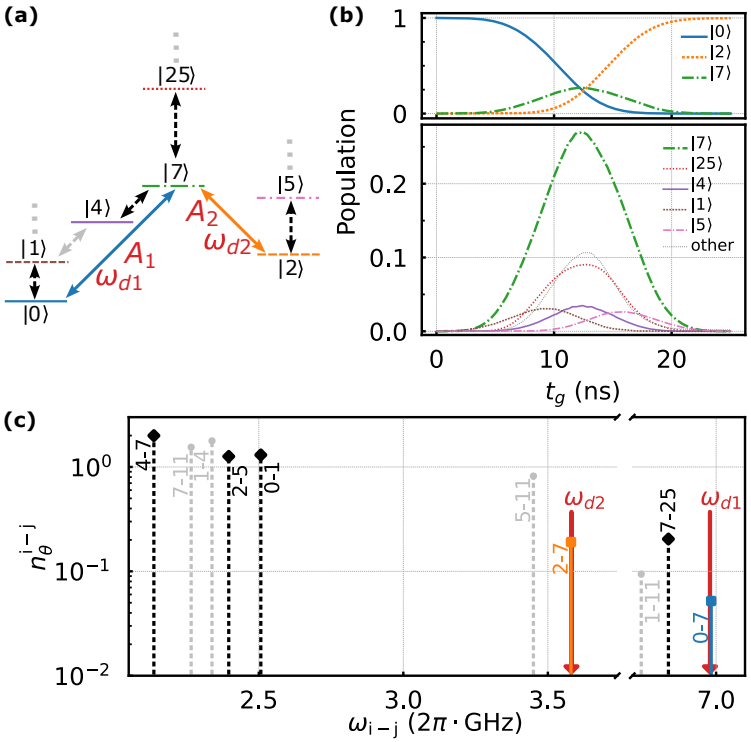


Figure 4.4: Numerically optimized X gate by driving n_θ, with $F = 99.758\%$ and gate time $t_g = 25.0$ ns. (a) Energy level diagram. Leakage pathways are denoted with dashed arrows. Black arrows show "level-1" leakage transitions involving an intentionally populated state ($|0\rangle$, $|2\rangle$, or $|7\rangle$). (b) Population transfer. "other" includes the total population of all states not explicitly shown. (c) Charge matrix elements $n_\theta^{i\text{-}j}$ versus transition frequency $\omega_{i\text{-}j}$. The two Raman transition frequencies are $\omega_{0\text{-}7}/2\pi = 6.983$ GHz and $\omega_{2\text{-}7}/2\pi = 3.580$ GHz, with $n_\theta^{0\text{-}7} = 0.052$ and $n_\theta^{2\text{-}7} = 0.191$. Pulse parameters can be found in Table 4.2.

elements are roughly an order of magnitude larger than those in the Λ system, leading to significant population transfer. Nonetheless, there is substantially less leakage overall when driving the θ mode compared to driving the ϕ mode.

We can also compare our results with those of Abdelhafez et al. [182]. In that work, the authors use optimal control to design an X gate for the $0-\pi$ qubit in a harder parameter regime (larger E_J, E_C^ϕ and smaller E_L, E_C^θ). The authors drive n_θ with an amplitude of 1.5 GHz $\approx 238 \times 2\pi$ MHz, which is comparable to the drive amplitude used in our gate; see Table 4.2. For a gate time $t_g = 60$ ns, they achieve an X gate with 98.6% coherent fidelity.[2] Although this indicates lower performance

[2]This is also consistent with Paolo et al. [37] when the fidelity and gate time are extracted from its results. That work uses the same parameters as Ref. [182].

than for our X gate, slower gates are expected in this more strongly protected $0 - \pi$ parameter regime. Indeed, when we simulate a Raman-style X gate in the same parameter regime, we see comparable gate fidelities for this gate time.

4.3.4 Single-Qubit Noise Model

To study the effects of noise on these gates, we use a Lindblad master equation to phenomenologically include energy relaxation and dephasing. The evolution of the state matrix is given by

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_{i,j} \mathcal{D}[L_{ij}^r]\rho + \sum_l \mathcal{D}[L_l^\varphi]\rho, \tag{4.9}$$

where $\mathcal{D}[X]\rho = X\rho X^\dagger - \frac{1}{2}(X^\dagger X\rho + \rho X^\dagger X)$ is the dissipation superoperator, and L_{ij}^r and L_l^φ are collapse operators for relaxation and dephasing, respectively. The relaxation and dephasing operators have the form

$$L_{ij}^r = \sqrt{\frac{1}{T_{ij}^r}} \left|i\right\rangle \left\langle j\right|, \tag{4.10a}$$

$$L_l^\varphi = \sqrt{\frac{2}{T_l^\varphi}} \left|l\right\rangle \left\langle l\right|, \tag{4.10b}$$

where T_{ij}^r is the relaxation time of the transition $|j\rangle \to |i\rangle$ and T_l^φ is the dephasing time of the state $|l\rangle$.

The main challenge in building a noise model is specifying the operators L_{ij}^r and L_l^φ. Our noise model is similar to the one considered in Ref. [182]. For energy relaxation, we consider the relevant charge operator n_{op} (with $\text{op} \in \{\theta, \phi\}$ depending on the drive) and compute the downward transition rates between states $|j\rangle \to |i\rangle$ as

$$\gamma_{j\text{-}i}^r = \Gamma \left|\left\langle i\right| n_{\text{op}} \left|j\right\rangle\right|^2. \tag{4.11}$$

The overall transition rate Γ is not fixed by the model and must be set. Because both gate schemes use an effective Λ system, we fix Γ by assigning a target lifetime T to the dominant decay channel from the intermediate state $|\lambda\rangle$. Specifically, we identify the largest matrix element connecting $|\lambda\rangle$ to a lower level $|M\rangle$, and choose Γ so that the corresponding decay time is $1/\gamma^r_{\lambda\text{-}M} = T$, where T is a target value (e.g., 3, 30, 170 μs). Once Γ is set, all other decay rates follow. For n_ϕ, the intermediate state is $|9\rangle$, and the dominant transition is $|9\rangle \leftrightarrow |4\rangle$; we set $1/\gamma^r_{9\text{-}4} = T$. For n_θ, the intermediate state is $|7\rangle$, and we fix Γ using $1/\gamma^r_{7\text{-}4} = T$.

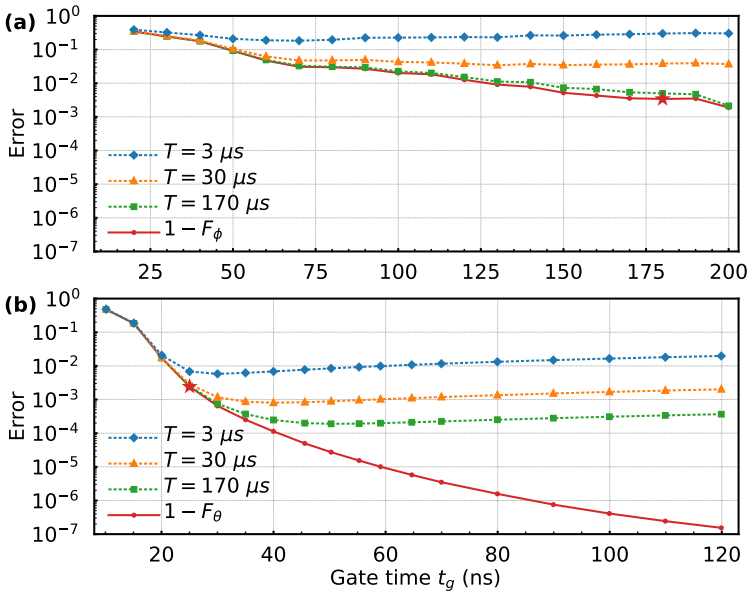


Figure 4.5: Simulated X gate error $1 - F$ versus gate time t_g for numerically optimized pulses driving (a) n_ϕ and (b) n_θ. Solid lines represent coherent gate error without added noise. Dashed lines show the performance including open-system evolution with energy relaxation and pure dephasing, with $T_1 = T_\phi = T \in \{3, 30, 170\}$ μs. The two stars correspond to the pulses considered in Figure 4.3 and Figure 4.4. The simulations here use a reduced model with 160 levels for the n_ϕ gate and 157 levels for the n_θ gate. See Section D.1 for further numerical implementation details.

Charge dispersion is negligible at the chosen operating point [Figure 4.1(d)], so we include only flux-induced pure dephasing. At $\phi_{\text{ext}} = 0$, the first-order flux sensitivity vanishes, and the dephasing

n_{op}	Intermediate states	$n_{\text{op}}^{0\text{-}i}$	$n_{\text{op}}^{2\text{-}i}$	Leakage states	t_g (ns)	$\Omega_{0\text{-}i}$ (2π·MHz)	$\Omega_{2\text{-}i}$ (2π·MHz)	δ_1 (2π·MHz)	δ_2 (2π·MHz)	F_{optimize}	F_{open} (T=30μs)
n_ϕ	$\|9\rangle$	0.049	0.020	$[\|10\rangle, \|19\rangle, \|33\rangle]$	180.0	66.08	10.81	335.73	360.59	99.659%	96.183%
n_θ	$\|7\rangle$	0.052	0.191	$\|25\rangle$	25.0	59.87	56.29	-3.79	1.54	99.758%	99.713%

Table 4.2: Numerically optimized parameters corresponding to the two highlighted red stars in Figure 4.5. The drive amplitudes and detunings $A_i/2\pi$, $\delta_i/2\pi$ are limited to be less than 500 MHz. The effective Rabi rates are given by $\Omega_{0\text{-}i} = A_1(e^2-1)n_{\text{op}}^{0\text{-}i}$ and $\Omega_{2\text{-}i} = A_2(e^2-1)n_{\text{op}}^{2\text{-}i}$.

rate is determined by the curvature of the transition frequency [184],

$$\frac{1}{T_{0\text{-}l}^{\varphi}} = A_{\phi_{\text{ext}}} \left| \frac{\partial^2 \omega_{0\text{-}l}}{\partial \phi_{\text{ext}}^2} \right|. \tag{4.12}$$

We choose $A_{\phi_{\text{ext}}}$ such that $T_{0\text{-}2}^{\varphi} = T$, after which Equation (4.12) determines the remaining dephasing times. We use the $|0\rangle \leftrightarrow |2\rangle$ transition for this calibration because it has the largest flux curvature among the transitions in the effective Λ system.

There are two key points to emphasize. First, when we specify a noise strength such as $T = 3\mu$s, we refer to both relaxation and dephasing times associated with the relevant transitions of the gate. Second, for simplicity, we adopt a family of noise models in which relaxation and dephasing rates are equal, i.e., $T^r = T^{\varphi} = T$.

4.3.5 Noisy gate fidelity vs. gate time

We evaluate gate performance using the average infidelity between the ideal X gate and its noisy implementation $\tilde{X}$, denoted $1 - F(X, \tilde{X})$ [161, 185]. For each gate duration t_g, we optimize the pulse (A, δ) parameters to minimize $\log_{10}(1-F)$ in a closed quantum system. For additional details, see Sections D.1.2 and D.1.3.

The optimization pipeline is as follows. For each gate time t_g, we optimize the two amplitudes $A_{1,2}$ and detunings $\delta_{1,2}$ in the closed-system model. The amplitudes are constrained to $A/2\pi \in [0, 500)$ MHz, and the detunings $\delta = \omega_d - \omega_{i\text{-}j}$ are varied within $\pm 2\pi \times 500$ MHz around the target transition $\omega_{i\text{-}j}$. The gate error is then computed by applying pulses optimized in the closed-system

model to the noisy evolution generated by the master equation in Equation (4.9). For more details, see Sections D.1.3 and D.1.4.

Figure 4.5(a) shows the simulated X-gate error $1 - F$ versus gate time t_g when driving n_ϕ, while Figure 4.5(b) shows the corresponding results for driving n_θ. The y-axis scales are matched to enable direct comparison. The key observation is that X gates implemented via n_θ driving achieve both shorter gate times and higher fidelities than those obtained with n_ϕ driving.

There are two main reasons for this advantage. First, in our parameter regime the relevant matrix elements of n_θ are significantly larger than those of n_ϕ (see Table 4.2). In particular, $n_\theta^{2\text{-}7}/n_\phi^{2\text{-}9} \sim 10$ is an order of magnitude larger than $n_\theta^{0\text{-}7}/n_\phi^{0\text{-}9} \sim 1$. Second, the n_θ drive has fewer leakage pathways than the n_ϕ drive, as seen in Figure 4.3(a)–(b) and Figure 4.4(a)–(b). With fewer leakage channels, stronger and shorter drives can be used without compromising fidelity. Moreover, the optimized drive frequencies for n_ϕ are often off-resonant (for the gate times considered), while those for n_θ are typically near-resonant, further reducing leakage.

The optimized gates are significantly faster than the experimentally demonstrated gate in Ref. [38]. In Section D.2, we give some explanation for this apparent discrepancy (e.g., experimental limitations on drive amplitude, mixed mode coupling, and decoherence not predicted by our phenomenological noise model). Ultimately, we show that our numerical models can roughly reproduce the behavior observed in the experiment.

4.4 Coupling for Two $0 - \pi$ Qubits

Our single-qubit analysis shows that coupling to the θ mode provides larger charge matrix elements and fewer leakage pathways than coupling to the ϕ mode. Motivated by this finding, we focus on designing two-qubit gates based on a charge-charge interaction between the two circuits' θ modes, i.e., an $n_{\theta 1} n_{\theta 2}$ interaction.

The θ mode involves all nodes of the $0 - \pi$ circuit, so producing an $n_{\theta 1} n_{\theta 2}$ interaction requires multi-node coupling. We show two such schemes in Figure 4.6. The Hamiltonian for these two

$0-\pi$ systems can be written as

$$H_{\text{cp}} = H_{0\text{-}\pi}^{(1)} + H_{0\text{-}\pi}^{(2)} + H_{\text{int}}, \tag{4.13}$$

where the superscript (i) refers to the ith qubit and H_{int} is the interaction Hamiltonian. The two circuits in Figure 4.6 (a) and (b) yield interactions

$$H_{\text{int}}^{(\text{a})} = g_\theta n_{\theta 1} n_{\theta 2} \ , \tag{4.14a}$$

$$H_{\text{int}}^{(\text{b})} = g_\theta n_{\theta 1} n_{\theta 2} - g_\phi n_{\phi 1} n_{\phi 2} \ . \tag{4.14b}$$

Circuit (a) implements a pure $n_{\theta 1} n_{\theta 2}$ interaction and forms the foundation of our two-qubit gate implementation. Circuit (b) does not produce a pure $n_{\theta 1} n_{\theta 2}$ interaction. Nonetheless, circuit (b) does have some advantages despite its mixed interaction. Notably, it requires fewer capacitors, making it easier to construct, and the renormalized $\tilde{E}_C^\phi$ remains light even with a large coupling capacitor. A full derivation of both interactions is provided in Section D.3. Since the remainder of the chapter focuses on circuit (a), we briefly outline its derivation below.

Because the coupling network in circuit (a) is purely capacitive, it modifies only the kinetic terms in the Lagrangian. As labeled in Figure 4.6 (a), nodes 1–4 correspond to qubit 1, nodes 5–8 to qubit 2, and nodes 9–10 to the coupler. With this convention, the interaction Lagrangian in node flux variables is

$$\begin{aligned} T_{\text{int}} = \frac{C_c}{2} \big[&(\dot{\varphi}_1 - \dot{\varphi}_{10})^2 + (\dot{\varphi}_4 - \dot{\varphi}_{10})^2 + (\dot{\varphi}_5 - \dot{\varphi}_{10})^2 \\ &+ (\dot{\varphi}_8 - \dot{\varphi}_{10})^2 + (\dot{\varphi}_2 - \dot{\varphi}_9)^2 + (\dot{\varphi}_3 - \dot{\varphi}_9)^2 \\ &+ (\dot{\varphi}_6 - \dot{\varphi}_9)^2 + (\dot{\varphi}_7 - \dot{\varphi}_9)^2 \big] \ . \end{aligned} \tag{4.15}$$

Here, C_c is the value of the coupling capacitors, shown in green in Figure 4.6. Equation (4.15) shows that both qubits couple to nodes 9 and 10.

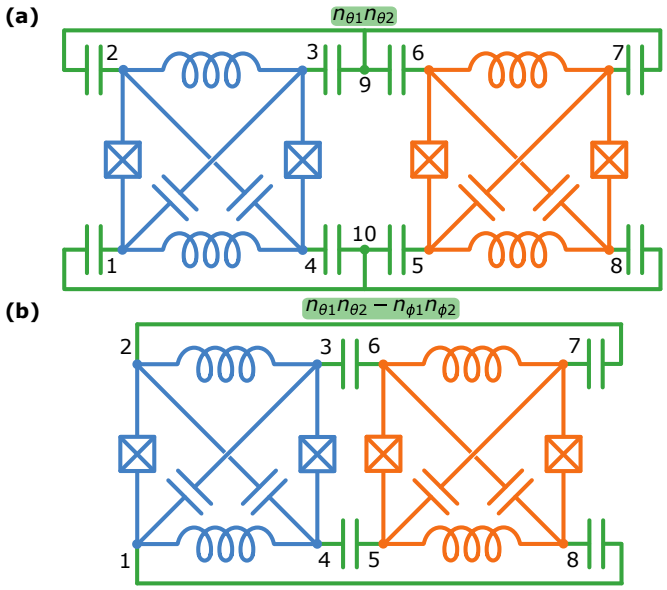


Figure 4.6: Two coupled $0-\pi$ qubits with (a) pure $g_\theta n_{\theta 1} n_{\theta 2}$ and (b) mixed $g_\theta n_{\theta 1} n_{\theta 2} - g_\phi n_{\phi 1} n_{\phi 2}$ interaction terms. The individual qubits are shown in blue and orange, while the coupling circuit is drawn in green.

After performing the variable transformation in Equation (4.1) and setting $\varphi_\pm = \varphi_9 \pm \varphi_{10}$, we obtain

$$T_{\text{int}} = \frac{C_c}{2} \sum_{i=1}^{2} \left[\dot{\phi}_i^2 + \dot{\zeta}_i^2 + (\dot{\varphi}_+ - \dot{\Sigma}_i)^2 + (\dot{\varphi}_- - \dot{\theta}_i)^2 \right] . \tag{4.16}$$

The φ_- mode is coupled directly to both θ_1 and θ_2. There are no potential energy terms corresponding to the Σ_1, Σ_2, φ_+, and φ_- modes, so they are "free modes" and can be removed from the Lagrangian [43, 44, 58].

After free mode removal and circuit quantization, the coupling Hamiltonian is

$$H_{\text{int}} = g_\theta \, n_{\theta 1} n_{\theta 2} \, , \tag{4.17}$$

with coupling energy

$$g_\theta = \frac{2e^2 C_c}{\tilde{C}_\theta (\tilde{C}_\theta - C_c)} \approx \frac{e^2 C_c}{2C^2} \, . \tag{4.18}$$

Here $\tilde{C}_\theta = 2C + 2C_J + C_0 + C_c = C_\theta + C_c$. The approximation is valid when $C \gg C_c, C_J, C_0$.

One consequence of capacitive coupling is the renormalization of each qubit's charging energy. For the circuit in Figure 4.6 (a), the charging energies become

$$\tilde{E}_C^\phi = \frac{e^2}{2\tilde{C}_\phi}, \quad \tilde{E}_C^\theta = \frac{e^2(2\tilde{C}_\theta - C_c)}{4\tilde{C}_\theta(\tilde{C}_\theta - C_c)}, \tag{4.19}$$

where $\tilde{C}_\phi = 2C_J + C_0 + C_c = C_\phi + C_c$. The coupling capacitance makes both modes heavier ($\tilde{E}_C^\phi < E_C^\phi$ and $\tilde{E}_C^\theta < E_C^\theta$), and these changes can be significant. Using the parameters from Table 4.1, $(C_J, C_0, C_c) \approx (2, 13, 19)$ fF, $\tilde{E}_C^\phi$ and $\tilde{E}_C^\theta$ decrease by approximately 50% and 4%, respectively. The hard $0-\pi$ regime requires $E_L, E_C^\theta \ll E_C^\phi$. This is difficult to achieve in circuit (a) because the coupling capacitance suppresses E_C^ϕ. Specifically, in a hard $0-\pi$ regime, we expect $C_c \gg C_J$ and $C_0 \to 0$, so the ϕ mode charging energy scales with $\tilde{E}_C^\phi \propto 1/2C_c$. Thus, a very large inductance and shunt capacitance would be required to achieve the hard $0-\pi$ regime. Circuit (b) avoids this limitation because $\tilde{E}_C^\phi \propto [1/C_J + 1/(C_J + C_c)]$, which approaches $1/C_J$ when $C_c \gg C_J$.

The coupling schemes in Figure 4.6 are experimentally demanding. They require capacitive access to all four nodes of each $0-\pi$ qubit, together with a multi-node coupling network and additional control and readout circuitry. The resulting layout is substantially more complex than that of conventional superconducting-qubit architectures. Suppressing parasitic capacitances remains an open challenge. Our analysis also assumes identical circuit elements, for which unwanted coupling terms cancel exactly. In a realistic device, disorder in the coupling capacitors and stray capacitances will break this symmetry and generate residual interactions. We nevertheless adopt the zero-disorder limit because it reduces the Hilbert-space dimension, simplifies the numerical analysis, and establishes a best-case benchmark for gate performance. A detailed treatment of disorder, parasitic couplings, and layout constraints is left for future work.

Before closing this section, we briefly comment on generalizations of this coupling scheme. Similar layouts can be used to connect any pair of $0-\pi$ modes, and we show the required connections in Table 4.3. As with the system in Figure 4.6 (a), these coupling circuits require many more

capacitors than are typically needed in transmon architectures, even before accounting for drive lines.

Interaction	Nodes coupled to 9	Nodes coupled to 10
$n_{\theta 1} n_{\theta 2}$	2,3 + 6,7	1,4 + 5,8
$n_{\phi 1} n_{\phi 2}$	1,3 + 5,7	2,4 + 6,8
$n_{\theta 1} n_{\phi 2}$	2,3 + 5,7	1,4 + 6,8
$n_{\phi 1} n_{\theta 2}$	1,3 + 6,7	2,4 + 5,8

Table 4.3: Capacitive connections that couple the specified modes of two $0-\pi$ qubits. Node numbers match Figure 4.6 (a).

4.5 Microwave-activated CZ gate

We now develop a CZ gate for two coupled $0-\pi$ qubits, using the coupling scheme from Section 4.4 and a microwave drive applied to one qubit. Inspired by similar CZ and CCZ gates in fluxonium [186–188], the gate drives a transition between a computational state and a higher non-computational state to accumulate a π phase. As in fluxonium, the $0-\pi$ spectrum is crowded, necessitating careful consideration of competing transitions when optimizing gate performance.

4.5.1 Driven Hamiltonian

The Hamiltonian of the coupled qubits with drive is

$$H_{\text{tot}} = H^{(1)}_{0\text{-}\pi} + H^{(2)}_{0\text{-}\pi} + H_{\text{int}} + H^{(2)}_{\text{drive}}, \tag{4.20}$$

where $H^{(2)}_{\text{drive}}$ is a microwave drive on $n_{\theta 2}$ of the form given in Equation (4.5). The full circuit diagram is shown in Figure 4.7(a), with the drive circuitry highlighted in red. It should be noted that the capacitive coupling to the voltage source induces additional energy renormalization, which we do not consider, as it is well studied in prior work [37].

We set device parameters so that both $0-\pi$ circuits are in the soft regime, while allowing

slight asymmetry. The first qubit, described by $H^{(1)}_{0\text{-}\pi}$, uses the same parameters as the experimental device in Ref. [38]. The second qubit, $H^{(2)}_{0\text{-}\pi}$, has E_J reduced by 10%, with all other parameters unchanged (see Table 4.1).

To analyze the gate operation, we establish the following notation. The eigenstates of the uncoupled system ($H_{\text{int}} = 0$) are denoted $|k,l\rangle_0$, where k and l label the eigenstates of the first and second qubits, respectively. These are commonly referred to as bare states. With finite coupling ($H_{\text{int}} \neq 0$), the "dressed" eigenstates are denoted $|k,l\rangle$ with eigenvalues $E_{k,l}$. Each dressed state is indexed by the bare state with which it has the largest overlap. For example, the dressed state $|5,0\rangle$ is defined by its large overlap with $|5,0\rangle_0$: $|{}_0\langle 5,0|5,0\rangle| = 0.91$. In addition, a key quantity is the transition frequency between dressed states. For a transition $|k,l\rangle$ - $|k',l'\rangle$, we denote the transition frequency as $\omega_{k,l\text{-}k',l'} = |\omega_{k,l} - \omega_{k',l'}|$ with $\omega_{k,l} = E_{k,l}/\hbar$.

4.5.2 Gate Concept and Transition Landscape

A two-qubit gate requires an interaction between the two $0-\pi$ circuits, as described by the total Hamiltonian in Equation (4.20). Because charge matrix elements within the logical subspace are strongly suppressed, direct microwave-driven transitions between logical states are impractical. To overcome this limitation, we use $H^{(2)}_{\text{drive}}$ to drive a transition between a logical state and a non-computational state, thereby avoiding direct population transfer within the logical subspace.

Many CPHASE-gate schemes selectively couple the computational state $|1,1\rangle_L$ to a non-computational auxiliary state, e.g., $|2,0\rangle$ for a transmon. The drive returns the population to $|1,1\rangle_L$ along a closed trajectory on the corresponding Bloch sphere. This cyclic evolution gives $|1,1\rangle_L$ a geometric phase relative to the other computational states, set by the solid angle enclosed by the trajectory [186, 187]. A complete resonant round trip produces a phase of π and therefore implements a CZ gate. Closed trajectories that enclose a smaller solid angle produce a smaller controlled phase.

Our two-qubit gates exploit the static qubit–qubit interaction H_{int} to lift the degeneracy of otherwise identical transition frequencies, while $H^{(2)}_{\text{drive}}$ selectively drives one of the resulting

transitions. For two coupled quantum systems, as the interaction strength g is increased from 0, degenerate states hybridize if they are coupled by the interaction. The hybridization gap grows with the magnitude of the matrix element $|_0 \langle n', m'| H_{\text{int}} |n, m\rangle_0|$. Corresponding transitions in the dressed basis are split. This splitting enables gates that selectively drive one of the previously degenerate transitions.[3]

We now identify a suitable transition for implementing the CZ gate. In the case of $0-\pi$, the interaction Hamiltonian most strongly couples states involving the logical states and the $|1\rangle$ and $|5\rangle$ non-computational states. As a result, four prominent pairs of conditionally split transitions emerge, which we show in Figure 4.7 (c) and (d):

- pink pair: $|2,0\rangle \leftrightarrow |5,0\rangle$ and $|2,2\rangle \leftrightarrow |5,2\rangle$,
- gray pair: $|0,0\rangle \leftrightarrow |1,0\rangle$ and $|0,2\rangle \leftrightarrow |1,2\rangle$,
- purple pair: $|0,2\rangle \leftrightarrow |0,5\rangle$ and $|2,2\rangle \leftrightarrow |2,5\rangle$,
- brown pair: $|0,0\rangle \leftrightarrow |0,1\rangle$ and $|2,0\rangle \leftrightarrow |2,1\rangle$.

Recall that the physical state $|2\rangle$ is our logical $|1\rangle_L$. Only the transition $|2,2\rangle \leftrightarrow |2,5\rangle$ is analogous to the $|1,1\rangle_L \leftrightarrow |2,0\rangle$ transition used in a transmon CZ. However, all such transitions result in a gate equivalent to a controlled-phase gate up to local single-qubit rotations.

To find the most suitable candidate, we additionally consider the frequency mismatch Δ, defined as the frequency difference between the two transitions within each pair. We expect that the most spectrally isolated transition will be the easiest to selectively drive. Among the four pairs, the pair $[|2,2\rangle \text{-} |5,2\rangle\,, |2,0\rangle \text{-} |5,0\rangle]$ has the largest $\Delta = |\omega_{2,0\text{-}5,0} - \omega_{2,2\text{-}5,2}|$.

The matrix elements of these transitions depend strongly on which qubit is driven. Transitions that primarily change the state of the first qubit, such as $|2,0\rangle \leftrightarrow |5,0\rangle$, have smaller matrix elements under an $n_{\theta 2}$ drive than under an $n_{\theta 1}$ drive. This behavior is illustrated in Figure 4.7(c). For the

[3]There is no direct coupling between logical states in the $0-\pi$, so the static ZZ rate $(\omega_{0,2-2,2} - \omega_{0,0-2,0})/2\pi = \Gamma_{ZZ} = 171$ Hz of the coupled $0-\pi$ system is substantially smaller than in transmon or fluxonium devices with comparable interaction strength.

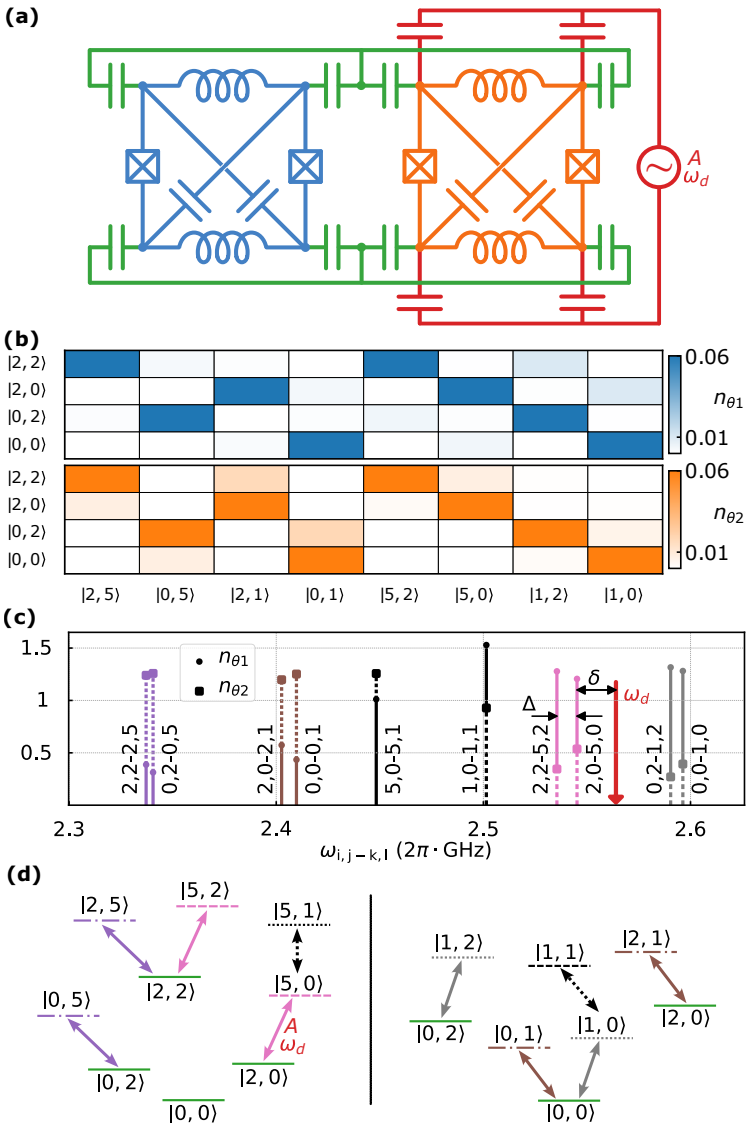


Figure 4.7: CZ gate implemented by driving $n_{\theta 2}$. (a) Circuit diagram. The coupling and drive circuits are depicted in green and red, respectively. (b) Dressed basis charge matrix elements $\langle i,j|\, n_{\theta r}\, |k,l\rangle$ for $r \in \{1,2\}$. The matrix elements are between two-qubit logical states (y-axis) and some non-logical states (x-axis). Four pairs of dominant matrix elements are present in both diagrams. (c) Charge matrix elements $\langle i,j|\, n_{\theta r}\, |k,l\rangle$ for $r \in \{1,2\}$ versus transition frequency $\omega_{i,j\text{-}k,l}$. Solid lines ending in circles denote $n_{\theta 1}$, and dashed lines ending in squares denote $n_{\theta 2}$. (d) Energy level diagram. Four transition pairs in the CZ gate are denoted by solid arrows, using the same color scheme as in (c). Other leakage transitions are represented by dashed arrows.

selected pair, the $|2,0\rangle \leftrightarrow |5,0\rangle$ transition is sufficiently strong, whereas its conditional counterpart $|2,2\rangle \leftrightarrow |5,2\rangle$ is weaker. The resulting matrix element asymmetry is greater than for an $n_{\theta 1}$ drive, making this transition pair well suited for selective excitation. We therefore choose $|2,0\rangle \leftrightarrow |5,0\rangle$ as the primary driven transition under an $n_{\theta 2}$ drive. The combination of a large conditional frequency splitting, an appreciable target matrix element, and a strong matrix element asymmetry makes this transition the most favorable choice for implementing the CZ gate.

In practice, we use the selected transition frequency as a starting point and numerically optimize pulse parameters to maximize gate fidelity. As seen in Figure 4.8, the optimized drive activates multiple transitions, rather than simply driving a round trip between $|2,0\rangle$ and $|5,0\rangle$. It is possible to recover this simpler scheme at the cost of increased gate time. Regardless of the intermediate populations, the final "candidate" gate is approximately diagonal in the computational basis and has the form

$$U_{\text{can}} = \text{diag}\left[e^{i\phi_{0,0}}, e^{i\phi_{0,2}}, e^{i\phi_{2,0}}, e^{i\phi_{2,2}}\right] . \tag{4.21}$$

To arrive at the standard CZ = diag$[1,1,1,-1]$ gate, we apply single-qubit z rotations $R_z(\phi) = \exp[-i\phi Z/2]$ to Equation (4.21). In particular, we transform U_{can} to $U_{\text{std}} = [R_z(\alpha_2) \otimes R_z(\alpha_1)]\, U_{\text{can}}$, where $\alpha_1 = \phi_{0,0} - \phi_{0,2}$ and $\alpha_2 = \phi_{0,0} - \phi_{2,0}$ cancel the local phases. When applying these phase corrections to U_{can}, we arrive at

$$U_{\text{std}} = e^{i\phi_g}\text{diag}\left[1,1,1,e^{i\chi}\right] . \tag{4.22}$$

Here, $\phi_g = (\phi_{0,2} + \phi_{2,0})/2$ is a global phase, and $\chi = \phi_{2,2} + \phi_{0,0} - \phi_{0,2} - \phi_{2,0}$ is the controlled phase. For an ideal CZ gate, $\chi = \pi$.

We now briefly discuss an alternative drive scheme that results in a CNOT gate. This CNOT gate uses a two-qubit Λ-type transition, inspired by the single-qubit X gates in Section 4.3. Thus it requires a two tone drive on one of the qubits. Full details are provided in Section D.4. In noiseless simulations, the CNOT gate achieves fidelities above 99.9%, but it is about two times slower than

the CZ gate at comparable fidelities. The slower operation results from the weaker transition matrix elements and smaller conditional frequency splittings involved. This CNOT construction serves as a proof of principle that the coupling architecture supports microwave-activated two-qubit gates beyond the CZ gate.

4.6 CZ Gate Performance with Decoherence

In this section, we present numerically optimized CZ pulses. We additionally extend the single-qubit noise model presented in Section 4.3.4 to two coupled qubits and examine the optimized pulses subject to this noise model. Performing these open-system simulations is computationally intensive and requires careful construction of the two-qubit Hamiltonian. We refer the interested reader to Section D.5 for additional numerical details.

Gate	n_{op}	Transitions	Associated n_{op}	t_g (ns)	Ω_1 ($2\pi\cdot$MHz)	δ_1 ($2\pi\cdot$MHz)	F_{optimize}	F_{open} ($T = 30\ \mu$s)
CZ	$n_{\theta 2}$	$\lvert 2,0\rangle$-$\lvert 5,0\rangle$	0.537	91.95	61.45	25.19	99.893%	99.830%
				182.01	27.04	13.13	99.993%	99.908%

Table 4.4: The parameters of the two CZ pulses corresponding to the red star and pentagon in Figure 4.9. $A_i/2\pi$ is limited to be less than 250 MHz. The effective Rabi rate is given by $\Omega_1 = A_1(e^2 - 1)n_{\text{op}}$.

4.6.1 Two-Qubit Noise Model

Our phenomenological two-qubit noise model is built as a tensor product of independent single-qubit noise channels. Let L_{ij}^r and L_l^φ denote single-qubit relaxation and dephasing Lindblad operators from Equation (4.10). The two-qubit operators are constructed as

$$\mathfrak{L}^{(1)} = L \otimes I \text{ and } \mathfrak{L}^{(2)} = I \otimes L\,, \tag{4.23}$$

with $L \in \{L_{ij}^{r}, L_{l}^{\varphi}\}$. Though each operator requires multiple indices to characterize the noise process, here we omit them for clarity. The decay and dephasing rates are chosen as in the single-qubit case (see Section 4.3.4). In particular, we choose $T_1 = T_\phi = T$ for both qubits. Dynamics are simulated using the master equation in Equation (4.9). The full two-qubit state is evolved to obtain a propagator and gate fidelity. We provide a more thorough description of the open-system model in Section D.5.2.

4.6.2 Optimized Gate

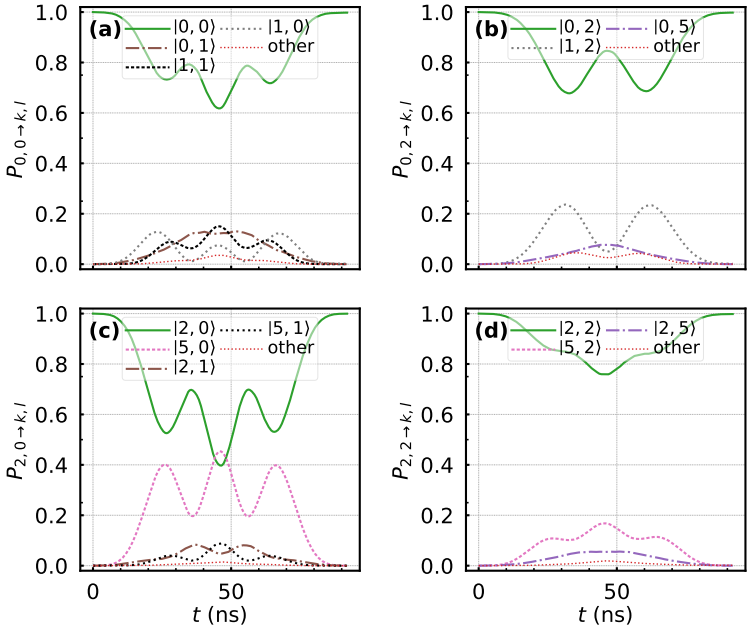


Figure 4.8: Noiseless simulation of population transfer during a CZ gate at $t_g = 91.95$ ns with $F = 99.893\%$. $P_{i,j\text{-}k,l}$ denotes the population of state $|k, l\rangle$ for an initial state $|i, j\rangle$. The initial states in panels (a)–(d) are $|0, 0\rangle$, $|0, 2\rangle$, $|2, 0\rangle$, and $|2, 2\rangle$, respectively. The "other" state refers to the total population of all states not explicitly shown within the lowest 1000 dressed states.

To understand the gate mechanism, we examine the noiseless population dynamics during an optimized CZ gate. In Figure 4.8, we show these dynamics for a gate with $t_g = 91.95$ ns and $F = 99.893\%$. The gate uses a single Gaussian pulse to simultaneously drive multiple transitions. The main CZ transition $|2, 0\rangle$ - $|5, 0\rangle$, shown in panel (c), reaches approximately 40% population in

$|5,0\rangle$. Additional non-computational states associated with the four identified transition pairs are also appreciably populated. For example, Figure 4.8(d) shows population transfer from $|2,2\rangle$ to both $|2,5\rangle$ and $|5,2\rangle$. Despite the nontrivial intermediate dynamics, leakage is small at the end of the gate, with the final population in all other states below 0.4%. These excursions are acceptable under the noise model considered in Section 4.6.1, although intermediate populations can be reduced by increasing the gate duration or utilizing smaller controlled-phase operations.

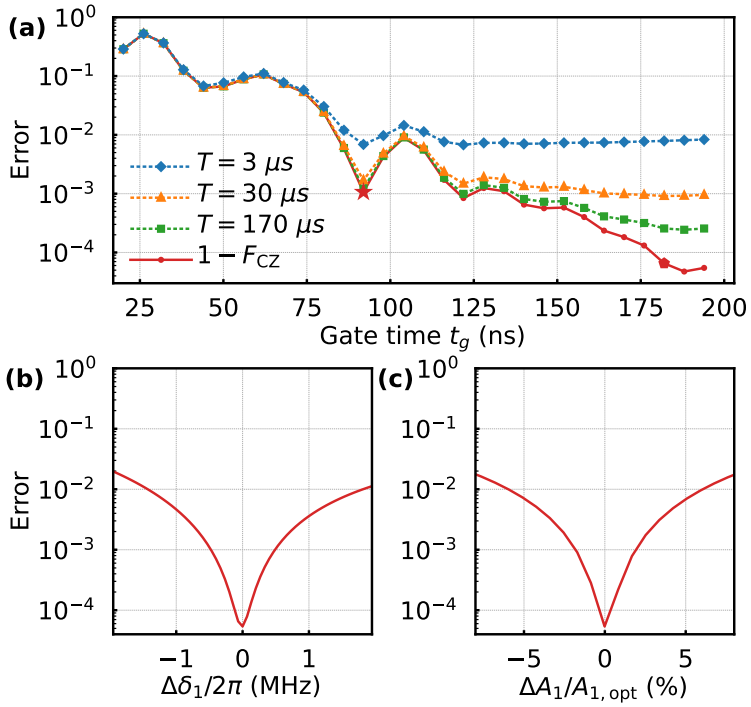


Figure 4.9: CZ gate performance. (a) Gate error $(1 - F)$ versus gate time t_g, using optimized pulse parameters for each t_g. Solid lines show errors for unitary (closed-system) evolution, while dashed lines include decoherence with $T_1 = T_\phi = T \in \{3, 30, 170\}\ \mu$s. The star marks the operating point illustrated in Figure 4.8. We use a reduced model with 60 states to simulate this gate. See Section D.5.2 for additional modeling details. (b) Coherent gate error as a function of the drive-detuning deviation $\Delta\delta_1 = \delta_1 - \delta_{1,\mathrm{opt}}$ at $t_g = 182$ ns, with the drive amplitude fixed at its optimized value. (c) Coherent gate error as a function of the relative drive-amplitude deviation $\Delta A_1/A_{\mathrm{opt}} = (A_1 - A_{1,\mathrm{opt}})/A_{1,\mathrm{opt}}$ at $t_g = 182$ ns, with the detuning fixed at its optimized value. Full pulse parameters for both highlighted pulses are given in Table 4.4.

Figure 4.9 shows how the gate error $1 - F$ scales with gate time t_g for the CZ gate under different coherence times. Panel (a) clearly shows that the CZ gate is coherence limited for shorter

lifetimes. For $T = 3\,\mu$s, for example, the minimum error is approximately 10^{-2}, whereas for $T = 30\,\mu$s it reaches approximately 10^{-3}. Within our phenomenological noise model, the dominant error arises from dephasing of the logical state $|1\rangle_L = |2\rangle$. Panels (b) and (c) show the coherent gate-error landscapes near the optimized operating point as the drive detuning and amplitude are varied. This allows us to assess the robustness of gates to control-parameter errors. The coherent fidelity remains above 99.9% for drive-frequency deviations of approximately 500 kHz and drive-amplitude deviations of about $\pm 2\%$. The resulting coherent errors are much smaller than those introduced by dissipation, indicating that the gate is reasonably robust to calibration errors. For experimentally relevant coherence times, fidelity is therefore limited by logical-qubit coherence, rather than by coherent control errors.

Finally, we investigate the robustness of the optimized CZ gate to device-fabrication imperfections. We independently vary each circuit parameter of the second qubit by up to $\pm 5\%$, re-optimize the gate, and compute the resulting infidelity. The results, summarized in Section D.6, show that the gate is most sensitive to variations in E_C, E_L, and E_J, but relatively insensitive to changes in ϕ_{ext}, n_g^θ, E_{Cc}, E_{CJ}, and E_{C0}. Nevertheless, the median degradation in gate performance is small over the parameter ranges considered, particularly for longer gate times.

4.7 Conclusion

In summary, we have proposed and analyzed a two-qubit CZ gate for capacitively coupled soft $0-\pi$ circuits. Using a phenomenological noise model, we find that modest improvements in experimental coherence times to the tens-of-microseconds regime could enable the CZ gate to achieve error rates of approximately 10^{-3}. These results position the soft $0-\pi$ as a promising intermediate platform between conventional superconducting qubits and more strongly protected qubits.

To make our numerical modeling tractable, we made several simplifying assumptions. Our analysis and simulations assume zero-disorder and neglect the ζ mode, neither of which is realistic. We also model decoherence phenomenologically rather than through a microscopic description of the

device environment. These approximations are sufficient to establish the feasibility of the proposed gate protocols, but the resulting fidelities should be interpreted as an upper bound, rather than as a quantitative prediction for a specific experimental device. Moving beyond these approximations is numerically demanding but would be a worthwhile direction for future work.

Looking forward, several important directions are worth pursuing. First, future theoretical work should quantify how disorder and additional harmonic modes, such as the ζ mode, affect gate performance. Second, experimental measurements of relaxation and dephasing in higher-energy $0-\pi$ states would enable more realistic noise models and pulse optimization. Third, the performance of our gate as devices approach the hard-$0-\pi$ regime is unclear. Recent work by Kolesnikow et al. [178] has proposed protected single- and two-qubit gates for hard-$0-\pi$ devices; incorporating realistic noise into that framework would be a valuable next step.

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Appendix A

Code Listings

The code of the numerical examples of Section 2.3 is collected here.

Listing A.1: Diagonalizing a fluxonium in numpy using a discretized phase basis

```
import numpy as np
import matplotlib.pyplot as plt
def fluxonium_phase(n_pts, EJ, EC, EL, phi_ext, phi_range=(-5*np.pi, 3*np.pi)):
    phi_vals = np.linspace(*phi_range, n_pts)
    dphi = phi_vals[1]-phi_vals[0]
    n2_op = (2*np.eye(n_pts) - np.diag(np.ones(n_pts-1),1) - np.diag(np.ones(n_pts-1),-1))/dphi**2
    H = 4*EC*n2_op + np.diag(0.5*EL*(phi_vals + phi_ext)**2 - EJ*np.cos(phi_vals))
    evals, evecs = np.linalg.eigh(H)
    return phi_vals, evals, evecs
EJ, EC, EL, phi_ext = 10, 0.5, 0.4, np.pi
phi_vals, evals, evecs = fluxonium_phase(201, EJ, EC, EL, phi_ext)
def fix_phase(psi):
    # global phase s.t. the largest point in the first half of the grid is real and positive
    half = psi[:len(psi)//2]
    ref = half[np.argmax(np.abs(half))]
    return psi*np.conj(ref)/np.abs(ref)
# Plotting
plt.figure()
plt.plot(phi_vals, 0.5*EL*(phi_vals + phi_ext)**2 - EJ*np.cos(phi_vals), "--", c="gray", alpha=0.5, label="potential")
n_plot = 4
for i in range(n_plot):
    plt.plot(phi_vals, evals[i] + 10*np.real(fix_phase(evecs[:,i])), label = rf'$|{i}\rangle$')
    plt.axhline(evals[i], ls=":", c=f"C{i}")
plt.legend()
plt.show()
```

Listing A.2: Diagonalizing the same fluxonium in numpy using a truncated Fock basis

```
import numpy as np
from scipy.linalg import expm
def fluxonium_fock(n_fock, EJ, EC, EL, phi_ext):
    a = np.diag(np.sqrt(np.arange(1, n_fock)), 1)
    ad = a.T
    phi_zpf = (2*EC/EL)**0.25
    n_zpf = (EL/(32*EC))**0.25
    phi_op = phi_zpf*(ad + a)
    n_op = 1j*n_zpf*(ad - a)
    # cos(phi) from displacement operators D(+-i phi_zpf)
    alpha = 1j*phi_zpf
    D = expm(alpha*ad - np.conj(alpha)*a)
    cos_phi_op = 0.5*(D + D.conj().T)
    shifted = phi_op + phi_ext*np.eye(n_fock)
    H = 4*EC*n_op@n_op + 0.5*EL*shifted@shifted - EJ*cos_phi_op
    return np.linalg.eigh(H)
EJ, EC, EL, phi_ext = 10, 0.5, 0.4, np.pi
evals, evecs = fluxonium_fock(60, EJ, EC, EL, phi_ext)
```

Listing A.3: Convergence of the lowest four eigenvalues in the Fock basis and in the phase basis with different finite-difference stencils

```
def fluxonium_phase_fd(n_pts, EJ, EC, EL, phi_ext, weights, phi_range
    =(-5*np.pi, 3*np.pi)):
    # as fluxonium_phase, with n^2 from a symmetric stencil weights =
        (w_0, w_1, ..., w_k)
    phi_vals = np.linspace(*phi_range, n_pts)
    dphi = phi_vals[1]-phi_vals[0]
    d2_op = weights[0]*np.eye(n_pts) + sum(w*(np.eye(n_pts, k=j) + np
        .eye(n_pts, k=-j)) for j, w in enumerate(weights) if j > 0)
    H = -4*EC*d2_op/dphi**2 + np.diag(0.5*EL*(phi_vals + phi_ext)**2
        - EJ*np.cos(phi_vals))
    return phi_vals, np.linalg.eigvalsh(H)
ref = fluxonium_fock(140, EJ, EC, EL, phi_ext)[0][:4]
# Fock basis: error vs. number of levels
fock_err = {n: np.abs(fluxonium_fock(n, EJ, EC, EL, phi_ext)[0][:4] -
    ref)
            for n in range(10, 101, 5)}
# Phase basis: error vs. grid spacing for the stencils of the table
stencils = {2: (-2, 1),
            4: (-5/2, 4/3, -1/12),
            6: (-49/18, 3/2, -3/20, 1/90),
            8: (-205/72, 8/5, -1/5, 8/315, -1/560)}
grid_err = {}
for order, weights in stencils.items():
    for n_pts in (85, 101, 151, 201, 301, 401, 601, 801):
        phi_vals, evals = fluxonium_phase_fd(n_pts, EJ, EC, EL,
            phi_ext, weights)
        grid_err[order, phi_vals[1] - phi_vals[0]] = np.abs(evals[:4]
```

```
            - ref).max()
```

Listing A.4: Diagonalizing a transmon in numpy using the charge basis

```
import numpy as np
def transmon_charge(n_cut, EJ, EC, ng):
    n_vals = np.arange(-n_cut, n_cut+1)
    dim = len(n_vals)
    # cos(phi) = (e^{i phi} + e^{-i phi})/2 shifts n by +-1
    cos_phi_op = 0.5*(np.eye(dim, k=1) + np.eye(dim, k=-1))
    H = np.diag(4*EC*(n_vals - ng)**2) - EJ*cos_phi_op
    return n_vals, *np.linalg.eigh(H)
# lowest three levels vs. offset charge
ng_vals = np.linspace(-0.5, 0.5, 51)
levels = np.array([transmon_charge(10, 5, 0.2, ng)[1][:3] for ng in
   ng_vals])
```

Listing A.5: Diagonalizing a transmon in numpy using a periodic phase grid with offset charge

```
import numpy as np
# central first-derivative stencils (w_1, ..., w_k): d/dphi psi_i =
    sum_j w_j (psi_{i+j} - psi_{i-j}) / dphi
stencils_d1 = {2: (1/2,),
               4: (2/3, -1/12),
               6: (3/4, -3/20, 1/60),
               8: (4/5, -1/5, 4/105, -1/280)}
def transmon_phase_fd(n_pts, EJ, EC, ng, order):
    # periodic grid: phi = -pi and pi are the same point, so leave
       out the endpoint
    phi_vals = np.linspace(-np.pi, np.pi, n_pts, endpoint=False)
    dphi = phi_vals[1]-phi_vals[0]
    # periodic gauge: psi(phi + 2 pi) = psi(phi), so the stencils
       wrap around cyclically
    shift = lambda j: np.roll(np.eye(n_pts), j, axis=0)  # (S psi)_i
       = psi_{i-j}
    w1, w2 = stencils_d1[order], stencils[order]
    d1_op = sum(w*(shift(-j) - shift(j)) for j, w in enumerate(w1, 1)
       )/dphi
    d2_op = (w2[0]*np.eye(n_pts) + sum(w*(shift(j) + shift(-j)) for j
       , w in enumerate(w2) if j > 0))/dphi**2
    n_op, n2_op = -1j*d1_op, -d2_op  # n = -i d/dphi
    # the offset charge enters only through the Hamiltonian, 4 EC (n
       - ng)^2 expanded
    H = 4*EC*(n2_op - 2*ng*n_op + ng**2*np.eye(n_pts)) - EJ*np.diag(
       np.cos(phi_vals))
    return phi_vals, *np.linalg.eigh(H)
```

Listing A.6: Hierarchical diagonalization of a capacitively coupled fluxonium and transmon

```
import numpy as np
```

```
def coupled_levels(E1, V1, O1, E2, V2, O2, g, N1, N2, n_levels=10):
    # hierarchical diagonalization: keep the lowest N1 (N2)
       eigenstates of each subsystem
    # and express the coupling g O1 (x) O2 in those eigenbases
    V1, V2 = V1[:, :N1], V2[:, :N2]
    H = np.kron(np.diag(E1[:N1]), np.eye(N2)) + np.kron(np.eye(N1),
       np.diag(E2[:N2]))
    H = H + g*np.kron(V1.conj().T @ O1 @ V1, V2.conj().T @ O2 @ V2)
    return np.linalg.eigvalsh(H)[:n_levels]
# subsystems: fluxonium (Fock basis) and transmon (charge basis),
   diagonalized separately
n_fock, n_cut = 60, 10
E_f, V_f = fluxonium_fock(n_fock, EJ, EC, EL, phi_ext)
n_vals, E_t, V_t = transmon_charge(n_cut, 5, 0.2, 0)
# charge operators in the bare bases
a = np.diag(np.sqrt(np.arange(1, n_fock)), 1)
n_f = 1j*(EL/(32*EC))**0.25*(a.T - a)
n_t = np.diag(n_vals)
# capacitive coupling H_int = g n_f n_t
g = 0.1
couple = lambda N1, N2: coupled_levels(E_f, V_f, n_f, E_t, V_t, n_t,
   g, N1, N2)
ref_cap = couple(n_fock, 2*n_cut + 1)  # no truncation of the
   subsystem bases
N_f_vals, N_t_vals = range(4, 31), range(4, 22)  # N_f N_t >=
   n_levels
err_cap = np.array([[np.abs(couple(N_f, N_t) - ref_cap).max() for N_t
    in N_t_vals]
                    for N_f in N_f_vals])
```

Appendix B

Circuit Quantization and Modeling

B.1 Assigning External Flux Using a Spanning Tree

A spanning tree supplies a convenient allocation and is the standard tool used to assign external fluxes self-consistently [23]. In the spanning tree construction, a subgraph is chosen that touches every vertex without closing a loop. The edges left out of the tree are the closure branches, and all external fluxes are assigned to them. In our vector notation, we can order the branches as (tree – t, closure – c). We then define

$$\mathrm{G} = \begin{pmatrix} \mathrm{G}_t & I \end{pmatrix} \quad \text{and} \quad \mathrm{B} = \begin{pmatrix} 0 \\ I \end{pmatrix} , \tag{B.1}$$

which solves $\mathrm{GB} = I$ immediately. The block G_t is also fixed. Expanding $\mathrm{G}\Omega^T = 0$ in the same ordering gives $\mathrm{G}_t\Omega_t^T + \Omega_c^T = 0$, so $\mathrm{G}_t = -\Omega_c^T \left(\Omega_t^T\right)^{-1}$, where Ω_t and Ω_c are the tree and closure blocks of Ω. A spanning tree on N_v vertices has $N_v - 1$ edges, so Ω_t is not of full row rank. For this construction we can simply delete a row. The cycle space does not depend on the choice, and the resulting G satisfies $\mathrm{G}\Omega^T = 0$ against the full Ω as well. Deleting the root of the tree is the convenient choice and is equivalent to the standard procedure of starting the spanning tree at the ground node.[1] Typically, the ground node is excluded from the branch analysis at the start, so the root-deletion step is implicit. Here we have included it explicitly as we treat grounded circuits on

[1] Ordering each tree edge by the vertex it feeds, and the vertices so that parents precede children, makes Ω_t triangular, and its inverse is then triangular with entries in $\{0, \pm 1\}$, given by whether an edge lies on the path from the root to a vertex. The entries of G thus come out in $\{0, \pm 1\}$ as required.

equal footing with floating ones.

B.2 Converting Between Physical and Frequency Units for Circuit Parameters

It is common to work with dimensionless phase φ and Cooper-pair number n, obtained from the dimensionful node flux Φ and node charge Q by

$$\varphi = \frac{\Phi}{\phi_0}, \quad \text{and} \quad n = \frac{Q}{2e}. \tag{B.2}$$

Here $e > 0$ is the elementary charge of an electron and $\phi_0 = \Phi_0/2\pi = \hbar/2e$ is the reduced flux quantum. External fluxes and offset charges rescale the same way, with

$$\varphi_{\text{ext}} = \frac{\Phi_{\text{ext}}}{\phi_0} \quad \text{and} \quad n_g = \frac{Q_g}{2e}. \tag{B.3}$$

This choice gives the canonical commutator $[\varphi_i, n_j] = i\delta_{ij}$ with no residual $\hbar$. It also removes ϕ_0 from the Josephson cosines, so compact directions are 2π-periodic in phase with integer-valued conjugate n.

With dimensionless flux and charge operators, circuit parameters are typically quoted as frequencies. For a single element the charging and inductive energies are related to the physical capacitance C and inductance L by

$$hE_C = \frac{e^2}{2C}, \quad \& \quad hE_L = \frac{\phi_0^2}{L}, \tag{B.4}$$

and the Josephson energy of a junction of critical current I_c by $hE_J = I_c\phi_0$. Of course, Equation (B.4) ensures that the energy stored in a capacitor and in an inductor,

$$4hE_C n^2 = \frac{Q^2}{2C}, \quad \& \quad \frac{1}{2}hE_L\varphi^2 = \frac{\Phi^2}{2L}, \tag{B.5}$$

is the same in either set of units. We follow the usual convention of writing E_J, E_C, E_L, and H for

the frequencies E_J/h, E_C/h, E_L/h, and H/h.

Applying Equation (B.2) to the multimode Hamiltonian defines the frequency-unit matrices

$$\mathcal{E}_C \equiv \frac{e^2}{2h} C^{-1}, \quad \mathcal{E}_L \equiv \frac{\phi_0^2}{h} L^*, \quad \text{and} \quad \mathcal{E}_{L_b} \equiv \frac{\phi_0^2}{h} L_b^{-1}, \tag{B.6}$$

which reduce to Equation (B.4) for a single element. In these units, Equation (2.13) becomes

$$\mathcal{H}(n, \varphi) = 4\big(n - \tilde{n}_g(t)\big)^T \mathcal{E}_C \big(n - \tilde{n}_g(t)\big) + \mathcal{U}(\varphi)\,, \tag{B.7}$$

where the potential energy is

$$\mathcal{U}(\varphi) = \frac{1}{2}\varphi^T \mathcal{E}_L \varphi + \varphi^T \tilde{\omega}_L \mathcal{E}_{L_b} \mathrm{B}_L \varphi_{\mathrm{ext}}(t) - \sum_{k \in \mathcal{S}_J} E_{J_k} \cos\big([\tilde{\omega}_J^T \varphi + \mathrm{B}_J \varphi_{\mathrm{ext}}(t)]_k\big)\,.$$

Here C and L^* are the retained $\tilde{C}$ and $\tilde{L}^*$ of the reduced Hamiltonian, so the inverse is taken on the invertible $\tilde{C}$ and never on the singular full node capacitance matrix. The energy of the system is then given by hH.

B.3 Product States and Mean-Field Hamiltonians

This appendix gives the construction behind the product-state coordinates of Section 2.4.2. This includes the product ansatz and its energy, the mean-field Hamiltonians, the outer optimization over S, and the hierarchical diagonalization using the mean-field eigenstates as a basis. We work through the $\cos(2\varphi)$ KITE of Figure 2.9 at each step.

B.3.1 The Hamiltonian in a transformed frame

In the published coordinates $(\varphi, \varphi_\Sigma, \varphi_\Delta)$ of Ref. [41], with conjugate charges (n, n_Σ, n_Δ), zero offset charge, and the loop phase $\pi + \delta$ shared equally over the four inductors, the KITE Hamiltonian

of Figure 2.9(a) is

$$H = 4\,n^T \mathcal{E}_C n + E_L\left(\varphi_\Sigma^2 + \varphi_\Delta^2\right) - E_L(\pi+\delta)\,\varphi_\Delta - E_J(1+\epsilon)\cos(\varphi - \varphi_\Sigma + \varphi_\Delta) - E_J(1-\epsilon)\cos(\varphi - \varphi_\Sigma - \varphi_\Delta)\,, \tag{B.8}$$

with the charging matrix

$$\mathcal{E}_C = \begin{pmatrix} E_{C_S} & E_{C_S} & 0 \\ E_{C_S} & E_{C_S} + \dfrac{E_{C_J}}{2(1-\epsilon^2)} & \dfrac{\epsilon E_{C_J}}{2(1-\epsilon^2)} \\ 0 & \dfrac{\epsilon E_{C_J}}{2(1-\epsilon^2)} & \dfrac{E_{C_J}}{2(1-\epsilon^2)} \end{pmatrix}.$$

Here E_L is the inductive energy of each arm, whose two inductors of energy $2E_L$ sit on either side of its junction, and the junction area disorder scales E_J and C_J together by $1 \pm \epsilon$. Each junction spans all three coordinates, while the quadratic coupling is only the E_{C_S} entry between φ and φ_Σ and the ϵE_{C_J} entry between φ_Σ and φ_Δ. We consider static external flux, no voltage sources, and zero offset charge.

Written in the coordinates $\varphi = S\varphi'$ of a transformation $S \in \mathcal{A}_{\text{relaxed}}$, Equation (B.8) becomes

$$H' = 4\,n'^T \mathcal{E}'_C n' + \tfrac{1}{2}\varphi'^T \mathcal{E}'_L \varphi' + b'^T \varphi' - \sum_{k \in \mathcal{S}_J} E_{J_k} \cos\left(\tilde{\omega}'^T_{J,k}\varphi' + \delta_k\right), \tag{B.9}$$

with $\mathcal{E}'_C = S^{-1}\mathcal{E}_C S^{-T}$, $\mathcal{E}'_L = S^T \mathcal{E}_L S$, $b' = S^T b$, and $\tilde{\omega}'_{J,k} = S^T \tilde{\omega}_{J,k}$. Here $b = \tilde{\omega}_L \mathcal{E}_{L_b} \mathrm{B}_L \varphi_{\text{ext}}$ collects the external flux assigned to the inductors, $\delta_k = [\mathrm{B}_J \varphi_{\text{ext}}]_k$ the flux assigned to junction k, and $\tilde{\omega}_{J,k}$ is the kth column of $\tilde{\omega}_J$, so that $\tilde{\omega}'^T_{J,k}\varphi'$ is the phase across junction k. For the KITE, Equation (B.8) has $\mathcal{E}_L = \mathrm{diag}(0, 2E_L, 2E_L)$, $b = (0, 0, -E_L(\pi+\delta))$, junction rows $\tilde{\omega}_{J,1} = (1, -1, 1)$ and $\tilde{\omega}_{J,2} = (1, -1, -1)$, and $\delta_k = 0$. At the parameters of Section 2.4.2, $E_J = 16.83$, $E_{C_J} = 4.82$, $E_{C_S} = 0.072$, and $E_L = 1.27$ GHz with $\epsilon = 0.03$ and $\delta = 0$, this gives $b = (0, 0, -3.99)$ GHz and a charging coupling of 0.072 GHz. In the optimized frame of Equation (2.32) the junction rows become

$\tilde{\omega}'_{J,1} = (1, -1.03, 0.13)$ and $\tilde{\omega}'_{J,2} = (1, 0.11, -1.13)$, at the price of off-diagonal quadratic terms,

$$\mathcal{E}'_C = \begin{pmatrix} 0.29 & -0.55 & -0.48 \\ -0.55 & 3.01 & -0.46 \\ -0.48 & -0.46 & 2.77 \end{pmatrix}, \quad \mathcal{E}'_L = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2.06 & 0.38 \\ 0 & 0.38 & 2.37 \end{pmatrix}$$

in GHz, with $b' = (0, 2.28, -2.51)$ GHz.

B.3.2 Optimization over S with a product state anzatz

The goal is a transformation S where the products of single-mode states represent the full Hamiltonian well, so that a few eigenstates per mode allow for accurate hierarchical diagonalization. As a proxy we use the best product approximation to the ground state, as in the optimized coordinates of vibrational structure theory [82]. The lower its energy in the full Hamiltonian, the closer the ground state is to a product, and the better a product basis built from the same single-mode states should represent the low-lying levels. Beginning with a product of normalized single-mode states, $|\Psi\rangle = \bigotimes_i |\psi_i\rangle$, its energy $E_{\text{prod}} = \langle\Psi| H' |\Psi\rangle$ is a variational upper bound on the ground energy E_0. Figure B.1 summarizes the two nested minimizations that follow.

Every term of Equation (B.9) is a product of single-mode operators, so its expectation value in a product state factorizes into single-mode expectation values. We iteratively minimize each mode's ground state as a function of the single mode eigenstates independently and continue until the overall ground state energy converges. For diagonalizing each mode, we use a uniform phase grid with eighth-order finite differences, wide enough that the inductive energy at its edge exceeds twice the junction energy the mode sees, fine enough to resolve the zero-point width and the shortest junction wavelength, and enlarged if a converged state reaches its edge. For the KITE this gives 33 charge states and about 120 grid points per extended mode.

Each step of this iteration diagonalizes a mean-field Hamiltonian, which we denote h_i. This is the full Hamiltonian Equation (B.9), with each cosine of a sum of phases expanded by the angle-addition formula, and every operator of a mode other than the one in question replaced by the

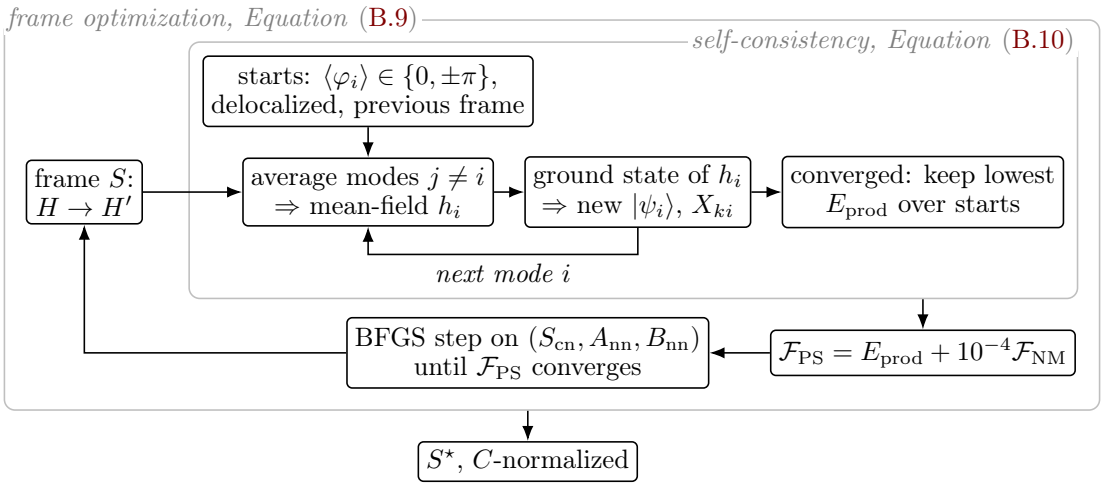


Figure B.1: **The two nested minimizations of the product-state search.** The inner loop converges a product state in a fixed frame: each mode in turn is replaced by the ground state of its mean-field Hamiltonian h_i, the full Hamiltonian averaged over the other modes, until E_{prod} stops changing, and the lowest of the solutions reached from the standard starts is kept. The outer loop minimizes the cost of Equation (B.12) over the frame S by BFGS.

scalar expectation value of the current product state. Mode i then sees the average potential the other modes create. We write the mean-field Hamiltonian h_i in terms of the expectation value $\langle\ \ \rangle$ of the current $|\psi_j\rangle$,

$$h_i = 4\mathcal{E}'_{C,ii}n_i^2 + \tfrac{1}{2}\mathcal{E}'_{L,ii}\varphi_i^2 + \Big(b'_i + \sum_{j\neq i}\mathcal{E}'_{L,ij}\langle\varphi_j\rangle\Big)\varphi_i + 8\sum_{j\neq i}\mathcal{E}'_{C,ij}\langle n_j\rangle\, n_i$$
$$- \sum_k E_{J_k}|z_{ki}|\cos\big(\tilde{\omega}'_{J,ki}\varphi_i + \arg z_{ki}\big)\,, \qquad z_{ki} = e^{i\delta_k}\prod_{j\neq i}X_{kj}\,. \tag{B.10}$$

Here $X_{kj} = \langle\cos(\tilde{\omega}'_{J,kj}\varphi_j)\rangle + i\langle\sin(\tilde{\omega}'_{J,kj}\varphi_j)\rangle$ collects the sine and cosine averages of mode j over one period of junction k. The final $E_{\mathrm{prod}} = \langle\psi_i|\, h_i\, |\psi_i\rangle + \mathrm{const}$. A mode spread over a full period of a junction has $X_{kj} = 0$ and hides that junction from the others, while a mode sharply localized in a junction's well has $|X_{kj}| = 1$. The ground state of h_i is the best choice of $|\psi_i\rangle$ while the other modes are held fixed, so cycling through the modes and updating each in turn ends at a self-consistent set of states, each the ground state of the mean field the others create.

A system can have multiple localized wells, and the converged solution depends on the starting

guess. As a result, we run the iteration from several starts and keep the lowest converged energy. A start is the set of scalars that builds the first mean-field Hamiltonians. A displacement start pins every mode at a phase $\varphi_j^{(0)} \in \{0, \pm\pi\}$,

$$\langle n_j \rangle = 0\,, \qquad \langle \varphi_j \rangle = \varphi_j^{(0)}\,, \qquad X_{kj} = \cos\!\big(\tilde{\omega}'_{J,kj}\varphi_j^{(0)}\big) + i\sin\!\big(\tilde{\omega}'_{J,kj}\varphi_j^{(0)}\big)\,, \tag{B.11}$$

which are the expectation values of a state perfectly localized at $\varphi^{(0)}$; the mean phase is only needed for the extended modes, since a compact mode has no inductive terms. Every junction then enters the first h_i with full amplitude, $|z_{ki}| = 1$, and with its cosine shifted by the phase the other modes contribute at $\varphi^{(0)}$, so the iteration relaxes from that classical configuration into the nearest well. We take all 3^N combinations of 0 and $\pm\pi$. The delocalized start instead sets $\langle n_j \rangle = \langle \varphi_j \rangle = 0$ and every $X_{kj} = 0$, so no junction is visible at the first step and each mode begins from the ground state of its quadratic terms alone. For the KITE, the relevant solutions are the two wells of the $\cos(2\varphi)$ potential, with φ near 0 or π and the extended modes displaced to match. In the published coordinates the wells are degenerate at $E_{\text{prod}} = -8.08$ GHz, with $(\varphi, \varphi_\Delta)$ near $(0, 0.18)$ or $(\pi, 2.96)$, and the delocalized start alone reaches a third solution, a saddle at $+3.83$ GHz with $\langle \varphi_\Delta \rangle = \pi/2$. In the optimized frame the same starts reach only the two wells, at -9.43 GHz. The ground doublet of the full circuit is the symmetric and antisymmetric combination of the two wells, so a product state localized in one well overlaps each doublet member with weight near one half. This is why Section 2.4.2 reports the weight outside the doublet, $1 - \sum_s |\langle s|\Psi\rangle|^2$ over the two doublet states, rather than the overlap with the ground state.

The outer optimization minimizes the converged product energy over the frame, with a small tie-break toward the normal-mode cost $\mathcal{F}_{\text{NM}}$ of Equation (2.31),

$$\mathcal{F}_{\text{PS}}(S) = E_{\text{prod}}(S) + 10^{-4}\,\mathcal{F}_{\text{NM}}(S)\,, \tag{B.12}$$

minimized over $S \in \mathcal{A}_{\text{relaxed}}$ in the parametrization of Equation (2.30) by BFGS with finite-difference

gradients, from the identity and from a few random starts. Each evaluation of E_{prod} runs the mean-field iteration from the standard starts and, as a warm start, from the expectation values of the previous frame. The discrete block S_{cc} is enumerated separately, and the KITE has one compact mode, so we consider $S_{\mathrm{cc}} = \pm 1$ and perform a continuous optimization over the six entries of $(S_{\mathrm{cn}}, A_{\mathrm{nn}}, B_{\mathrm{nn}})$. We fix the scale afterwards by normalizing each noncompact column to $(S^T \mathcal{E}_C^{-1} S)_{jj} = 1$ in GHz^{-1} (i.e., the C-normalization of Equation (2.32)).

B.3.3 Hierarchical diagonalization with mean-field leaves

Hierarchical diagonalization trades the full single-mode grids for a small basis per mode. For each mode i one keeps m single-mode eigenstates states, which we call leaves, and represent the full Hamiltonian in the m^N products of leaves. Every single-mode operator A in Equation (B.9) becomes the $m \times m$ matrix $U_i^\dagger A U_i$, the products of these matrices are assembled into an $m^N \times m^N$ matrix, and its lowest levels are found by dense diagonalization. The usual leaves are the eigenstates of each mode's own Hamiltonian with the coupling dropped. For the KITE every junction couples all three modes, so these bare leaves are oscillator and charge states that know nothing of the junctions, and many of them are needed to build a state localized in a junction well.

The mean-field Hamiltonians describe instead what each mode sees in the ground state, so their eigenstates already contain each junction, at the amplitude and phase the other modes give it. They are a natural single-mode basis for the hierarchical diagonalization of Section 2.4.2, but they also require more states than a single mean field would. Each well w has its own $h_i^{(w)}$, and we draw the leaves from both, since the low-lying levels are doublets of the two wells while each mean-field solution describes only one of them. For each well w and each mode i we take the m lowest eigenvectors of $h_i^{(w)}$ on the single-mode grid. In the published coordinates the φ_Σ leaves of the two wells coincide and only the other two modes differ, while in the optimized frame all three differ, with an overlap of 0.37 between the wells' extended-mode ground states. We merge the two sets level by level and orthonormalize them by Gram–Schmidt, discarding any vector whose remainder is

below 10^{-6}, until m vectors U_i remain per mode.

In the optimized frame these two-well leaves give a largest error of the four lowest levels of 2×10^{-3} GHz at $m = 12$ and 2×10^{-5} GHz at $m = 20$. With leaves from one well only, the second doublet member is poorly represented at small m. Bare leaves, the eigenstates of each mode's own quadratic terms with no mean field, treat both wells alike but see no junction at all, since every junction of the KITE involves all three modes in either frame. They are an order of magnitude worse than the two-well leaves, at 3×10^{-2} and 5×10^{-4} GHz.

B.4 Notation

Table B.1 collects the symbols of Section 2.2, with their dimensions, units, and how they transform with Z.

Table B.1: **Notation of Section 2.2.** The circuit has N internal nodes (excluding ground and voltage-source nodes), N_b branches, N_ℓ independent loops, and N_V voltage sources. N_L and N_J count the inductive and Josephson branches, N_c the compact coordinates, and N_d the dynamical coordinates. The last column gives the behavior under $\Phi = Z\tilde{\Phi}$ if applicable.

Symbol	Object [Units]	Description	Under $\Phi = Z\tilde{\Phi}$
Constants and circuit parameters			
$2e$	constant	Cooper-pair charge, with $e > 0$ the elementary charge.	invariant
Φ_0, ϕ_0	constant [Wb]	Superconducting flux quantum $\Phi_0 = h/2e$ and reduced flux quantum $\phi_0 = \Phi_0/2\pi$.	invariant
C_k, L_k	$\mathbb{R}$ [F], [H]	Capacitance and inductance of branch k.	invariant
E_{J_k}	$\mathbb{R}$ [GHz]	Josephson energy of the junction on branch k. Quoted as the frequency E_{J_k}/h.	invariant
$\mathcal{S}_L$, $\mathcal{S}_J$	index sets	Branches carrying an inductor or a Josephson junction, of sizes N_L and N_J.	invariant
Coordinates and conjugate variables			
Φ	$\mathbb{R}^N$ [Wb]	Node-flux vector, the generalized coordinate of the Lagrangian, with $\Phi_i = \int_{-\infty}^{t} V_i(t')\,dt'$ for the voltage V_i of node i relative to ground.	$\tilde{\Phi} = Z^{-1}\Phi$
$\dot{\Phi}$	$\mathbb{R}^N$ [V]	Node voltages, the generalized velocities.	$\dot{\tilde{\Phi}} = Z^{-1}\dot{\Phi}$
Q	$\mathbb{R}^N$ [C]	Node charge conjugate to Φ, with $Q = C\dot{\Phi} + Q_g$ and $[\Phi_i, Q_j] = i\hbar\delta_{ij}$.	$\tilde{Q} = Z^T Q$
Φ_b	$\mathbb{R}^{N_b}$ [Wb]	Branch-flux vector, $\Phi_b = \Omega^T\Phi + \mathrm{B}\Phi_{\mathrm{ext}}$. It is physical rather than a coordinate choice.	invariant
Φ_{ext}	$\mathbb{R}^{N_\ell}$ [Wb]	External fluxes threading the N_ℓ independent loops, including loops broken by capacitors. Split as $\Phi_{\mathrm{ext}}(t) = \Phi_{\mathrm{ext}}^{\mathrm{DC}} + \Phi_{\mathrm{ext}}^{\mathrm{AC}}(t)$ into static DC and time-dependent AC parts.	invariant
$V(t)$	$\mathbb{R}^{N_V}$ [V]	Voltages of the classical source nodes.	invariant
Q_g	$\mathbb{R}^N$ [C]	Offset charge, $Q_g = -C_V V(t) + \Omega C_b \mathrm{B}\dot{\Phi}_{\mathrm{ext}}(t)$, entering the Hamiltonian as $Q - Q_g$.	$\tilde{Q}_g = Z^T Q_g$
φ, n	$\mathbb{R}^N$	Dimensionless phase $\varphi = \tilde{\Phi}/\phi_0$ and Cooper-pair number $n = \tilde{Q}/2e$.	as Φ and Q
$\tilde{n}_g$	$\mathbb{R}^N$	Dimensionless offset charge $\tilde{n}_g = \tilde{Q}_g/2e$.	as Q_g
φ_{ext}	$\mathbb{R}^{N_\ell}$	Dimensionless external flux $\varphi_{\mathrm{ext}} = \Phi_{\mathrm{ext}}/\phi_0$.	invariant
Circuit matrices			
Ω	$\{0, \pm 1\}^{N \times N_b}$	Branch incidence matrix. Column k has $+1$ at the head node and -1 at the tail node of branch k, with only the $+1$ for branches to ground or a voltage source.	$\tilde{\Omega}^T = \Omega^T Z$
ω_L	$\{0, \pm 1\}^{N \times N_L}$	Inductor incidence matrix, the columns of Ω for $k \in \mathcal{S}_L$.	$\tilde{\omega}_L^T = \omega_L^T Z$
ω_J	$\{0, \pm 1\}^{N \times N_J}$	Junction incidence matrix, the columns of Ω for $k \in \mathcal{S}_J$.	$\tilde{\omega}_J^T = \omega_J^T Z$
G	$\{0, \pm 1\}^{N_\ell \times N_b}$	Loop matrix, whose rows list signed branches around each independent loop, $\mathrm{G}\Omega^T = 0$.	invariant
G_{JL}	rows of G	Loops whose branches all carry an inductor or junction, which obey fluxoid quantization $\mathrm{G}_{JL}\Phi_b = \Phi_{\mathrm{ext}}$. The remaining loops are broken by a capacitor and obey Faraday's law.	invariant
B	$\mathbb{R}^{N_b \times N_\ell}$	Flux-assignment matrix distributing the external fluxes over branches, constrained by $\mathrm{GB} = I$. The irrotational choice $\mathrm{B}^{\mathrm{irr}}$, with $Z^T\Omega C_b \mathrm{B}^{\mathrm{irr}} = 0$, is Equation (2.20).	invariant
B^{DC}	$\mathbb{R}^{N_b \times N_\ell}$	Separate assignment of the DC flux $\Phi_{\mathrm{ext}}^{\mathrm{DC}}$, constrained only by $\mathrm{GB}^{\mathrm{DC}} = I$.	invariant
B_L	$\mathbb{R}^{N_L \times N_\ell}$	Rows of B for $k \in \mathcal{S}_L$.	invariant
B_J	$\mathbb{R}^{N_J \times N_\ell}$	Rows of B for $k \in \mathcal{S}_J$.	invariant
C_b	$\mathbb{R}^{N_b \times N_b}$ [F]	Diagonal branch capacitance matrix with entries C_k.	invariant
L_b^{-1}	$\mathbb{R}^{N_b \times N_b}$ $[\mathrm{H}^{-1}]$	Diagonal branch inverse-inductance matrix with entries $1/L_k$, and zero on branches without an inductor.	invariant
C	$\mathbb{R}^{N \times N}$ [F]	Node capacitance matrix, $C = \Omega C_b \Omega^T$.	$\tilde{C} = Z^T C Z$
L^*	$\mathbb{R}^{N \times N}$ $[\mathrm{H}^{-1}]$	Node susceptance matrix, $L^* = \Omega L_b^{-1}\Omega^T$.	$\tilde{L}^* = Z^T L^* Z$
C_V	$\mathbb{R}^{N \times N_V}$ [F]	Source-coupling matrix, with entry (i, s) the capacitance between node i and source s.	$\tilde{C}_V = Z^T C_V$
Z	$\mathrm{GL}(N, \mathbb{R})$	Coordinate transformation, with $\Phi = Z\tilde{\Phi}$ and $\tilde{\Phi} = Z^{-1}\Phi$.	—
Mode identification and compact coordinates			
C, L, J	subspaces of $\mathbb{R}^N$	Null spaces $\mathrm{null}(C)$, $\mathrm{null}(L^*)$, and $\mathrm{null}(\omega_J^T)$, which define the six mode types of Figure 2.2.	$Z^{-1}\mathsf{C}$, etc.
$\mathcal{P}$	lattice in $\mathbb{R}^{N_c}$	Periodicity lattice of the compact coordinates, $\mathcal{P} = \{a : \tilde{\omega}_{J,c}^T a \in 2\pi\mathbb{Z}^{N_J}\}$.	—
$\mathcal{K}$	index set	N_c junctions with linearly independent rows of $\tilde{\omega}_{J,c}^T$. Fixes coordinates in Equation (2.17).	—

Frequency units (Section B.2)

Appendix C

Circuit Enumeration

C.1 Subtleties in the Enumeration

Enumerating all possible circuits is a formidable task. Although we have carried out this enumeration, it relies on several assumptions. In this appendix, we discuss subtleties that fall outside the scope of the main study and may affect the circuits we enumerate. These include stray capacitances and inductances, readout resonators, transmission lines, and Purcell filters. The list is not exhaustive, and no single study could address all such complications.

C.1.1 Ground Placement and Circuit Equivalence

Our enumeration procedure does not assign any nodes to ground, although the choice of ground can significantly change a circuit's Hamiltonian. When modeling circuits in Section 3.6, we choose to include capacitive coupling between each node and ground.

Ground placement can also change the effective circuit topology. Consider the two circuits in Figure C.1(a) and (b), which have three and four nodes, respectively. If we ground one node in Figure C.1(a) and two nodes in Figure C.1(b), both reduce to a circuit that resembles galvanically coupled transmons. As a result, including grounds can cause our procedure to over-count unique circuits. For a graph with N nodes, there are $\sum_{k=1}^{N-1} \binom{N}{k} = 2^N - 2$ ways to ground one or more nodes. One could enumerate these possibilities and identify equivalent configurations by adding a ground "color" to the graph-isomorphism procedure described in Section 3.3. However, grounding

multiple nodes effectively reduces the number of nodes in the circuit. These reduced circuits are therefore not new configurations and already appear in the enumeration of smaller circuits.

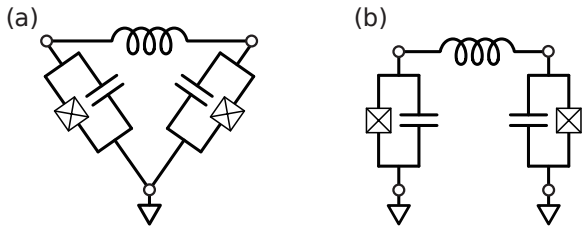


Figure C.1: Illustration of how the number and placement of ground nodes can induce circuit isomorphism. Circuit (a) has three nodes, while circuit (b) has four. The two circuits become equivalent when the indicated one node in (a) and two nodes in (b) are grounded.

C.1.2 Ambiguities Related to SQUIDs and Linearization

We did not include superconducting quantum interference devices (SQUIDs) or tunable inductors built from flux-biased SQUIDs in our enumeration. In a sense, any Josephson junction in our enumeration could be viewed as a split junction, i.e., a SQUID. Likewise, any linear inductor could be viewed as a linearized SQUID; see Figure C.2. These distinctions matter at the Hamiltonian level because SQUID-based elements introduce additional tunable terms, such as $(\phi - \phi_{\text{ext}})^2$ and $\cos(\phi - \phi_{\text{ext}})$.

Flux bias can also be used to engineer effective cubic or quartic terms [189, 190]. A standard example is the SNAIL [189]. The SNAIL consists of several Josephson junctions in a series-parallel arrangement. By tuning the external flux, one can isolate selected powers of the phase operator, ϕ^n. To include such effects, one could introduce new effective dipole elements into the enumeration. This is a direction that could reveal new circuit classes, but it lies beyond the scope of our present procedure.

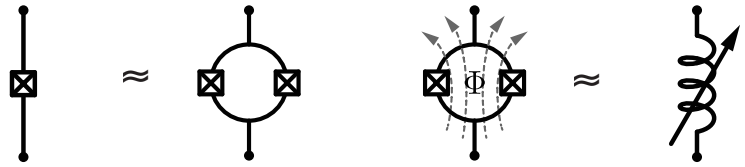


Figure C.2: Flux-tunable elements not included explicitly in our enumeration.

C.1.3 Charge and Flux Bias

We do not include charge and flux biases in the enumeration itself, although they do appear in the optimization over unique circuits; see Figure C.3. These biases can distinguish circuits that otherwise belong to the same Hamiltonian class, as defined in Section 3.4.

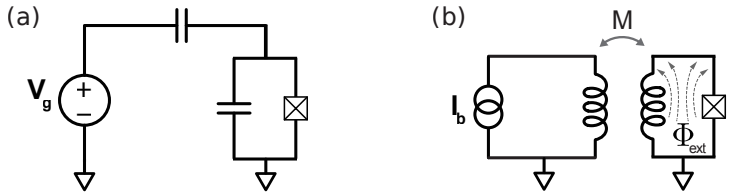


Figure C.3: Examples of (a) a charge-bias circuit and (b) a flux-bias circuit, neither of which is included explicitly in the enumeration.

To see this, consider the circuit Hamiltonian

$$H = \frac{1}{2}(Q - Q_g)^T C^{-1}(Q - Q_g) + U(\Phi, \Phi_{\text{ext}}), \tag{C.1}$$

where Q is the vector of node charges, Φ is the vector of node fluxes, C^{-1} is the inverse capacitance matrix, and Q_g is a vector of gate charges. We can identify a separate circuit and coupling Hamiltonian:

$$H_{\text{circuit}} = \frac{1}{2}Q^T C^{-1} Q + U(\Phi) \tag{C.2}$$

$$H_{\text{coupling}} = Q^T C^{-1} C_V V + U(\Phi, \Phi_{\text{ext}}). \tag{C.3}$$

In Section 3.4, we classify circuits using only H_{circuit}. This is equivalent to setting $Q_g = 0$ and $\Phi_{\text{ext}} = 0$ in the full Hamiltonian. Circuits in the same Hamiltonian class may still differ in H_{coupling}. Those differences determine how the system can be controlled and may also affect decoherence. A classification that also incorporates these bias-dependent coupling Hamiltonians is left to future work.

C.1.4 Circuit Elements Not Considered

We did not include several elements that appear in the superconducting-circuit literature, such as resistors and three-terminal devices like circulators. Two cases deserve further comment: quantum phase-slip junctions and multiport nonreciprocal devices; see Figure C.4.

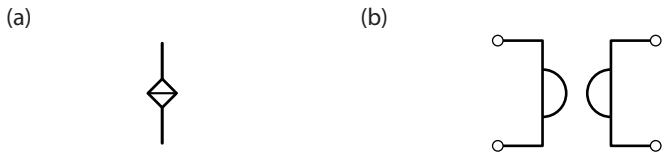


Figure C.4: QPS elements and multiport/multiterminal devices.

A quantum phase-slip junction (QPS) [191] is effectively a nonlinear capacitor and is the electrical dual of the Josephson junction. Although QPS has not yet been realized unequivocally in experiment, we excluded it from the main study for a more practical reason: modern superconducting-circuit analysis packages do not support it. That would sharply limit the analysis of the resulting circuits. Even so, adding a new two-terminal device is straightforward at the level of *circuit* enumeration. To begin exploring circuits with QPS, we carried out a secondary enumeration of circuits with up to four nodes using all fifteen combinations of capacitor (`C`), inductor (`L`), Josephson junction (`J`), and QPS (`Q`).

`C`, `J`, `L`, `Q`, `JC`, `CL`, `QC`, `JL`, `JQ`, `QL`,

`JCL`, `JQC`, `QCL`, `JQL`, and `JQCL`.

Table C.1 gives the resulting number of unique circuits.

Nodes	Circuits	Unique Nonlinear Circuits
2	15	12
3	3600	736
4	12,258,000	712,682

Table C.1: Number of enumerated circuits as a function of node count for circuits built from `C`, `L`, `J`, and `Q`. Column 1 gives the number of nodes. Column 2 gives the naive count, 15^{N_E}, summed over all base graphs for that node count (see Figure 3.3). Column 3 gives the number of unique nonlinear circuits containing at least one Josephson junction or one QPS element.

Nonreciprocal devices such as circulators and gyrators raise two separate issues: nonreciprocity and multiport structure. The first is that consistent quantization of nonreciprocal devices has only recently been developed [55], and current software packages do not yet support it. The second is that multiport devices would add substantial overhead to the enumeration itself. Their circuit topologies would need to be represented by directed hypergraphs rather than undirected graphs. That would increase the number of possible configurations and prevent us from using widely available precomputed lists of nonisomorphic simple graphs, since comparable lists for hypergraphs are not generally available.

C.1.5 Eliminating Single-Node Capacitive and Inductive Islands

We do not consider circuits that contain a single node connected only to capacitors or only to inductors. In such cases, one can construct an equivalent circuit with one fewer node and the same equations of motion. We show below how to identify the required effective capacitance or inductance from the circuit Lagrangian. This construction is equivalent to a generalized delta-Y, or star-mesh, transformation [151].

Consider a subcircuit of n capacitors, $[C_1, C_2, \ldots, C_n]$, that meet at a central node. Additional capacitive connections may already exist between the outer terminals.

The central-node variable is a free variable, but it is coupled to the outer nodes, which may themselves be dynamical. We now follow the procedure in Section C.2.2.2 to decouple this free variable and derive reduced dynamics that do not depend on the central node.

We first split the node flux variables into two parts,

$$\phi = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}.$$

where ϕ_1 contains the outer nodes and ϕ_2 contains the inner node variable. We likewise partition the capacitance matrix into blocks for the outer nodes C_{11}, inner nodes C_{22}, and their couplings C_{21}, C_{12}:

$$C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}.$$

Given the structure of the circuit,

$$\begin{aligned} C_{11} &= \mathrm{diag}([\mathrm{C}_1, \mathrm{C}_2, \ldots, \mathrm{C}_\mathrm{n}]) + \mathrm{C}_0 \\ C_{22} &= \sum_{i=1}^{n} C_i = C_\Sigma \\ C_{21} &= C_{12}^T = \mathrm{row}([\mathrm{C}_1, \mathrm{C}_2, \ldots, \mathrm{C}_\mathrm{n}]), \end{aligned} \tag{C.4}$$

where diag denotes a diagonal matrix and row a row vector. Here C_i is the capacitance between outer node i and the central node, while C_0 contains the capacitances that connect the outer ports to one another or to the rest of the circuit.

The modified capacitance matrix

$$\tilde{C} = C_{11} - C_{12} C_{22}^{-1} C_{21} = C/C_{22} \tag{C.5}$$

defines a reduced system dynamics. To calculate $\tilde{C}$, first consider the entries

$$[C/C_{22}]_{ij} = [\begin{pmatrix} C_1 \\ \vdots \\ C_n \end{pmatrix} \frac{1}{C_\Sigma} \begin{pmatrix} C_1 & \dots & C_n \end{pmatrix}]_{ij} = \frac{C_i C_j}{C_\Sigma}.$$

The diagonal entries are therefore

$$[\tilde{C}]_{ii} = [C_0]_{ii} + C_i - \frac{C_i^2}{C_\Sigma} = [C_0]_{ii} + \frac{\sum_{i \neq j} C_i C_j}{C_\Sigma} \tag{C.6}$$

and off-diagonal entries are

$$[\tilde{C}]_{ij} = [C_0]_{ij} - \frac{C_i C_j}{C_\Sigma}. \tag{C.7}$$

It follows that the same Lagrangian can be realized by a circuit with no inner node and an added capacitance $\frac{C_i C_j}{C_\Sigma}$ between outer nodes i and j. In the special case of one inner node and one outer node, that is, a dangling capacitor or inductor attached to a single node, Equation (C.6) reduces to $[\tilde{C}]_{ii} = [C_0]_{ii} + C_i - C_i^2/C_\Sigma = [C_0]_{ii}$. The dangling element therefore does not affect the dynamics, as no current can flow through that branch.

We omit the corresponding derivation for a node connected only to inductors, as it is identical to the capacitive case.

C.2 Variable Transformation Details

C.2.1 Choice of "Reference" Coordinates

The goal of this section is to define a set of "reference" coordinates for a general superconducting circuit made of capacitors, inductors, and Josephson junctions. Our method generates a transformation to bring any circuit Lagrangian into a standard form, which we use to categorize circuits into "Hamiltonian classes." Hamiltonian classes consist of Hamiltonians with the same

functional form but different parameter values.

The definition of our reference coordinates is motivated by a desire for minimally coupled circuit modes, without a dependence on underlying circuit parameters. The underlying transformation algorithm is flexible and could be used to generate different choices of "reference" coordinates.

For a general circuit, we impose the following requirements.

(1) Maximize the number of non-extended modes, identifying compact, harmonic, and non-dynamical modes wherever possible.

(2) Non-dynamical modes must not couple to dynamical modes.

(3) All dynamical coordinates must be parameter-independent linear combinations of node variables.

(4) Compact coordinates form an n-dimensional cubic unit cell, with charges quantized in units of Cooper pairs. Thus, the lattice vectors of the periodic potential are the standard basis vectors [45].

Within the constraints of the above requirements, some freedom in the choice of coordinates remains. We fix a unique choice by imposing the following criteria, in order of priority:

(a) Minimize nonlinear inter-mode coupling, measured by the off-diagonal support of $J = \sum_i E_{J,i}\, \omega_{J,i} \omega_{J,i}^T$, where $\omega_{J,i}$ denotes the ith junction incidence vector.

(b) Minimize linear inter-mode coupling, defined by the number of nonzero off-diagonal entries in the transformed inverse capacitance (C^{-1}) and susceptance (L^*) matrices. In case of a tie, prioritize minimizing inductive coupling. As a further tiebreaker, prefer transformations with fewer off-diagonal terms when all circuit parameters are taken to be identical.

(c) Ensure integer arguments in cosine potentials of extended modes (e.g., $\cos(2\varphi)$, not $\cos(1.45\varphi)$). Coefficients are chosen to minimize the ℓ_1 norm of the transformed junction incidence matrix.

Finally, we impose a preferred ordering of variable labels. Specifically, we order the coordinate vector as $\Phi = (\Phi^c, \Phi^e, \Phi^h, \Phi^{\text{fr,fz,cy}})^T$, with a vector of compact variables first, followed by extended, harmonic, and non-dynamical variables.

C.2.2 Generating a "Reference" Transformation

The requirements of our "reference" coordinates motivate a sequence of three linear transformations. The first transformation splits modes into different types and addresses requirements (1) and (4). The second transformation enforces requirement (2) by removing any residual coupling between dynamical and non-dynamical variables. Both steps can be carried out without violating the parameter-independence requirement of (3). Finally, the third transformation performs the minimization described in (a) and (b), again subject to the parameter dependence requirement of (3). An implementation of this transformation is included in our code [155].

C.2.2.1 Separate Mode Types

For a circuit with capacitance matrix C, susceptance matrix L^*, and junction incidence matrix ω_J, the Lagrangian is written in terms of node flux variables Φ as

$$\mathcal{L}(\Phi, \dot{\Phi}) = \frac{1}{2}\dot{\Phi}^T C \dot{\Phi} - \frac{1}{2}\Phi^T L^* \Phi + \sum_i E_{J,i} \cos\left(\frac{2\pi}{\Phi_0}\omega_{J,i}^T \Phi\right), \tag{C.8}$$

neglecting charge offset and external flux. Here Φ_0 is the superconducting magnetic flux quantum. $E_{J,i}$ and $\omega_{J,i}$ are the Josephson energy and incidence matrix column corresponding to junction i. Each ω_J column represents a Josephson junction, with a ± 1 at entries corresponding to connected nodes. For example, a three-node circuit with a junction connecting nodes two and three would have $\omega_J = \{0, 1, -1\}$ or $\omega_J = \{0, -1, 1\}$. Because cosine is an even function, the choice of head/tail does not matter, though consistency is key when considering time-dependent flux drives [45]. Note that the linear terms can also be expressed with incidence matrices, such as $\dot{\Phi}^T C \dot{\Phi} = \sum_i (C_i \omega_{C,i}^T \dot{\Phi})^2$.

We define a new set of coordinates $\tilde{\Phi}$ by a linear transformation of the node flux variables, $\tilde{\Phi} = Z^{-1}\Phi$. Although the inverse may appear unconventional, this choice ensures that the capacitance matrix C, the susceptance matrix L^*, and the junction incidence vectors ω_J transform with Z. This is seen by substituting $\Phi = Z\tilde{\Phi}$ into Equation (C.8), which yields

$$\begin{aligned}\mathcal{L}(\tilde{\Phi}, \dot{\tilde{\Phi}}) = \frac{1}{2}\dot{\tilde{\Phi}}^T Z^T C Z \dot{\tilde{\Phi}} - \frac{1}{2}\tilde{\Phi}^T Z^T L^* Z \tilde{\Phi} \\ + \sum_i E_{J,i} \cos\left(\frac{2\pi}{\Phi_0}\omega_{J,i}^T Z \tilde{\Phi}\right)\end{aligned} \tag{C.9}$$

and can now be written in terms of transformed matrices

$$\tilde{C} = Z^T C Z, \quad \tilde{L}^* = Z^T L^* Z, \quad \text{and} \quad \tilde{\omega}_J^T = \omega_J^T Z \ . \tag{C.10}$$

For circuits with capacitors, inductors, and Josephson junctions with intrinsic capacitance, six distinct modes arise.[1] Three are dynamical: compact (cosine potential), extended (cosine + quadratic potential), and harmonic (quadratic potential). Three are non-dynamical: free (kinetic only), frozen (potential only), and cyclic (absent from the Lagrangian).

We define coordinates $\Phi_1 = \{\Phi_1^{\mathrm{c}}, \Phi_1^{\mathrm{e}}, \Phi_1^{\mathrm{h}}, \Phi_1^{\mathrm{fr}}, \Phi_1^{\mathrm{fz}}, \Phi_1^{\mathrm{cy}}\}^T$ that contain blocks corresponding to different mode types.[2] The corresponding transformation can be written as $\Phi = Z_1\Phi_1$, or

$$\begin{pmatrix} \phi_0 \\ \phi_1 \\ \vdots \\ \phi_{n-1} \\ \phi_n \end{pmatrix} = Z_1 \begin{pmatrix} \Phi_1^{\mathrm{c}} \\ \Phi_1^{\mathrm{e}} \\ \Phi_1^{\mathrm{h}} \\ \Phi_1^{\mathrm{fr}} \\ \Phi_1^{\mathrm{fz}} \\ \Phi_1^{\mathrm{cy}} \end{pmatrix}. \tag{C.11}$$

Here ϕ_i denotes a circuit node flux variable. The columns of the transformation Z_1 impose restrictions

[1] Consistently treating Josephson junctions without intrinsic capacitance can lead to multi-valued constraints and remains an area of active research [192–194].

[2] A circuit has as many modes as active circuit nodes, so for any given circuit, many of these blocks will be of dimension zero.

on the nature of the corresponding variables. To see this, consider Z_1 as blocks corresponding to different variable types

$$Z_1 \begin{pmatrix} \Phi_1^{\mathrm{c}} \\ \Phi_1^{\mathrm{e}} \\ \Phi_1^{\mathrm{h}} \\ \Phi_1^{\mathrm{fr}} \\ \Phi_1^{\mathrm{fz}} \\ \Phi_1^{\mathrm{cy}} \end{pmatrix} = \begin{pmatrix} Z_1^{\mathrm{c}} & Z_1^{\mathrm{e}} & Z_1^{\mathrm{h}} & Z_1^{\mathrm{fr}} & Z_1^{\mathrm{fz}} & Z_1^{\mathrm{cy}} \end{pmatrix} \begin{pmatrix} \Phi_1^{\mathrm{c}} \\ \Phi_1^{\mathrm{e}} \\ \Phi_1^{\mathrm{h}} \\ \Phi_1^{\mathrm{fr}} \\ \Phi_1^{\mathrm{fz}} \\ \Phi_1^{\mathrm{cy}} \end{pmatrix}$$

$$= (Z_1^{\mathrm{c}}\Phi_1^{\mathrm{c}} + Z_1^{\mathrm{e}}\Phi_1^{\mathrm{e}} + Z_1^{\mathrm{h}}\Phi_1^{\mathrm{h}} +$$

$$Z_1^{\mathrm{fr}}\Phi_1^{\mathrm{fr}} + Z_1^{\mathrm{fz}}\Phi_1^{\mathrm{fz}} + Z_1^{\mathrm{cy}}\Phi_1^{\mathrm{cy}}). \tag{C.12}$$

For example, consider the requirements to define a harmonic coordinate. It must not have a cosine potential, which requires Φ_1^{h} never appear in any cos term of Equation (C.8). Equivalently, $\omega_J^T Z \Phi_1 = \omega_J^T (Z_1^{\mathrm{c}}\Phi_1^{\mathrm{c}} + Z_1^{\mathrm{e}}\Phi_1^{\mathrm{e}} + Z_1^{\mathrm{h}}\Phi_1^{\mathrm{h}} + Z_1^{\mathrm{fr}}\Phi_1^{\mathrm{fr}} + Z_1^{\mathrm{fz}}\Phi_1^{\mathrm{fz}} + Z_1^{\mathrm{cy}}\Phi_1^{\mathrm{cy}})$ must not depend on Φ_1^{h}. This condition is satisfied if and only if

$$\omega_J^T Z_1^{\mathrm{h}} = 0 \quad \text{or} \quad \mathrm{col}(Z_1^{\mathrm{h}}) \in \mathrm{null}(\omega_J^T), \tag{C.13}$$

where the notation $\mathrm{col}(A)$ denotes the set of all linear combinations of the columns of A, and $\mathrm{null}(A)$ means the set of vectors x for which $Ax = 0$. So Equation (C.13) means all columns of Z_1^{h} are inside the nullspace of ω_J^T. A harmonic mode must also have a quadratic potential and momentum, which requires

$$\mathrm{col}(Z_1^{\mathrm{h}}) \notin \mathrm{null}(\omega_L^T) \quad \text{and} \quad \mathrm{col}(Z_1^{\mathrm{h}}) \notin \mathrm{null}(\omega_C^T)\ , \tag{C.14}$$

for inductor and capacitor incidence matrices ω_L and ω_C.

Our algorithm incorporates the analogous requirements on $\mathrm{null}(\omega_J)$, $\mathrm{null}(\omega_L)$, and $\mathrm{null}(\omega_C)$ shown in Figure C.5, which enable the automatic construction of Z_1. We describe the procedure

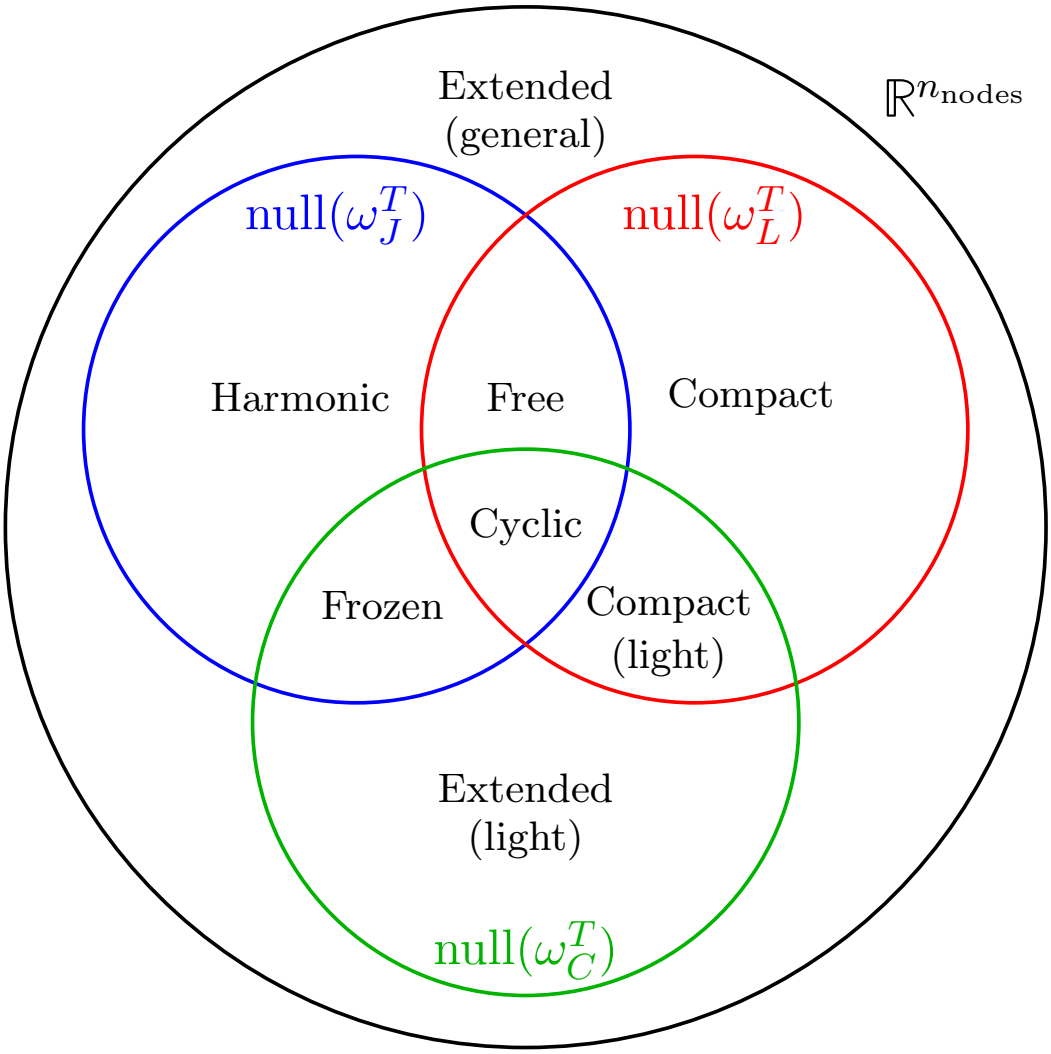


Figure C.5: Types of modes in superconducting circuits made of pure capacitors, pure inductors, and Josephson junctions with intrinsic capacitance. The columns of Z_1^m for a given mode type m are placed within the null spaces of the incidence matrices ω_J^T, ω_L^T, and ω_C^T. Light indicates a small capacitance or "mass" from only the intrinsic junction capacitance.

below.

(1) Construct incidence matrices ω_J, ω_C, and ω_L.

(2) Determine the dimension of each zone in Figure C.5 to identify the modes present.

(3) For non-compact modes m, select any valid choice of Z_1^m. These columns will be redefined later to minimize inter-mode coupling.

(4) For compact modes c, the desired lattice vectors are achieved when a subset of rows in $\omega_J^T Z_1^c$ is equal to the identity matrix. Beginning with any choice of Z_1^c that respects Figure C.5, all Z_1^c with desired lattice vectors can be obtained by inverting different linearly-independent sets of rows of $\omega_J^T Z_1^c$.

C.2.2.2 Decouple Non-Dynamical Modes

After the first transformation, dynamical modes may be linearly coupled to non-dynamical modes. It is always possible to define a transformation that decouples them [58]. We can define this secondary transformation $\Phi_1 = Z_2 \Phi_2$ in terms of the dynamical (c, e, h) and non-dynamical (fr,fz,cy) blocks of Φ_1

$$\begin{pmatrix} \Phi_1^{\text{c,e,h}} \\ \Phi_1^{\text{fr,fz,cy}} \end{pmatrix} = Z_2 \begin{pmatrix} \Phi_2^{\text{c,e,h}} \\ \Phi_2^{\text{fr,fz,cy}} \end{pmatrix}. \tag{C.15}$$

The only choice of Z_2 that does not alter the definitions of dynamical variables is

$$Z_2 = \begin{pmatrix} I & 0 \\ M & I \end{pmatrix}, \tag{C.16}$$

with $M = -C_{22}^{-1} C_{21}$ or $M = -(L_{22}^*)^{-1} L_{21}^*$ depending on whether the dynamical modes are coupled by charge or flux. Block 1 denotes dynamical modes (c, e, h), and block 2 denotes non-dynamical modes (fr, fz, cy). If coupling occurs through both, the two transformations are applied in series. It

can be shown that the decoupling transformation Z_2 is unique up to redefining the non-dynamical modes.

It is worth noting that while the inverse transformation

$$\begin{pmatrix} \Phi_2^{\mathrm{c,e,h}} \\ \Phi_2^{\mathrm{fr,fz,cy}} \end{pmatrix} = \begin{pmatrix} I & 0 \\ -M & I \end{pmatrix} \begin{pmatrix} \Phi_1^{\mathrm{c,e,h}} \\ \Phi_1^{\mathrm{fr,fz,cy}} \end{pmatrix} \tag{C.17}$$

does leave the definition of $\Phi_1^{\mathrm{c,e,h}}$ unchanged, it can introduce parameter dependence into the definitions of non-dynamical coordinates.

C.2.2.3 Reduce Intermode Coupling

The final transformation Z_3 accounts for all transformations that preserve the mode structure while leaving compact variables unchanged. A new Z_3 is generated for each choice of compact variable considered in Z_1. The transformation can be written as $\Phi_2 = Z_3\Phi_3$, or

$$\begin{pmatrix} \Phi_2^{\mathrm{c}} \\ \Phi_2^{\mathrm{e}} \\ \Phi_2^{\mathrm{h}} \\ \Phi_2^{\mathrm{fr,fz,cy}} \end{pmatrix} = \begin{pmatrix} I & Z_3^{\mathrm{ce}} & 0 & 0 \\ 0 & Z_3^{\mathrm{e}} & 0 & 0 \\ 0 & Z_3^{\mathrm{he}} & Z_3^{\mathrm{h}} & 0 \\ 0 & 0 & 0 & I \end{pmatrix} \begin{pmatrix} \Phi_3^{\mathrm{c}} \\ \Phi_3^{\mathrm{e}} \\ \Phi_3^{\mathrm{h}} \\ \Phi_3^{\mathrm{fr,fz,cy}} \end{pmatrix}. \tag{C.18}$$

Note that Z_3 acts only on dynamical modes, ensuring they remain decoupled from non-dynamical modes. The blocks $Z_3^{\mathrm{ce}}, Z_3^{\mathrm{e}}, Z_3^{\mathrm{he}}$, and Z_3^{h} are selected to minimize nonzero inter-mode coupling terms while maintaining parameter independence. Using the computer algebra systems `Sage` [195] and `Singular` [69] and the SMT solver `Z3` [72], we are able to obtain a provably optimal solution over rational numbers [155, 196].

A final ambiguity arises because multiple choices of Z_3 can produce the same number of

coupling terms. For example, consider two inductive couplings among three modes

$$L_3^{(1)} = \begin{pmatrix} 1/L_a & -1/L_aL_b & 0 \\ -1/L_aL_b & 1/L_b & 0 \\ 0 & 0 & 1/L_c \end{pmatrix} \quad \text{and} \tag{C.19}$$

$$L_3^{(2)} = \begin{pmatrix} 1/L_a & 0 & 0 \\ 0 & 1/L_b & -1/L_bL_c \\ 0 & -1/L_bL_c & 1/L_c \end{pmatrix}. \tag{C.20}$$

Which one should be preferred? Both are equivalent, differing only by an interchange of the first and third modes. We make the arbitrary choice to favor $L_3^{(2)}$.

We choose the column ordering that minimizes a binary string representing the nonzero off-diagonal entries of a square matrix. For example, $L_3^{(1)} \to 100$, while $L_3^{(2)} \to 001$. Columns must respect the ordering of Equation (C.18) (compact, extended, harmonic, free, frozen, cyclic), so swaps are only allowed within like modes.

After the third transformation, non-dynamical modes are discarded. The circuit Hamiltonian then follows from a Legendre transform, and its structure is used to group circuits with similar mode patterns.

C.2.3 Limitations

This method relies on the standard node flux formalism [22, 49, 76, 197] to obtain a circuit Hamiltonian. Node flux methods are well defined for circuits that contain reciprocal elements and a single type of nonlinear element (e.g., they cannot contain both a nonlinear capacitor and nonlinear inductor), as shown in Sec. IV.H of [52]. If a quantum phase slip junction (nonlinear capacitor) or non-reciprocal element were included alongside the Josephson junction, more modern geometric circuit quantization methods would be required to systematically obtain circuit Hamiltonians [52, 55].

We additionally do not consider any emergent dynamics from considering circuit singularities derived from Josephson junctions without intrinsic capacitance. A circuit is considered "singular"

when its capacitance matrix is not invertible (e.g., when it has a nontrivial null vector). In reality there is always some capacitance between every pair of nodes in a physical circuit, although it will be many orders of magnitude smaller than a dedicated capacitor. Considering these small capacitances as zero allows us to construct a reduced model with fewer variables. Each zero-capacitance island introduces a constraint upon the circuit variables, effectively reducing the degrees of freedom present. Each island also corresponds to a nontrivial null vector in the capacitance matrix. In the case of linear inductors without a parallel capacitance, this constraint is resolved straightforwardly by removing the "frozen" variable. However, if the island is shunted by a pure nonlinear inductor, this constraint can be multi-valued, and how to accurately resolve the singularity remains an open question [192, 193]. To avoid this inconsistency, we assume that Josephson junctions have an intrinsic capacitance, which we treat as a small ever-present capacitor in parallel with their nonlinear inductance.

C.3 Pseudocode for Constructing a Nonisomorphic Set of Colored Graphs

To construct a nonisomorphic set of colored graphs from a collection that may contain isomorphic graphs, we build the set one element at a time, following Algorithm 3 of [86]. For each candidate graph, we test isomorphism against the graphs already in the "unique" set. If it is not isomorphic to any of them, we add it. We perform these isomorphism checks with the Python package `networkx`. Details of the implementation are available in our GitHub repository [155].

For clarity, we also include pseudocode for this procedure.

Algorithm 1 Constructing a Nonisomorphic Set of Colored Graphs

Require: A set of colored graphs $G = \{G_1, G_2, \ldots, G_n\}$

Require: A function `isomorphic`(G_1, G_2) that tests graph isomorphism

Ensure: A subset $S \subseteq G$ containing one representative from each isomorphism class

```
S ← {}
for each graph G_i ∈ G do
    u ← true
    for each graph S_j ∈ S do
        if isomorphic(G_i, S_j) then
            u ← false
            break
        end if
    end for
    if u then
        S ← S ∪ {G_i}
    end if
end for
return S
```

C.4 Additional Enumeration Results by Number of Junctions, Loops, and Compact Variables

In this appendix, we provide three additional tables that summarize the numbers of circuits and Hamiltonian classes by number of Josephson junctions (Table C.2), number of compact variables (Table C.3), and number of flux loops (Table C.4).

For example, there are 2306 unique four-node circuits with two Josephson junctions. This set includes the $0-\pi$ qubit [31] and the Difluxmon [141]. Both arise from the fully connected four-node

base graph shown in Figure 3.3, which can also be drawn as a triangle with an interior node.

As the number of nodes increases, compact modes become less common among unique circuits. Indeed, 50%, 45%, 38%, and 26% of unique circuits with 2, 3, 4, and 5 nodes, respectively, contain at least one compact mode. By contrast, the fraction of Hamiltonian classes containing at least one compact variable increases with node count.

Nodes	JJs	Unique Circuits	Classes
2	1	4	2
3	1	20	6
	2	40	9
	3	20	4
4	1	744	46
	2	2306	61
	3	3588	59
	4	2852	41
	5	1216	15
	6	276	10
5	1	52012	614
	2	270895	846
	3	819532	838
	4	1584528	826
	5	1952812	616
	6	1745973	508
	7	1044878	374
	8	401711	170
	9	93920	73
	10	10688	28

Table C.2: Number of unique circuits and Hamiltonian classes, grouped by node count and number of Josephson junctions.

Nodes	Compact	Unique Circuits	Classes
2	0	2	1
	1	2	1
3	0	44	10
	1	26	6
	2	10	3
4	0	6824	122
	1	3108	74
	2	892	28
	3	158	8
5	0	5871560	2869
	1	1669398	1366
	2	363929	509
	3	66683	125
	4	5379	24

Table C.3: Number of unique circuits and Hamiltonian classes for circuits with the specified numbers of nodes and compact variables. As node count increases, compact variables appear in fewer unique circuits but in a larger fraction of Hamiltonian classes.

Two points follow from Table C.4. First, circuits with different numbers of flux loops can belong to the same Hamiltonian class. Second, in three-node circuits, inductors appear to correspond to extended or harmonic modes, up to the total number of modes present, while a Josephson junction without an accompanying inductor is needed to produce a compact mode. We further explore the impact of the inductive portion of the circuit in the next section.

C.5 Inductive Skeletons with Delocalized Modes

In this appendix, we define the delocalized coordinate used in Section 3.5.2 and derive necessary and sufficient conditions for its existence. The criteria depend only on the *inductive skeleton*, the

Nodes	Loops	Unique Circuits	Classes
2	0	2	1
	1	2	1
3	0	12	4
	1	28	8
	2	24	7
	3	12	3
	4	4	1
4	0	340	18
	1	1163	62
	2	2098	94
	3	2585	87
	4	2292	62
	5	1491	39
	6	708	20
	7	238	8
	8	56	2
	9	11	1
5	0	12826	84
	1	68702	501
	2	227894	1273
	3	477011	1811
	4	897840	1972
	5	1230312	1831
	6	1431401	1546
	7	1365488	1191
	8	1062982	844
	9	666862	532
	10	338594	305
	11	138860	150
	12	44599	68
	13	11144	25
	14	2104	7
	15	296	2
	16	34	1

Table C.4: Number of unique circuits and Hamiltonian classes by number of flux loops. Circuits with different numbers of flux loops can belong to the same Hamiltonian class.

subcircuit of junctions and inductors, so can be applied before choosing any capacitances.

C.5.1 Conditions for Delocalized Modes

Throughout this section, an admissible coordinate transformation satisfies the mode-separation and compact-lattice requirements of Section C.2. The results below apply only to such transformations.

We call an extended coordinate $\tilde{\Phi}_i$ *localized* if it can be chosen as the phase across a single Josephson junction. That junction phase is $\pm\tilde{\Phi}_i$ and contains no other coordinate. An extended coordinate is *delocalized* if no admissible transformation allows such a choice. Our goal is to determine whether a given inductive skeleton requires delocalized coordinates and, if so, how many.

We first count the compact and extended coordinates. Let ω_J and ω_L denote the junction and inductor incidence matrices for an N-node circuit, and define

$$r_J = \operatorname{rank}\omega_J, \quad r_L = \operatorname{rank}\omega_L\,, \tag{C.21a}$$

$$r_{JL} = \operatorname{rank}\left[\,\omega_J \;\; \omega_L\,\right] \le r_J + r_L\,. \tag{C.21b}$$

Here the rank is the number of linearly independent columns. We also define two nullspaces,

$$J = \operatorname{null}(\omega_J^T)\,, \qquad L = \operatorname{null}(\omega_L^T)\,. \tag{C.22}$$

Thus J contains the node-flux directions that change no junction phase, while L contains those that produce no inductive energy.

A compact coordinate lies in L, because it has no quadratic inductive energy. The intersection $J \cap L$ contains directions seen by neither junctions nor inductors. We therefore define the compact subspace $\mathcal{C} \equiv L \cap (J \cap L)^{\perp}$, where $A^{\perp}$ denotes the directions orthogonal to every vector in A. Thus $\mathcal{C}$ is the part of L orthogonal to these shared null directions, as illustrated in Figure C.5. Since

$\dim J = N - r_J$, $\dim L = N - r_L$, and $\dim(J \cap L) = N - r_{JL}$, the number of compact coordinates is

$$N_{\mathrm{c}} = \dim \mathcal{C} = r_{JL} - r_L \,. \tag{C.23}$$

The total number of nonlinear coordinates equals the number of independent junction branches. Thus $N_{\mathrm{c}} + N_{\mathrm{e}} = r_J$ where N_{e} is the number of extended nonlinear coordinates, giving

$$N_{\mathrm{e}} = r_J + r_L - r_{JL} \,. \tag{C.24}$$

This count assumes the intrinsic junction capacitance used throughout this chapter. The inductive skeleton fixes the compact/extended split, and added capacitances may change the remaining $N - r_J$ coordinates among harmonic, free, frozen, and cyclic types.

We call a junction *inductively shunted* if an inductor-only path connects its two nodes. Equivalently, its incidence column lies in $\operatorname{col} \omega_L$, the column space of ω_L. Let $\mathcal{S}$ be the set of inductively shunted junctions, and $\omega_{J,\mathcal{S}}$ be the matrix formed from their incidence columns.

To express the localization condition algebraically, write the node fluxes as $\Phi = Z\tilde{\Phi}$, where the columns z_j of Z define the new coordinates. The phase across junction k is

$$\omega_{J,k}^T \Phi = \sum_j \left(\omega_{J,k}^T z_j\right) \tilde{\Phi}_j \,. \tag{C.25}$$

The coordinate $\tilde{\Phi}_i$ is localized to junction k if this sum contains only $\pm\tilde{\Phi}_i$. Equivalently,

$$\omega_{J,k}^T Z = \pm e_i^T \quad \Longleftrightarrow \quad \omega_{J,k} = \pm (Z^{-1})_i^T \,, \tag{C.26}$$

where e_i is the ith standard basis vector and $(Z^{-1})_i$ is the ith row of Z^{-1}. The columns of Z corresponding to compact, free, and cyclic coordinates span L. By Equation (C.26), $\omega_{J,k}$ is orthogonal to all of these columns. Therefore $\omega_{J,k} \in L^\perp = \operatorname{col}(\omega_L)$, and junction k must be inductively shunted. Distinct localized extended coordinates correspond to linearly independent rows of Z^{-1}, so at most

$\operatorname{rank}\omega_{J,\mathcal{S}}$ can be localized simultaneously.

This upper bound is always attainable. Let $d = \operatorname{rank}(\omega_{J,\mathcal{S}})$ and select d independent shunted-junction incidence columns. Each selected column is orthogonal to both J and L. The N_{e} extended rows of Z^{-1}, after transposition, form a basis for the N_{e}-dimensional subspace orthogonal to both J and L. We may therefore choose the selected incidence columns as the first d transposed extended rows and complete them to a basis. This changes the basis only within the extended sector and therefore neither mixes mode types nor changes the compact lattice. By Equation (C.26), the first d extended coordinates are then phases across the selected junctions. The bound is therefore attained, and all extended coordinates can be localized if and only if $\operatorname{rank}\omega_{J,\mathcal{S}} = N_{\mathrm{e}}$. The number that must remain delocalized is

$$\delta = N_{\mathrm{e}} - \operatorname{rank}\omega_{J,\mathcal{S}}\,. \tag{C.27}$$

The index δ counts only extended coordinates. Compact coordinates are handled by the lattice construction of Section C.2.

The criterion also has a simple graphical interpretation. An extended coordinate arises from a loop containing both junctions and inductors. If such a mixed loop contains only one junction, the remaining inductors form an inductor-only path across that junction, so the coordinate can be localized to that junction. A delocalized coordinate therefore requires a mixed loop with at least two junctions. If no junction is inductively shunted, such a loop is also sufficient.

A nonzero δ means that the circuit must contain at least one delocalized nonlinear coordinate. It does not imply delocalized eigenstates or protection from noise; these require parameter-dependent spectral and noise analysis. Nevertheless, protected circuits motivate using $\delta > 0$ as a topology-level design filter [31, 156]. The inductive skeleton fixes the nonlinear mode structure, but not the protected regime; Figure 3.9 shows that circuits in the same Hamiltonian class need not share the same protected limit. Circuits with localized modes can be protected. For example the Josephson rhombus underlying the protected chains of Ref. [39] consists of nonlinearly coupled compact modes.

Similarly, our classification identifies the Difluxmon qubit [141] as relying on coupled localized modes to balance protection and control. Thus δ screens for collective nonlinear structure, not qubit protection.

C.5.2 Catalog and Construction of Skeletons

We evaluate Equation (C.27) for every enumerated inductive skeleton and show the unique irreducible-skeletons in Figure 3.11. We call a delocalized skeleton irreducible if $\delta = N_e$ and it meets the following conditions. First, its circuit graph is biconnected: removing any single node does not disconnect the remaining graph. Next, we exclude skeletons with an inductor directly in parallel to a junction or an internal node connected only by inductors. We also exclude all-junction or all-inductor loop additions that leave δ unchanged. A non-biconnected skeleton separates into smaller blocks, and both terms in Equation (C.27) split across these blocks, so their contributions to δ add. Similarly, branches attached through a single connection can add localized compact, harmonic, or non-dynamical coordinates, but do not create a new delocalized mode. Capacitors between different blocks may still couple their dynamics. The figure is therefore a catalog of irreducible-skeletons, not of every skeleton with $\delta \geq 1$.

One irreducible-skeleton appears at three nodes: the skeleton of class $3 - 16$ shown in Figure 3.10(d). Four appear at four nodes and seventeen at five nodes, giving the twenty-two examples in Figure 3.11. The same filters yield 149 six-node irreducible-skeletons, so we stop the catalog at five nodes and provide the general criterion instead.

Larger skeletons can be built from the irreducible-skeletons in three ways. First, a branch can be extended in series by inserting a node and an additional element. For example, the KITE element [34] can be constructed from the J–L–J–L loop of Figure 3.11. This loop is also the inductive skeleton of the $0 - \pi$ circuit. The KITE is obtained by replacing each inductor branch with two inductors in series. Second, a branch or chain can be added between existing nodes to create a new loop. Repeating these two operations generates larger skeletons. Each candidate must still be checked

using Equation (C.27), because a new inductive path can shunt a junction and thereby localize a coordinate. Third, two skeletons can be joined at a single node. Their values of δ then add block by block. Such joined structures are composite rather than irreducible and are therefore omitted from the catalog, although capacitances may still couple the dynamics of the different blocks.

All skeletons are nevertheless present in the full enumeration database. Through an inverse lookup as demonstrated in Section 3.5.1, all derived capacitive configurations could be identified and used as a starting point for an explicit protected qubit search.

C.6 Modeling Implementation Details

In modeling previously unstudied circuits, we are forced to make several key assumptions. Many are taken for granted in transmon-like systems, but they can have a much larger effect in multimode circuits. We list the main assumptions below; the list is not exhaustive.

Selection of computational states. The computational states are taken to be the ground state and first excited states: $|0\rangle_L = |0\rangle$ and $|1\rangle_L = |1\rangle$. As outlined in Section C.8, the dominant state for leakage errors is assumed to be the second excited state $|2\rangle$.

$1/f$ **Noise Amplitudes.** We use the `SQcircuit` default values of 10^{-4}, 10^{-6}, and 10^{-7} for charge, flux, and critical current noise respectively. These values align with the lower bounds laid out in [158].

Hilbert-space cutoff. `SQcircuit` represents compact modes in a charge basis and extended modes in a Fock basis. To choose the cutoff for each mode, we randomly initialize 200 circuits within the parameter search space. For each circuit, we increase the cutoff on each mode by a fixed increment until the fractional change in the first five eigenvalues falls below 10^{-6}. To balance speed and accuracy, we then choose, for each circuit, the 90th-percentile cutoff over the sampled points. We finally re-evaluate each optimum using a cutoff chosen for that specific parameter set. Only 2 of the 84 circuits showed a meaningful change in the predicted N_{gate} after re-evaluation.

Optimizing offsets. Because circuit performance can depend strongly on external inputs,

we treat flux and charge offsets as discrete variables. They may take the values 0, 0.25, 0.5, and 0.75 [ϕ_0 for flux], shifted slightly by 10^{-4} for charge and 10^{-6} for flux. These shifts avoid numerical instabilities at exactly 0 or 0.5 and represent a worst-case DC offset error equal to the $1/f$ noise amplitudes used in the T_ϕ calculation.

External-flux gauge. For each enumerated circuit, we construct an independent cycle basis and associate an `SQcircuit` loop object with each cycle. Given these loops and the full capacitance network, `SQcircuit` determines the parameter-dependent irrotational distribution of external flux over the circuit branches [45, 49, 62]. This satisfies the loop-flux constraints while eliminating the spurious coupling proportional to $\dot{\Phi}_{\text{ext}}$, and is recalculated for the circuit parameters used in each numerical analysis.

Coupling. We assume that each mode in a multimode circuit can be coupled independently through its offset charge or flux. We make this assumption so that circuit performance can be optimized over those offsets. We also ignore coupling capacitances associated with charge offsets when constructing the Hamiltonian.

Parameter values. We limit the underlying circuit parameters to slightly beyond experimentally realized values. We consider Josephson junctions to have a fixed small capacitance. See Table 3.3 for exact values.

Ground placement. The choice of ground can significantly affect circuit performance. We model each node as weakly capacitively coupled to ground. An alternative would be to ground individual nodes explicitly and then use the component-graph isomorphism method of Section 3.3.2 to identify unique ground placements.

Computation Time. To optimize all three-node circuits, we used about 30,000 core-hours on a server with dual AMD EPYC 7713 processors and 2 TB of RAM. Most of this time was spent on circuits with two extended modes, which require much larger Hilbert-space cutoffs. Optimizing a single three-node circuit should be feasible on a typical workstation, although optimizing the full set requires greater computational resources. The full optimization pipeline is included in our repository

[155].

C.7 Converting Between Physical and Frequency Units for Circuit Parameters

Consider dimensionful (Φ, Q) and dimensionless (φ, q) flux and charge operators

$$q = \frac{Q}{2e}, \quad \text{and} \quad \varphi = \frac{2\pi}{\Phi_0}\Phi. \tag{C.28}$$

Here e is the charge of a single electron, and Φ_0 is the magnetic flux quantum. E_C & E_L for a single element are related to physical capacitance C and inductance L by

$$hE_C = \frac{e^2}{2C}, \quad \& \quad hE_L = \frac{(\Phi_0/2\pi)^2}{L}\,. \tag{C.29}$$

These definitions ensure that the energy stored in a capacitor with capacitance C and an inductor with inductance L

$$4E_C q^2 = \frac{1}{2C}Q^2, \quad \& \quad \frac{1}{2}E_L\varphi^2 = \frac{1}{2L}\Phi^2 \tag{C.30}$$

is consistent between dimensionless and dimensionful units.

C.8 Gate Time Estimates

We estimate the gate time by modeling an arbitrary superconducting qubit as a three-level system. We consider either a direct transition, $|0\rangle \leftrightarrow |1\rangle$, or an indirect transition through the higher-lying state $|2\rangle$, that is, a Λ-type configuration; see Figure C.6. In larger circuits, either transition may require difficult multi-node couplings. The main purpose of this appendix is to treat leakage in a simple and consistent way for both direct and Raman-type transitions. We therefore compare the time required to transfer population from $|0\rangle$ to $|1\rangle$ using both schemes, subject to leakage and drive constraints, and take the faster of the two as the gate-time estimate.

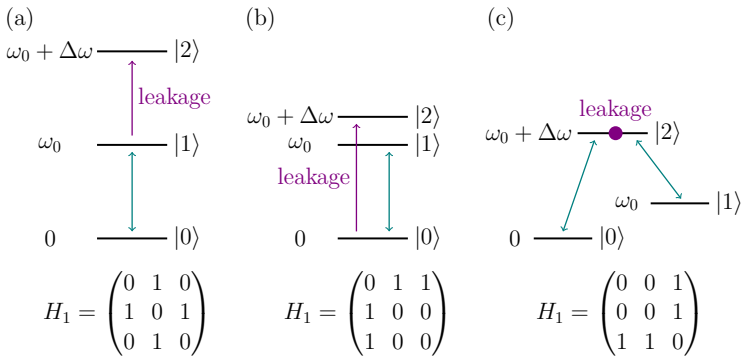


Figure C.6: Transitions and leakage channels included in the gate-time estimate. (a) A weakly anharmonic qubit driven directly on the $|0\rangle \to |1\rangle$ transition, with leakage through $|1\rangle \to |2\rangle$. (b) A qubit with a nearly degenerate second excited state, again driven directly on $|0\rangle \to |1\rangle$, with leakage through $|0\rangle \to |2\rangle$. (c) A Raman-type transition $|0\rangle \to |2\rangle \to |1\rangle$, where population in $|2\rangle$ is treated as leakage.

C.8.1 Direct Transition

We model the qubit as a three-level system with Hamiltonian

$$H = H_0 + f(t)H_1, \tag{C.31}$$

where the free Hamiltonian is

$$H_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \omega_0 & 0 \\ 0 & 0 & \omega_0 + \Delta\omega \end{pmatrix}. \tag{C.32}$$

Here $|\Delta\omega| \geq 0$ and $\omega_0 \geq 0$, and we set the ground-state energy to zero. In Figure C.6, case (a) uses the drive Hamiltonian

$$H_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \tag{C.33}$$

The upper-left 2×2 block drives the qubit σ_X transition, while the remaining term provides a leakage channel. The relevant leakage path depends on which transition, $|0\rangle \leftrightarrow |2\rangle$ or $|1\rangle \leftrightarrow |2\rangle$, lies closer in frequency to the qubit transition $|0\rangle \leftrightarrow |1\rangle$.

We take the drive to be a single-tone field at frequency ω_0 with a Gaussian pulse envelope $f(t) = V_0 e^{-t^2/2\tau^2} \cos(\omega_0 t)$. Here V_0 is the peak drive amplitude. In general, the matrix elements of H_1, taken here to be unity, depend on the coupling scheme of the circuit.

We expect population to transfer between $|0\rangle$ and $|1\rangle$ when the pulse area is π. This corresponds to a pulse half-width of $\tau = \sqrt{\pi/2}/V_0$. By setting a maximum value for V_0, we obtain a minimum gate time. Faster gates, however, also increase leakage. A shorter pulse has a broader frequency spectrum, which drives the leakage transition more strongly. For weakly anharmonic systems such as the transmon, this is the main source of leakage. The DRAG protocol can suppress this effect [198, 199].

To estimate leakage, we study the effective Hamiltonian:

$$H_{\text{2-tone}} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & \lambda \\ 0 & \lambda & 0 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 0 & 1 & \lambda \\ 1 & 0 & 0 \\ \lambda & 0 & 0 \end{pmatrix}, \tag{C.34}$$

with $\lambda = |F(\omega_0 + \Delta\omega)/F(\omega_0)|$ or $\lambda = |F(\Delta\omega)/F(\omega_0)|$, the amplitude ratio of the Fourier transform of the drive at the leakage transition compared with the qubit transition. In chapter 9.5 of Ref. [200], it is shown that for our choice of $f(t)$, we have

$$F(\omega) = \sqrt{\frac{\pi}{2}} \tau V_0 (e^{-(\omega-\omega_0)^2\tau^2/2} + e^{-(\omega+\omega_0)^2\tau^2/2}) \,. \tag{C.35}$$

The amplitude ratio is thus $\lambda = e^{-\delta^2\tau^2/2}$, where δ is the detuning between the leakage transition and the qubit transition (either $\Delta\omega - \omega_0$ or $\Delta\omega$ depending on the level structure).

Solving the time-dependent Schrödinger equation with $|\psi(t=0)\rangle = |0\rangle$, the maximum proba-

bility to be in the $|1\rangle$ state (in both cases) is

$$P_{\max,|1\rangle} = \frac{1}{1+\lambda^2} = \frac{1}{1+e^{-\delta^2\tau^2}}\,. \tag{C.36}$$

To estimate the gate time, we substitute $\tau = \sqrt{\pi/2}/V_0$ corresponding to a π pulse and solve for the V_0 required to achieve a population transfer of 99.9%

$$\frac{1}{1+e^{-\delta^2(\sqrt{\pi/2}/V_0)^2}} = 0.999 \rightarrow V_0 \approx \frac{\delta}{2}\,. \tag{C.37}$$

Finally, we impose two drive limits. First, we assume an absolute lower bound of about 1 ns for the π-pulse width, which gives a maximum drive amplitude $V_{\max} = 200$ MHz. Second, we require $10V_0 \leq \omega_0$, so that the direct drive remains within the regime described above. This gives

$$\tau_{g,\text{direct}} = \begin{cases} \sqrt{2\pi}/\delta, & \delta/2 \leq V_{\text{lim}} \\ \sqrt{\pi/2}/V_{\max}, & \delta/2 > V_{\text{lim}} \end{cases}, \tag{C.38}$$

with δ again being the detuning between ω_0 and the leakage transition ($|0\rangle \leftrightarrow |2\rangle$ or $|1\rangle \leftrightarrow |2\rangle$) and $V_{\text{lim}} = \min(V_{\max}, \omega_0/10)$. Note that reported gate times are set to 5τ, which corresponds to a pulse area of $\approx 99\%$ of the Gaussian envelope. We see three distinct limiting factors emerge from Equation (C.38): (1) τ increases when δ is small, (2) τ increases when ω_0 is small, and (3) the absolute speed limit (i.e., $\tau_{\min}$ is set by $V_{\max}$). In the case of transmon-like systems, factor (1) reproduces the conventional wisdom that gate time is inversely proportional to anharmonicity.

C.8.2 Raman-Type Gate

For transitions via $|2\rangle$, we take H_0 to be identical to the direct transition case, but we now consider

$$H_1 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \tag{C.39}$$

and drive the $|0\rangle \leftrightarrow |2\rangle$ and $|1\rangle \leftrightarrow |2\rangle$ transitions with

$$f(t) = e^{-t^2/2\tau^2}[V_1 \cos(\omega_0 t - \Delta) + V_2 \cos(\Delta\omega t - \Delta)]\,. \tag{C.40}$$

When the state begins in $|\psi(t=0)\rangle = |0\rangle$, it experiences Rabi oscillations between $|0\rangle$ and $|1\rangle$ with an effective Rabi frequency of $\Omega_R = V_1 V_2/2\Delta$ [31]. The intermediate state population experiences oscillations with a maximum value of

$$P_{\max,|2\rangle} = \frac{V_1^2}{\Delta^2 + V_1^2 + V_2^2}\,. \tag{C.41}$$

Taking this maximum intermediate state population as a proxy for leakage, we find the τ required to limit leakage to 0.1% (equivalent to the 99.9% population transfer considered in the direct case). Taking $V_1 = V_2 = V$ and $\Delta = 30V$ yields an effective Rabi frequency of $\Omega_R = V/60$, with corresponding $\tau = 60\sqrt{\pi/2}/V$ for a π pulse. We assume here that there exists a spectrally isolated higher level state, so our only constraint is that $V \leq V_{\max}$. This gives τ for a Raman-style transition of

$$\tau_{g,\text{raman}} = 60\sqrt{\pi/2}/V_{\max} \tag{C.42}$$

Taking $V_{\max} = 200$ MHz as in the direct transition, we get $\tau_{g,\text{raman}} \approx 60$ ns. Again, the gate time is taken to be 5τ to capture most of the pulse area. While $V_{\max}$ in the direct transition established a

minimum gate time, the same constraint here establishes a maximum gate time of approximately 300 ns.

The Raman-type gate will be preferred over the direct transition only in the cases of nearly degenerate ground states and very small anharmonicity. Due to the slow nature of these estimates, we do not consider leakage due to spectral broadening for the Raman-style transition.

C.9 Comparing Model with Published Results

To validate our model of qubit performance (e.g., Equation (3.8)), we compare our predicted optimal operating points for the transmon (`JC`) and fluxonium (`JCL`) circuits with published experimental values. Including charge offset n_g and external flux φ_{ext}, the Hamiltonians for the two circuits are

$$H_{\texttt{JC}} = 4E_C(n - n_g)^2 - E_J \cos(\theta) \tag{C.43}$$

$$H_{\texttt{JCL}} = 4E_C q^2 + \frac{1}{2}E_L\varphi^2 - E_J \cos(\varphi - \varphi_{\text{ext}})\,. \tag{C.44}$$

We visualize N_{gate} as a 2-D parameter sweep in Figure C.7. In the case of fluxonium, E_C is fixed at 500 MHz, with E_L and E_J expressed in units of E_C. For the transmon, $n_g = 0.25$ is chosen to represent a worst-case offset when a DC offset is not included. For fluxonium, the standard $\varphi_{\text{ext}} = \pi$ is used. To provide a fair comparison with the published experimental values, we restrict our parameter ranges slightly relative to the ranges in Table 3.3 and consider only capacitors and inductors of E_C and $E_L \geq 0.1$ GHz. We use two popular software packages which have different noise models, namely `SQcircuit` [45, 64] and `scQubits` [43, 44].

In Figure C.7 (a-b) the published transmon parameters (E_J and E_C) from the literature fall close to the 50% contour of the numerically optimal point. The agreement is remarkable, perhaps reflecting that noise in the transmon is well understood. In the case of fluxonium, shown in Figure C.7 (c-d), the situation is more complicated. When using the `SQcircuit` package fewer of the published

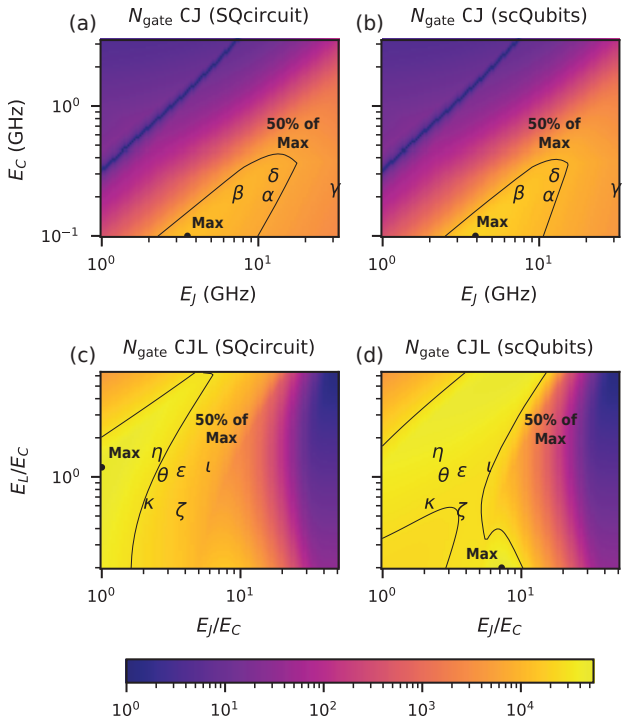


Figure C.7: Qualitative estimate of N_{gate} calculated for the `JC` (a-b) and `JCL` (c-d) circuits. The left column (a,c) uses the `SQcircuit` noise model, which uses a second-order approximation for pure dephasing, while the right column (b,d) uses the `scQubits` noise model, which uses a first-order approximation for pure dephasing. For the `JC` circuit, $n_g = 0.25$ is chosen to represent a worst-case when a DC offset is not included. For the `JCL` circuit, the standard $\varphi_{\text{ext}} = \pi$ is used, and E_C is fixed at 500 MHz. The black dot marks the maximum point, and the black contour shows the region that is at least 50% of the maximum value. The maximum value of N_{gate} is approximately 12,500 for the transmon (`JC`) and 54,500 for fluxonium (`JCL`). The streak with low N_{gate} for the `JC` circuit is due to a zero crossing of the anharmonicity. The bottom right represents the Transmon regime ($E_J/E_C \gg 1$), while the top left represents the Cooper Pair Box, or charge qubit, regime (small E_J/E_C). To gauge the discrepancies between our model and experimental data, we extracted the circuit parameters from various published qubit designs. Each letter corresponds to parameters from a published paper: transmon α–[28], β–[201], γ–[171], and δ–[202]; fluxonium ϵ–[170], ζ–[203], η–[204], θ–[205], ι–[18], and κ–[206]. Qubits from the literature fall near or within the best performing regions for both the transmon and fluxonium circuits.

experiments fall within 50% of our numerically optimized point. When using the `scQubits` package, the agreement improves dramatically. We believe the difference between the packages' models arises from a difference in pure dephasing rates. `SQcircuit` uses a second-order approximation, while `scQubits` uses a first-order approximation. As a result, the `SQcircuit` noise model penalizes larger E_J values more due to second-order critical current noise (see Figure C.8 for a breakdown of the individual noise channels). Nevertheless, all points except ι fall above 25% of the maximum value for the `SQcircuit` noise model. The agreement is still impressive considering neither the noise nor gate speed models have been fine-tuned for fluxonium.

As an additional validation of the gate-time estimate in Section C.8, we compare our predicted gate times with the experimentally reported gate times of the reference transmon and fluxonium devices. The data are listed in Table C.5 and Table C.6. The model reproduces the ordering of the reported gate times for all devices that report one, although the variation is much smaller for the transmons. It does not predict the times exactly, but it appears to serve its intended role as a high-throughput speed limit for the optimization procedure. Somewhat surprisingly, the transmon estimates disagree more with experiment than the fluxonium estimates. One likely reason is that our target single-qubit gate coherent fidelity of 99.9% is lower than the 99.99% target relevant for the transmon devices. The context also differs: the transmon devices were integrated into a larger architecture (perhaps with e.g. DRAG pulses), whereas the fluxonium devices were more isolated research devices.

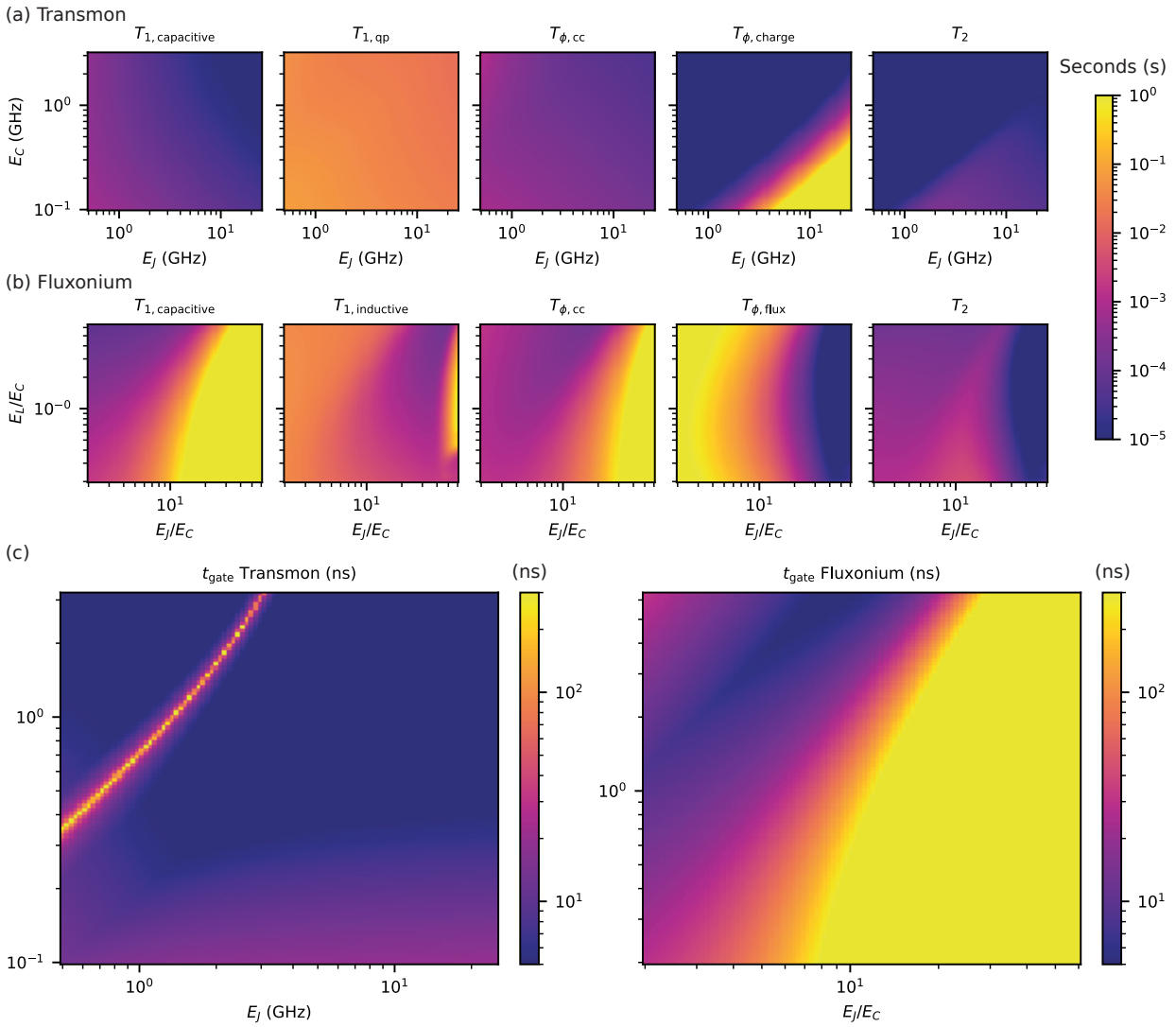


Figure C.8: Underlying coherence times and gate times estimates for the number of gates N_{gate} estimates in Figure C.7. Noise estimates from `SQcircuit` are shown for (a) transmon and (b) fluxonium. Channels not shown did not contribute meaningfully for the respective systems. Gate time estimates based on the analysis given in Section C.8 are shown in (c). The gate time estimates for the transmon seem consistent with gate times from the literature. It is important to note that removing noises or reducing the noise magnitude will change the conclusion of our numerics.

Group or Device	E_J (GHz)	E_C (GHz)	f_{01}, η (GHz)	T_1 (μs)	T_2 (μs)	T_2^{echo} (μs)	T_{gate} (ns)	$T_{\text{gate,est}}$ (ns)	N_{gate}
α–[28] ETH Zurich (2022)	10.5	0.18	3.7, -0.18	60	78	89	40	11	1950
β–[201] Rigetti (2021)	6.8	0.18	3.4, -0.21	51	10	81	20	9	500
γ–[171] Google Sycamore (2019)	28.3	0.21	6.7, -0.21	16	–	–	20	9	1280
δ–[202] IBM Brooklyn (2022)	11.5	0.25	–	–	–	–	–	–	–

Table C.5: Transmon: estimates of device parameters shown in Figure C.7, extracted from published results. If incomplete parameters are given, we infer missing values by approximating $\eta \approx -E_C$ and $f_{01} \approx \sqrt{8E_C E_J} - E_C$. For tunable qubits, if no idle frequency is given, we use the middle of the given range. N_{gate} is calculated by $N_{\text{gate}} = T_2/T_{\text{gate}}$. If no T_2 is provided, we take $1/T_2 = 1/2T_1$ – which implies (incorrectly) that T_ϕ is infinite. For α, we choose device D6, for β we choose device q0, and for γ we use the mean device parameters.

Group or Device	E_J (GHz)	E_C (GHz)	E_L (GHz)	f_{01}, η (GHz)	T_1 (μs)	T_2 (μs)	T_2^{echo} (μs)	T_{gate} (ns)	$T_{\text{gate,est}}$ (ns)	N_{gate}
ϵ–[170] Manucharyan (2019)	3.43	1.00	0.58	0.39, 2.84	500	510	–	–	25	–
ζ–[203] Schuster (2021)	3.40	0.48	0.13	0.01, 2.96	315	300	100	120	299	2500
η–[204] Alibaba (2022)	2.26	1.40	0.52	1.09, 1.93	80	35	30	10	9	3500
θ–[205] SEEQC (2024)	2.50	0.89	1.14	1.25, 1.42	77	29.5	38	6	8	4900
ι–[18] Manucharyan (2023)	5.57	1.08	0.64	0.16, 5.57	1200	1160	1550	80	62	18,500
κ–[206] Besedin (2022)	1.99	0.56	0.86	0.75, 1.05	172	113	146	13	13	8500

Table C.6: Fluxonium: device parameters shown in Figure C.7, extracted from published results. N_{gate} is calculated by $N_{\text{gate}} = T_2/T_{\text{gate}}$. $T_{\text{gate,est}}$ represents our estimated gate time, calculated using the model described in Section C.8. For ϵ, we choose device I, for η we choose device A, for θ we choose device 3, and for κ we choose device S.

C.10 Optimization Results by Number of Junctions, Loops, and Compact Variables

In Tables C.7 to C.9 we split by JJs, number of charge modes, number of flux loops, respectively. Broadly speaking, these splits tell us about the relative importance of charge and flux noise when modeling the number of gates. From Table C.7, we infer that fewer JJs are preferred by our model. From Table C.8, we infer that 1 or 2 flux loops are preferred over 0, 3, or 4 loops. These preferences may be a reflection of the limited design space of three-node circuits. More flux loops means a higher sensitivity to flux noise, and with only three nodes, it is difficult to add more junctions without creating new flux loops. In Table C.9, we additionally see that compact modes are heavily penalized. Again, this may be because of the limited configurations available in three-node circuits.

N_{JJ}	Circuit	N_{gate} (Mean)	N_{gate} (Std)
1	J, L, CL	$2.77 \cdot 10^5$	$1.6 \cdot 10^4$
	J, CL, CL	$2.50 \cdot 10^5$	$1.1 \cdot 10^4$
	CL, CL, CJ	$2.38 \cdot 10^5$	$1.1 \cdot 10^4$
2	J, L, JL	$2.39 \cdot 10^5$	$1.8 \cdot 10^4$
	L, JL, CJ	$2.36 \cdot 10^5$	$1.1 \cdot 10^4$
	C, CJL, CJL	$1.90 \cdot 10^5$	$5.5 \cdot 10^3$
3	CJ, CJL, CJL	$1.86 \cdot 10^5$	$3.1 \cdot 10^3$
	JL, JL, CJ	$1.35 \cdot 10^5$	$2.0 \cdot 10^4$
	JL, CJ, CJL	$1.23 \cdot 10^5$	$3.4 \cdot 10^3$

Table C.7: Optimized number of gates for all three-node circuits, split by number of junctions N_{JJ}. Circuits with fewer junctions perform better than those with more junctions.

C.11 Circuit Quantization Details for Top Performing Circuit

Here we derive the Hamiltonian for coupling the top-performing three-node circuit from Section 3.6 to an AC voltage source. The circuit diagram appears in Figure C.9, with node labels

N_{loop}	Circuit	N_{gate} (Mean)	N_{gate} (Std)
0	C, J, CL	$1.24 \cdot 10^5$	$4.8 \cdot 10^3$
	C, J, L	$1.22 \cdot 10^5$	$5.6 \cdot 10^3$
1	J, L, CL	$2.77 \cdot 10^5$	$1.6 \cdot 10^4$
	J, CL, CL	$2.50 \cdot 10^5$	$1.1 \cdot 10^4$
2	J, L, JL	$2.39 \cdot 10^5$	$1.8 \cdot 10^4$
	L, JL, CJ	$2.36 \cdot 10^5$	$1.1 \cdot 10^4$
3	CJ, CJL, CJL	$1.86 \cdot 10^5$	$3.1 \cdot 10^3$
	JL, JL, CJ	$1.35 \cdot 10^5$	$2.0 \cdot 10^4$
4	CJL, CJL, CJL	$7.04 \cdot 10^4$	$10.0 \cdot 10^2$
	JL, JL, CJL	$6.56 \cdot 10^4$	$6.7 \cdot 10^2$

Table C.8: Optimized number of gates for all three-node circuits, split by number of flux loops N_{loop}. Circuits with one or two loops score the highest.

N_{compact}	Circuit	N_{gate} (Mean)	N_{gate} (Std)
0	J, L, CL	$2.77 \cdot 10^5$	$1.6 \cdot 10^4$
	J, CL, CL	$2.50 \cdot 10^5$	$1.1 \cdot 10^4$
	J, L, JL	$2.39 \cdot 10^5$	$1.8 \cdot 10^4$
1	C, J, CJL	$1.88 \cdot 10^5$	$4.6 \cdot 10^3$
	C, CJ, CJL	$1.87 \cdot 10^5$	$3.8 \cdot 10^3$
	C, JL, CJ	$1.87 \cdot 10^5$	$5.1 \cdot 10^3$
2	J, J, J	$6.74 \cdot 10^4$	$1.9 \cdot 10^3$
	J, J, CJ	$6.03 \cdot 10^4$	$1.7 \cdot 10^3$
	CJ, CJ	$4.53 \cdot 10^4$	$6.2 \cdot 10^2$

Table C.9: Optimized number of gates for all three-node circuits, split by number of compact modes N_{compact}. Circuits with fewer compact modes perform better.

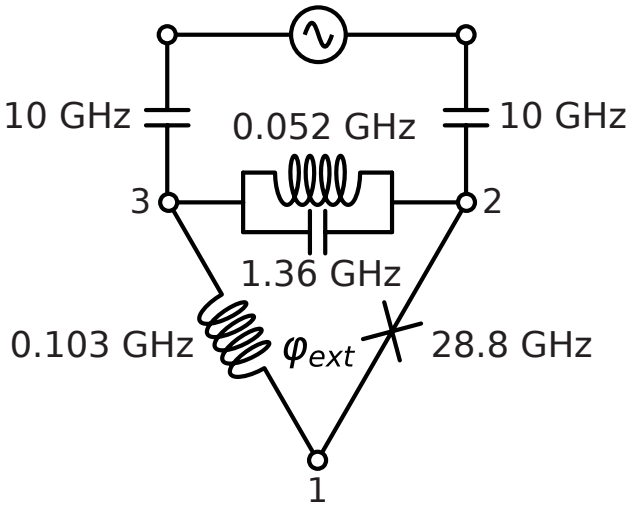


Figure C.9: Circuit diagram for the top performing circuit, coupled to an AC voltage source. Each circuit node is labeled 1, 2, or 3, corresponding to node flux variables ϕ_i.

corresponding to node-flux variables ϕ_i. In the node variables, the circuit has a capacitance matrix

$$C = \begin{pmatrix} C_J & -C_J & 0 \\ -C_J & C_J + C_C + C_{23} & -C_{23} \\ 0 & -C_{23} & C_{23} + C_C \end{pmatrix} \tag{C.45}$$

and inverse inductance matrix

$$L^* = \begin{pmatrix} 1/L_{13} & 0 & -1/L_{13} \\ 0 & 1/L_{23} & -1/L_{23} \\ -1/L_{13} & -1/L_{23} & 1/L_{13} + 1/L_{23} \end{pmatrix}, \tag{C.46}$$

where L_{ij} and C_{ij} are the inductance and capacitance between nodes i and j. C_J is the intrinsic junction capacitance and C_C is the coupling capacitance to the voltage source. We do not include the effects of stray capacitance to ground, although this is included in the gate simulations of Figure 3.14. Nodes 2 and 3 are connected to fixed-voltage nodes, giving an offset charge of

$$Q_V = C_V V = \begin{pmatrix} 0 & 0 \\ C_c & 0 \\ 0 & C_c \end{pmatrix} \begin{pmatrix} v/2 \\ -v/2 \end{pmatrix} = \begin{pmatrix} 0 \\ C_c v/2 \\ -C_c v/2 \end{pmatrix}. \tag{C.47}$$

Here we set the two nodes with fixed voltage to have values of $v/2$ and $-v/2$.

We choose new variables $\varphi_1 = \frac{1}{2}(\phi_3 - \phi_2)$, $\varphi_2 = \phi_2 - \phi_1$, and $\varphi_3 = (\phi_1 + \phi_2 + \phi_3)$, which corresponds to a transformation matrix $\phi = Z\varphi$ of

$$Z = \begin{pmatrix} -2/3 & -2/3 & 1/3 \\ -2/3 & 1/3 & 1/3 \\ 4/3 & 1/3 & 1/3 \end{pmatrix}, \tag{C.48}$$

selected by the coordinate generation algorithm. After decoupling non-dynamical modes, the transformed capacitance, inductance, and charge offset matrices are

$$\tilde{C} = Z^T C Z = \begin{pmatrix} 4C_{23} + 2C_c & 0 & 0 \\ 0 & C_J & 0 \\ 0 & 0 & 2C_c \end{pmatrix} \tag{C.49a}$$

$$\tilde{L}^* = Z^T L^* Z = \begin{pmatrix} \frac{4(L_{13}+L_{23})}{L_{13}L_{23}} & -\frac{2}{L_{13}} & 0 \\ -\frac{2}{L_{13}} & \frac{1}{L_{13}} & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{C.49b}$$

$$\tilde{Q}_V = Z^T C_V V = \begin{pmatrix} -C_c v \\ 0 \\ 0 \end{pmatrix}. \tag{C.49c}$$

The third variable is a non-dynamical free variable, uncoupled from the rest of the system, and can be ignored. The remaining two-variable $\tilde{C}$ is diagonal and can therefore be inverted trivially, yielding

$$\tilde{C}^{-1} = \begin{pmatrix} \frac{1}{4C_{23}+2C_c} & 0 \\ 0 & \frac{1}{C_J} \end{pmatrix}. \tag{C.50}$$

After performing a Legendre transformation, we obtain the circuit Hamiltonian

$$\begin{aligned} H =& \frac{1}{2}(Q-\tilde{Q}_V)^T\tilde{C}^{-1}(Q-\tilde{Q}_V) \\ &+ \frac{1}{2}\Phi^T\tilde{L}^*\Phi + U_L(\Phi,\Phi_{\text{ext}}) + U_{JJ}(\Phi) \\ =& \frac{(Q_1+C_cV)^2}{8C_{23}+4C_c} + \frac{Q_2^2}{2C_J} + \frac{2(L_{13}+L_{23})\Phi_1^2}{L_{13}L_{23}} \\ &- \frac{2\Phi_1\Phi_2}{L_{13}} + \frac{\Phi_2^2}{2L_{13}} \\ &+ \frac{2\Phi_1-\Phi_2}{L_{13}}\Phi_{\text{ext}} - E_J\cos\left(\frac{2\pi}{\Phi_0}\Phi_2\right), \end{aligned} \tag{C.51}$$

with dimension-full charge and flux variables Q_i and Φ_i, where Φ_0 is the flux quantum. We have chosen to assign all external flux to the inductor between nodes 1 and 3 to satisfy the irrotational constraint [49, 62]. We separate the external (drive) Hamiltonian coupled to the voltage source as

$$H_{\text{drive}} = \frac{2C_cVQ_1}{8C_{23}+4C_c} = \frac{8e^2n_gq_1}{8C_{23}+4C_c} \tag{C.52}$$

for a dimensionless charge offset $n_g = C_cV/2e$ and charge operator $q_i = Q_i/2e$. Here we have dropped the constant offset from the $(C_cV)^2$ term.

By examining the Hamiltonian, we see a few desirable features of this coupling scheme. First, in the ideal case, the coupling circuit drives only the first mode, whose matrix elements are large enough to enable a direct transition between $|0\rangle$ and $|1\rangle$. Second, the coupling capacitance C_c adds mass to the already heavy mode while keeping the second mode light, so it only weakly perturbs the uncoupled Hamiltonian. Lastly, coupling to the voltage source is strong, so gates can be driven with small values of C_cV. For the optimized pulses in Figure 3.14, the amplitude of the control pulse V is on the order of μV. Taking into account the effects of stray capacitance does mix the coupling between the two modes, but the majority of the effect remains in the first mode. Using the circuit parameters from Figure 3.14, we have $H_{\text{drive}} = n_g(g_1q_1 + g_2q_2)$, with $g_1 \approx 2.45$ GHz and $g_2 \approx 0.87$ GHz.

Appendix D

Two-Qubit Gates for the Soft $0-\pi$ Qubit

D.1 Single-Qubit Simulation Details

The software used to generate the figures in this chapter is available in our open-source GitHub repository [183]. Note that we use the "scQubits" Python package to construct the single- and two-qubit Hamiltonians [43, 44]. This section outlines the numerical techniques used in the single-qubit simulations. For more details, please refer to the source code.

D.1.1 Noiseless simulation of single-qubit gates

To simulate the $0-\pi$ qubit, we numerically diagonalize the Hamiltonian in Equation (4.2) and keep the lowest $N_{\text{max}}+1$ energy levels. Before diagonalization, we construct operators for the θ mode in the charge basis and for the ϕ mode in the phase basis with $\phi \in [-6\pi, 6\pi]$. We set the dimensions of these bases to 181 and 300, respectively. Numerical simulations with larger truncations show that the qubit frequency (the eigenvalue of $|2\rangle$) converges to within $\sim 10^{-8}$ GHz when either basis is reduced by 5. For the highest retained state ($N_{\text{max}} = 299$), the corresponding maximum change is $\sim 10^{-5}$ GHz. In the diagonal basis, the $0-\pi$ Hamiltonian is

$$H_{0\text{-}\pi} = \sum_{p=0}^{N_{\text{max}}} \omega_{0\text{-}p} \left|p\right\rangle \left\langle p\right|, \tag{D.1}$$

where $|p\rangle$ is an eigenvector with eigenvalue $\hbar\omega_{0\text{-}p}$ (with $\hbar = 1$). To add the drive, we express the operator n_{op} with $\text{op} \in \{\theta, \phi\}$ in the diagonal basis of the $0-\pi$ Hamiltonian as $n_{\text{op}} = \sum_{i,j=0}^{N_{\text{max}}} n_{\text{op}}^{i\text{-}j} \left|i\right\rangle \left\langle j\right|$,

where $n_{\text{op}}^{i\text{-}j} = \langle i| n_{\text{op}} |j\rangle$ and $|i\rangle, |j\rangle \in \{|p\rangle\}$.

Leakage from the computational subspace into higher excited states is a known issue for the $0-\pi$ qubit [180, 181]. We therefore verify that our finite-level truncation faithfully approximates the dynamics of a converged larger model. We first increase the truncation size n in increments of 100 and calculate the corresponding gate fidelity F_n. We assess convergence from the change in infidelity between successive truncations by requiring $|(1-F_n)-(1-F_{n-100})| \leq (1-F_n)/10$, such that the truncation-induced change is at least one order of magnitude smaller than the infidelity itself. For example, if $1-F_n = 10^{-3}$, we require the change relative to the calculation with $n-100$ states to be no greater than 10^{-4}. The largest model, with $N_{\max} = 1000$ states, satisfies this convergence criterion and serves as the reference. We then compare calculations with smaller truncations against the $N_{\max} = 1000$ result to identify the smallest model that reproduces the reference infidelity to the same level of accuracy. Using this procedure, we find that $N_{\max} = 300$ states is sufficient for the single-qubit gate simulations.

We also control integration error using QuTiP's propagator routine. The time-step convergence criterion is similar to that for truncation. We chose the maximum time step so that increasing it by approximately a factor of three changed the infidelity by less than one tenth of the infidelity itself. Thus we used a maximum step size of $\Delta t_{\max} = 3 \times 10^{-4}$ ns for n_θ drives and 1×10^{-3} ns for n_ϕ drives in our simulations.

D.1.2 Model reduction for noiseless pulse optimization

Modeling the $0-\pi$ requires a large Hilbert space, making simulations slow. Optimizing pulse parameters requires solving the dynamics many times. To balance speed and accuracy, we construct a reduced model for pulse optimization in the $0-\pi$ qubit. The reduced model includes only states accessible from the logical subspace under the chosen drive.

We build the reduced model by first including the logical states $|0\rangle$ and $|2\rangle$, and the intended intermediate state ($|7\rangle$ for n_θ, $|9\rangle$ for n_ϕ). The drive operator n_{op} with $\text{op} \in \{\theta, \phi\}$ couples the states

in the current reduced model to other states in the Hilbert space. Next we compute matrix elements $n_{\text{op}}^{i\text{-}j} = \langle i | \, n_{\text{op}} \, | j \rangle$ for all $|i\rangle$ in the reduced model and $|j\rangle$ outside it. Any state $|j\rangle$ with $|n_{\text{op}}^{i\text{-}j}| \geq n_{\text{thresh}}$ is absorbed into the model. Repeating this procedure over the lowest 300 states yields a stable reduced model once no further states are added. We call the resulting set the selected states. For thresholds $n_{\text{thresh}} \in [10^{-8}, 0.2]$, the same reduced models emerge, containing 157 states for n_θ and 160 for n_ϕ.

Even with a reduced Hilbert space, simulations remain costly. We gain further efficiency by evolving only the logical states instead of the full propagator. By default, QuTiP v4.7 evolves all basis states, which is unnecessary for gate fidelity optimization. Restricting to logical states greatly speeds up simulations, allowing a 100-ns X gate in a ∼200-state space to be optimized in about 1 hour using 30 AMD EPYC 7713 cores.

D.1.3 Single-qubit pulse optimization

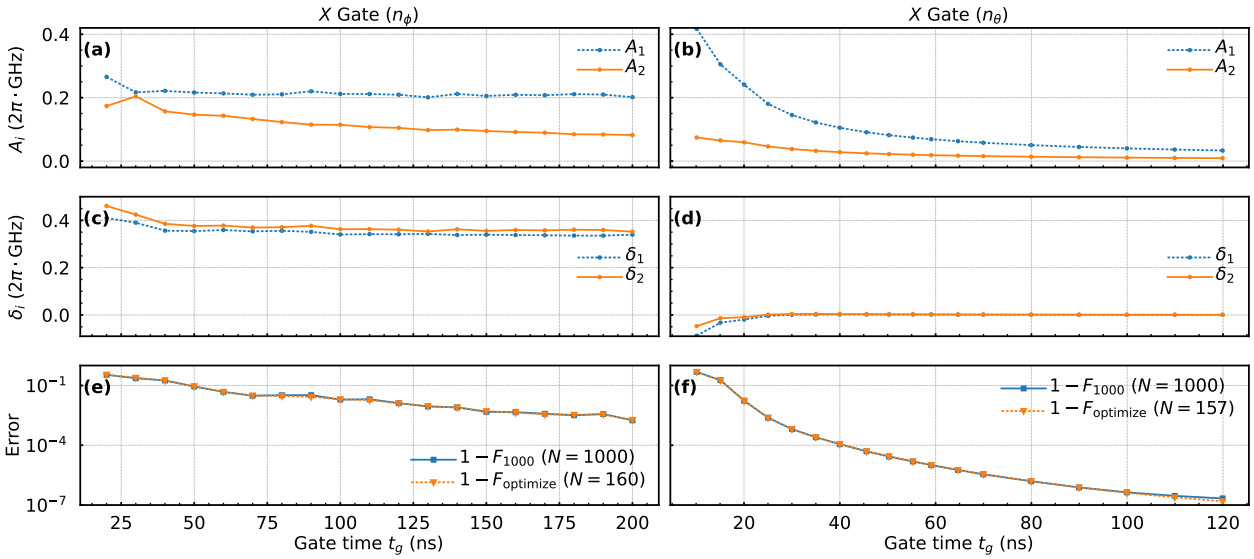


Figure D.1: Optimized pulse parameters and gate infidelities for an X gate driven through n_{op}, with op $\in \{\phi, \theta\}$. For each gate time t_g, panels (a,b) show the optimized amplitudes (A_1, A_2), panels (c,d) show the optimized detunings (δ_1, δ_2), and panels (e,f) show the corresponding infidelities. The pulses are optimized in a closed-system truncated model with $N_{\text{optimize}} = 160$ levels for the n_ϕ drive and $N_{\text{optimize}} = 157$ levels for the n_θ drive. The resulting fidelity is F_{optimize}. The same pulses are then tested in a closed-system 1000-level model, giving F_{1000}.

To construct an X gate, we use an effective Λ system driven on both legs, which unavoidably introduces some leakage. The drive Hamiltonian in Equation (4.5), with Gaussian envelopes defined in Equation (4.6), is specified by five parameters: gate time t_g, amplitudes $A_{1,2}$, and detunings $\delta_{1,2}$. With t_g fixed (tolerance ± 0.1 ns), only four parameters are optimized.

Given the pulse parameters, we numerically solve the Schrödinger equation,

$$i\frac{\partial V(t)}{\partial t} = \left(H_{0\text{-}\pi} + H_{\text{drive}}\right) V(t), \tag{D.2}$$

with $V(0) = I$, to obtain the evolution operator $V(t)$. Gate quality is measured by the average fidelity between the target unitary U_{targ} $(= X)$ and the simulated propagator V in the logical subspace $\{|0\rangle, |2\rangle\}$:

$$F(U_{\text{targ}}, V) = \frac{\text{Tr}(V^\dagger V) + \left|\,\text{Tr}(U_{\text{targ}}^\dagger V)\right|^2}{d(d+1)}, \tag{D.3}$$

with $d = 2$. For improved numerical behavior, we optimize over the quantity $\log_{10}(1 - F)$. Due to leakage, V is generally not unitary, although it is close to unitary for high-fidelity operations.

The optimization is constrained by $A_i/2\pi \in (0,\ 500)$ MHz and $\delta_i/2\pi \in (-500,\ 500)$ MHz. We first optimize in the reduced model from Section D.1.2, then re-run the search using the result as an initial guess to avoid local minima. The optimization uses "differential evolution", a gradient-free stochastic method [207] that efficiently explores large parameter spaces in parallel. Finally, we validate the optimized pulse by simulating the system in a larger Hilbert space spanned by the lowest 1000 eigenstates.

After each optimization, we record the infidelity and use the result as the initial guess for the next iteration to improve convergence. This process is repeated for all gate times. The optimized pulse parameters obtained for the previous gate time are used as the initial guess for the optimization at the next gate time. This warm-start strategy significantly accelerates the optimization.

In Figure D.1, we show the optimized pulse parameters and gate infidelity for the X gate with n_θ and n_ϕ drives. Panels (a)–(b) display the drive amplitudes. The amplitude ratio A_1/A_2 is largely

set by the ratio of charge matrix elements $n_{\text{op}}^{2\text{-}i}/n_{\text{op}}^{0\text{-}i}$. In an ideal three-level Λ system these ratios coincide, giving the fastest transfer. Due to transitions nearby in a large Hilbert space, we need to optimize A_1 and A_2 independently. For the optimized pulse parameters shown in Figure D.1, the amplitude ratio A_1/A_2 agrees well with the matrix element ratio, $n_{\text{op}}^{2\text{-}i}/n_{\text{op}}^{0\text{-}i}$ for n_θ driving. This agreement does not hold for n_ϕ driving. However, the ratio of the drive amplitudes converges closely to the ratio of the charge matrix elements as the gate time increases beyond 200 ns for n_ϕ driving. This is also the case for the ∼800 ns X gate, achieved via driving n_ϕ, presented in Section D.2. Also visible in (b) is the decrease in amplitudes with gate time, which correlates with reduced gate error in (f). Panels (c)–(d) show the detunings, indicating off-resonant vs. resonant driving for n_ϕ and n_θ. Panels (e)–(f) show gate infidelities for the full model with 1000 states $(1 - F_{n_{\text{op}}})$ and two reduced models.

To get an idea of the order of magnitude of the error incurred by truncation, we also compute the relative gate-error difference $|F_{1000} - F_{\text{optimize}}|/(1 - F_{1000})$ in Figure D.1(e)–(f). For n_θ driving, the average and maximum differences are 1.16×10^{-2} and 5.56×10^{-2}, respectively. For n_ϕ driving, they are 4.59×10^{-2} and 1.47×10^{-1}.

D.1.4 Noisy simulation of single-qubit gates

We simulate noise using the master equation $\dot{\rho} = \mathcal{L}[\rho]$, whose solution is $\rho(t) = e^{\mathcal{L}t}[\rho(0)]$, with $e^{\mathcal{L}t}$ the propagator from $\rho(0)$ to $\rho(t)$. The noise model includes energy relaxation (T_1) and pure dephasing (T_ϕ), with equal characteristic times $T_1 = T_\phi = T$. We consider $T \in \{3, 30, 170\}\,\mu$s. These values are chosen to represent noise regimes corresponding approximately to gate fidelities of $\sim 99\%$, $\sim 99.9\%$, and above 99.9%, respectively.

To compute gate fidelities, we fix a target gate time and use the noiselessly optimized pulse parameters (drive frequencies and amplitudes). We then integrate the master equation to obtain the superoperator propagator $\tilde{\mathcal{X}} = e^{\mathcal{L}t}$, with Liouvillian $\mathcal{L}$. At the final time, $\tilde{\mathcal{X}}$ is converted into Kraus operators V_k, defining the noisy channel $V(\rho) = \sum_k V_k \rho V_k^\dagger$. The average channel fidelity relative to

the target X gate is then

$$F(U_{\text{targ}}, V) = \frac{1}{d(d+1)} \sum_k \left[\text{Tr}(V_k^\dagger V_k) + \left| \text{Tr}(U_{\text{targ}}^\dagger V_k) \right|^2 \right], \tag{D.4}$$

with $U_{\text{targ}} = X$ and $d = 2$. Because energy-relaxation rates scale with charge matrix elements, they vanish when the matrix element is negligible. Because open-system simulations are computationally more demanding, we employ a smaller Hilbert-space truncation than in the corresponding closed-system simulations. For noisy single-qubit simulations, we evolve the optimized pulse in a reduced subspace of N selected states, with $N = 160$ for the n_ϕ drive and $N = 157$ for the n_θ drive. The resulting superoperator is then used to compute the noisy channel fidelity relative to the ideal X gate.

D.2 Comparison to Experimental Single-Qubit Gate Performance

To validate our methods, we simulate a gate similar to the X gate demonstrated in Ref. [38]. It used a Gaussian pulse with detuning $\delta \approx -4$ MHz and $\sigma = 200$ ns, so $t_g \approx 800$ ns. That gate achieved a fidelity $\leq 90\%$ [208].

We already use the same device parameters (such as E_J, E_C, etc.) as in the experiment. To replicate their resonator-pin drive, we model the drive operator as $n_{\text{op}} \propto b_\phi n_\phi + b_\theta n_\theta$, where the coupling rates are $b_\phi = 0.27$ and $b_\theta = 6.6 \times 10^{-3}$ [38]. With this model, we numerically optimize a pulse and find a gate with $t_{\text{g}} \approx 830$ ns, which is of the same order of magnitude as the gate duration in Ref. [38]. This pulse was optimized on the closed system model, not the open system model. It would likely be improved by directly optimizing the open system model.

Figure D.2 shows the noiseless population transfer for this gate. With parameters $A_1 = 13.56$ MHz, $A_2 = 34.96$ MHz, $\delta_1 = -3.03$ MHz, $\delta_2 = -3.18$ MHz, and $t_g = 828.76$ ns, we obtain a noiseless fidelity of $F = 99.85\%$. Including noise with $T_1 = T_\phi = 3$ μs reduces the fidelity to $F = 85.72\%$, which is comparable to the experimental value. This demonstrates that our model captures key features of the experimental gate.

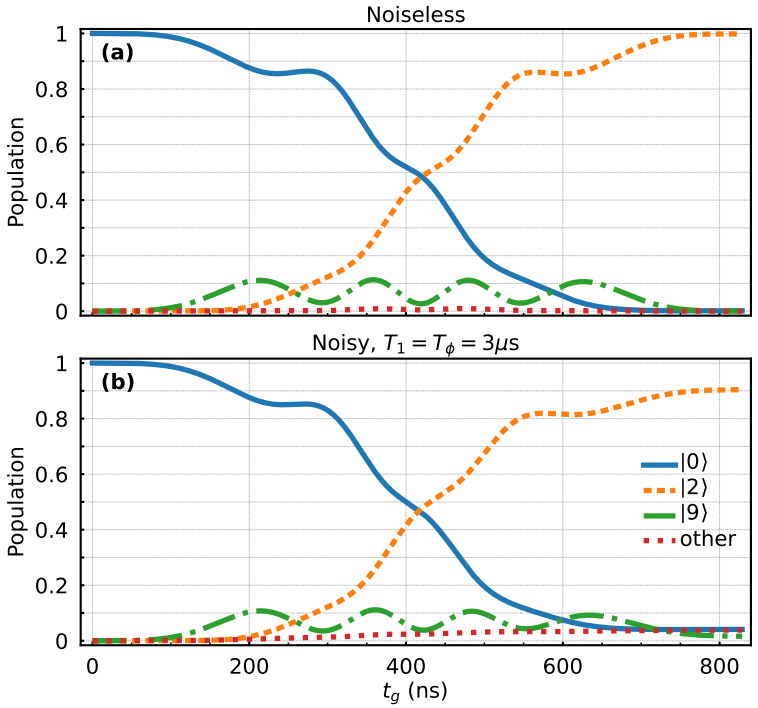


Figure D.2: Population transfer during an X gate with $t_g = 828.76$ ns, driven by $n_{op} \propto b_\theta n_\theta + b_\phi n_\phi$. (a) Noiseless simulation with fidelity $F = 99.85\%$. (b) Noisy simulation with $T_1 = T_\phi = 3\ \mu$s, reducing fidelity to $F = 85.72\%$. The experimental drive coefficients were $b_\theta = 6.6 \times 10^{-3}$ and $b_\phi = 0.27$. For consistency with the n_ϕ and n_θ driving in the main text, we normalize these coefficients while preserving their ratio, giving $n_{op} = 0.976\, n_\phi + 0.024\, n_\theta$ in the simulations.

There is an apparent discrepancy between the gate times studied here and those in the main text. The gate in this section is significantly slower, and its closest counterpart in the main text is the n_ϕ-driven gate.

The main reason is the much larger effective Rabi rates achievable in our protocol. Ref. [38] demonstrated Raman control with effective Rabi frequencies of only a few MHz. However, this limitation was not intrinsic to the soft 0-π qubit itself. Rather, as noted in Ref. [38], the control tones were applied through the readout cavity, whose strong filtering substantially reduced the drive power reaching the qubit. In contrast, our simulations assume direct microwave control and exploit intermediate states with relatively large transition matrix elements. The optimized parameters in Table 4.2 yield effective single-photon Rabi rates up to $\Omega/2\pi \approx 60$ MHz, approximately an order of

magnitude larger than those reported experimentally. These larger effective Rabi rates are a major factor behind the significantly shorter gate times found in the main text.

A second difference is the drive operator. In this section, the drive has the form $b_\phi n_\phi + b_\theta n_\theta$, whereas the main text studies gates driven by n_ϕ or n_θ alone. Simulations not shown here indicate that driving the combined operator $b_\phi n_\phi + b_\theta n_\theta$ produces more leakage than driving either operator alone. These additional leakage pathways reduce the gate fidelity and likely contribute to the slower gate in Ref. [38].

Finally, there are likely differences associated with noise and leakage modeling. Our model captures key experimental features, but it is not a complete device-level model. In particular, our model does not include a ζ mode, and higher excited states may decohere more strongly in the experiment than assumed in our simulations. The main-text n_ϕ gate also reaches a larger intermediate-state population than the gate in this appendix, with $P_9 \sim 0.22$ compared with $P_9 \sim 0.12$. Reducing this population is possible, but generally requires longer gate times. For comparison, the simulation in Appendix G of Ref. [38] shows an intermediate-state population of approximately 5.5% in Fig. 16(e), likely due to the use of an effective three-level Hilbert space in that calculation.

After submission, we became aware of the thesis by Premkumar [181]. We do not directly compare our model with the results in Fig. 4.9 of the thesis because the simulations differ in several important respects. In particular, the simulated device uses parameters that differ substantially from those of Ref. [38], the gate employs a flat-top Gaussian pulse rather than the pulse considered here, and the two Raman tones are constrained to have a common detuning from their respective transitions. By contrast, we optimize the two drive detunings independently.

D.3 Quantization of two coupled $0-\pi$ qubits

In Section D.3.1 we first quantize the circuit in Figure 4.6(a), which produces an $n_{\theta 1} n_{\theta 2}$ coupling. We then quantize the circuit in Figure 4.6(b), which produces an $n_{\theta 1} n_{\theta 2} - n_{\phi 1} n_{\phi 2}$ coupling.

D.3.1 Pure coupling scheme: $n_{\theta 1} n_{\theta 2}$

Our goal is to quantize the 10-node circuit in Figure 4.6 (a). The derivation below assumes equal capacitances for both $0-\pi$ circuits. If we instead allowed different capacitances for the two qubits, the coupling strength and θ mode capacitance would be slightly modified. Crucially, this does not change the circuit quantization procedure, variable transformation, or the form of the effective Hamiltonian.

Let $T^{(i)}_{0\text{-}\pi}$ and $U^{(i)}_{0\text{-}\pi}$ denote the kinetic (capacitive) and potential terms of the Lagrangian for the ith $0-\pi$ qubit, respectively, where $i \in \{1,2\}$. Let T_{int} represent the interaction between the two qubits. Then, the full system Lagrangian is

$$\mathcal{L} = \sum_{i=1,2} (T^{(i)}_{0\text{-}\pi} - U^{(i)}_{0\text{-}\pi}) + T_{\text{int}}\,. \tag{D.5}$$

Here, the superscript $i \in \{1,2\}$ labels the qubit, and T_{int} represents the interaction between the two $0-\pi$ qubits. $T^{(i)}_{0\text{-}\pi}$ and $U^{(i)}_{0\text{-}\pi}$ are the kinetic and potential parts of the single-qubit Lagrangian, respectively. The kinetic terms for the two $0-\pi$ qubits are

$$\begin{aligned} T^{(1)}_{0\text{-}\pi} &= (C+C_J)\dot{\theta}_1^2 + C_J\dot{\phi}_1^2 + C\dot{\zeta}_1^2 + \tfrac{1}{2}C_0\sum_{i=1}^{4}\dot{\varphi}_i^2 \\ &= (C+C_J+\tfrac{1}{2}C_0)\dot{\theta}_1^2 + (C_J+\tfrac{1}{2}C_0)\dot{\phi}_1^2 + (C+\tfrac{1}{2}C_0)\dot{\zeta}_1^2 + \tfrac{1}{2}C_0\dot{\Sigma}_1^2, \end{aligned} \tag{D.6}$$

$$\begin{aligned} T^{(2)}_{0\text{-}\pi} &= (C+C_J)\dot{\theta}_2^2 + C_J\dot{\phi}_2^2 + C\dot{\zeta}_2^2 + \tfrac{1}{2}C_0\sum_{i=5}^{8}\dot{\varphi}_i^2 \\ &= (C+C_J+\tfrac{1}{2}C_0)\dot{\theta}_2^2 + (C_J+\tfrac{1}{2}C_0)\dot{\phi}_2^2 + (C+\tfrac{1}{2}C_0)\dot{\zeta}_2^2 + \tfrac{1}{2}C_0\dot{\Sigma}_2^2\,. \end{aligned} \tag{D.7}$$

In these expressions, the top line has the node variables and the bottom line is re-expressed with respect to our variable transformation in Equation (4.1). The additional terms relative to the single $0-\pi$ circuit, proportional to C_0, arise from the ground capacitors. The coupling term is more

complicated and can be written as

$$\begin{aligned} T_{\text{int}} &= \frac{C_c}{2}[(\dot{\varphi}_1 - \dot{\varphi}_{10})^2 + (\dot{\varphi}_4 - \dot{\varphi}_{10})^2 + (\dot{\varphi}_5 - \dot{\varphi}_{10})^2 + (\dot{\varphi}_8 - \dot{\varphi}_{10})^2 \\ &\quad + (\dot{\varphi}_2 - \dot{\varphi}_9)^2 + (\dot{\varphi}_3 - \dot{\varphi}_9)^2 + (\dot{\varphi}_6 - \dot{\varphi}_9)^2 + (\dot{\varphi}_7 - \dot{\varphi}_9)^2] \\ &= \frac{C_c}{2}\Big[\sum_{i=1}^{8} \dot{\varphi}_i^2 + 4(\dot{\varphi}_9^2 + \dot{\varphi}_{10}^2) - 2\dot{\varphi}_9(\dot{\varphi}_2 + \dot{\varphi}_3 + \dot{\varphi}_6 + \dot{\varphi}_7) - 2\dot{\varphi}_{10}(\dot{\varphi}_1 + \dot{\varphi}_4 + \dot{\varphi}_5 + \dot{\varphi}_8)\Big] \\ &= \frac{C_c}{2}\sum_{i=1}^{2}[\dot{\phi}_i^2 + \dot{\zeta}_i^2 + (\dot{\varphi}_+ - \dot{\Sigma}_i)^2 + (\dot{\varphi}_- - \dot{\theta}_i)^2]\,. \end{aligned} \tag{D.8}$$

The final line is written using the variable transformation in Equation (4.1) and the sum and difference variables $\varphi_\pm = \varphi_9 \pm \varphi_{10}$. Finally, the potential term is

$$U_{0\text{-}\pi}^{(i)} = -2E_{Ji}\cos\theta_i \cos\left(\phi_i - \phi_{\text{ext},i}/2\right) + E_{Li}\phi_i^2 + E_{Li}\zeta_i^2. \tag{D.9}$$

After doing the Legendre transform, the vector of conjugate momenta $Q_\mu = \partial\mathcal{L}/\partial\dot{\mu}$ (μ =

$\theta_j, \phi_j, \zeta_j, \Sigma_j, +, -;\ j = 1, 2)$ is

$$\begin{pmatrix} Q_{\theta 1} \\ Q_{\Sigma 1} \\ Q_{\phi 1} \\ Q_{\zeta 1} \\ Q_{\theta 2} \\ Q_{\Sigma 2} \\ Q_{\phi 2} \\ Q_{\zeta 2} \\ Q_{+} \\ Q_{-} \end{pmatrix} = \begin{pmatrix} C_\theta & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -C_c \\ 0 & C_\Sigma & 0 & 0 & 0 & 0 & 0 & 0 & -C_c & 0 \\ 0 & 0 & C_\phi & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & C_\zeta & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & C_\theta & 0 & 0 & 0 & 0 & -C_c \\ 0 & 0 & 0 & 0 & 0 & C_\Sigma & 0 & 0 & -C_c & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & C_\phi & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & C_\zeta & 0 & 0 \\ 0 & -C_c & 0 & 0 & 0 & -C_c & 0 & 0 & 2C_c & 0 \\ -C_c & 0 & 0 & 0 & -C_c & 0 & 0 & 0 & 0 & 2C_c \end{pmatrix} \begin{pmatrix} \dot{\theta}_1 \\ \dot{\Sigma}_1 \\ \dot{\phi}_1 \\ \dot{\zeta}_1 \\ \dot{\theta}_2 \\ \dot{\Sigma}_2 \\ \dot{\phi}_2 \\ \dot{\zeta}_2 \\ \dot{\varphi}_+ \\ \dot{\varphi}_- \end{pmatrix}. \tag{D.10}$$

In the above equation, the capacitances C_μ of the $0 - \pi$ qubit modes for $\mu \in \{\theta,\ \phi,\ \zeta,\ \Sigma\}$ are specified by

$$\begin{pmatrix} C_\theta \\ C_\phi \\ C_\zeta \\ C_\Sigma \end{pmatrix} = \begin{pmatrix} 2 & 2 & 1 & 1 \\ 0 & 2 & 1 & 1 \\ 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} C \\ C_J \\ C_c \\ C_0 \end{pmatrix} \tag{D.11}$$

where C_0 is the capacitance between the node and ground. With a less intuitive choice of intermediate variables,

$$\varphi'_+ = \varphi_+ - (\Sigma_1 + \Sigma_2)/2 \quad \text{and} \quad \varphi'_- = \varphi_- - (\theta_1 + \theta_2)/2, \tag{D.12}$$

the coupling between the θ and Σ modes of the two qubits can be seen directly:

$$\begin{pmatrix} Q_{\theta 1} \\ Q_{\Sigma 1} \\ Q_{\phi 1} \\ Q_{\zeta 1} \\ Q_{\theta 2} \\ Q_{\Sigma 2} \\ Q_{\phi 2} \\ Q_{\zeta 2} \\ Q_{+'} \\ Q_{-'} \end{pmatrix} = \begin{pmatrix} C_\theta - \frac{C_c}{2} & 0 & 0 & 0 & -\frac{C_c}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & C_\Sigma - \frac{C_c}{2} & 0 & 0 & 0 & -\frac{C_c}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & C_\phi & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & C_\zeta & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{C_c}{2} & 0 & 0 & 0 & C_\theta - \frac{C_c}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{C_c}{2} & 0 & 0 & 0 & C_\Sigma - \frac{C_c}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & C_\phi & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & C_\zeta & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2C_c & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2C_c \end{pmatrix} \begin{pmatrix} \dot{\theta}_1 \\ \dot{\Sigma}_1 \\ \dot{\phi}_1 \\ \dot{\zeta}_1 \\ \dot{\theta}_2 \\ \dot{\Sigma}_2 \\ \dot{\phi}_2 \\ \dot{\zeta}_2 \\ \dot{\varphi}'_+ \\ \dot{\varphi}'_- \end{pmatrix}. \tag{D.13}$$

Calculating the inverse of the capacitance matrix in Equation (D.13) gives the total quantized Hamiltonian $H = \frac{1}{2}Q^T C^{-1} Q + U$, which can be decomposed as

$$H = H^{(1)}_{0\text{-}\pi} + H^{(2)}_{0\text{-}\pi} + H_{\text{int}} + H_{\text{coupler}}, \tag{D.14}$$

where H_{int} represents the interaction between the two qubit circuits and the coupling circuit, while H_{coupler} is the coupler self-Hamiltonian. The Hamiltonian of the individual $0-\pi$ qubits is

$$H_{0\text{-}\pi}^{(i)} = 4\tilde{E}_C^{\theta} n_{\theta i}^2 + 4\tilde{E}_C^{\phi} n_{\phi i}^2 + 4\tilde{E}_C^{\zeta} n_{\zeta i}^2 + 4\tilde{E}_C^{\Sigma} n_{\Sigma i}^2 + E_{Li}\phi_i^2 + E_{Li}\zeta_i^2 - 2E_{Ji}\cos\theta_i \cos\left(\phi_i - \frac{\phi_{\text{ext}}}{2}\right), \tag{D.15}$$

where the associated energies are

$$\tilde{E}_C^{\theta} = \frac{e^2(2C_\theta - C_c)}{4C_\theta(C_\theta - C_c)}, \qquad \tilde{E}_C^{\phi} = \frac{e^2}{2C_\phi}, \tag{D.16a}$$

$$\tilde{E}_C^{\Sigma} = \frac{e^2(2C_\Sigma - C_c)}{4C_\Sigma(C_\Sigma - C_c)}, \qquad \tilde{E}_C^{\zeta} = \frac{e^2}{2C_\zeta}. \tag{D.16b}$$

The two remaining parts of Equation (D.14) are

$$H_{\text{coupler}} = 4E_C^- n_m^2 + 4E_C^+ n_p^2, \quad \& \quad H_{\text{int}} = g_\theta n_{\theta_1} n_{\theta_2} + g_\Sigma n_{\Sigma_1} n_{\Sigma_2}, \tag{D.17a}$$

with the coefficients

$$g_\theta = \frac{2e^2 C_c}{C_\theta(C_\theta - C_c)} \approx \frac{e^2 C_c}{2C^2}, \qquad E_C^+ = \frac{e^2}{4C_c}, \tag{D.18a}$$

$$g_\Sigma = \frac{2e^2 C_c}{C_\Sigma(C_\Sigma - C_c)} = \frac{2e^2 C_c}{C_0(C_0 + C_c)}, \qquad E_C^- = \frac{e^2}{4C_c}. \tag{D.18b}$$

The approximate expressions are valid in the limit of $C \gg C_c, C_J, C_0$. In the above expressions, the modes ζ_1 and ζ_2 are decoupled from the effective modes of the $0-\pi$ qubit and can thus be considered separately. The modes Σ_1, Σ_2, φ_+, and φ_- do not have corresponding potential energy terms and are considered free modes. Because they are decoupled from the θ, ϕ, and ζ degrees of freedom, the free modes may be ignored without affecting the system dynamics. Thus the final Hamiltonian can

be written as

$$H_{\text{tot}} = H^{(1)}_{0\text{-}\pi} + H^{(2)}_{0\text{-}\pi} + H_{\text{int}}, \tag{D.19a}$$

$$H^{(i)}_{0\text{-}\pi} = 4\tilde{E}^{\theta}_{C} n^2_{\theta i} + 4\tilde{E}^{\phi}_{C} n^2_{\phi i} + E_{Li}\phi_i^2 - 2E_{Ji}\cos\theta_i \cos\left(\phi_i - \frac{\phi_{\text{ext}}}{2}\right), \tag{D.19b}$$

$$H_{\text{int}} = g_\theta n_{\theta_1} n_{\theta_2}. \tag{D.19c}$$

D.3.2 Mixed coupling scheme: $n_{\theta 1}n_{\theta 2} - n_{\phi 1}n_{\phi 2}$

Now we quantize the circuit in Figure 4.6(b). We use the variable transformation in Equation (4.1) on both $0-\pi$ qubits. The Lagrangian is expressed, as before, in Equation (D.5). Recall that the superscript $i \in \{1,2\}$ labels the qubit; T_{int} is the qubit–qubit interaction; $T^{(i)}_{0\text{-}\pi}$ and $U^{(i)}_{0\text{-}\pi}$ are the kinetic (capacitive) and potential parts of the Lagrangian, respectively. The kinetic part of the Lagrangian for the two $0-\pi$ qubits can be written compactly as $T^{(i)}_{0\text{-}\pi} = (C + C_J + \frac{C_0}{2})\dot{\theta}_i^2 + (C_J + \frac{C_0}{2})\dot{\phi}_i^2 + (C + \frac{C_0}{2})\dot{\zeta}_i^2 + \frac{C_0}{2}\dot{\Sigma}_i^2$. The coupling term is more complicated and can be written as

$$\begin{aligned} T_{\text{int}} &= \frac{C_c}{2}[(\dot{\varphi}_1 - \dot{\varphi}_8)^2 + (\dot{\varphi}_2 - \dot{\varphi}_7)^2 + (\dot{\varphi}_3 - \dot{\varphi}_6)^2 + (\dot{\varphi}_4 - \dot{\varphi}_5)^2] \\ &= \frac{C_c}{2}[(\dot{\theta}_1 - \dot{\theta}_2)^2 + (\dot{\phi}_1 + \dot{\phi}_2)^2 + (\dot{\zeta}_1 + \dot{\zeta}_2)^2 + (\dot{\Sigma}_1 - \dot{\Sigma}_2)^2]. \end{aligned} \tag{D.20}$$

Finally, the potential term is

$$U^{(i)}_{0\text{-}\pi} = -2E_{Ji}\cos\theta_i \cos\left(\phi_i - \phi_{\text{ext},i}/2\right) + E_{Li}\phi_i^2 + E_{Li}\zeta_i^2\,. \tag{D.21}$$

After doing the Legendre transform, the vector of conjugate momenta $Q_\mu = \partial\mathcal{L}/\partial\dot{\mu}$ ($\mu = \theta_j, \phi_j, \zeta_j, \Sigma_j;\ j = 1, 2$) is

$$\begin{pmatrix} Q_{\theta 1} \\ Q_{\Sigma 1} \\ Q_{\phi 1} \\ Q_{\zeta 1} \\ Q_{\theta 2} \\ Q_{\Sigma 2} \\ Q_{\phi 2} \\ Q_{\zeta 2} \end{pmatrix} = \begin{pmatrix} C_\theta & 0 & 0 & 0 & -C_c & 0 & 0 & 0 \\ 0 & C_\Sigma & 0 & 0 & 0 & -C_c & 0 & 0 \\ 0 & 0 & C_\phi & 0 & 0 & 0 & C_c & 0 \\ 0 & 0 & 0 & C_\zeta & 0 & 0 & 0 & C_c \\ -C_c & 0 & 0 & 0 & C_\theta & 0 & 0 & 0 \\ 0 & -C_c & 0 & 0 & 0 & C_\Sigma & 0 & 0 \\ 0 & 0 & C_c & 0 & 0 & 0 & C_\phi & 0 \\ 0 & 0 & 0 & C_c & 0 & 0 & 0 & C_\zeta \end{pmatrix} \begin{pmatrix} \dot{\theta}_1 \\ \dot{\Sigma}_1 \\ \dot{\phi}_1 \\ \dot{\zeta}_1 \\ \dot{\theta}_2 \\ \dot{\Sigma}_2 \\ \dot{\phi}_2 \\ \dot{\zeta}_2 \end{pmatrix}. \tag{D.22}$$

Here, the capacitances C_μ of the $0-\pi$ qubit modes $\theta,\ \phi,\ \zeta,\ \Sigma$ are specified by Equation (D.11).

Calculating the inverse of the capacitance matrix in Equation (D.22) gives the total quantized Hamiltonian $H = \frac{1}{2}Q^T C^{-1} Q + U$, which can be decomposed as

$$H = H^{(1)}_{0\text{-}\pi} + H^{(2)}_{0\text{-}\pi} + H_{\text{int}}, \tag{D.23}$$

where H_{int} represents the interaction between the two qubit circuits.

The Hamiltonian of the individual $0-\pi$ qubits is

$$H^{(i)}_{0\text{-}\pi} = 4\tilde{E}^\theta_C n^2_{\theta i} + 4\tilde{E}^\phi_C n^2_{\phi i} + 4\tilde{E}^\zeta_C n^2_{\zeta i} + 4\tilde{E}^\Sigma_C n^2_{\Sigma i} + E_{Li}\phi_i^2 + E_{Li}\zeta_i^2 - 2E_{Ji}\cos\theta_i \cos\left(\phi_i - \frac{\phi_{\text{ext}}}{2}\right), \tag{D.24}$$

where the associated energies are

$$\tilde{E}_C^{\theta} = \frac{e^2 C_\theta}{2(C_\theta^2 - C_c^2)}, \qquad \tilde{E}_C^{\phi} = \frac{e^2 C_\phi}{2(C_\phi^2 - C_c^2)}, \tag{D.25a}$$

$$\tilde{E}_C^{\Sigma} = \frac{e^2 C_\Sigma}{2(C_\Sigma^2 - C_c^2)}, \qquad \tilde{E}_C^{\zeta} = \frac{e^2 C_\zeta}{2(C_\zeta^2 - C_c^2)} \tag{D.25b}$$

The approximate expressions are valid in the limit of $C \gg C_c, C_J, C_0$. The remaining part of Equation (D.23) is

$$H_{\text{int}} = g_\theta n_{\theta_1} n_{\theta_2} - g_\phi n_{\phi_1} n_{\phi_2} - g_\zeta n_{\zeta_1} n_{\zeta_2} + g_\Sigma n_{\Sigma_1} n_{\Sigma_2}, \tag{D.26}$$

where the coefficients are given by

$$g_\theta = \frac{4e^2 C_c}{C_\theta^2 - C_c^2} \approx \frac{e^2 C_c}{C^2}, \qquad g_\phi = \frac{4e^2 C_c}{C_\phi^2 - C_c^2} = \frac{4e^2 C_c}{(2C_J + C_0 + 2C_c)(2C_J + C_0)}, \tag{D.27a}$$

$$g_\zeta = \frac{4e^2 C_c}{C_\zeta^2 - C_c^2} \approx \frac{e^2 C_c}{C^2}, \qquad g_\Sigma = \frac{4e^2 C_c}{C_\Sigma^2 - C_c^2} = \frac{4e^2 C_c}{C_0(C_0 + 2C_c)}. \tag{D.27b}$$

The approximate expressions are valid in the limit of $C \gg C_c, C_J, C_0$. One important point about Equation (D.26) is that all modes of the two $0-\pi$ circuits are included in this coupling scheme. This is because the coupling circuit does not address any particular mode of the circuits. From Equation (D.27) it is clear that to increase or decrease g_θ relative to g_ϕ, we need to vary C, which may contradict moving towards a hard $0-\pi$ regime.

Here, the modes ζ_1 and ζ_2 are still decoupled from the effective modes of their respective $0-\pi$ qubits and can thus be removed from the Hamiltonian. Again, the "free modes" Σ_1 and Σ_2 do not have corresponding potential energy terms and can be ignored without affecting the system

dynamics. Thus the final Hamiltonian can be written as

$$H = H_{0\text{-}\pi}^{(1)} + H_{0\text{-}\pi}^{(2)} + H_{\text{int}}, \tag{D.28a}$$

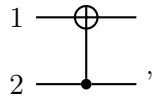
$$H_{0\text{-}\pi}^{(i)} = 4\tilde{E}_C^{\theta} n_{\theta i}^2 + 4\tilde{E}_C^{\phi} n_{\phi i}^2 + E_{Li}\phi_i^2 - 2E_{Ji}\cos\theta_i \cos\left(\phi_i - \frac{\phi_{\text{ext}}}{2}\right), \tag{D.28b}$$

$$H_{\text{int}} = g_\theta n_{\theta_1} n_{\theta_2} - g_\phi n_{\phi_1} n_{\phi_2}. \tag{D.28c}$$

D.4 CNOT gate via a two-tone Raman pulse

In this section, we introduce a two-qubit CNOT gate for the $0-\pi$ qubit. The gate uses the $n_{\theta 1} n_{\theta 2}$ coupling derived in Section 4.4 to realize an effective Λ system involving a non-computational state. While its operating principle resembles that of the single-qubit gates in Section 4.3, it is significantly slower than both the single-qubit gates and the CZ gate of Section 4.5. A potential advantage of this scheme is its compatibility with stimulated Raman adiabatic passage (STIRAP), which can suppress occupation of the intermediate state and thereby mitigate leakage.

The CNOT we construct is controlled on qubit two



, (D.29)

with a corresponding unitary

$$\text{CNOT}_{21} = I \otimes |0_L\rangle\langle 0_L| + X \otimes |1_L\rangle\langle 1_L| \ . \tag{D.30}$$

The subscript of CNOT_{21} indicates the control and target qubits. For comparison, $\text{CNOT}_{12} = |0_L\rangle\langle 0_L| \otimes I + |1_L\rangle\langle 1_L| \otimes X$ places the control on qubit 1. CNOT_{12} can be implemented by driving $n_{\theta 2}$.

D.4.1 Driven Hamiltonian

The driven coupled circuit from Section 4.4 is shown in Figure D.3(a), with total Hamiltonian

$$H_{\text{tot}} = H^{(1)}_{0\text{-}\pi} + H^{(2)}_{0\text{-}\pi} + H_{\text{int}} + H^{(1)}_{\text{drive}}, \tag{D.31}$$

where $H^{(1)}_{\text{drive}}$ represents two microwave drives on $n_{\theta 1}$, each with the form in Equation (4.5).

Gate	n_{op}	Transitions	Associated n_{op}	t_g (ns)	Ω_1 (2π·MHz)	Ω_2 (2π·MHz)	δ_1 (2π·MHz)	δ_2 (2π·MHz)	F_{optimize}
CNOT	$n_{\theta 1}$	[$\|0,2\rangle$-$\|8,2\rangle$, $\|2,2\rangle$-$\|8,2\rangle$]	$[0.037, 0.057]$	170.00	8.28	8.28	-8.23	-7.89	99.229%

Table D.1: The parameters of the CNOT gate corresponding to the red star in Figure D.4. $A_i/2\pi$ is limited to be less than 100 MHz. The effective Rabi rate is given by $\Omega_i = A_i(e^2 - 1)n^i_{\text{op}}$ for $i = 1, 2$.

D.4.2 Transition selection and gate principle

To identify a suitable Raman pathway, we examine the matrix elements of $n_{\theta 1}$ in Figure D.3(b). The rows correspond to computational states and the columns to low-energy non-computational states. A column containing two appreciable matrix elements identifies a candidate intermediate state for an effective Λ system. Desirable pathways combine large matrix elements, which enable faster gates, with sufficient spectral isolation to suppress off-resonant transitions.

Figure D.3(c) shows the relevant matrix elements as a function of transition frequency. Although the pathway through $|4,5\rangle$ is the most spectrally isolated, numerical optimization gives better gate performance using $|8,2\rangle$. This indicates that spectral isolation alone does not determine the fidelity: the complete set of matrix elements, detunings, and off-resonant couplings must also be considered. We therefore use $|8,2\rangle$ as the intermediate state.[1]

The inter-qubit coupling spectrally separates the Raman pathways through $|8,0\rangle$ and $|8,2\rangle$, allowing the latter to be driven selectively. Two tones couple $|0,2\rangle \leftrightarrow |8,2\rangle$ and $|2,2\rangle \leftrightarrow |8,2\rangle$, as

[1]The single-qubit n_θ gate involves $|7\rangle$. In the coupled circuit, charging-energy renormalization produces an avoided crossing and exchanges the roles of $|7\rangle$ and $|8\rangle$, so that $|8\rangle$ plays the analogous role here. See Section D.4.5.

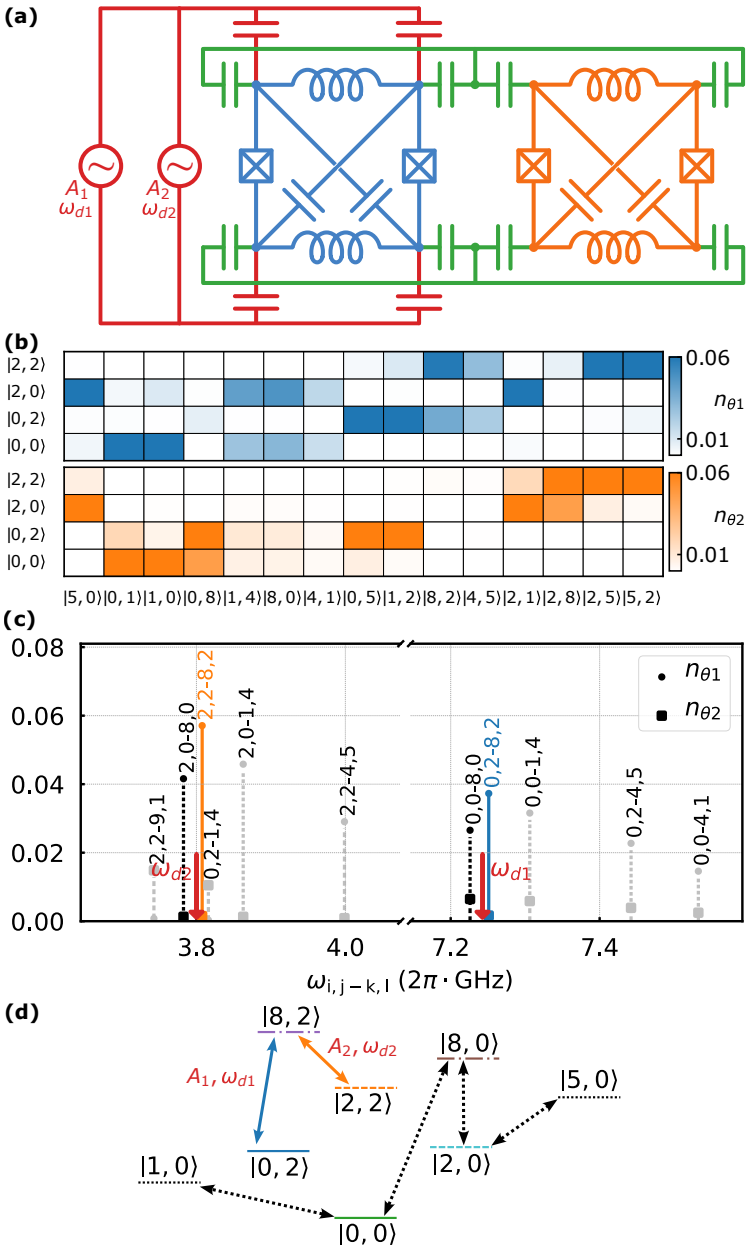


Figure D.3: CNOT gate implemented by driving $n_{\theta 1}$. (a) Circuit diagram. The coupling and drive circuits are depicted in green and red, respectively. (b) Charge matrix elements of $n_{\theta 1}$ and $n_{\theta 2}$ between two-qubit logical states and some non-logical states. Note that the matrix elements used in the CNOT gate correspond to the $n_{\theta 1}$ matrix elements for the $|2,2\rangle$-$|8,2\rangle$ and $|0,2\rangle$-$|8,2\rangle$ transitions. (c) Charge matrix elements $n_{\theta 1}$ and $n_{\theta 2}$ versus transition frequency $\omega_{i,j\text{-}k,l}$ for a transition $|i,j\rangle$-$|k,l\rangle$. The two Raman transitions are depicted in blue and orange, while the two dominant leakage transitions are depicted in green and red. (d) Energy level diagram. The Raman and leakage transitions are plotted using the same color scheme as in panel (c).

shown in Figure D.3(d). The first qubit is therefore flipped only when the second qubit is in $|2\rangle$, implementing a CNOT with qubit 2 as the control and qubit 1 as the target. Residual phases are removed using single-qubit Z rotations, as shown in Section D.4.4 [209].

Other choices of Raman pathway produce different two-qubit gates. Coupling $|0,0\rangle$ and $|2,2\rangle$ gives a bSWAP gate, while coupling $|0,2\rangle$ and $|2,0\rangle$ gives an iSWAP gate.

D.4.3 Gate operation and noiseless fidelity

Panels (a)–(d) of Figure D.4 show the noiseless time evolution of various two-qubit state populations during the CNOT. The drive pulses are optimized in the same way as in the single-qubit case, with details given in Section D.5.

The optimized gate has duration $t_{\mathrm{g}} = 170$ ns and fidelity $F \sim 99\%$. As expected, only cases where the control qubit is in $|2\rangle$ show population transfer at the final time, as seen in Figure D.4(b) and (d). The intermediate state $|8,2\rangle$ is substantially populated, reaching $> 40\%$.

From Figure D.4(a) and (c), we identify $|8,0\rangle$ as the dominant leakage state, from which we infer that the main leakage transitions are $|0,0\rangle$-$|8,0\rangle$ and $|2,0\rangle$-$|8,0\rangle$. These are plotted in green and red in Figure D.3(c) and (d). The leakage during the CNOT gate is small, which can be deduced from the population in unintentionally populated states ("other"). The total population in these states is approximately 0.66%, with no single state's contribution exceeding 0.15%. This indicates only minimal leakage during the gate.

Finally, Figure D.4(e) shows the average gate error $(1 - F)$ versus t_g in the absence of noise. Compared to the CZ gate discussed in the main text, this gate is much slower. Similar to the single-qubit lambda gates, it also exhibits some leakage. In principle, STIRAP could protect the gate, but competing factors such as decoherence arise. Ultimately, the slow operation time is the main limitation. For this reason we do not study decoherence for the CNOT gate.

Now we compare the CNOT gate with the CZ from the main text. The CZ gate achieves both shorter duration and higher fidelity. This advantage arises for two reasons: (i) CNOT requires driving

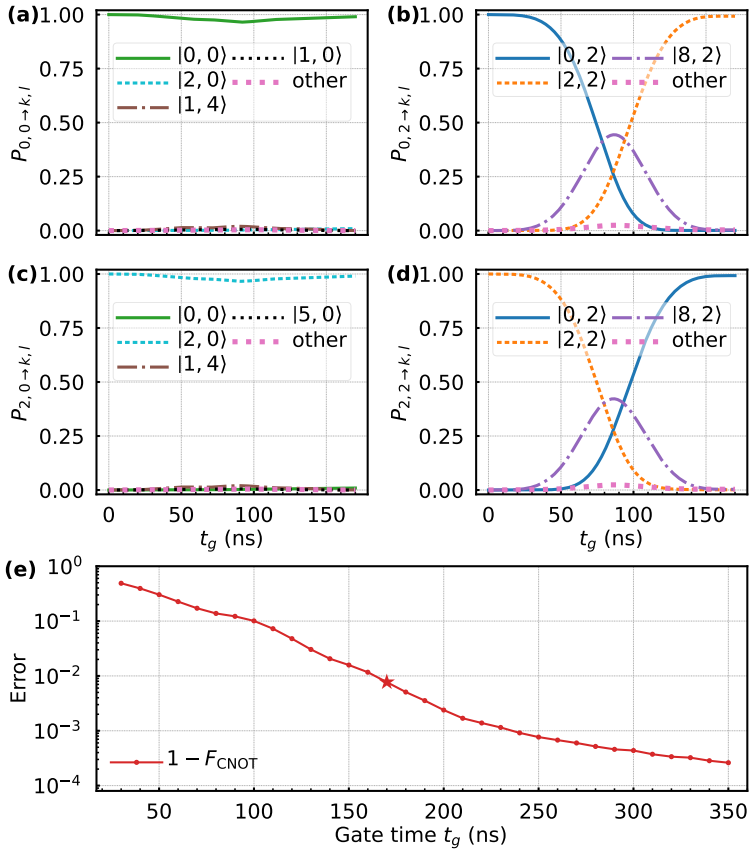


Figure D.4: Noiseless time evolution of the populations of various states during gate operation. In subfigures (a)–(d), the symbol $P_{i,j\to k,l}$ denotes the population transfer that starts at the state $|i,j\rangle$ and ends in the state $|k,l\rangle$. The "other" state refers to the total population of all states not explicitly shown within the lowest 1000 dressed states. (e) Average gate error $(1-F)$ versus t_g in the absence of noise. Each point corresponds to optimized pulse parameters for the given t_g. Star markers indicate the population transfer shown in (a)–(d).

two transitions in a Λ system, while CZ uses direct transitions; and (ii) the relevant CZ transition has a charge matrix element an order of magnitude larger, with $\langle 2,0|\, n_{\theta 2}\, |5,0\rangle = 0.537$ compared to $\langle 0,2|\, n_{\theta 1}\, |8,2\rangle = 0.037$ and $\langle 2,2|\, n_{\theta 1}\, |8,2\rangle = 0.057$ for CNOT (see Table 4.4 and Table D.1).

D.4.4 Single-qubit Z corrections for the CNOT gate

For a bipartite Hilbert space $\mathcal{H}_1 \otimes \mathcal{H}_2$, a controlled-NOT with control on subsystem i and target on j is denoted CNOT_{ij}. The ideal CNOT_{21} is

$$\mathrm{CNOT}_{21} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}. \tag{D.32}$$

Our numerics yield a candidate gate U_{can} by projecting the simulated propagator (from the master equation) into the logical space. We parameterize its principal matrix elements as $z_{ij} = |z_{ij}|e^{i\phi_{ij}}$ and denote the small error terms collectively by y. The resulting form is

$$U_{\mathrm{can}} = \begin{bmatrix} |z_{00}|e^{i\phi_{00}} & y & y & y \\ y & y & y & |z_{02}|e^{i\phi_{02}} \\ y & y & |z_{20}|e^{i\phi_{20}} & y \\ y & |z_{22}|e^{i\phi_{22}} & y & y \end{bmatrix}. \tag{D.33}$$

The elements $|z_{ij}|$ are equal to 1 in the ideal operator, and the phases ϕ_{ij} capture the phase accumulated in time evolution that must be corrected.

To facilitate a comparison between this operator and the ideal CNOT, we apply additional single-qubit Z rotations $R_z(\phi) = \exp[-i\phi Z/2] = \mathrm{diag}[e^{-i\phi/2}, e^{i\phi/2}]$ both before and after the gate,

$$U_{\mathrm{f}} = e^{-i\phi_{\mathrm{gb}}}[R_z(\phi_2) \otimes R_z(\phi_1)] \cdot U_{\mathrm{can}} \cdot [R_z(\phi_3) \otimes I]\,. \tag{D.34}$$

Here, $\phi_{\rm gb} = \phi_{00} - (\phi_1 + \phi_2 + \phi_3)/2$ is a global phase. To cancel out all phases of the key matrix elements, we solve linear equations to find the $R_z(\phi_i)$ rotation angles

$$\begin{aligned} \phi_1 &= \frac{1}{2}\left(\phi_{00} - \phi_{02} + \phi_{20} - \phi_{22}\right) , \\ \phi_2 &= \frac{1}{2}\left(\phi_{00} + \phi_{02} - \phi_{20} - \phi_{22}\right) , \\ \phi_3 &= \frac{1}{2}\left(\phi_{00} - \phi_{02} - \phi_{20} + \phi_{22}\right) . \end{aligned} \tag{D.35}$$

After using those angles, we arrive at

$$U_{\rm f} = \begin{bmatrix} |z_{00}| & y' & y' & y' \\ y' & y' & y' & |z_{02}| \\ y' & y' & |z_{20}| & y' \\ y' & |z_{22}| & y' & y' \end{bmatrix} , \tag{D.36}$$

where the entries denoted by y' are complex numbers and satisfy $|y'| \ll 1$ in the high-fidelity limit. We then compute the gate fidelity between $U_{\rm f}$ and the ideal CNOT_{21}. A similar derivation for a CNOT_{12} is presented in Ref. [209].

D.4.5 Level Crossing in Coupled Circuits

This section examines the level crossing between $|7\rangle$ and $|8\rangle$ as two $0-\pi$ circuits are coupled. Figure D.5 shows eigenvalues of the first $0-\pi$ qubit versus coupling rate g, with $g = 0$ for uncoupled systems. Eigenvalue shifts arise solely from the renormalized charging energies in Equation (D.16a). The coupling rate depends on the capacitance C_c (Equation (D.18)); larger C_c yields stronger coupling. For our gates, we use $E_{C_c}/h = 1.0$ GHz (black dashed line), giving $g/2\pi \approx 0.031$ GHz.

There are two important level crossings at $g/2\pi \approx 0.01$ GHz, between $|8\rangle$ and $|9\rangle$, and at $g/2\pi \approx 0.02$ GHz, between $|7\rangle$ and $|8\rangle$. These level crossings arise because the coupling capacitances

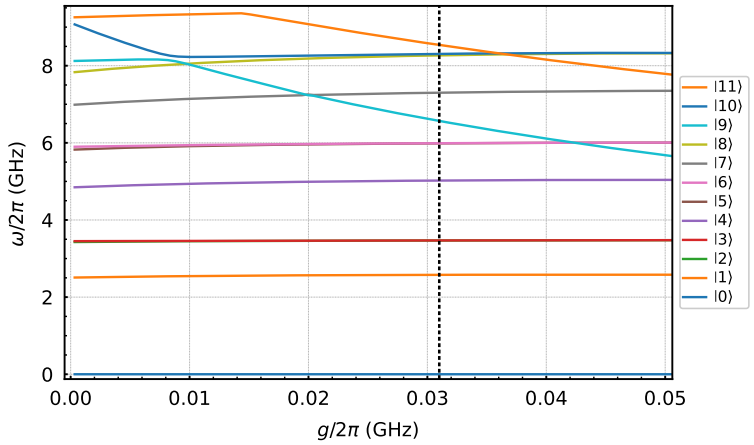


Figure D.5: Energy spectrum of the first $0-\pi$ qubit including coupling capacitances C_c versus coupling rate g. The black dashed line is the operating point of the coupled qubits, $g/2\pi \approx 0.031$ GHz. Levels $|7\rangle$ and $|8\rangle$ cross near $g/2\pi \approx 0.02$ GHz.

renormalize the charging energies. Because $|7\rangle$ is used as an intermediate state in the X gate, we expect $|7,2\rangle$ to be an important candidate state for two-qubit gates when there is no level crossing. However, due to the level crossing, $|8,2\rangle$ ends up being the intermediate state in the CNOT gate, while $|7,2\rangle$ has a nearly zero charge matrix element with respect to the logical states (not shown). We therefore expect the best CNOT performance near $g/2\pi \approx 0.031$ GHz. This is the region where the intermediate state $|8,2\rangle$ is most isolated.

D.5 Two-Qubit Gate Simulation Details

The software used to generate the two-qubit gate simulation results is available in the repository [183]. This section provides a high-level overview of the numerical techniques used to simulate CZ and CNOT gates, along with key implementation details. The goal is to help readers understand the core methods without requiring a deep dive into the source code.

D.5.1 Noiseless simulation of two-qubit gates

To simulate two-qubit gates, we first diagonalize the single-qubit Hamiltonians $H_{0\text{-}\pi}^{(1)}$ and $H_{0\text{-}\pi}^{(2)}$ to obtain eigenstates $|p\rangle^{(j)}$ and eigenvalues $E_p^{(j)}$, with $j = 1, 2$. We retain the lowest 300 eigenstates

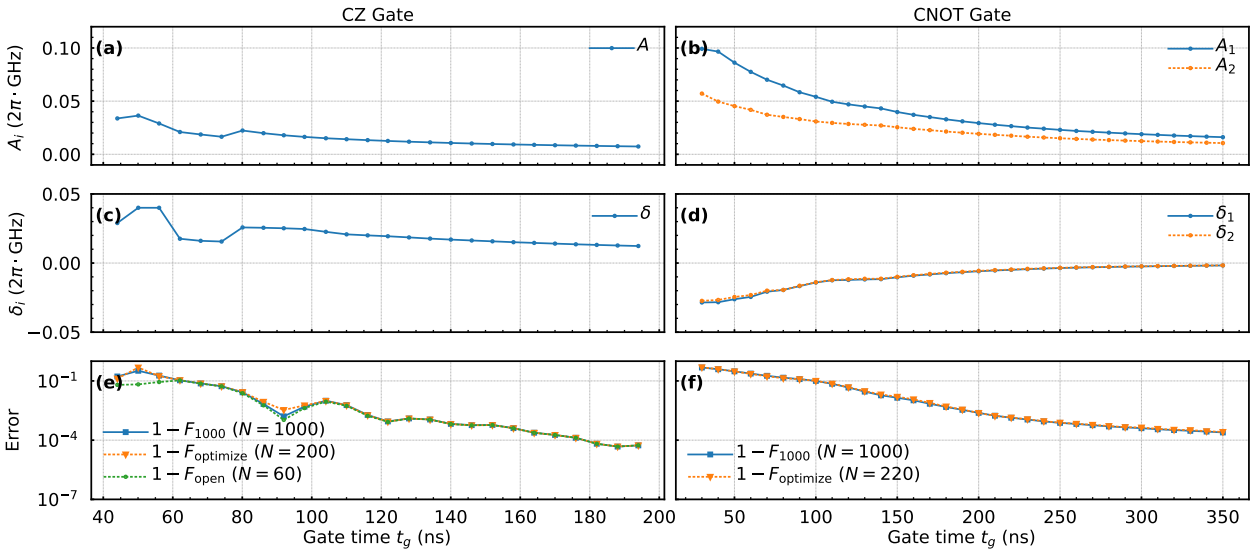


Figure D.6: Optimized pulse parameters and gate infidelities for the CZ and CNOT gates. For each gate time t_g, panels (a,b) show the optimized drive amplitude A, panels (c,d) show the optimized detuning δ, and panels (e,f) show the corresponding infidelities. For each fixed t_g, the Gaussian pulse is optimized in a truncated closed-system model of two coupled $0-\pi$ qubits with $N_{\text{optimize}} = 200$ or 220 selected levels, depending on the gate. The resulting fidelity is F_{optimize}. The same optimized pulse is then tested in a closed-system model with 1000 selected levels, giving F_{1000}, and in a smaller "selected" N-level model, giving F_{open}. The "selected" N-level model is used for open-system simulation including energy relaxation and pure dephasing. In this figure we only compare the closed-system fidelity for these three models. Note that we were unable to simulate the CNOT gate in an open-system due to the large model needed for convergence ($N_{\text{open}} = 240$). The plotted infidelities are $1 - F_{\text{optimize}}$, $1 - F_{1000}$, and $1 - F_{\text{open}}$.

for each qubit. The eigenstates of the uncoupled system ($H_{\text{int}} = 0$) are denoted $|k\rangle_0 \otimes |l\rangle_0 = |k, l\rangle_0$, where k and l label the eigenstates of the first and second qubits, respectively.

Next, we express the interaction term H_{int}, e.g., $H_{\text{int}} \approx g_\theta n_{\theta_1} n_{\theta_2}$, in the uncoupled basis:

$$H_{\text{int}} = \sum_{p,q,r,s=0}^{M_{\max}} c_{pqrs} \, |p\rangle_0 \langle q| \otimes |r\rangle_0 \langle s| \,, \tag{D.37}$$

where $c_{pqrs} = g_{\theta\,0} \langle p, r| \left(n_{\theta 1} \otimes n_{\theta 2}\right) |q, s\rangle_0$. We then diagonalize the full coupled Hamiltonian $H_{\text{cp}} = H^{(1)}_{0\text{-}\pi} + H^{(2)}_{0\text{-}\pi} + H_{\text{int}}$ to obtain dressed eigenstates $|p, q\rangle$ and eigenvalues $E_{p,q}$. The resulting time-independent Hamiltonian H_{cp} is supplemented with a drive Hamiltonian and used to evolve the

system under the Schrödinger equation:

$$i\frac{\partial V(t)}{\partial t} = \left(H_{\text{cp}} + H_{\text{drive}}\right) V(t). \tag{D.38}$$

The evolution operator $V(t_g)$ at the final time t_g is evaluated on the logical subspace spanned by $\{|0,0\rangle, |0,2\rangle, |2,0\rangle, |2,2\rangle\}$ to form the 4×4 effective gate matrix $\bar{V}$, with elements

$$\bar{V}_{k,l;k',l'} = \langle k,l| V(t_g) |k',l'\rangle . \tag{D.39}$$

This operator is generally non-unitary due to leakage, although it will be nearly unitary for a high-fidelity operation.

D.5.2 Noisy simulation of two-qubit gates

We use a master equation with local (to each $0-\pi$ system) energy relaxation (L_{ij}^{r}) and dephasing (L_{l}^{φ}). The Lindblad operators for each $0-\pi$ are constructed as

$$\mathfrak{L}^{(1)} = L \otimes I \text{ and } \mathfrak{L}^{(2)} = I \otimes L\,, \tag{D.40}$$

where $L \in \{L_{ij}^{r}, L_{l}^{\varphi}\}$ and we have suppressed the indices needed to label all the operators. Note that the operators in Equation (D.40) are defined in the bare basis, but our simulations use the dressed basis. These operators must be transformed into the dressed basis before solving the master equation.

Due to computational limitations, we cannot compute the noisy dynamics for systems with more than approximately 150 levels. As seen in Figure D.6, we identify a subset of states (60 for the CZ gate) that accurately represents the coherent dynamics of these operations. As in the single-qubit gate case, we evolve the system in these reduced subspaces to obtain the propagator $e^{\mathcal{L}t}$ that maps the initial density matrix $\rho(0)$ to $\rho(t)$. The resulting superoperator is projected into

the computational subspace and corrected with single-qubit Z rotations. Finally, we compute the average channel fidelity with the target CZ, according to Equation (D.4).

D.6 Sensitivity Analysis of Circuit Parameters

Parameter	Sensitivity (decades)	
	$t_g = 92$ ns	$t_g = 182$ ns
E_C	0.445	0.244
E_{CJ}	0.000	0.000
E_J	0.549	0.336
E_L	0.242	0.000
E_{Cc} (coupling)	0.000	0.075
E_{C0} (ground)	0.000	0.000
ϕ_{ext}	0.010	0.002
n_g^θ	-0.001	-0.001

Table D.2: Robustness of the optimized CZ gates against fabrication-induced parameter variations and operating-point offsets. Each circuit parameter of the second 0-π qubit is independently varied by $\pm 5\%$ around its nominal value while all other parameters are held fixed. For circuit parameters, we take 21 uniformly sampled points for the statistics. The offset parameters n_g^θ and ϕ_{ext} are sampled uniformly in log-space over the ranges $[10^{-6}, 10^{-2}]$ and $[10^{-8}, 10^{-4}]$, respectively. The sensitivity is the median change in coherent gate error, measured in decades relative to the nominal-point gate error. Negative values indicate that the median gate error over the sweep is slightly smaller than the nominal-point error. Both optimized gates are robust against most fabrication variations, while the shorter 92-ns gate exhibits increased sensitivity to variations in E_C, E_J, and E_L.

In this section, we investigate the robustness of the optimized gates against circuit-parameter variations and operating-point offsets. Each circuit parameter of the second $0-\pi$ qubit is sampled uniformly in a range $\pm 5\%$ around its nominal value, while all other parameters are held fixed. Note that the offset parameters n_g^θ and ϕ_{ext} are sampled uniformly in log-space over the ranges $[10^{-6}, 10^{-2}]$ and $[10^{-8}, 10^{-4}]$, respectively. The pulse is then re-optimized, and the corresponding coherent gate error is evaluated. We use 21 points to generate the statistics in Table D.2.

We examine the two pulses highlighted in Figure 4.9, whose parameters are summarized in Table 4.4. The shorter pulse, with $t_g \approx 92$ ns, lies near a sharp minimum in the gate-error curve and is therefore expected to be sensitive to variations in gate time and other parameters. By contrast,

the 182-ns pulse is not in a minimum, suggesting greater robustness to such variations.

The results are summarized in Table D.2. The reported sensitivity is defined as the median value of $\log_{10}[(1-F)/(1-F_0)]$ over the parameter sweep, where F_0 is the coherent fidelity at the nominal parameter point; thus, one decade corresponds to a tenfold increase in coherent gate error. The 182-ns gate is generally insensitive to these variations, with all parameters producing a median increase in coherent gate error of less than one order of magnitude. By contrast, the faster 92-ns gate is more sensitive to variations in E_C, E_J, and E_L, while remaining comparatively robust against changes in E_{CJ}, ϕ_{ext}, n_g^θ, the coupling energy E_{Cc}, and the ground-capacitance energy E_{C0}. These results demonstrate the trade-off between gate speed and robustness, with the longer gate exhibiting substantially improved tolerance to parameter variations.